

Project Data Page:

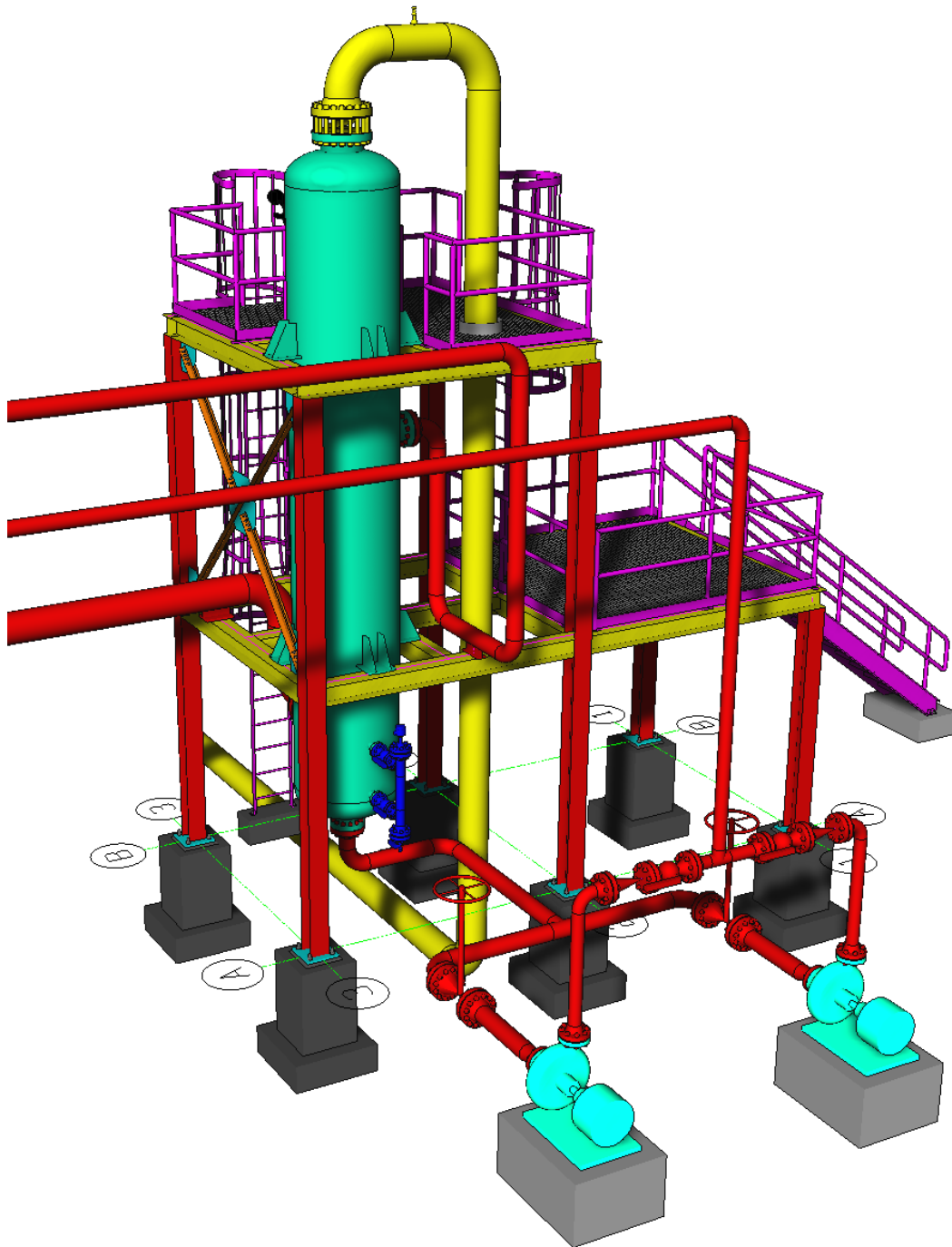




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\401-E-31003Waste Heat Boiler-RP.pvdb

Date of Analysis : Aug 12,2026 9:32am

PV Elite 28, January 2026



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 FileName : 401-E-31003_Waste Heat Boiler-RP Page 2 of 308
 Vessel Design Summary: Step: 39 9:32am Aug 12,2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 1300.000 x 2300.000 mm ID
 Vessel Design Length, Tangent to Tangent 928.60 cm
 Specified Datum Line Distance 0.00 cm
 Shell Side Design Temperature 270 °C
 Channel Side Design Temperature 450 °C
 Shell Side Design Pressure 49.967 kgf/cm²
 Channel Side Design Pressure 163.158 kgf/cm²
 Shop Shell Side Test Pressure 63.500 kgf/cm²
 Shop Channel Side Test Pressure 93.493 kgf/cm²
 Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 22	2	...	K21590	
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	+N
Cone	SA-516 70	...	...	K02700	+N
Flange	SA-336 F22	3	...	K21590	
Nozzle	SA-336 F22	3	...	K21590	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-213 T22	...	...	K21590	
Tubesheet	SA-336 F22	3	...	K21590	
Flg Bolting	SA-193 B16	...	64 < t <= 100	K14072	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Chnl. cover	163.158	0.00	170.000	0.0000	N/A	No
Chnl. flange	163.162	0.00	169.000	0.0000	N/A	No
Channel	163.162	0.00	166.000	1.5000	N/A	No
Shell	50.098	1.03	72.200	3.0000	N/A	No
Shell cone	50.197	1.03	64.400	3.0000	N/A	No
Shell-2	50.197	1.03	63.600	3.0000	N/A	No
Head	50.197	1.03	64.300	3.0000	No	No
Head	49.967	1.03	74.800	3.0000	No	No

Liquid Level: 230.00 cm Dens.: 999.542 kg/m³ Sp. Gr.: 1.000



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 Vessel Design Summary: Step: 39 9:32am Aug 12,2026

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	32.5	32.0	74.0	320.0	313.7	...	1.00	1.00
Body Flg	64.1	31.6	74.0	205.0	199.0	...	1.00	1.00
Cylinder	214.1	150.0	74.0	74.0	72.8	...	1.00	1.00
Cylinder	268.6	30.0	30.0	30.0	21.6	...	1.00	1.00
Conical	443.6	175.0	52.0	52.0	40.9	...	1.00	1.00
Cylinder	923.6	480.0	45.0	45.0	35.9	...	1.00	1.00
Ellipse	928.6	5.0	48.0	40.8	32.4	...	1.00	1.00
Ellipse	628.6	0.0	35.0	29.8	20.8	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	320	132.791	...	No Calc
20	30	205	126.975	...	No Calc
30	Tsht	74	No Calc	...	174
Tsht	40	74	No Calc	...	No Calc
40	50	30	5.20793	1.033	63.2
50	60	52	10.4319	1.033	40.6
60	70	45	12.9732	1.033	28.8
70	80	40.8	8.57211	1.033	31.8
80	90	29.75	6.39382	1.033	39

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	149.997	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	30.003	2539997	No Calc	No Calc
50	60	175	175	No Calc	No Calc
60	70	504.167	2539997	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1- Channel in	WRC-297	0.988	Shell	Passed
N2- Channel ou	WRC-297	0.988	Shell	Passed
N3- Shell inle	WRC-297	0.786	Shell	Passed
N4- Shell outl	WRC-297	0.783	Shell	Passed
N5- Blow down	WRC-297	0.786	Shell	Passed
N8- MP steam	WRC-297	0.786	Shell	Passed
K1- PSV	WRC-297	0.783	Shell	Passed
K2- PSV	WRC-297	0.783	Shell	Passed

Foundation/Support Summary

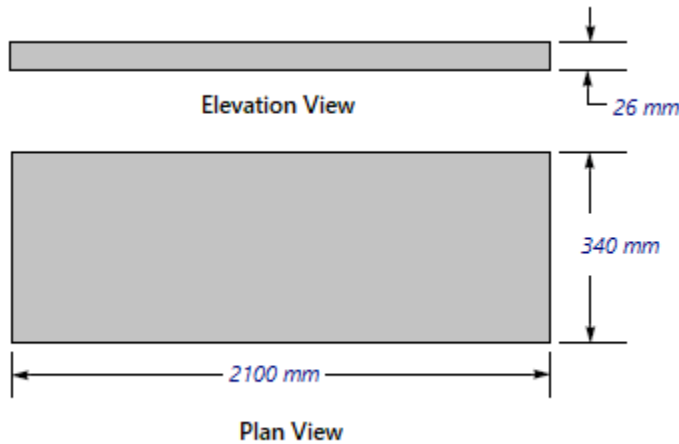


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 Vessel Design Summary: Step: 39 9:32am Aug 12,2026

Saddle Parameters:

Saddle Width	204.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1520.000	mm
Wear Pad Width	306.000	mm
Wear Pad Thickness	10.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	1275.000	mm
Baseplate Length	2100.000	mm
Baseplate Thickness	26.000	mm
Baseplate Width	340.000	mm
Number of Ribs (including outside ribs)	6	
Rib Thickness	10.000	mm
Web Thickness	10.000	mm
Height of Center Web	306.000	mm
Number of Bolts in Baseplate	4	

Baseplate Sketch



Maximum Tensile Bolt Load 60. kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	95297.16	kgf
Maximum Transverse Saddle Shear Load	4286.54	kgf
Maximum Longitudinal Saddle Shear Load	8573.08	kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	89670.28	kgf
Maximum Transverse Saddle Shear Load	632.08	kgf
Maximum Longitudinal Saddle Shear Load	428.09	kgf

Weights:

Fabricated - Bare w/o removable internals	67084.6	kg
Shop Test - Fabricated + water (full)	95546.4	kg
Shipping - Fab. + rem. intls.+ shipping app.	67084.6	kg
Erected - Fab. + rem. intls.+ insul. (etc)	67084.6	kg
Empty - Fab. + intls. + details + wghts.	68551.0	kg
Operating - Empty + operating liquid (no ca)	92192.3	kg



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 Nozzle Summary: Step: 37 9:32am Aug 12,2026

Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	Tb1 4.5.2	Weld	Pass/Fail
N1- Channel inl	166	...	...	OK	OK	Passed
N2- Channel out	166	...	...	OK	OK	Passed
N4- Demister	50.8	OK	...	OK	OK	Passed
N3- Shell inlet	63.3	OK	...	OK	OK	Passed
N4- Shell outle	62.6	OK	...	OK	OK	Passed
N5- Blow down	63.2	OK	...	OK	OK	Passed
N8- MP steam	63.2	OK	...	OK	OK	Passed
K1- PSV	63.5	OK	...	OK	OK	Passed
K2- PSV	63.6	OK	...	OK	OK	Passed
K9- Temperature	63.3	OK	...	OK	OK	Passed
N10- Chemical a	63.4	OK	...	OK	OK	Passed
MH1- Manhole	64.4	OK	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 50.8 Nozzle : N4- Demister [Shellside]
 Minimum MAWP Nozzles : 166 Nozzle : N2- Channel out [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Inside Radius mm
30	N1- Channel inle	1391.000	270.000	227.500
30	N2- Channel outl	1391.000	90.000	227.500
60	N4- Demister	6536.001	270.000	702.897
60	N3- Shell inlet	5336.000	90.000	68.900
60	N4- Shell outlet	4936.001	270.000	110.950
60	N5- Blow down	5836.000	90.000	24.425
60	N8- MP steam	6436.001	90.000	36.325
60	K1- PSV	7936.001	270.000	24.425
60	K2- PSV	8436.000	270.000	68.900
60	K9- Temperature	7936.001	90.000	19.993
60	N10- Chemical ad	8936.001	0.000	10.607
70	MH1- Manhole	0.000	0.000	261.775

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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 Nozzle Schedule: Step: 36 9:32am Aug 12,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N10- Chemical ad	1.000 in	XXS	WNF	1.315	9.093	133.40	15.00	345.12	600
K9- Temperature	1.500 in	160	WNF	1.900	7.137	148.26	15.00	345.25	600
N5- Blow down	2.000 in	160	WNF	2.375	8.738	373.05	20.00	345.40	600
K1- PSV	2.000 in	160	WNF	2.375	8.738	160.32	15.00	345.40	600
N8- MP steam	3.000 in	160	WNF	3.500	11.125	288.90	20.00	345.86	600
N3- Shell inlet	6.000 in	160	WNF	6.625	18.237	368.27	20.00	348.08	600
K2- PSV	6.000 in	160	WNF	6.625	18.237	368.27	15.00	348.08	600
N4- Shell outlet	10.000 in	160	WNF	10.750	28.575	373.05	20.00	353.13	600
N1- Channel inle	20.000 in	Actual	WNF	20.000	28.000	...	...	453.74	2500
N2- Channel outl	20.000 in	Actual	None	20.000	28.000	...	...	453.74	...
MH1- Manhole	24.000 in	120	WNF	24.000	46.025	909.60	34.00	263.01	600
N4- Demister	55.110 in	Actual	None	62.354	92.000	1883.79	38.00	1061.09	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N10- Chemic	SA-106 B	38.000	20.000	15.000	15.000	...
K9- Tempera	SA-106 B	38.000	20.000	15.000	15.000	...
N5- Blow do	SA-106 B	38.000	20.000	20.000	20.000	...
K1- PSV	SA-106 B	38.000	20.000	15.000	15.000	...
N8- MP stea	SA-106 B	38.000	20.000	20.000	20.000	...
N3- Shell i	SA-106 B	38.000	20.000	20.000	20.000	...
K2- PSV	SA-106 B	38.000	20.000	15.000	15.000	...
N4- Shell o	SA-106 B	38.000	20.000	20.000	20.000	...
N1- Channel	SA-336 F22	74.000	20.000	...	...	...
N2- Channel	SA-336 F22	74.000	20.000	...	...	...
MH1- Manhol	SA-106 B	34.000	20.000	26.000	34.000	...
N4- Demiste	SA-516 70	38.000	30.000	30.000	38.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N10- Chemical ad	893.600	0.0	300.00	0.00	Shell-2
K9- Temperature	793.600	90.0	300.00	0.00	Shell-2
N5- Blow down	583.600	90.0	300.00	0.00	Shell-2
K1- PSV	793.600	270.0	300.00	0.00	Shell-2
N8- MP steam	643.600	90.0	300.00	0.00	Shell-2
N3- Shell inlet	533.600	90.0	300.00	0.00	Shell-2
K2- PSV	843.600	270.0	300.00	0.00	Shell-2
N4- Shell outlet	493.600	270.0	300.00	0.00	Shell-2



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Nozzle Schedule: Step: 36 9:32am Aug 12,2026

N1- Channel inle	139.100	270.0	280.00	0.00	Channel
N2- Channel outl	139.100	90.0	280.00	0.00	Channel
MH1- Manhole	...	0.0	200.00	0.00	Head
N4- Demister	653.600	270.0	700.00	0.00	Shell-2

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Bill of Material: Step: 35 9:32am Aug 12,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 320.0mm THK X 1930.0mm OD	SA-336 F22
1	WELD NECK FLANGE: 205.0mm THK X 1930.0mm OD	SA-336 F22
1	ELLIPTICAL HEAD: 2.0 X 1, 48.0mm THK X 2300.0mm ID X 5.0cm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 35.0mm THK X 1400.0mm ID X 5.0cm	SA-516 70
1	INSULATION: 1500mm X 145mm THK	Hot
8	BAFFLES: 1250.0mm DIA X 8.0mm THK	SA-516 70
1	TUBESHEET: 1448.0mm X 245.0mm THK	SA-336 F22
888	TUBES: 6600.0mm X 25.0mm DIA X 2.5mm THK	SA-213 T22
1	INSULATION: 1749mm X 145mm THK	Hot
2	SADDLE: 204.0mm X 120 DEG	SA-516 70
1	INSULATION: 4800mm X 145mm THK	Hot
1	INSULATION: 625mm X 145mm THK	Hot
1	CLASS 600 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	GASKET: 1326.0mm OD X 1300.0mm ID	...
28	BODY FLANGE BOLTS: 88.9mm DIA	SA-193 B16
56	NUTS FOR BODY FLANGE BOLTS: 88.9mm DIA	...
2	CYLINDRICAL SEGMENT 74.0mm THK X 1622.3mm X 150.0cm	SA-387 22
1	CYLINDRICAL SEGMENT 74.0mm THK X 839.5mm X 150.0cm	SA-387 22
1	CYLINDRICAL SEGMENT 30.0mm THK X 3051.7mm X 30.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 30.0mm THK X 1032.3mm X 30.0cm	SA-516 70
1	ECC. CONICAL SEGMENT 52.0mm THK X 5399.1mm X 175.0cm X 3051.7mm	SA-516 70
1	ECC. CONICAL SEGMENT 52.0mm THK X 1826.5mm X 175.0cm X 1032.3mm	SA-516 70
1	CYLINDRICAL SEGMENT 45.0mm THK X 5399.1mm X 480.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 45.0mm THK X 1826.5mm X 480.0cm	SA-516 70
1	NAMEPLATE	...



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 MDMT Summary: Step: 38 9:32am Aug 12,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Shell	[8]	B	17	5	No	0.698	30.000	1.00
Shell cone	[8]	D	-26	-38	Yes	0.783	52.000	1.00
Shell-2	[8]	D	-29	-40	Yes	0.793	45.000	1.00
Head	[10]	D	-30	-42	Yes	0.786	40.800	1.00
Head	[7]	D	-27	-42	Yes	0.740	48.000	1.00
Left Saddle	[24]	D	-48	-48	Yes	1.000	10.000	1.00
Right saddle	[24]	D	-48	-48	Yes	1.000	10.000	1.00
N4- Demister	[1]	B	4	1	Yes	0.936	38.000	1.00
N3- Shell inle	[1]	B	-9	-9	Yes	1.000	20.000	1.00
Nozzle Flg	[5]	A	-18	-48	Yes			
N4- Shell outl	[1]	B	-9	-9	Yes	1.000	20.000	1.00
Nozzle Flg	[5]	A	-18	-48	Yes			
N5- Blow down	[1]	B	-9	-9	Yes	1.000	20.000	1.00
Nozzle Flg	[5]	A	-18	-48	Yes			
N8- MP steam	[1]	B	-9	-9	Yes	1.000	20.000	1.00
Nozzle Flg	[5]	A	-18	-48	Yes			
K1- PSV	[1]	B	-14	-14	Yes	1.000	15.000	1.00
Nozzle Flg	[5]	A	-18	-48	Yes			
K2- PSV	[1]	B	-14	-14	Yes	1.000	15.000	1.00
Nozzle Flg	[5]	A	-18	-48	Yes			
K9- Temperatur	[1]	B	-14	-14	Yes	1.000	15.000	1.00
Nozzle Flg	[5]	A	-18	-48	Yes			
N10- Chemical	[1]	B	-14	-14	Yes	1.000	15.000	1.00
Nozzle Flg	[5]	A	-18	-48	Yes			
MH1- Manhole	[1]	B	2	-1	Yes	0.945	34.000	1.00
Nozzle Flg	[5]	A	-18	-48	Yes			
Tubesheet: SS	[13]	B	34	13	No	0.487	61.250	1.00

Warmest MDMT: 34 13

Chnl. cover	[11]	B	18	-35	Yes	0.422	80.000	1.00
Chnl. flange	[11]	B	16	-37	Yes	0.336	74.000	1.00
Channel	[8]	A	37	-16	Yes	0.324	74.000	1.00
Head	[10]	D	-18	-18	No	1.000	29.750	1.00
Head	[7]	D	-15	-18	No	0.940	35.000	1.00
N1- Channel in	[1]	A	37	22	Yes	0.722	74.000	1.00
Nozzle Flg	[5]	B	-46	-48	Yes			
N2- Channel ou	[1]	A	37	22	Yes	0.722	74.000	1.00
Tubesheet: CS	[14]	A	37	18	Yes	0.664	74.000	1.00
Bolting	[21]		-29					

Warmest MDMT: 37 22

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	13.0	11.4	Fail
Channel/Tube	22.0	11.4	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.



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MDMT Summary: Step: 38 9:32am Aug 12,2026

- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

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Warnings and Errors: Step: 0 9:32am Aug 12,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.

Temperatures of one or more elements, internal or external was greater than the maximum allowed by Table 4.4.1. Please determine if this limitation pertains to your particular design.

Maximum Temperatures per Table 4.4.1 are:

Materials	°C	°F
• Carbon and Low Allow Steels	425	800
• High Alloy Steels	425	800
• Quenched and Tempered Steels	370	700
• Aluminum and Aluminum Alloys	150	300
• Copper and Copper Alloys	65	150
• Nickel and Nickel Alloys	480	900
• Titanium and Titanium Alloys	315	600



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.

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Units used in this Analysis (Kgfcm2):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	49.967	kgf/cm ²
Channel Side Design Pressure	163.16	kgf/cm ²
Shell Side Design Temperature	270.0	°C
Channel Side Design Temperature	450.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Sour	
MDMT (CET), Shell Side	11.4	°C
MDMT (CET), Tube Side	11.4	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070



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0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test

Hydrostatic

Pressure Test Position

Horizontal

Test Performed in Corroded Condition

No

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000

Seismic Design Code

IS 1893-RSM

Importance Factor for IS-1893

2.000



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Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/8
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Chnl. cover
Distance "FROM" to "TO"	32 cm
Flange Outside Diameter	1930 mm
Element Thickness	320 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	74 mm
External Corrosion Allowance	0 mm



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Design Internal Pressure	163.16	kgf/cm ²
Design Temperature Internal Pressure	450	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-336 F22	
Allowable Stress, Ambient	2110.7	kgf/cm ²
Allowable Stress, Operating	1570.3	kgf/cm ²
Allowable Stress, Hydrotest	3002.9	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	2365.7	kgf/cm ²
MDMT Curve	B	
External Pressure Chart Name	CS-3	
UNS Number	K21590	
Class / Thickness / Grade	3 ::	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	2/8	
Element From Node	20	
Element To Node	30	
Element Type	Flange	
Description	Chnl. flange	
Distance "FROM" to "TO"	31.6	cm
Flange Inside Diameter	1300	mm
Element Thickness	205	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	74	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	163.16	kgf/cm ²
Design Temperature Internal Pressure	450	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-336 F22	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	130	cm
Liquid Density	29.986	kg/m ³

Element#	3/8	
Element From Node	30	
Element To Node	40	
Element Type	Cylinder	
Description	Channel	
Distance "FROM" to "TO"	150	cm
Inside Diameter	1300	mm
Element Thickness	74	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	74	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	163.16	kgf/cm ²
Design Temperature Internal Pressure	450	°C
Design External Pressure	0	kgf/cm ²



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Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 22	
Allowable Stress, Ambient	2110.7	kgf/cm ²
Allowable Stress, Operating	1570.3	kgf/cm ²
Allowable Stress, Hydrotest	3002.9	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	2365.7	kgf/cm ²
MDMT Curve	A	
External Pressure Chart Name	CS-3	
UNS Number	K21590	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	130	cm
Liquid Density	29.986	kg/m ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Hot insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	150	cm
Thickness of Insulation	145	mm
Density	0.00015	kg/cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N1- Channel inle	
Dist. from "FROM" Node / Offset dist	75	cm
Nozzle Diameter	20	in.
Nozzle Schedule	None	
Nozzle Class	2500	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	573.22	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-336 F22	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N2- Channel outl	
Dist. from "FROM" Node / Offset dist	75	cm
Nozzle Diameter	20	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	573.22	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F22	

Element#	4/8
Element From Node	40
Element To Node	50
Element Type	Cylinder



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Description	Shell
Distance "FROM" to "TO"	30 cm
Inside Diameter	1300 mm
Element Thickness	30 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	30 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	49.967 kgf/cm ²
Design Temperature Internal Pressure	270 °C
Design External Pressure	1.033 kgf/cm ²
Design Temperature External Pressure	270 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	1784.4 kgf/cm ²
Allowable Stress, Operating	1784.4 kgf/cm ²
Allowable Stress, Hydrotest	2518.6 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	0 mm
Yield Stress, Operating	2671.5 kgf/cm ²
MDMT Curve	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	130 cm
Liquid Density	999.54 kg/m ³

Element#	5/8
Element From Node	50
Element To Node	60
Element Type	Conical
Description	Shell cone
Distance "FROM" to "TO"	175 cm
Inside Diameter	1300 mm
Element Thickness	52 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	52 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	49.967 kgf/cm ²
Design Temperature Internal Pressure	270 °C
Design External Pressure	1.033 kgf/cm ²
Design Temperature External Pressure	270 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Cone Diameter at "To" End	2300 mm
Design Length of Cone	1750 mm
Half Apex Angle of Cone	29.744886 degrees
Toriconical (Y/N)	N
Weld is pre-Heated	No
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 cm



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Height/Length of Liquid	230	cm
Liquid Density	999.54	kg/m ³
Element From Node	50	
Detail Type	Insulation	
Detail ID	Hot insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	175	cm
Thickness of Insulation	145	mm
Density	0.00015	kg/cm ³

Element#	6/8	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	Shell-2	
Distance "FROM" to "TO"	480	cm
Inside Diameter	2300	mm
Element Thickness	45	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	45	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	49.967	kgf/cm ²
Design Temperature Internal Pressure	270	°C
Design External Pressure	1.033	kgf/cm ²
Design Temperature External Pressure	270	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+N
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	40	cm
Width of Saddle	204	mm
Height of Saddle at Bottom	1520	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	306	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	60	
Detail Type	Saddle	
Detail ID	Right saddle	
Dist. from "FROM" Node / Offset dist	430	cm
Width of Saddle	204	mm
Height of Saddle at Bottom	1520	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	306	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	



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Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	230	cm
Liquid Density	999.54	kg/m ³

Element From Node	60	
Detail Type	Insulation	
Detail ID	Hot	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	480	cm
Thickness of Insulation	145	mm
Density	0.00015	kg/cm ³

Element From Node	60	
Detail Type	Nozzle	
Detail ID	N4- Demister	
Dist. from "FROM" Node / Offset dist	210	cm
Nozzle Diameter	55.11	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2730.2	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-516 70	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	N3- Shell inlet	
Dist. from "FROM" Node / Offset dist	90	cm
Nozzle Diameter	6	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	73.319	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	N4- Shell outlet	
Dist. from "FROM" Node / Offset dist	50	cm
Nozzle Diameter	10	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	161.28	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	N5- Blow down	
Dist. from "FROM" Node / Offset dist	140	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	



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Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	25.185	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	N8- MP steam	
Dist. from "FROM" Node / Offset dist	200	cm
Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K1- PSV	
Dist. from "FROM" Node / Offset dist	350	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.7	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K2- PSV	
Dist. from "FROM" Node / Offset dist	400	cm
Nozzle Diameter	6	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	70.053	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K9- Temperature	
Dist. from "FROM" Node / Offset dist	350	cm
Nozzle Diameter	1.5	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	N10- Chemical ad	
Dist. from "FROM" Node / Offset dist	450	cm
Nozzle Diameter	1	in.
Nozzle Schedule	XXS	



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Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	60
Detail Type	Weight
Detail ID	Extra weight fo
Dist. from "FROM" Node / Offset dist	240 cm
Miscellaneous Weight	9500 kgf
Offset from Element Centerline	0 mm

Element#	7/8
Element From Node	70
Element To Node	80
Element Type	Elliptical
Description	Head
Distance "FROM" to "TO"	5 cm
Inside Diameter	2300 mm
Element Thickness	40.8 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	48 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	49.967 kgf/cm ²
Design Temperature Internal Pressure	270 °C
Design External Pressure	1.033 kgf/cm ²
Design Temperature External Pressure	270 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	230 cm
Liquid Density	999.54 kg/m ³

Element From Node	70
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	62.5 cm
Thickness of Insulation	145 mm
Density	0.00015 kg/cm ³

Element From Node	70
Detail Type	Nozzle
Detail ID	MH1- Manhole
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	120
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	1272.1 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B



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Element#	8/8
Element From Node	80
Element To Node	90
Element Type	Elliptical
Description	Head
Distance "FROM" to "TO"	5 cm
Inside Diameter	1400 mm
Element Thickness	29.75 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	35 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	49.967 kgf/cm ²
Design Temperature Internal Pressure	270 °C
Design External Pressure	1.033 kgf/cm ²
Design Temperature External Pressure	270 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

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XY Coordinate Calculations: Step: 2 9:32am Aug 12,2026

XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
Chnl. cover		32.5	...	32	...
Chnl. flange		64.1	...	31.6	...
Channel		214.1	...	150	...
Shell		268.6	...	30	...
Shell cone		443.6	...	175	...
Shell-2		923.6	...	480	...
Head		928.6	...	5	...
Head		628.6	185	...	5

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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 FileName : 401-E-31003_Waste Heat Boiler-RP Page 24 of 308
 Flg Calc [Int P]: New Flange Flng: 5 9:32am Aug 12,2026

Flange Input Data Values

Description: New Flange :

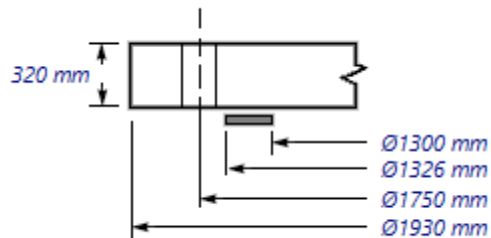
Chnl. cover

Flange Type		Blind	
Design Pressure	P	163.16	kgf/cm ²
Design Temperature		450	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1930.000	mm
Flange Thickness	t	320.0000	mm
Flange Material		SA-336 F22	
Flange Material UNS number		K21590	
Flange Allowable Stress At Temperature	Sfo	1570.34	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	2110.70	kgf/cm ²
Bolt Material		SA-193 B16	
Bolt Allowable Stress At Temperature	Sb	1498.93	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1549.91	kgf/cm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Perimeter along the Center of the Bolts	L	5497.787	mm
Diameter of Bolt Circle	C	1750.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		28	
Flange Face Outside Diameter	Fod	1329.000	mm
Flange Face Inside Diameter	Fid	1300.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1326.000	mm
Gasket Inside Diameter	Gi	1300.000	mm
Gasket Factor	m	0.0000	
Gasket Design Seating Stress	y	0.00	kgf/cm ²
Gasket Thickness	tg	5.0000	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	N = Self-Energizing	0.000	mm
Basic Gasket Width,	bo = N/2	0.000	mm
Effective Gasket Width,	b = bo	0.000	mm



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Gasket Reaction Diameter, G = Go (Self-Energizing) 1326.000 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1326^2 \cdot 163$$

$$= 2253127.500 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 0 \cdot \pi \cdot 1326 \cdot 0 \cdot 163$$

$$= 0.000 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(2253127 + 0 + 0, 0)$$

$$= 2253127.500 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 0 \cdot 0 \cdot \pi \cdot 1326 + 0 \cdot 0 \cdot 0$$

$$= 0.000 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((2253127 + 0 + 4 \cdot 0/1326)/1499, 0/1550)$$

$$= 1503.158 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1503.158	1580.461	
Radial Distance between Bolts and Edge:	82.550	90.000	
Circ. Spacing between the Bolts:	180.975	195.938	...

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 1550 (1503 + 1580)/2$$

$$= 2389670.75 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 2253127.50 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1750 - 1326)/2$$

$$= 212.0000 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc/(t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2/t^2$$

$$= 1.9 \cdot 2253127 \cdot 212 \cdot 1/(320^2 \cdot 1326) + 0.3 \cdot 1 \cdot 163 \cdot 1326^2/320^2$$

$$= 1508.86 \text{ kgf/cm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc/(t^2 \cdot G)$$

$$= 1.9 \cdot 2389670 \cdot 212 \cdot 1/(320^2 \cdot 1326)$$

$$= 708.90 \text{ kgf/cm}^2$$



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Bolt Stress, Operating [BS_o]:

= $Wm1 / Ab$
 = 2253127/1580
 = 1425.62 kgf/cm²

Bolt Stress, Seating [BS_a]:

= ($Wm2 / Ab$)
 = (0/1580)
 = 0.00 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	1508.86	1570.34	708.90	2110.70
Bolting	1425.62	1498.93	0.00	1549.91

Reqd. Blind Flange Thickness at Center 313.673 mm
 Reqd. Blind Flange Thickness at Gasket 208.770 mm

Estimated M.A.W.P. (Operating) 170 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 7255.7 kg
 Estimated Unfinished Weight of Forging at given Thk 7255.7 kg

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -35 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 18 °C
 Temperature reduction per Graph 3.13 : 52 °C
 The computed stress ratio is : 0.422

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	1508.86	2110.70	697.53	2110.70
Bolting	1425.62	1549.91	0.00	1549.91

Minimum Req'd Thickness, ambient, no Corrosion: 270.561 mm

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 Flg Calc [Int P]: FLANGE Flng: 6 9:32am Aug 12,2026

Flange Input Data Values

Description: FLANGE :

Chnl. flange

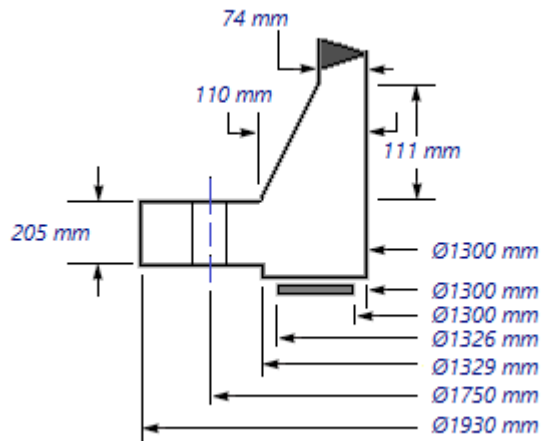
Flange Type	Integral Weld Neck		
Design Pressure	P	163.16	kgf/cm ²
Design Temperature		450	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	1300.000	mm
Flange Outside Diameter	A	1930.000	mm
Flange Thickness	t	205.0000	mm
Thickness of Hub at Small End	go	74.0000	mm
Thickness of Hub at Large End	g1	110.0000	mm
Length of Hub	h	111.0000	mm
Flange Material	SA-336 F22		
Flange Material UNS number	K21590		
Flange Allowable Stress At Temperature	Sfo	1570.34	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	2110.70	kgf/cm ²
Bolt Material	SA-193 B16		
Bolt Allowable Stress At Temperature	Sb	1498.93	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1549.91	kgf/cm ²

Warning:
It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	1750.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	28		
Flange Face Outside Diameter	Fod	1329.000	mm
Flange Face Inside Diameter	Fid	1300.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1326.000	mm
Gasket Inside Diameter	Gi	1300.000	mm
Gasket Factor	m	0.0000	
Gasket Design Seating Stress	y	0.00	kgf/cm ²
Gasket Thickness	tg	5.0000	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P(R + Ca) / (S - 0.5 \cdot P)) + (Ca + Ca_{ext})$$

$$= (163(1300+0) / (1570-0.5 \cdot 163)) + (0)$$

$$= 71.2375 \text{ mm}$$

Small End Hub MAWP:

$$= S \cdot \log((2 \cdot T + R) / R)$$

$$= 1570 \cdot \log((2 \cdot 74 + 650) / 650)$$

$$= 169.313 \text{ kgf/cm}^2$$

Small End Hub MAPnc:

$$= S_a \cdot \log((2 \cdot T + R) / R)$$

$$= 2111 \cdot \log((2 \cdot 74 + 650) / 650)$$

$$= 227.574 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	1300.000 mm
Corroded Large Hub,	g1Cor = g1-ci	110.000 mm
Corroded Small Hub,	g0Cor = go-ci	74.000 mm
Code R Dimension,	R = (C-B)/2 - g1	115.000 mm

Gasket Contact Width,	N = Self-Energizing	0.000 mm
Basic Gasket Width,	bo = N/2	0.000 mm
Effective Gasket Width,	b = bo	0.000 mm
Gasket Reaction Diameter,	G = Go (Self-Energizing)	1326.000 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1326^2 \cdot 163$$

$$= 2253181.250 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 0 \cdot \pi \cdot 1326 \cdot 0 \cdot 163$$

$$= 0.000 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1300^2 \cdot 163$$

$$= 2165687.750 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:



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= H - Hd
 = 2253181 - 2165687
 = 87493.586 kgf

Design Bolt Load for the Operating Condition [Wo]:
 = max(Wo,1 + Wo,2 + H'p, 0)
 = max(2253181 + 0 + 0, 0)
 = 2253181.250 kgf

Design Bolt Load for the Gasket Seating Condition [Wg]:
 = y • b • π • G + yPart • bPart • lp
 = 0•0•π•1326+0•0•0
 = 0.000 kgf

Required Bolt Area [Am]:
 = max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)
 = max((2253181 + 0 + 4•0/1326)/1499, 0/1550)
 = 1503.194 cm²

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
 = 2a + 6t/(m + 0.5)
 = 2 • 88.9 + 6 • 205/(0 + 0.5)
 = 2637.800 mm

Actual Circumferential Bolt Spacing [Bs]:
 = C • sin(π / n)
 = 1750 • sin(π/28)
 = 195.938 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
 = max(√(Bs/(2a + t)), 1)
 = max(√(196/(2 • 88.9 + 205)), 1)
 = 1.0000

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1503.194	1580.461	
Radial Distance between Hub and Bolts:	104.775	115.000	
Radial Distance between Bolts and Edge:	82.550	90.000	
Circ. Spacing between the Bolts:	180.975	195.938	2637.800

Flange Design Bolt Load, Gasket Seating [W or Wgs]:
 = Sa(Am + Ab) / 2
 = 1550 (1503 + 1580)/2
 = 2389698.75 kgf

Gasket Load for the Operating Condition [HG]:
 = Wo - H
 = 2253181 - 2253181
 = 0.00 kgf

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 = (C - G) / 2
 = (1750 - 1326)/2
 = 212.0000 mm

Distance to Face Pressure Reaction [ht]:
 = 0.5((C - B)/2 + hg)
 = 0.5((1750 - 1300)/2 + 212)
 = 218.5000 mm



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Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (1750 - 1300 - 110) / 2$
 $= 170.0000 \text{ mm}$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	2165688.	170.0000	1.0000	368167.
Face Pressure, Mt	87494.	218.5000	1.0000	19117.
Gasket Load, Mg	0.	212.0000	1.0000	0.
Gasket Seating, Matm	2389699.	212.0000	1.0000	506617.

Total Moment for Operation, Mo 387285. kgf-m.
 Total Moment for Gasket seating, Mg 506617. kgf-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 310.161 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 0.358
 Thickness Ratio, $g1/g_o = (g1_{cor}/g_{oCor})$ 1.486

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.871
 Factor V per 4.16.5 0.354
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 1930/1300$
 $= 1.485$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169(1.48^2 \cdot \log_{10}(1.48))) / (1.48^2 - 1)) / (1.48 - 1)$
 $= 5.085$

Stress factor from Table 4.16.4 [T]:

$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.48^2(1 + 8.55246 \cdot \log_{10}(1.48)) - 1) / ((1.0472 + 1.9448 \cdot 1.48^2) \cdot (1.48 - 1))$
 $= 1.717$

Stress factor from Table 4.16.4 [U]:

$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
 $= (1.48^2(1 + 8.55246 \cdot \log_{10}(1.48)) - 1) / (1.36136(1.48^2 - 1) \cdot (1.48 - 1))$
 $= 5.588$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.48^2 + 1) / (1.48^2 - 1)$
 $= 2.661$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_o^2 \cdot h_o / V$
 $= 5.59 \cdot 74^2 \cdot 310 / 0.35$
 $= 26798538.000 \text{ mm}^3$

Stress factor from Table 4.16.4 for integral flanges [e]:

$= F / h_o$
 $= 0.87/310$
 $= 0.003 \text{ 1/mm}$

Stress factor from Table 4.16.4 [L]:

$= (t \cdot e + 1) / T + t^3/d$
 $= (205 \cdot 0.0028 + 1) / 1.72 + 205^3/26798538$



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= 1.239

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$$

$$= (1 \cdot 387285 / 1300) / (1.24 \cdot 110^2)$$

$$= 1987.04 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$$

$$= (1 \cdot 506617 / 1300) / (1.24 \cdot 110^2)$$

$$= 2599.29 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.77 \cdot 387285 / 1300) / (1.24 \cdot 205^2)$$

$$= 1011.39 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.77 \cdot 506617 / 1300) / (1.24 \cdot 205^2)$$

$$= 1323.02 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (5.09 \cdot 387285 / (205^2 \cdot 1300)) - 2.66 \cdot 1011$$

$$= 913.53 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (5.09 \cdot 506617 / (205^2 \cdot 1300)) - 2.66 \cdot 1323$$

$$= 1195.01 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (1987 + \max(1011, 914)) / 2$$

$$= 1499.21 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (2599 + \max(1323, 1195)) / 2$$

$$= 1961.16 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 2253181 / 1580$$

$$= 1425.65 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (0 / 1580)$$

$$= 0.00 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	1987.04	2355.51	2599.29	3166.05
Radial Flange	1011.39	1570.34	1323.02	2110.70
Tangential Flange	913.53	1570.34	1195.01	2110.70
Maximum Average	1499.21	1570.34	1961.16	2110.70
Bolting	1425.65	1498.93	0.00	1549.91



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Minimum Required Flange Thickness 199.034 mm

Estimated M.A.W.P. (Operating) 169 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 2885.4 kg
 Estimated Unfinished Weight of Forging at given Thk 3914.2 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 506617 / 1 \cdot 100000 \cdot 0.35 / (1.2 \cdot 2151394 \cdot 74^2 \cdot 310 \cdot 0.3)$
 $= 0.710$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 387285 / 1 \cdot 100000 \cdot 0.35 / (1.2 \cdot 1828437 \cdot 74^2 \cdot 310 \cdot 0.3)$
 $= 0.638$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 506617 / 1 \cdot 100000 \cdot 0.35 / (1.24 \cdot 2151394 \cdot 74^2 \cdot 310 \cdot 0.3)$
 $= 0.688$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 387285 / 1 \cdot 100000 \cdot 0.35 / (1.24 \cdot 1828437 \cdot 74^2 \cdot 310 \cdot 0.3)$
 $= 0.619$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -37 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B
From Figure 3.8M- with PWHT, the MDMT is	: 16 °C
Temperature reduction per Graph 3.13	: 53 °C
The computed stress ratio is	: 0.336

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	1987.04	3166.05	2557.61	3166.05
Radial Flange	1011.39	2110.70	1301.80	2110.70
Tangential Flange	913.53	2110.70	1175.85	2110.70
Maximum Average	1499.21	2110.70	1929.71	2110.70
Bolting	1425.65	1549.91	0.00	1549.91

Minimum Req'd Thickness, ambient, no Corrosion: 193.523 mm



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 Internal Pressure Calculations: Step: 5 9:32am Aug 12,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Chnl. cover		163.16	74	...	1930	1570.3	1
Chnl. flange		163.16	74	...	1300	1570.3	2
Channel		163.16	74	1.5	1300	1570.3	3
Shell		50.098	30	3	1300	1784.4	4
Shell cone		50.197	52	3	2300	1784.4	5
Shell-2		50.197	45	3	2300	1784.4	6
Head		50.197	48	3	2300	1784.4	7
Head		49.967	35	3	1400	1784.4	8

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Chnl. cover		163.158	170	177	320	313.673	1
Chnl. flange		163.158	169	177	205	199.034	2
Channel		163.158	166	228	74	72.8342	3
Shell		49.9673	72.2	80.5	30	21.5928	4
Shell cone		49.9673	64.4	68.7	52	40.8876	5
Shell-2		49.9673	63.6	68.5	45	35.8956	6
Head		49.9673	64.3	69.7	40.8	32.3644	7
Head		49.9673	74.8	83.3	29.75	20.808	8

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 50.8 kgf/cm²
 Shell Side MAPnc = 68.5 kgf/cm²
 Channel Side MAWP = 74.8 kgf/cm²
 Channel Side MAPnc = 83.3 kgf/cm²

Elements Suitable for Design Internal Pressure.

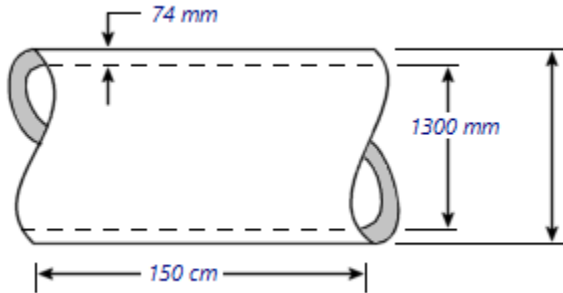
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 Element: Channel Step: 6 9:32am Aug 12,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Channel

Cylindrical Shell Design Information:			
Design Pressure P	163.162 kgf/cm ²	Int. Design Temperature	450 °C
Material	SA-387 22	Ext. Design Temperature	40 °C
Length	150.000 cm	External Pressure Chart	CS-3
		UNS Number	K21590
Int. Corr. All. c	1.5000 mm	Allowable Stress (ope) S	1570.3 kgf/cm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	2110.7 kgf/cm ²
Inside Diameter Di	1300.000 mm	Actual Stress	1466.2 kgf/cm ²
Spec. Min. Thk t _e , or sa	74.0000 mm	Spec. Nominal Thk.	74.000 mm
Surface Area	68235.4 cm ²	SG of Contents	0.030
Element Volume	1990988.9 cm ³	Weight of Contents	59.7 kgf
Empty Weight	3713.5 kgf	Operating Weight	3773.2 kgf
		Comp. All. Ope. Stress	753.5 kgf/cm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1303 \cdot (\exp(163/(1570 \cdot 1)) - 1) + 1.5 + 0$
 $= 71.3 + 1.5 + 0 = 72.8 \text{ mm}$ Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1570 \cdot 1 \cdot \ln((2 \cdot 72.5 + 1303) / 1303) - 0.0039$
 $= 166 - 0.0039 = 166 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2111 \cdot 1 \cdot \ln((2 \cdot 74 + 1300) / 1300)$
 $= 228 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [S_i]:
 $= P \cdot D / (E (D_o - D))$
 $= 163 \cdot 1303 / (1 (1448 - 1303))$
 $= 1466.2107 \text{ kgf/cm}^2$



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Element: Channel Step: 6 9:32am Aug 12,2026

Computed Membrane Stress at Design Pressure [Sm]:

$$\begin{aligned} &= P / (E \cdot \ln((2 \cdot t + D) / D)) \\ &= 163 / (1 \cdot \ln((2 \cdot 72.5 + 1303) / 1303)) \\ &= 1546.3573 \text{ kgf/cm}^2 \end{aligned}$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 5.386 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-16 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	37 °C
Temperature reduction per Graph 3.13	:	53 °C
The computed stress ratio is	:	0.324

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext}) \\ &= (23.5 \cdot 1) / (74 - 1.5 - 0) \\ &= 0.324 \end{aligned}$$

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

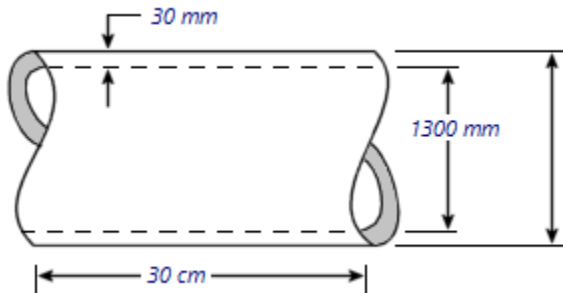
MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	74.795 kgf/cm ²
Governing Thk.	74.000 mm	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	0.324	Basic MDMT	37 °C
		Temperature Difference	53 °C
		Computed Min. Temp.	-16 °C
		Joint Efficiency E*	100 %



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 Element: Shell Step: 7 9:32am Aug 12,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	50.098 kgf/cm ²	Int. Design Temperature	270 °C
Material	SA-516 70	Ext. Design Temperature	270 °C
Length	30.000 cm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	1784.4 kgf/cm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	1784.4 kgf/cm ²
Inside Diameter Di	1300.000 mm	Actual Stress	1211.6 kgf/cm ²
Spec. Min. Thk t,e, or sa	30.0000 mm	Spec. Nominal Thk.	30.000 mm
Surface Area	12817.7 cm ²	SG of Contents	1.000
Element Volume	267428.7 cm ³	Weight of Contents	267.3 kgf
Empty Weight	291.5 kgf	Operating Weight	558.8 kgf
		Comp. All. Ope. Stress	926.2 kgf/cm ²
Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1306 \cdot (\exp(50.1/(1784 \cdot 1)) - 1) + 3 + 0$
 $= 18.6 + 3 + 0 = 21.6 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1784 \cdot 1 \cdot \ln((2 \cdot 27 + 1306) / 1306) - 0.13$
 $= 72.3 - 0.13 = 72.2 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1784 \cdot 1 \cdot \ln((2 \cdot 30 + 1300) / 1300)$
 $= 80.5 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 50.1 \cdot 1306 / (1 (1360 - 1306))$
 $= 1211.6174 \text{ kgf/cm}^2$



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Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 50.1 / (1 \cdot \ln((2 \cdot 27 + 1306) / 1306))$$

$$= 1236.4972 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 2.256 \%$$

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	5 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.7M- Non PWHT, the MDMT is	:	17 °C
Temperature reduction per Graph 3.12	:	12 °C
The computed stress ratio is	:	0.698

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (18.9 \cdot 1) / (30 - 3 - 0)$$

$$= 0.698$$

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	50.800 kgf/cm ²
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	30.000 mm	Basic MDMT	17 °C
Stress/Thk. Ratio	0.698	Temperature Difference	12 °C
		Computed Min. Temp.	5 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 270 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{ (R_o \cdot t) }$$

$$= 300 / \sqrt{ (680 \cdot 27) }$$

$$= 2.214$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (2.21 - 0.579)$$

$$= 0.5626$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.56 \cdot 1909280 (27 / 1360)$$

$$= 34120.480 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 34120.5 / 2671.5$$

$$= 12.772$$

Predicted Buckling Stress [Fic]:



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Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2672/4921$$

$$= 0.543$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 34120/1909280$$

$$= 0.017871$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.54) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$$

$$= 0.133$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 2672 (1 + 0.0020) / \ln(1 + 0.0020) 0.13312802$$

$$= 6123.387 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2^2 - R)$$

$$= 0.6 (1 - 0.54)$$

$$= 0.274$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (4921 \cdot \exp(0.27)) / (0.27 \cdot 0.27)$$

$$= 9232.526 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.541.5 - 0.5 \cdot 0.542.5 - 0.543.5$$

$$= 0.374$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-2.502	0.162E-05	0.157E-05	0.597E-05	-0.564E-05	0.247E+06	0.792E-02
2	3691	0.426	0.227E-04	-0.314E-04	0.175E-04	0.423E-04	0.194E+05	0.173E+00
3	3075	-1.038	0.693E-05	0.271E-05	0.108E-04	0.205E-06	0.473E+05	0.472E-01
4	2768	-1.770	0.349E-05	0.296E-05	0.816E-05	-0.608E-05	0.111E+06	0.717E-02
5	2614	-2.136	0.240E-05	0.223E-05	0.701E-05	-0.617E-05	0.167E+06	0.218E-02
6	2691	-1.953	0.290E-05	0.260E-05	0.757E-05	-0.624E-05	0.136E+06	0.192E-02
7	2652	-2.044	0.264E-05	0.241E-05	0.729E-05	-0.623E-05	0.151E+06	0.253E-03
8	2672	-1.998	0.277E-05	0.251E-05	0.743E-05	-0.624E-05	0.143E+06	0.801E-03
9	2662	-2.021	0.271E-05	0.246E-05	0.736E-05	-0.624E-05	0.147E+06	0.266E-03
10	2657	-2.033	0.268E-05	0.244E-05	0.732E-05	-0.623E-05	0.149E+06	0.439E-05
11	2655	-2.038	0.266E-05	0.242E-05	0.731E-05	-0.623E-05	0.150E+06	0.125E-03
12	2656	-2.036	0.267E-05	0.243E-05	0.731E-05	-0.623E-05	0.149E+06	0.604E-04
13	2657	-2.034	0.267E-05	0.243E-05	0.732E-05	-0.623E-05	0.149E+06	0.280E-04
14	2657	-2.033	0.267E-05	0.243E-05	0.732E-05	-0.623E-05	0.149E+06	0.118E-04
15	2657	-2.033	0.267E-05	0.244E-05	0.732E-05	-0.623E-05	0.149E+06	0.372E-05
16	2657	-2.033	0.268E-05	0.244E-05	0.732E-05	-0.623E-05	0.149E+06	0.333E-06
17	2657	-2.033	0.268E-05	0.244E-05	0.732E-05	-0.623E-05	0.149E+06	0.169E-05

Factor of Safety [FS]:

Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 2658/2672$$

$$= 1.670$$

Allowable Circumferential Compressive Stress [Fha]:



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Element: Shell

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= Fic/FS
= 2658/1.67
= 1591.583 kgf/cm²

Allowable External Pressure at the given Thickness [Pa]:

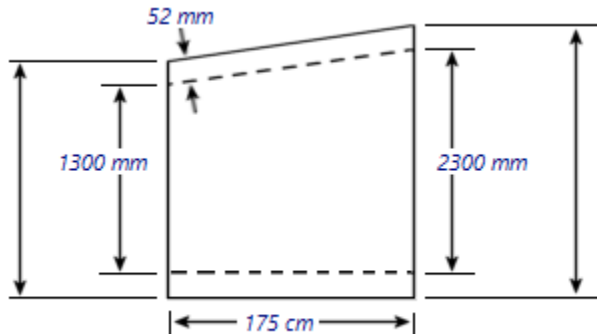
= 2 • Fha(t / Do)
= 2 • 1592 (27/1360)
= 63.195 kgf/cm²

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell cone

Conical Head/Section Design Information:					
Design Pressure	P	50.197 kgf/cm²	Int. Design Temperature	270 °C	
Material		SA-516 70	Ext. Design Temperature	270 °C	
Length		175.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1784.4 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	2300.000 mm	Actual Stress	1360.4 kgf/cm²	
Spec. Min. Thk t,e, or sa		52.0000 mm	Spec. Nominal Thk.	52.000 mm	
Surface Area		109769.0 cm²	SG of Contents	1.000	
Element Volume		3804936.2 cm³	Weight of Contents	3803.2 kgf	
Empty Weight		4746.1 kgf	Operating Weight	8549.3 kgf	
			Comp. All. Ope. Stress	1026.0 kgf/cm²	
			To End Diameter	1300.000 mm	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Thickness due to Internal Pressure: Equation 4.3.4.1 [tr]:

$$= 0.5 \cdot D_c / \cos(\alpha) \cdot (\exp(P / (S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2306 / \cos(29.7) \cdot (\exp(50.2 / (1784 \cdot 1)) - 1) + c_i + c_o$$

$$= 37.9 + 3 + 0 = 40.9 \text{ mm}$$

Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln(2 \cdot t_{cor} \cdot \cos(\alpha) + D_{cor}) / D_{cor}$$

$$= 1784 \cdot 1 \cdot \ln(2 \cdot 49 \cdot \cos(29.7) + 2306) / 2306$$

$$= 64.7 \text{ kgf/cm}^2$$

Maximum Allowable Pressure New and Cold [MAPnc]:

$$= S_a \cdot E \cdot \ln(2 \cdot t \cdot \cos(\alpha) + D) / D$$

$$= 1784 \cdot 1 \cdot \ln(2 \cdot 52 \cdot \cos(29.7) + 2300) / 2300$$

$$= 68.7 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D) \cos(\alpha))$$



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$$= 50.2 \cdot 2306 / (1(2404 - 2306) \cos(29.7))$$

$$= 1360.422 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln ((2 \cdot t_{cor} \cdot \cos(\alpha) + D_{cor}) / D_{cor}))$$

$$= 50.2 / (1 \cdot \ln((2 \cdot 49 \cdot \cos(29.7) + 2306) / 2306))$$

$$= 1385.367 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cones formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 3.824 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -38 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -26 °C
 Temperature reduction per Graph 3.13 : 12 °C
 The computed stress ratio is : 0.783

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (38.3 \cdot 1) / (52 - 3 - 0)$$

$$= 0.783$$

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	50.800 kgf/cm ²
		Min. Des. Mtl. Temp.	11 °C
Governing Thk.	52.000 mm	Basic MDMT	-26 °C
Stress/Thk. Ratio	0.783	Temperature Difference	12 °C
		Computed Min. Temp.	-38 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: Shell cone

Cone From 50 to 60 Ext. Chart: CS-2 at 270 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{ (R_o \cdot t) }$$

$$= 2016 / \sqrt{ (1393 \cdot 49) }$$

$$= 7.713$$

Compute the Factor [Ch]:

Since Mx > 1.5 and Mx < 13, Ch = 0.92 / (Mx - 0.579)

$$= 0.92 / (7.71 - 0.579)$$

$$= 0.1290$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.13 \cdot 1909280 (49 / 2787)$$

$$= 6925.973 \text{ kgf/cm}^2$$



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Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 6926.0/2671.5$$

$$= 2.593$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2672/4921$$

$$= 0.543$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 6926/1909280$$

$$= 0.003628$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.54) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$$

$$= 0.133$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{psp}) m_1$$

$$= 2672 (1 + 0.0020) / \ln(1 + 0.0020) 0.13312802$$

$$= 6123.386 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2 f_2 - R)$$

$$= 0.6 (1 - 0.54)$$

$$= 0.274$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (4921 \cdot \exp(0.27)) / (0.27 \cdot 0.27)$$

$$= 9232.526 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R^{1.5} - 0.5 \cdot R^{2.5} - R^{3.5}$$

$$= 1.5 \cdot 0.541.5 - 0.5 \cdot 0.542.5 - 0.543.5$$

$$= 0.374$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-2.502	0.162E-05	0.157E-05	0.597E-05	-0.564E-05	0.247E+06	0.632E-02
2	1230	-5.430	0.178E-07	0.178E-07	0.954E-06	-0.954E-06	0.179E+07	0.294E-02
3	1845	-3.966	0.249E-06	0.249E-06	0.279E-05	-0.278E-05	0.973E+06	0.173E-02
4	2153	-3.234	0.679E-06	0.674E-06	0.420E-05	-0.415E-05	0.519E+06	0.521E-03
5	1999	-3.600	0.419E-06	0.418E-06	0.345E-05	-0.343E-05	0.725E+06	0.870E-03
6	2076	-3.417	0.536E-06	0.533E-06	0.381E-05	-0.378E-05	0.616E+06	0.257E-03
7	2114	-3.325	0.604E-06	0.600E-06	0.400E-05	-0.396E-05	0.566E+06	0.109E-03
8	2095	-3.371	0.569E-06	0.566E-06	0.390E-05	-0.387E-05	0.591E+06	0.793E-04
9	2105	-3.348	0.586E-06	0.583E-06	0.395E-05	-0.392E-05	0.578E+06	0.135E-04
10	2100	-3.359	0.578E-06	0.574E-06	0.393E-05	-0.389E-05	0.584E+06	0.333E-04
11	2102	-3.354	0.582E-06	0.579E-06	0.394E-05	-0.390E-05	0.581E+06	0.997E-05
12	2103	-3.351	0.584E-06	0.581E-06	0.395E-05	-0.391E-05	0.580E+06	0.174E-05
13	2103	-3.352	0.583E-06	0.580E-06	0.394E-05	-0.391E-05	0.580E+06	0.412E-05
14	2103	-3.352	0.584E-06	0.580E-06	0.394E-05	-0.391E-05	0.580E+06	0.120E-05
15	2103	-3.351	0.584E-06	0.581E-06	0.395E-05	-0.391E-05	0.580E+06	0.270E-06
16	2103	-3.351	0.584E-06	0.580E-06	0.394E-05	-0.391E-05	0.580E+06	0.463E-06
17	2103	-3.351	0.584E-06	0.580E-06	0.395E-05	-0.391E-05	0.580E+06	0.967E-07



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Element: Shell cone

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Factor of Safety [FS]:

Since $F_{ic} > 0.55 F_y$ and $F_{ic} < F_y$

$= 2.407 - 0.741 \cdot F_{ic} / F_y$

$= 2.407 - 0.741 \cdot 2104/2672$

$= 1.823$

Allowable Circumferential Compressive Stress [Fha]:

$= F_{ic}/FS$

$= 2104/1.82$

$= 1153.664 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:

$= 2 \cdot F_{ha} (t / D_o)$

$= 2 \cdot 1154 (49/2787)$

$= 40.567 \text{ kgf/cm}^2$

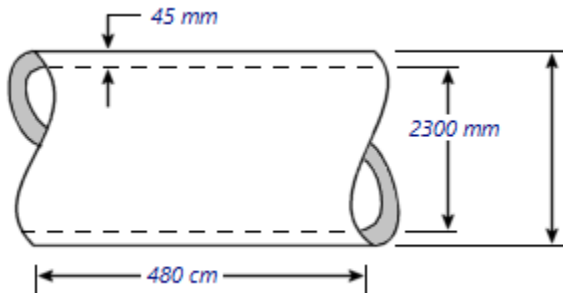
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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:			
Design Pressure P	50.197 kgf/cm ²	Int. Design Temperature	270 °C
Material	SA-516 70	Ext. Design Temperature	270 °C
Length	480.000 cm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	1784.4 kgf/cm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	1784.4 kgf/cm ²
Inside Diameter Di	2300.000 mm	Actual Stress	1378.0 kgf/cm ²
Spec. Min. Thk t,e, or sa	45.0000 mm	Spec. Nominal Thk.	45.000 mm
Surface Area	360403.5 cm ²	SG of Contents	1.000
Element Volume	17643536.0 cm ³	Weight of Contents	17635.6 kgf
Empty Weight	12332.4 kgf	Operating Weight	29968.0 kgf
		Comp. All. Ope. Stress	908.4 kgf/cm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Material treatment options: +N

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 2306 \cdot (\exp(50.2/(1784 \cdot 1)) - 1) + 3 + 0$
 $= 32.9 + 3 + 0 = 35.9 \text{ mm}$ Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1784 \cdot 1 \cdot \ln((2 \cdot 42 + 2306) / 2306) - 0.23$
 $= 63.8 - 0.23 = 63.6 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1784 \cdot 1 \cdot \ln((2 \cdot 45 + 2300) / 2300)$
 $= 68.5 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$



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$$= 50.2 \cdot 2306 / (1 \cdot (2390 - 2306))$$

$$= 1378.0414 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$
 $= 50.2 / (1 \cdot \ln((2 \cdot 42 + 2306) / 2306))$
 $= 1402.9907 \text{ kgf/cm}^2$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.919 %

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -40 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -29 °C
 Temperature reduction per Graph 3.13 : 11 °C
 The computed stress ratio is : 0.793

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (33.3 \cdot 1) / (45 - 3 - 0)$
 $= 0.793$

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	50.800 kgf/cm ²
Governing Thk.	45.000 mm	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	0.793	Basic MDMT	-29 °C
		Temperature Difference	11 °C
		Computed Min. Temp.	-40 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 270 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:
 $= L / \sqrt{R_o \cdot t}$
 $= 5042 / \sqrt{1195 \cdot 42}$
 $= 22.504$

Compute the Factor [Ch]:
 Since $M_x > 13$ and $M_x < 2 \cdot (D_o/t)^{0.94}$, $Ch = 1.12 \cdot M_x^{-1.058}$
 $= 1.12 \cdot 22.5^{-1.058}$
 $= 0.0415$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot E_y (t / D_o)$
 $= 1.6 \cdot 0.042 \cdot 1909280 (42 / 2390)$
 $= 2230.294 \text{ kgf/cm}^2$



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Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 2230.3/2671.5$$

$$= 0.835$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2672/4921$$

$$= 0.543$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 2230/1909280$$

$$= 0.001168$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.54) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$$

$$= 0.133$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{psp}) m_1$$

$$= 2672(1 + 0.0020) / \ln(1 + 0.0020) 0.13312802$$

$$= 6123.386 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2 f_2 - R)$$

$$= 0.6(1 - 0.54)$$

$$= 0.274$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (4921 \cdot \exp(0.27)) / (0.27 \cdot 0.27)$$

$$= 9232.526 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R^{1.5} - 0.5 \cdot R^{2.5} - R^{3.5}$$

$$= 1.5 \cdot 0.541.5 - 0.5 \cdot 0.542.5 - 0.543.5$$

$$= 0.374$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-2.502	0.162E-05	0.157E-05	0.597E-05	-0.564E-05	0.247E+06	0.878E-02
2	1230	-5.430	0.178E-07	0.178E-07	0.954E-06	-0.954E-06	0.179E+07	0.480E-03
3	1845	-3.966	0.249E-06	0.249E-06	0.279E-05	-0.278E-05	0.973E+06	0.729E-03
4	1537	-4.698	0.759E-07	0.759E-07	0.172E-05	-0.172E-05	0.148E+07	0.128E-03
5	1691	-4.332	0.141E-06	0.141E-06	0.222E-05	-0.221E-05	0.124E+07	0.200E-03
6	1614	-4.515	0.104E-06	0.104E-06	0.196E-05	-0.196E-05	0.136E+07	0.169E-04
7	1576	-4.606	0.892E-07	0.892E-07	0.184E-05	-0.184E-05	0.142E+07	0.596E-04
8	1595	-4.560	0.965E-07	0.965E-07	0.190E-05	-0.190E-05	0.139E+07	0.224E-04
9	1605	-4.538	0.100E-06	0.100E-06	0.193E-05	-0.193E-05	0.138E+07	0.304E-05
10	1610	-4.526	0.102E-06	0.102E-06	0.194E-05	-0.194E-05	0.137E+07	0.687E-05
11	1607	-4.532	0.101E-06	0.101E-06	0.194E-05	-0.194E-05	0.137E+07	0.190E-05
12	1606	-4.535	0.101E-06	0.101E-06	0.193E-05	-0.193E-05	0.138E+07	0.575E-06
13	1607	-4.533	0.101E-06	0.101E-06	0.193E-05	-0.193E-05	0.137E+07	0.661E-06
14	1606	-4.534	0.101E-06	0.101E-06	0.193E-05	-0.193E-05	0.138E+07	0.426E-07
15	1606	-4.534	0.101E-06	0.101E-06	0.193E-05	-0.193E-05	0.138E+07	0.266E-06
16	1606	-4.534	0.101E-06	0.101E-06	0.193E-05	-0.193E-05	0.138E+07	0.112E-06
17	1606	-4.534	0.101E-06	0.101E-06	0.193E-05	-0.193E-05	0.138E+07	0.346E-07



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Element: Shell-2

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Factor of Safety [FS]:

Since $F_{ic} > 0.55 F_y$ and $F_{ic} < F_y$

$= 2.407 - 0.741 \cdot F_{ic} / F_y$

$= 2.407 - 0.741 \cdot 1607/2672$

$= 1.961$

Allowable Circumferential Compressive Stress [Fha]:

$= F_{ic}/FS$

$= 1607/1.96$

$= 819.191 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:

$= 2 \cdot F_{ha} (t / D_o)$

$= 2 \cdot 819 (42/2390)$

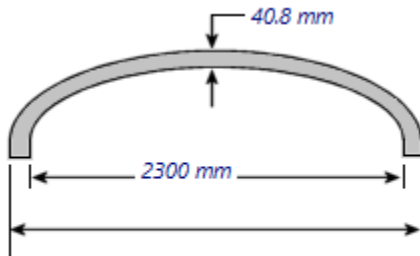
$= 28.792 \text{ kgf/cm}^2$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Head

Elliptical Head Design Information:							
Design Pressure	P	50.197	kgf/cm ²	Int. Design Temperature		270	°C
Material		SA-516	70	Ext. Design Temperature		270	°C
Length		5.000	cm	External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.	c	3.0000	mm	Allowable Stress (ope)	S	1784.4	kgf/cm ²
Ext. Corr. All.	ce	0.0000	mm	Allowable Stress (amb)	Sa	1784.4	kgf/cm ²
Inside Diameter	Di	2300.000	mm	Actual Stress		1389.0	kgf/cm ²
Spec. Min. Thk t _e , or s _a		40.8000	mm	Spec. Nominal Thk.		48.000	mm
Surface Area		65993.2	cm ²	SG of Contents		1.000	
Element Volume		1800398.8	cm ³	Weight of Contents		1799.6	kgf
Empty Weight		2591.8	kgf	Operating Weight		4391.4	kgf
				Comp. All. Ope. Stress		908.4	kgf/cm ²
				Aspect Ratio		2.00	
Radiography Information:							
...				...			
Circ. Joint Efficiency		100	%	Long. Joint Eff.	E or v	100	%

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.375 mm

Material treatment options: +N

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:
 tr = 29.4 mm

$$\begin{aligned}
 t &= t_r + c_i + c_o \\
 &= 29.4 + 3 + 0 \\
 &= 32.4 \text{ mm - see below for the derivation}
 \end{aligned}$$

For Elliptical heads: Compute [h, k, r_i] per 4.3.7.1

$$\begin{aligned}
 h &= D / (2 \cdot \text{Aspect Ratio}) \\
 &= 2300 / (2 \cdot 2) \\
 &= 575.000 \text{ mm}
 \end{aligned}$$



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Element: Head

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$$\begin{aligned} k &= D/(2 \cdot h) \\ &= 2300/(2 \cdot 575) \\ &= 2.000 \end{aligned}$$

$$\begin{aligned} r &= D(0.5/k - 0.08) + ca \\ &= 2300(0.5/2 - 0.08) + 3 \\ &= 394.000 \text{ mm} \end{aligned}$$

$$\begin{aligned} L &= D(0.44 \cdot k + 0.02) + ca \\ &= 2300(0.44 \cdot 2 + 0.02) + 3 \\ &= 2073.000 \text{ mm} \end{aligned}$$

Step 3: Geometric Constants [βTh , ϕTh , Rth]:

Equations 4.3.8 to 4.3.11 as appropriate

t (corroded) and P for the calc.: $t = 29.4 \text{ mm}$, $P = 50.2 \text{ kgf/cm}^2$

Angle [βTh]:

$$\begin{aligned} &= \arccos[(0.5 \cdot D - r)/(L - r)] \\ &= \arccos[(0.5 \cdot 2306 - 394)/(2073 - 394)] \\ &= 1.1 \text{ radians - Equation 4.3.8} \end{aligned}$$

$$\begin{aligned} \phi Th &= \sqrt{(L \cdot t) / r} \\ &= \sqrt{(2073 \cdot 29.4) / 394} \\ &= 0.63 \text{ radians - Equation 4.3.9} \end{aligned}$$

$\phi Th < \beta Th$, Thus:

$$\begin{aligned} Rth &= (0.5 \cdot D - r) / \cos(\beta Th - \phi Th) + r \\ &= (0.5 \cdot 2306 - 394) / \cos(1.1 - 0.63) + 394 \\ &= 1248 \text{ mm - Equation 4.3.10} \end{aligned}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [C1 and C2]:

$$\begin{aligned} r/D &> 0.08, \text{ Thus:} \\ C1 &= 0.692(r/D) + 0.605 \\ &= 0.692(394/2306) + 0.605 \\ &= 0.72 \text{ - Equation 4.3.13} \end{aligned}$$

$$\begin{aligned} C2 &= 1.46 - 2.6(r/D) = 1.46 - 2.6(394/2306) \\ &= 1.02, \text{ Equation 4.3.15} \end{aligned}$$

Step 5: Pressure to produce knuckle elastic buckling [P_{eth}]:

$$\begin{aligned} &= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r)) \\ &= 0.72 \cdot 1909280 \cdot 29.4^2 / (1.02 \cdot 1248(0.5 \cdot 1248 - 394)) \\ &= 4087 \text{ kgf/cm}^2, \text{ Equation 4.3.16} \end{aligned}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [P_y]:

$C3 = S_y = 2672 \text{ kgf/cm}^2$, Not in the creep range - 4.3.6.1(f)

$$\begin{aligned} P_y &= C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1)) \\ &= 2672 \cdot 29.4 / (1.02 \cdot 1248(0.5 \cdot 1248/394 - 1)) \\ &= 106 \text{ kgf/cm}^2, \text{ Equation 4.3.17} \end{aligned}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and P_{ck}]:

$$G = P_{eth}/P_y = 4087/106 = 38.5$$

$G > 1$, Thus:

Buckling Pressure [P_{ck}]:

$$\begin{aligned} &= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) / \\ &\quad (1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) P_y \\ &= ((0.77508 \cdot 38.5 - 0.20354 \cdot 38.5^2 + 0.019274 \cdot 38.5^3) / \\ &\quad (1 + 0.19014 \cdot 38.5 - 0.089534 \cdot 38.5^2 + 0.0093965 \cdot 38.5^3)) \cdot 106 \\ &= 213 \text{ kgf/cm}^2, \text{ Equation 4.3.19} \end{aligned}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [P_{ak}]:



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$$= Pck/1.5 = 213/1.5 = 142 \text{ kgf/cm}^2 - \text{Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$\begin{aligned} &= 2 \cdot S \cdot E / (L / t + 0.5) \\ &= 2 \cdot 1784 \cdot 1 / (2073/29.4 + 0.5) \\ &= 50.2 \text{ kgf/cm}^2, \text{Equation 4.3.22} \end{aligned}$$

Step 10: Compute final pressure [Pa]:

$$\begin{aligned} &= \min(Pak, Pac) = \min(142, 50.2) \\ &= 50.2 \text{ kgf/cm}^2, \text{Equation 4.3.23: (same as original design pressure)} \end{aligned}$$

Allowable pressure in Crown at given thickness [Pac]:

$$\begin{aligned} &= 2 \cdot S \cdot E / (L / t + 0.5) \\ &= 2 \cdot 1784 \cdot 1 / (2073/37.8 + 0.5) \\ &= 64.488 \text{ kgf/cm}^2 \end{aligned}$$

Computed Maximum Allowable Pressure [MAWP]:

$$= \text{MaxPressure} - \text{Pliq} = 64.5 - 0.23 = 64.3 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New and Cold [MAPnc]:

$$= 69.7 \text{ kgf/cm}^2$$

Computed Membrane Stress Intensity at Design Pressure [Sm]:

$$= 1389.0 \text{ kgf/cm}^2$$

Required Thickness of Straight Flange = 35.896 mm

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$	8.675 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b/(D_f - 2t))$	23.136 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 65993 / 3.14159) \\ &= 2898.710 \text{ mm} \end{aligned}$$

MDMT Calculations in the Knuckle Portion:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -42 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: D
From Figure 3.8M- with PWHT, the MDMT is	: -30 °C
Temperature reduction per Graph 3.13	: 11 °C
The computed stress ratio is	: 0.786

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (t_{nom} - ci - c_{ext}) \\ &= (29.7 \cdot 1) / (40.8 - 3 - 0) \\ &= 0.786 \end{aligned}$$

MDMT Calculations in the Head Straight Flange:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -42 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: D
From Figure 3.8M- with PWHT, the MDMT is	: -27 °C
Temperature reduction per Graph 3.13	: 14 °C
The computed stress ratio is	: 0.740

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (t_{nom} - ci - c_{ext}) \\ &= (33.3 \cdot 1) / (48 - 3 - 0) \end{aligned}$$



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= 0.740

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	50.800 kgf/cm ²
Governing Thk.	40.800 mm	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	0.786	Basic MDMT	-30 °C
		Temperature Difference	11 °C
		Computed Min. Temp.	-42 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.375 mm

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: Head

Elliptical Head From 70 to 80 Ext. Chart: CS-2 at 270 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y(t / R_o)$$

$$= 0.075 \cdot 1909280 (37.8/2075)$$

$$= 2609.073 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 2609.1 / 2671.5$$

$$= 0.977$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2672/4921$$

$$= 0.543$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 2609/1909280$$

$$= 0.001367$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.54) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$$

$$= 0.133$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2672 (1 + 0.0020) / \ln(1 + 0.0020)^{0.13312802}$$

$$= 6123.386 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6 (1 - 0.54)$$



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= 0.274

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (4921 \cdot \exp(0.27)) / (0.27 \cdot 0.27)$$

$$= 9232.526 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.541.5 - 0.5 \cdot 0.542.5 - 0.543.5$$

$$= 0.374$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-2.502	0.162E-05	0.157E-05	0.597E-05	-0.564E-05	0.247E+06	0.858E-02
2	1230	-5.430	0.178E-07	0.178E-07	0.954E-06	-0.954E-06	0.179E+07	0.678E-03
3	1845	-3.966	0.249E-06	0.249E-06	0.279E-05	-0.278E-05	0.973E+06	0.531E-03
4	1537	-4.698	0.759E-07	0.759E-07	0.172E-05	-0.172E-05	0.148E+07	0.326E-03
5	1691	-4.332	0.141E-06	0.141E-06	0.222E-05	-0.221E-05	0.124E+07	0.148E-05
6	1614	-4.515	0.104E-06	0.104E-06	0.196E-05	-0.196E-05	0.136E+07	0.181E-03
7	1653	-4.423	0.122E-06	0.122E-06	0.209E-05	-0.208E-05	0.130E+07	0.954E-04
8	1672	-4.377	0.131E-06	0.131E-06	0.215E-05	-0.215E-05	0.127E+07	0.484E-04
9	1682	-4.355	0.136E-06	0.136E-06	0.218E-05	-0.218E-05	0.125E+07	0.238E-04
10	1686	-4.343	0.139E-06	0.139E-06	0.220E-05	-0.220E-05	0.124E+07	0.113E-04
11	1689	-4.337	0.140E-06	0.140E-06	0.221E-05	-0.221E-05	0.124E+07	0.492E-05
12	1690	-4.335	0.141E-06	0.141E-06	0.221E-05	-0.221E-05	0.124E+07	0.173E-05
13	1691	-4.333	0.141E-06	0.141E-06	0.221E-05	-0.221E-05	0.124E+07	0.125E-06
14	1691	-4.332	0.141E-06	0.141E-06	0.222E-05	-0.221E-05	0.124E+07	0.677E-06
15	1691	-4.333	0.141E-06	0.141E-06	0.221E-05	-0.221E-05	0.124E+07	0.276E-06
16	1691	-4.333	0.141E-06	0.141E-06	0.221E-05	-0.221E-05	0.124E+07	0.755E-07
17	1691	-4.333	0.141E-06	0.141E-06	0.221E-05	-0.221E-05	0.124E+07	0.248E-07

Factor of Safety [FS]:

Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 1691/2672$$

$$= 1.938$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic/FS$$

$$= 1691/1.94$$

$$= 872.692 \text{ kgf/cm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha[t/Ro]$$

$$= 2 \cdot 873 [37.8/2075]$$

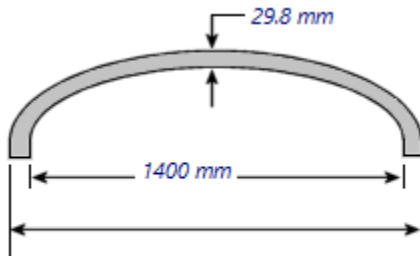
$$= 31.801 \text{ kgf/cm}^2$$



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Internal Pressure Calculation Results:

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Internal Pressure Results for: Head

Elliptical Head Design Information:					
Design Pressure	P	49.967 kgf/cm ²	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		270 °C
Length		5.000 cm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1784.4 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm ²
Inside Diameter	Di	1400.000 mm	Actual Stress		1192.1 kgf/cm ²
Spec. Min. Thk t _e , or s _a		29.7500 mm	Spec. Nominal Thk.		35.000 mm
Surface Area		25732.9 cm ²	SG of Contents		0.000
Element Volume		436158.8 cm ³	Weight of Contents		0.0 kgf
Empty Weight		729.7 kgf	Operating Weight		729.7 kgf
			Comp. All. Ope. Stress		893.5 kgf/cm ²
			Aspect Ratio		2.00
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 17.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 8.750 mm

Material treatment options: +N

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:
 tr = 17.8 mm

$$\begin{aligned}
 t &= t_r + c_i + c_o \\
 &= 17.8 + 3 + 0 \\
 &= 20.8 \text{ mm - see below for the derivation}
 \end{aligned}$$

For Elliptical heads: Compute [h, k, r_i] per 4.3.7.1

$$\begin{aligned}
 h &= D / (2 \cdot \text{Aspect Ratio}) \\
 &= 1400 / (2 \cdot 2) \\
 &= 350.000 \text{ mm}
 \end{aligned}$$



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Element: Head

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$$\begin{aligned} k &= D/(2 \cdot h) \\ &= 1400/(2 \cdot 350) \\ &= 2.000 \end{aligned}$$

$$\begin{aligned} r &= D(0.5/k - 0.08) + ca \\ &= 1400(0.5/2 - 0.08) + 3 \\ &= 241.000 \text{ mm} \end{aligned}$$

$$\begin{aligned} L &= D(0.44 \cdot k + 0.02) + ca \\ &= 1400(0.44 \cdot 2 + 0.02) + 3 \\ &= 1263.000 \text{ mm} \end{aligned}$$

Step 3: Geometric Constants [βTh , ϕTh , Rth]:

Equations 4.3.8 to 4.3.11 as appropriate

t (corroded) and P for the calc.: $t = 17.8 \text{ mm}$, $P = 50 \text{ kgf/cm}^2$

Angle [βTh]:

$$\begin{aligned} &= \arccos[(0.5 \cdot D - r)/(L - r)] \\ &= \arccos[(0.5 \cdot 1406 - 241)/(1263 - 241)] \\ &= 1.1 \text{ radians - Equation 4.3.8} \end{aligned}$$

$$\begin{aligned} \phi Th &= \sqrt{(L \cdot t) / r} \\ &= \sqrt{(1263 \cdot 17.8) / 241} \\ &= 0.62 \text{ radians - Equation 4.3.9} \end{aligned}$$

$\phi Th < \beta Th$, Thus:

$$\begin{aligned} Rth &= (0.5 \cdot D - r) / \cos(\beta Th - \phi Th) + r \\ &= (0.5 \cdot 1406 - 241) / \cos(1.1 - 0.62) + 241 \\ &= 762 \text{ mm - Equation 4.3.10} \end{aligned}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [$C1$ and $C2$]:

$$\begin{aligned} r/D &> 0.08, \text{ Thus:} \\ C1 &= 0.692(r/D) + 0.605 \\ &= 0.692(241/1406) + 0.605 \\ &= 0.72 \text{ - Equation 4.3.13} \end{aligned}$$

$$\begin{aligned} C2 &= 1.46 - 2.6(r/D) = 1.46 - 2.6(241/1406) \\ &= 1.01, \text{ Equation 4.3.15} \end{aligned}$$

Step 5: Pressure to produce knuckle elastic buckling [P_{eth}]:

$$\begin{aligned} &= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r)) \\ &= 0.72 \cdot 1909280 \cdot 17.8^2 / (1.01 \cdot 762(0.5 \cdot 762 - 241)) \\ &= 4055 \text{ kgf/cm}^2, \text{ Equation 4.3.16} \end{aligned}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [P_y]:

$C3 = S_y = 2672 \text{ kgf/cm}^2$, Not in the creep range - 4.3.6.1(f)

$$\begin{aligned} P_y &= C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1)) \\ &= 2672 \cdot 17.8 / (1.01 \cdot 762(0.5 \cdot 762 / 241 - 1)) \\ &= 106 \text{ kgf/cm}^2, \text{ Equation 4.3.17} \end{aligned}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and P_{ck}]:

$$G = P_{eth}/P_y = 4055/106 = 38.2$$

$G > 1$, Thus:

Buckling Pressure [P_{ck}]:

$$\begin{aligned} &= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) / \\ &\quad (1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) P_y \\ &= ((0.77508 \cdot 38.2 - 0.20354 \cdot 38.2^2 + 0.019274 \cdot 38.2^3) / \\ &\quad (1 + 0.19014 \cdot 38.2 - 0.089534 \cdot 38.2^2 + 0.0093965 \cdot 38.2^3)) \cdot 106 \\ &= 213 \text{ kgf/cm}^2, \text{ Equation 4.3.19} \end{aligned}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [P_{ak}]:



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$$= Pck/1.5 = 213/1.5 = 142 \text{ kgf/cm}^2 - \text{Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$\begin{aligned} &= 2 \cdot S \cdot E / (L / t + 0.5) \\ &= 2 \cdot 1784 \cdot 1 / (1263/17.8 + 0.5) \\ &= 50 \text{ kgf/cm}^2, \text{Equation 4.3.22} \end{aligned}$$

Step 10: Compute final pressure [Pa]:

$$\begin{aligned} &= \min(Pak, Pac) = \min(142, 50) \\ &= 50 \text{ kgf/cm}^2, \text{Equation 4.3.23: (same as original design pressure)} \end{aligned}$$

Allowable pressure in Crown at given thickness [Pac]:

$$\begin{aligned} &= 2 \cdot S \cdot E / (L / t + 0.5) \\ &= 2 \cdot 1784 \cdot 1 / (1263/26.8 + 0.5) \\ &= 74.795 \text{ kgf/cm}^2 \end{aligned}$$

Computed Maximum Allowable Pressure [MAWP]:

$$= \text{MaxPressure} - \text{Pliq} = 74.8 - 0 = 74.8 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New and Cold [MAPnc]:

$$= 83.3 \text{ kgf/cm}^2$$

Computed Membrane Stress Intensity at Design Pressure [Sm]:

$$= 1192.1 \text{ kgf/cm}^2$$

Required Thickness of Straight Flange = 22.964 mm

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$	10.274 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b/(D_f - 2t))$	25.690 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 25733 / 3.14159) \\ &= 1810.087 \text{ mm} \end{aligned}$$

MDMT Calculations in the Knuckle Portion:

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -18 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: D
From Figure 3.7M- Non PWHT, the MDMT is	: -18 °C
Temperature reduction per Graph 3.12	: 0 °C
The computed stress ratio is	: 1.000

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (t_{nom} - ci - c_{ext}) \\ &= (26.8 \cdot 1) / (29.8 - 3 - 0) \\ &= 1.000 \end{aligned}$$

MDMT Calculations in the Head Straight Flange:

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -18 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: D
From Figure 3.7M- Non PWHT, the MDMT is	: -15 °C
Temperature reduction per Graph 3.12	: 3 °C
The computed stress ratio is	: 0.940

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (t_{nom} - ci - c_{ext}) \\ &= (30.1 \cdot 1) / (35 - 3 - 0) \end{aligned}$$



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= 0.940

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	74.795 kgf/cm ²
Governing Thk.	35.000 mm	Min. Des. Mtl. Temp.	11 °C
Stress/Thk. Ratio	0.940	Basic MDMT	-15 °C
		Temperature Difference	3 °C
		Computed Min. Temp.	-18 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 17.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 8.750 mm

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: Head

Elliptical Head From 80 to 90 Ext. Chart: CS-2 at 270 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y(t / R_o)$$

$$= 0.075 \cdot 1909280 (26.8/1264)$$

$$= 3031.447 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 3031.4 / 2671.5$$

$$= 1.135$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 2672/4921$$

$$= 0.543$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 3031/1909280$$

$$= 0.001588$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.54) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$$

$$= 0.133$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 2672 (1 + 0.0020) / \ln(1 + 0.0020)^{0.13312802}$$

$$= 6123.386 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6 (1 - 0.54)$$

$$= 0.274$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$



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$$= (4921 \cdot \exp(0.27)) / (0.27 \cdot 0.27)$$

$$= 9232.526 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.541.5 - 0.5 \cdot 0.542.5 - 0.543.5$
 $= 0.374$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-2.502	0.162E-05	0.157E-05	0.597E-05	-0.564E-05	0.247E+06	0.836E-02
2	1230	-5.430	0.178E-07	0.178E-07	0.954E-06	-0.954E-06	0.179E+07	0.900E-03
3	1845	-3.966	0.249E-06	0.249E-06	0.279E-05	-0.278E-05	0.973E+06	0.310E-03
4	1537	-4.698	0.759E-07	0.759E-07	0.172E-05	-0.172E-05	0.148E+07	0.547E-03
5	1691	-4.332	0.141E-06	0.141E-06	0.222E-05	-0.221E-05	0.124E+07	0.220E-03
6	1768	-4.149	0.189E-06	0.188E-06	0.249E-05	-0.249E-05	0.110E+07	0.129E-04
7	1730	-4.240	0.164E-06	0.163E-06	0.235E-05	-0.235E-05	0.117E+07	0.110E-03
8	1749	-4.194	0.176E-06	0.176E-06	0.242E-05	-0.242E-05	0.114E+07	0.506E-04
9	1759	-4.172	0.182E-06	0.182E-06	0.246E-05	-0.245E-05	0.112E+07	0.193E-04
10	1763	-4.160	0.185E-06	0.185E-06	0.248E-05	-0.247E-05	0.111E+07	0.333E-05
11	1766	-4.154	0.187E-06	0.187E-06	0.248E-05	-0.248E-05	0.111E+07	0.477E-05
12	1765	-4.157	0.186E-06	0.186E-06	0.248E-05	-0.248E-05	0.111E+07	0.713E-06
13	1764	-4.159	0.186E-06	0.186E-06	0.248E-05	-0.247E-05	0.111E+07	0.131E-05
14	1764	-4.158	0.186E-06	0.186E-06	0.248E-05	-0.247E-05	0.111E+07	0.299E-06
15	1764	-4.158	0.186E-06	0.186E-06	0.248E-05	-0.248E-05	0.111E+07	0.207E-06
16	1764	-4.158	0.186E-06	0.186E-06	0.248E-05	-0.247E-05	0.111E+07	0.458E-07
17	1764	-4.158	0.186E-06	0.186E-06	0.248E-05	-0.247E-05	0.111E+07	0.806E-07

Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$
 $= 2.407 - 0.741 \cdot Fic / F_y$
 $= 2.407 - 0.741 \cdot 1765/2672$
 $= 1.917$

Allowable Circumferential Compressive Stress [Fha]:
 $= Fic/FS$
 $= 1765/1.92$
 $= 920.411 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha[t/Ro]$
 $= 2 \cdot 920 [26.8/1264]$
 $= 38.970 \text{ kgf/cm}^2$



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 Shop Pressure Test Results: Step: 12 9:32am Aug 12,2026

Hydrostatic Test Pressure Results:

Hydrotest Pressure Calculation [Shell Side]:
 = Test Factor • ShellSide MAWP(St/S)
 = 1.25 • 50.8 (1)
 = 63.500 kgf/cm²

Hydrotest Pressure Calculation [Channel Side]:
 = Test Factor • Channel Side MAWP(St/S)
 = 1.25 • 74.8 (1)
 = 93.493 kgf/cm²

Horizontal Test performed per: per VIII-2, Section 8.2

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Channel	868.3	3002.9	0.289	93.62
Shell	1410.2	2518.6	0.560	63.63
Shell cone	1654.9	2518.6	0.657	63.73
Shell-2	1660.3	2518.6	0.659	63.73
Head	1632.6	2518.6	0.648	63.73
Head	2010.4	2518.6	0.798	93.83

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1- Channel inle	Nozzle	2110.70	1570.34	1.344
N2- Channel outl	Nozzle	2110.70	1570.34	1.344
N4- Demister	Nozzle	1784.41	1435.72	1.243
N4- Demister	Pad	1784.41	1435.72	1.243
N3- Shell inlet	Nozzle	1641.66	1321.51	1.242
N3- Shell inlet	Pad	1784.41	1435.72	1.243
N4- Shell outlet	Nozzle	1641.66	1321.51	1.242
N4- Shell outlet	Pad	1784.41	1435.72	1.243
N5- Blow down	Nozzle	1641.66	1321.51	1.242
N5- Blow down	Pad	1784.41	1435.72	1.243
N8- MP steam	Nozzle	1641.66	1321.51	1.242
N8- MP steam	Pad	1784.41	1435.72	1.243
K1- PSV	Nozzle	1641.66	1321.51	1.242
K1- PSV	Pad	1784.41	1435.72	1.243
K2- PSV	Nozzle	1641.66	1321.51	1.242
K2- PSV	Pad	1784.41	1435.72	1.243
K9- Temperature	Nozzle	1641.66	1321.51	1.242
K9- Temperature	Pad	1784.41	1435.72	1.243
N10- Chemical ad	Nozzle	1641.66	1321.51	1.242
N10- Chemical ad	Pad	1784.41	1435.72	1.243
MH1- Manhole	Nozzle	1641.66	1321.51	1.242
MH1- Manhole	Pad	1784.41	1435.72	1.243

Minimum 1.242

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Chnl. cover	2110.70	1570.34	1.344



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Shop Pressure Test Results: Step: 12 9:32am Aug 12,2026

Chnl. flange	2110.70	1570.34	1.344
Channel	2110.70	1570.34	1.344
Shell	1784.41	1784.41	1.000
Shell cone	1784.41	1784.41	1.000
Shell-2	1784.41	1784.41	1.000
Head	1784.41	1784.41	1.000
Head	1784.41	1784.41	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	1407.13	1233.81	1.140
Tubesheet Material	2110.70	1570.34	1.344

Minimum 1.140

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N1- Channel inle	801.57	3002.91	0.267
N2- Channel outl	801.57	3002.91	0.267
N4- Demister	516.04	2518.57	0.205
N3- Shell inlet	260.86	2324.83	0.112
N4- Shell outlet	271.37	2324.83	0.117
N5- Blow down	186.32	2324.83	0.080
N8- MP steam	221.23	2324.83	0.095
K1- PSV	186.32	2324.83	0.080
K2- PSV	260.86	2324.83	0.112
K9- Temperature	181.73	2324.83	0.078
N10- Chemical ad	81.05	2324.83	0.035
MH1- Manhole	389.32	2324.83	0.167

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 9:32a Aug 12,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		63.50 kgf/cm ²
Shell Thickness	ts	30.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1300.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	1784.41 kgf/cm ²
Shell Allowable Stress at Ambient		1784.41 kgf/cm ²

Channel Data:

Channel Description: Channel

Channel Type:		Cylinder
Channel Test Pressure		93.49 kgf/cm ²
Channel Thickness	tc	74.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1300.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 22

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K21590
Channel Allowable Stress at Temperature	Sc	2110.70 kgf/cm ²
Channel Allowable Stress at Ambient		2110.70 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	888
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	25.0000 mm
Total Straight Tube Length	Lt	6600.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6355.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 T22
Tube Material UNS Number		K21590
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		1407.13 kgf/cm ²
Tube Allowable Stress At Ambient		1407.13 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	2090.31 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	34.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	617.500 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm

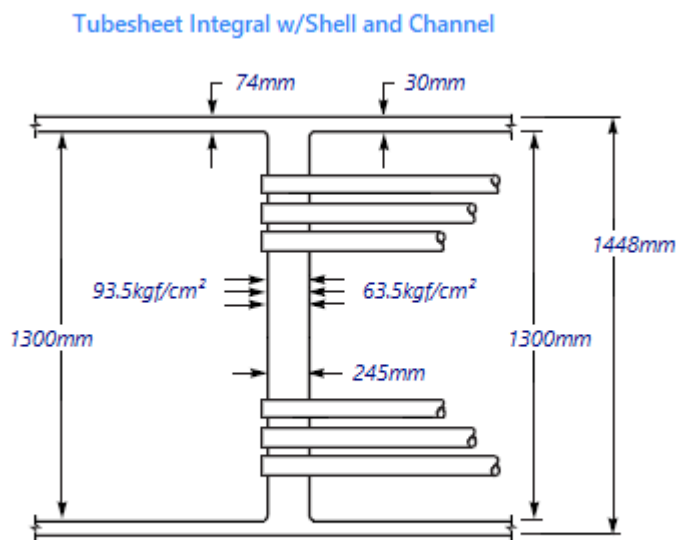
Tubesheet Data:



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Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-336 F22
Tubesheet Material UNS Number		K21590
Tubesheet Allowable Stress at Temperature	S	2110.70 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	2110.70 kgf/cm ²
Thickness of Tubesheet	h	245.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1448.000 mm
Area of the Untubed Lanes	AL	0.0 cm ²
Junction Stress Reduction option		Perform Plastic Calculation
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TM-1 Material Group D

Elastic Mod. at Design Temperature	21.1 °C	0.21514E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21514E+07 kgf/cm ²

Tubes - TM-1 Material Group D



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Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.21514E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21514E+07 kgf/cm²

TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	21.1 °C	0.21514E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21514E+07 kgf/cm²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c$ Equation 4.3.1
 $= 0.5 \cdot 20 \cdot (\exp(93.5/(1407 \cdot 1)) - 1) + 0$
 $= 0.69 + 0$
 $= 0.69 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 6972 / \sqrt{(12.5 \cdot 2.5)}$
 $= 1247.279$

Compute the Factor [Ch]:

Since $M_x > 2 \cdot (D_o/t)^{0.94}$, $Ch = 0.55 / (D_o/t)$
 $= 0.55 / (25/2.5)$
 $= 0.0550$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y (t / D_o)$
 $= 1.6 \cdot 0.055 \cdot 2151394 (2.5/25)$
 $= 18932.266 \text{ kgf/cm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 18932.3/2090.3$
 $= 9.057$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_y / \sigma_{uts}$
 $= 2090/4221$
 $= 0.495$

Ratio [AE]:

$= Fhe / E$
 $= 18932/2151394$
 $= 0.008800$

Factor [m1]:

$= (\ln(r) + (\epsilon_{sp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{sp}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.5) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.153$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_y (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 2090 (1 + 0.0020) / \ln(1 + 0.0020)^{0.15304112}$
 $= 5422.479 \text{ kgf/cm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f / (m2f2 - R)$
 $= 0.6 / (1 - 0.5)$
 $= 0.303$



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Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$
 $= (4221 \cdot \exp(0.3)) / (0.3 \cdot 0.3)$
 $= 8203.450 \text{ kgf/cm}^2$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.51.5 - 0.5 \cdot 0.52.5 - 0.53.5$
 $= 0.351$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2110	-1.947	0.325E-05	0.290E-05	0.884E-05	-0.730E-05	0.123E+06	0.840E-02
2	1055	-4.769	0.701E-07	0.701E-07	0.179E-05	-0.179E-05	0.165E+07	0.816E-02
3	1582	-3.358	0.661E-06	0.657E-06	0.456E-05	-0.452E-05	0.549E+06	0.592E-02
4	1846	-2.652	0.155E-05	0.151E-05	0.650E-05	-0.624E-05	0.264E+06	0.181E-02
5	1978	-2.299	0.227E-05	0.216E-05	0.762E-05	-0.698E-05	0.181E+06	0.213E-02
6	1912	-2.476	0.188E-05	0.182E-05	0.705E-05	-0.664E-05	0.219E+06	0.623E-04
7	1945	-2.388	0.207E-05	0.198E-05	0.733E-05	-0.682E-05	0.199E+06	0.973E-03
8	1928	-2.432	0.198E-05	0.190E-05	0.719E-05	-0.673E-05	0.209E+06	0.440E-03
9	1920	-2.454	0.193E-05	0.186E-05	0.712E-05	-0.669E-05	0.214E+06	0.185E-03
10	1916	-2.465	0.191E-05	0.184E-05	0.708E-05	-0.667E-05	0.216E+06	0.607E-04
11	1914	-2.470	0.190E-05	0.183E-05	0.706E-05	-0.665E-05	0.218E+06	0.105E-05
12	1915	-2.468	0.190E-05	0.183E-05	0.707E-05	-0.666E-05	0.217E+06	0.298E-04
13	1915	-2.469	0.190E-05	0.183E-05	0.707E-05	-0.666E-05	0.217E+06	0.143E-04
14	1914	-2.470	0.190E-05	0.183E-05	0.706E-05	-0.665E-05	0.217E+06	0.665E-05
15	1914	-2.470	0.190E-05	0.183E-05	0.706E-05	-0.665E-05	0.218E+06	0.280E-05
16	1914	-2.470	0.190E-05	0.183E-05	0.706E-05	-0.665E-05	0.218E+06	0.877E-06

Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$
 $= 2.407 - 0.741 \cdot Fic / F_y$
 $= 2.407 - 0.741 \cdot 1915/2090$
 $= 1.728$

Allowable Circumferential Compressive Stress [Fha]:
 $= Fic/FS$
 $= 1915/1.73$
 $= 1107.764 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha (t / Do)$
 $= 2 \cdot 1108 (2.5/25)$
 $= 221.553 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.0773 mm
Allowable Internal Pressure at Corroded thickness	313.99 kgf/cm²
Required Internal Design Pressure	93.49 kgf/cm²
Allowable External Pressure at Corroded thickness	221.55 kgf/cm²
Required External Design Pressure	63.50 kgf/cm²
Required Thickness due to Shell Side pressure	1.0773 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :



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ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + d_t \\ &= 2 \cdot 618 + 25 \\ &= 1260.000 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - d_t) / p \\ &= (34 - 25) / 34 \\ &= 0.265 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 1300 / 1260 \\ &= 1.0317 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 1300 / 1260 \\ &= 1.0317 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0 / 245 \\ &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned} &= \max(d_t - 2t_t (E_t T / E) \cdot (S_t T / S) \cdot (\rho); d_t - 2t_t) \\ &= \max(25 - 2 \cdot 2.5 (2151394 / 2151394) \cdot (1407 / 2111) \cdot (0); 25 - 2 \cdot 2.5) \\ &= 25.0000 \text{ mm} \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned} &= p / \sqrt{(1 - 4 \cdot \min(AI; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2))} \\ &= 34 / \sqrt{(1 - 4 \cdot \min(0; 4 \cdot 1260 \cdot 34) / (\pi \cdot 1260^2))} \\ &= 34.0000 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (34 - 25) / 34 \\ &= 0.2647 \end{aligned}$$

$h/p = 7.205882$; $\mu^* = 0.264706$; $E^* = 506316 \text{ kgf/cm}^2$

$E^*/E = 0.235343$; $v^* = 0.382088$

Note: As h/p (7.206) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - v_s^2))^{1/4}) / \sqrt{(D_s + t_s)t_s)} \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1300 + 30)30)} \\ &= 0.0091 \text{ 1/mm} \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned} &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\ &= 0.0091 \cdot 2067025 \cdot 30^3 / (6(1 - 0.3^2)) \\ &= 93022120.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$



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Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1300 + 30) \cdot 93022120 / 245^3 \cdot (1 + 245 \cdot 0.0091 + (245 \cdot 0.0091)^2 / 2)$$

$$= 288489.2 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 1300^2 / (4 \cdot 2067025 \cdot 30) \cdot (1 - 1300 \cdot 0.3 / (2(1300 + 30)))$$

$$= 0.58144\text{E-}02 \text{ mm/kgf/cm}^2$$

Parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.03 \cdot 93022120 \cdot 0.0091 \cdot 0.005814385 (1 + 245 \cdot 0.0091)$$

$$= 16401.6074 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (1300 + 74) 74 }$$

$$= 0.0057 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.0057 \cdot 2151394 \cdot 74^3 / (6(1 - 0.3^2))$$

$$= 910267264.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1300 + 74 / \cos(0)) \cdot 910267264 / 245^3 \cdot (1 + 245 \cdot 0.0057 \cdot (1 + 1) / 2 + (245 \cdot 0.0057)^2 / 2 \cdot 1)$$

$$= 1720745.2 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1300^2 / (4 \cdot 2151394 \cdot 74) \cdot (1 - 1300 \cdot 0.3 / (2(1300 + 74)))$$

$$= 0.22772\text{E-}02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.03 \cdot 910267264 [0.002277205 (1 + 0.0057 \cdot 1 \cdot 245) - 0.000 (1 / 245 + 1 \cdot 0.0057 / 2)]$$

$$= 29221.9590 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1448 / 1260$$

$$= 1.1492$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.38) / 506316 \cdot (288489 + 1720745 + 2151394 \cdot \ln(1.15))$$

$$= 2.8172$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:



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Moment on tubesheet due to pressures (Ps, Pt) [Mts]:
 $= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$
 $= 1260^2/16 \cdot [(1.03 - 1) \cdot (1.03^2 + 1) \cdot 63.5 -$
 $(1.03 - 1) \cdot (1.03^2 + 1) \cdot 93.5]$
 $= -195054.7188 \text{ kgf/cm}^2 \cdot \text{mm}^2$

Moment acting on unperforated tubesheet rim [M*]:
 $= Mts + \omega c \cdot Pt - \omega s \cdot Ps$
 $= -195055 + 29222 \cdot 93.5 - 16402 \cdot 63.5$
 $= 1495509.1250 \text{ kgf/cm}^2 \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:
 $= ((M*) - Do^2/32 \cdot F (Ps - Pt)) / (1 + F)$
 $= ((1495509) - 1260^2/32 \cdot 2.82 (63.5 - 93.5)) / (1 + 2.82)$
 $= 1490015.8750 \text{ kgf/cm}^2 \cdot \text{mm}^2$

Maximum bending moment acting on center of tubesheet [Mo]:
 $= Mp + Do^2/64 \cdot (3 + \nu*) \cdot (Ps - Pt)$
 $= 1490015 + 1260^2/64 \cdot (3 + 0.38) \cdot (63.5 - 93.5)$
 $= -1026369.0625 \text{ kgf/cm}^2 \cdot \text{mm}^2$

Maximum bending moment acting on tubesheet [M]:
 $= \max(| Mp | ; | Mo |)$
 $= \max(| 1490015 | ; | -1026369 |)$
 $= 1490015.8750 \text{ kgf/cm}^2 \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:
 $= 6 \cdot M / (\mu* (h - hg')^2)$
 $= 6 \cdot 1490015 / (0.26 (245 - 8)^2)$
 $= 601.2870 \text{ kgf/cm}^2$

Allowable Tubesheet Bending Stress [σ_{All}]:
 $= 2 \cdot \beta T \cdot Sy$
 $= 2 \cdot 0.95 \cdot 3161$
 $= 6005.81 \text{ kgf/cm}^2$

Tubesheet bending stress at final thickness [σ]:
 $= 6 \cdot M / (\mu* (h - hg')^2)$
 $= 6 \cdot 1469289 / (0.26 (82.5 - 8)^2)$
 $= 6004.3530 \text{ kgf/cm}^2$

Required tubesheet thickness for Bending Stress [HreqB]:
 $= h + Cats + Catc$
 $= 82.5 + 0 + 0$
 $= 82.4757 \text{ mm}$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:
 $= \max(HreqB; HreqS)$
 $= \max(82.5 ; 14.9)$
 $= 82.4757 \text{ mm}$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:
 $= | Ps - Pt |$
 $= | 63.5 - 93.5 |$
 $= 29.994 \text{ kgf/cm}^2$

Shear Stress check [τ_{limit}]:
 $= 4 \cdot \mu \cdot h/Do \cdot 0.8 \cdot \beta T \cdot Sy$
 $= 4 \cdot 0.26 \cdot 245/1260 \cdot 0.8 \cdot 0.95 \cdot 3161$



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= 494.596 kgf/cm²

| Ps - Pt | is ≤ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:
 = $1 / (4 \cdot \mu) \cdot (Do / (0.8 \cdot \beta T \cdot Sy)) \cdot |Ps - Pt| + Cats + Catc$
 = $1 / (4 \cdot 0.26) \cdot (1260 / 2402) \cdot |63.5 - 93.5| + 0$
 = 14.8575 mm

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 30.000mm
 Min. Shell len. adjacent to the tubesheet is: 355.472 mm

Note: For a given Channel thickness of: 74.000 mm
 Min. Channel len. adjacent to the tubesheet is: 558.290mm

Tube Metal Area [At]:
 = $\pi (do - t) t$
 = $\pi (25 - 2.5) 2.5$
 = 1.767 cm²

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:
 = $\pi \cdot t (do - t) S$
 = $\pi \cdot 2.5 (25 - 2.5) 1986$
 = 3509.180 kgf

Groove Weld Strength [Fg]:
 = $0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw$ (but not > Ft)
 = $0.85 \cdot \pi \cdot 2 (25 + 0.67 \cdot 2) 1986$
 = 2793.4949 kgf

Weld Strength Factor [fw]:
 = S / Sw where $Sw = \min(S \text{ and } St)$
 = $1986 / 1986$
 = 1.000

Design Strength Ratio [fd]:
 = 1 for full strength welds

Fillet Weld Strength [Ff]:
 = $0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw$ (but not > Ft)
 = $0.55 \cdot \pi \cdot 2 (25 + 0.67 \cdot 2) 1986$
 = 1807.5555 kgf

Minimum Length of Weld [ar]:
 = $2 (\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd)} - 0.75 \cdot do)$
 = $2 (\sqrt{((0.75 \cdot 25)^2 + 1.07 \cdot 2.5 \cdot (25 - 2.5) 1 \cdot 1)} - 0.75 \cdot 25)$
 = 3.083 mm

Minimum Required Fillet Weld Leg [afr]: 1.5416 mm
 Minimum Required Groove Weld Leg [agr]: 1.5416 mm

Tube-Tubesheet Jt allowable, 3509 is >= tube strength 3509 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
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Tubesheet bend. stress	601.3	<=	6005.8 kgf/cm ²	Ok
Tubesheet shear stress	145.7	<=	2402.3 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	82.476	245.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.542	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.542	2.000	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	135.673	...	1606	6006 454 2402	Ps+Pt D1	Ok
D2uc	113.552	...	1145	6006 308 2402	Ps+Pt D2	Ok
D3uc	82.476	...	601	6006 146 2402	Ps+Pt D3	Ok
Max:	135.6731	... mm	0.267	0.189 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	3778	...	4504 Ok
D2uc	...	...	...	3778	...	4504 Ok
D3uc	...	...	...	3778	...	4504 Ok
Max	...	...	...	...	...	

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	135.673	245.000	Ok
Tube thickness :	1.077	2.500	Ok
Tube-tubesheet fillet weld leg :	1.542	2.000	Ok
Tube-tubesheet groove weld leg :	1.542	2.000	Ok



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ASME TS Calc: Pressure Test

Case: 1 9:32a Aug 12, 2026

Given Shell Thickness: 30.0000 mm

Given Channel Thickness: 74.0000 mm

Min Shell length of thk, (30.000) adj. to tubesheet: 355.472 mm

Min Channel length of thk, (74.000) adj. to tubesheet: 558.290 mm

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	...	320	No Calc	No Calc
20	30	No Calc	...	205	No Calc	No Calc
30	Tsht	149.997	1448	72.5	No Calc	No Calc
Tsht	40	No Calc	...	72.5	No Calc	No Calc
40	50	30.003	1360	27	No Calc	No Calc
50	60	175	2419.78	49	No Calc	No Calc
60	70	504.167	2390	42	No Calc	No Calc
70	80	No Calc	2381.6	37.8	No Calc	No Calc
80	90	No Calc	1459.5	26.75	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	320	132.791	...	No Calc
20	30	205	126.975	...	No Calc
30	Tsht	74	No Calc	...	174
Tsht	40	74	No Calc	...	No Calc
40	50	30	5.20793	1.033	63.2
50	60	52	10.4319	1.033	40.6
60	70	45	12.9732	1.033	28.8
70	80	40.8	8.57211	1.033	31.8
80	90	29.75	6.39382	1.033	39

Minimum

28.8

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	149.997	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	30.003	2539997	No Calc	No Calc
50	60	175	175	No Calc	No Calc
60	70	504.167	2539997	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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 Element and Detail Weights: Step: 14 9:32am Aug 12,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	7255.66	...	7255.66	...	725.566
20	30	2885.39	419435	2885.39	419435	288.539
30	40	3713.49	1990988	3642.19	2000188	371.349
40	50	291.452	267429	262.898	271113	29.1452
50	60	4746.1	3804936	4480.6	3839194	474.61
60	70	12332.4	17643536	11524.9	17747722	1233.24
70	80	2591.79	1800398	2429.8	1813980	259.179
80	90	729.659	436159	667.117	441458	72.9659
Total		34545	26362884.00	33148	26533092.00	3454



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
20	Liqd	12.5774	15.8	...	...	Liquid: 20
30	Liqd	59.7028	75	...	...	Liquid: 30
30	Insl	163.462	75	...	...	Hot insulation
30	Nozl	630.539	75	90.4	...	N1- Channel inle
30	Nozl	630.539	75	-90.4	...	N2- Channel outl
40	Liqd	267.308	15	...	...	Liquid: 40
50	Liqd	3803.21	72.8874	-25	...	Liquid: 50
50	Insl	277.924	87.5	...	...	Hot insulation
60	Sadl	424.719	40	130.75	...	Left Saddle
60	Sadl	424.719	430	130.75	...	Right saddle
60	Liqd	17635.6	240	...	...	Liquid: 60
60	Insl	832.394	240	...	...	Hot
60	Nozl	3805.84	210	184.99	...	N4- Demister
60	Nozl	80.6509	90	-121.59	...	N3- Shell inlet
60	Nozl	177.407	50	125.795	...	N4- Shell outlet
60	Nozl	27.7037	140	-117.143	...	N5- Blow down
60	Nozl	28.3901	200	-118.332	...	N8- MP steam
60	Nozl	11.7705	350	117.143	...	K1- PSV
60	Nozl	77.0583	400	121.59	...	K2- PSV
60	Nozl	9.058	350	-116.699	...	K9- Temperature
60	Nozl	6.01206	450	...	115.761	N10- Chemical ad
60	Wght	9500	240	...	...	Extra weight fo
70	Liqd	1799.59	24.1667	0.14534E-04	...	Liquid: 70
70	Insl	192.651	31.25	...	...	Hot Insulation
70	Nozl	1399.29	62.5	...	...	MH1- Manhole
30	FTsh	2529.13	162.25	...	...	Tubesheet
30	Tube	9806.26	480	...	...	Tubes
40	Baf1	39.7	200	...	...	Baffles
40	Baf1	39.7	250	...	...	Baffles
40	Baf1	39.7	300	...	...	Baffles
40	Baf1	39.7	350	...	...	Baffles
40	Baf1	39.7	400	...	...	Baffles
40	Baf1	39.7	450	...	...	Baffles
40	Baf1	39.7	500	...	...	Baffles
40	Baf1	39.7	550	...	...	Baffles



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 Element and Detail Weights: Step: 14 9:32am Aug 12,2026

Total Weight of Each Detail Type:

Saddles	849.4
Liquid	23578.0
Insulation	1466.4
Nozzles	6884.3
Weights	9500.0
Exchanger Components	12653.0
Liquid in Tubes	63.3
Sum of the Detail Weights	54994.4 kg

Note:
 Because some nozzles were attached to a head or flange, their weights include the attached head or flange.

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	37197.9	37197.9	37197.9	37197.9	37197.9	37197.9
Saddles	849.4	849.4	849.4	849.4	849.4	849.4
Nozzles	6884.3	6884.3	6884.3	6884.3	6884.3	6884.3
Wld Weights	9500.0	9500.0	9500.0	9500.0	9500.0	9500.0
Exchanger	12653.0	12653.0	12653.0	12653.0	12653.0	12653.0
Insulation	...	...	...	...	1466.4	1466.4 *
Ope. Liquid	...	...	...	...	...	23578.0
Tube Ope Lqd	...	...	...	...	...	63.3
Test Liquid	...	26351.0	...	...	...	...
Tube Tst Lqd	...	2110.9	...	...	...	...
Totals	67084.6	95546.4	67084.6	67084.6	68551.0	92192.3

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	67084.6 kg
Shop Test Wt.	- Fabricated weight + water (full)	95546.4 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	67084.6 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	67084.6 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	68551.0 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	92192.3 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	90724.0 kg

Exchanger Tube Data

Volume of Exchanger tubes :	2111825.0 cm ³
Weight of Ope Liq in tubes :	63.3 kg
Weight of Water in tubes :	2110.9 kg

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	48657.8	...



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Element and Detail Weights: Step: 14 9:32am Aug 12,2026

20	30	33712.3	12905.7
30	40	68235.4	61261.1
40	50	12817.7	12252.2
50	60	109769	102920
60	70	360404	346832
70	80	65993.2	60955.7
80	90	25732.9	23445.2

Total	725321.9	620571.8 cm ²
Total	780.7	668.0 ft ²

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 FileName : 401-E-31003_Waste Heat Boiler-RP Page 74 of 308
 Nozzle Flange MAWP: Step: 15 9:32am Aug 12,2026

Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N1- Channel in	195.20	433.49	450	2500	GR 1.1	...	...	...	...
N3- Shell inle	83.88	104.01	270	600	GR 1.1	42.593	83.234	83.885	83.234
N4- Shell outl	83.88	104.01	270	600	GR 1.1	32.421	83.885	83.885	83.885
N5- Blow down	83.88	104.01	270	600	GR 1.1	67.584	58.243	16.300	58.243
N8- MP steam	83.88	104.01	270	600	GR 1.1	54.086	71.741	83.885	71.741
K1- PSV	83.88	104.01	270	600	GR 1.1	67.584	58.243	16.300	58.243
K2- PSV	83.88	104.01	270	600	GR 1.1	42.593	83.234	83.885	83.234
K9- Temperatur	83.88	104.01	270	600	GR 1.1	...	...	...	...
N10- Chemical	83.88	104.01	270	600	GR 1.1	...	...	...	...
MH1- Manhole	83.88	104.01	270	600	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 58.243 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 104.014 kgf/cm²

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 195.198 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 433.491 kgf/cm²

Selected Method for Derating ASME B16.5 Flange MAWP: ASME 4.16.12

The 4.16.12 Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the 4.16.12 method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm²	Maximum Pressure Pmax kgf/cm²	Pass Fail
N3- Shell inlet	192.151	0.50	569	510	50.800	83.234	Pass
N4- Shell outlet	296.037	0.50	1588	857	50.800	83.885	Pass
N5- Blow down	77.851	0.50	59	163	50.800	58.243	Pass
N8- MP steam	111.125	0.50	139	255	50.800	71.741	Pass
K1- PSV	77.851	0.50	59	163	50.800	58.243	Pass
K2- PSV	192.151	0.50	569	510	50.800	83.234	Pass

Evaluating Flange on Nozzle [N3- Shell inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(569) + 4 \cdot \text{abs}(510) \cdot 192.151 \leq \pi \cdot 192.151^3 \cdot ((83.885 - 50.800) + .500 \cdot 83.885)$$

$$= 0.94932E+04 \leq 0.16722E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3- Shell inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(83.885, 0.500 \cdot 83.885 + 83.885 - 9493.2 / (\pi \cdot 192.151^3))$$

$$= \min(83.885, 83.234)$$

$$= 83.234 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [N4- Shell outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1588) + 4 \cdot \text{abs}(857) \cdot 296.037 \leq \pi \cdot 296.037^3 \cdot ((83.885 - 50.800) + .500 \cdot 83.885)$$

$$= 0.26425E+05 \leq 0.61152E+05 \text{ Check Passes}$$



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 Nozzle Flange MAWP: Step: 15 9:32am Aug 12,2026

Maximum Pressure for Flange on Nozzle N4- Shell outlet [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(83.885, 0.500 \cdot 83.885 + 83.885 - 26425.0 / (\pi \cdot 296.037^3))$
 $= \min(83.885, 93.406)$
 $= 83.885 \text{ kgf/cm}^2$

Evaluating Flange on Nozzle [N5- Blow down]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$
 $= 16(59) + 4 \cdot \text{abs}(163)77.851 \leq \pi \cdot 77.851^3 [(83.885 - 50.800) + .500 \cdot 83.885]$
 $= 0.10018E+04 \leq 0.11121E+04 \text{ Check Passes}$

Maximum Pressure for Flange on Nozzle N5- Blow down [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(83.885, 0.500 \cdot 83.885 + 83.885 - 1001.8 / (\pi \cdot 77.851^3))$
 $= \min(83.885, 58.243)$
 $= 58.243 \text{ kgf/cm}^2$

Evaluating Flange on Nozzle [N8- MP steam]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$
 $= 16(139) + 4 \cdot \text{abs}(255)111.125 \leq \pi \cdot 111.125^3 [(83.885 - 50.800) + .500 \cdot 83.885]$
 $= 0.23317E+04 \leq 0.32345E+04 \text{ Check Passes}$

Maximum Pressure for Flange on Nozzle N8- MP steam [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(83.885, 0.500 \cdot 83.885 + 83.885 - 2331.7 / (\pi \cdot 111.125^3))$
 $= \min(83.885, 71.741)$
 $= 71.741 \text{ kgf/cm}^2$

Evaluating Flange on Nozzle [K1- PSV]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$
 $= 16(59) + 4 \cdot \text{abs}(163)77.851 \leq \pi \cdot 77.851^3 [(83.885 - 50.800) + .500 \cdot 83.885]$
 $= 0.10018E+04 \leq 0.11121E+04 \text{ Check Passes}$

Maximum Pressure for Flange on Nozzle K1- PSV [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(83.885, 0.500 \cdot 83.885 + 83.885 - 1001.8 / (\pi \cdot 77.851^3))$
 $= \min(83.885, 58.243)$
 $= 58.243 \text{ kgf/cm}^2$

Evaluating Flange on Nozzle [K2- PSV]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$
 $= 16(569) + 4 \cdot \text{abs}(510)192.151 \leq \pi \cdot 192.151^3 [(83.885 - 50.800) + .500 \cdot 83.885]$
 $= 0.94932E+04 \leq 0.16722E+05 \text{ Check Passes}$

Maximum Pressure for Flange on Nozzle K2- PSV [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(83.885, 0.500 \cdot 83.885 + 83.885 - 9493.2 / (\pi \cdot 192.151^3))$
 $= \min(83.885, 83.234)$
 $= 83.234 \text{ kgf/cm}^2$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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 Wind Load Calculation: Step: 16 9:32am Aug 12,2026

Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Use Gust Response Factor (Dynamic Analysis)	No	

Calculated Values:

Vessel Length to Diameter ratio: 2.000

Design Wind Speed [Vz]:
 = Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)
 = 50 • 1 • 1 • 1 • 1.15
 = 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:
 = max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)
 = max(0.7 • 1984, 1 • 0.9 • 1 • 0.6 • 57.5²)
 = 1789.5 N/m²

Element Wind Load (operating condition):
 = Wind Area • Cf • pd
 = 0.74 • 0.7 • 1789
 = 928.3 N

From	Height m	k1	k2	k3	Vz m/sec	Cf
10	1.82	1.0000	1.0000	1.0000	57.50	0.7000
20	1.82	1.0000	1.0000	1.0000	57.50	0.7000
30	1.82	1.0000	1.0000	1.0000	57.50	0.7000
40	1.82	1.0000	1.0000	1.0000	57.50	0.7000
50	1.82	1.0000	1.0000	1.0000	57.50	0.7000
60	1.82	1.0000	1.0000	1.0000	57.50	0.7000
70	1.82	1.0000	1.0000	1.0000	57.50	0.7000
80	2.35	1.0000	1.0000	1.0000	57.50	0.7000

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm²	Elevation cm	Pressure kgf/m²	Force kgf
Extra weight	0.00	182.00	182.55	0.00

Wind Load Calculation:

From	To	Wind Height	Wind Diameter	Wind Area	Wind Pressure	Element Wind Load
------	----	----------------	------------------	--------------	------------------	----------------------



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Wind Load Calculation: Step: 16 9:32am Aug 12,2026

		cm	cm	cm ²	kgf/m ²	kgf
10	20	182	231.6	7411.2	182.55	104.137
20	30	182	156	4929.6	182.55	69.2671
30	40	182	208.56	31284	182.55	439.58
40	50	182	163.2	4896	182.55	68.795
50	60	182	263.28	46074	182.55	647.398
60	70	182	321.6	154368	182.55	2169.07
70	80	182	320.592	17566.6	182.55	246.833
80	90	234.531	175.14	6099.34	182.55	85.7034

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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 FileName : 401-E-31003_Waste Heat Boiler-RP Page 78 of 308
 Earthquake Load Calculation: Step: 17 9:32am Aug 12,2026

Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
 Zone Number 2
 Soil Type Rocky or Hard Soil
 Response reduction factor (Force factor) R 2.0000
 Time Period of Vibration (program or user) 0.0303 Seconds
 Value of Sa/g from user Spectrum Data 0.0939
 Percent Damping 2.0 %
 Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.094 \cdot 1) \cdot 1$
 $= 0.094$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.03 secs
 Computed Sa/g = 0.094

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
 The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 10149 \cdot 0.094 \cdot 1$
 $= 952.6 \text{ kgf}$

Vertical load per 8.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.094 \cdot 91343 \cdot 1$



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Earthquake Load Calculation: Step: 17 9:32am Aug 12,2026

= 5715.4 kgf

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf
10	20	96.5	10149.2	952.564
20	30	65	10149.2	952.564
30	40	65	10149.2	952.564
40	50	65	10149.2	952.564
50	60	65	10149.2	952.564
60	Sad1	115	10149.2	952.564
Sad1	70	115	10149.2	952.564
60	70	115	10149.2	952.564
70	80	115	10149.2	952.564

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 18 9:32am Aug 12,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	678.600 cm
Center of Gravity of the Liquid	641.986 cm
Center of Gravity of the Insulation	596.471 cm
Center of Gravity of the Nozzles	623.844 cm
Center of Gravity of the Added Weights (Operating)	683.600 cm
Center of Gravity of the Added Weights (Empty)	683.600 cm
Center of Gravity of the Tubesheet(s)	226.350 cm
Center of Gravity of the Tubes	544.100 cm
Center of Gravity of the Baffles	613.600 cm
Center of Gravity of Bare Shell New and Cold	403.345 cm
Center of Gravity of Bare Shell Corroded	394.433 cm
Vessel CG in the Operating Condition	523.700 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	483.089 cm
Vessel CG in the Test Condition	516.224 cm

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 Saddle Calcs: Operating Case: Step: 19 9:32am Aug 12,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

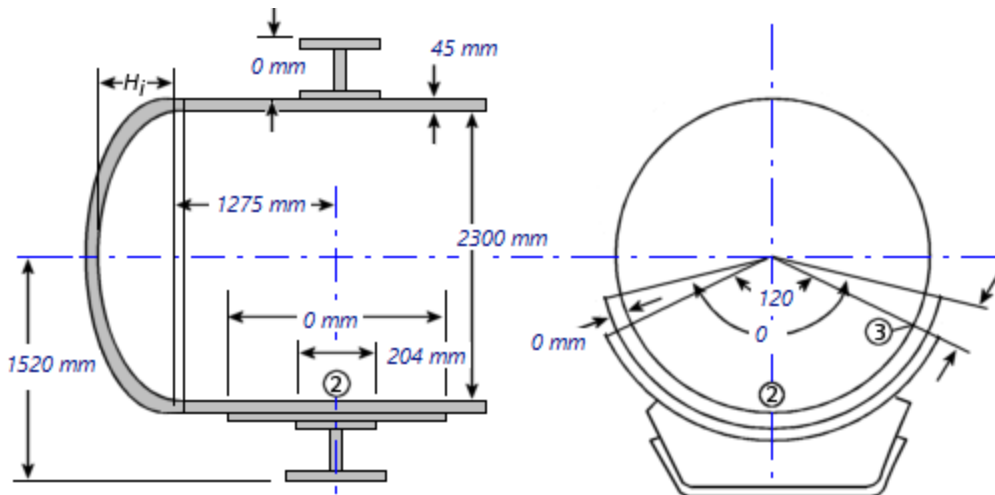
Horizontal Vessel Stress Calculations : Operating Case

Note:
 Wear Pad Width (306.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$
 $= \min(204 + 1.56 \cdot \sqrt{ 1174 \cdot 42 }, 2 \cdot 1275)$
 $= 550.4043 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	1174.00	mm
Shell Thickness used in this Case	t	45.000	mm
Stiffened Vessel Length per 4.15.6	L	690.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1275.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		1784.41	kgf/cm ²
Head Allowable Stress used in Calculation		1570.34	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1520.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		94872.44	kgf
Pressure used in Analysis	P	50.082	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	671.27	1784.41
Long. Stress at Bottom of Midspan	728.64	1784.41
Long. Stress at Top of Saddles	766.37	1784.41
Long. Stress at Bottom of Saddles	663.15	1784.41



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 Saddle Calcs: Operating Case: Step: 19 9:32am Aug 12,2026

Tangential Shear in Shell	142.01	1423.92
Circ. Stress at Horn of Saddle	682.97	2230.52
Circ. Compressive Stress in Shell	31.20	1784.41

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (3831/2 + 0) \cdot 1520/2087$
 $= 4184.8\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(1297, 0) \cdot 1520/3900$
 $= 505.6\ kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(8573, 0) \cdot 1520/3900$
 $= 3341.3\ kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (8573/2 + 0) \cdot 1520/2087$
 $= 9365.3\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 85507 + \max(506, 4185, 3341, 9365)$
 $= 94872.4\ kgf$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1790 (0 + \pi/4(2.39 \cdot 1.2)^2)$
 $= 12721.669\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	95297.16	kgf
Transverse Shear Load	4286.54	kgf
Longitudinal Shear Load	8573.08	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(2300/2 + 45 + 10) \sin(120/2), 2 \cdot 1174)$
 $= 2087.121\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -94872 \cdot 127 [1 - (1 - 127/690 + (117^2 - 0^2)/(2 \cdot 127 \cdot 690)) / (1 + (4 \cdot 0)/(3 \cdot 690))]$
 $= -12876.4\ kgf \cdot m$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$



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$$= 94872 \cdot 690 / 4 \cdot ((1 + 2(117^2 - 0^2) / (690^2)) / (1 + (4 \cdot 0) / (3 \cdot 690))) - 4 \cdot 127 / 690$$

$$= 52168.1 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 50.1 \cdot 1174 / (2 \cdot 42) - 52168 / (\pi \cdot 1174^2 \cdot 42)$$

$$= 671.27 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 50.1 \cdot 1174 / (2 \cdot 42) + 52168 / (\pi \cdot 1174^2 \cdot 42)$$

$$= 728.64 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 50.1 \cdot 1174 / (2 \cdot 42) - 12876 / (0.11 \cdot \pi \cdot 1174^2 \cdot 42)$$

$$= 766.37 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 50.1 \cdot 1174 / (2 \cdot 42) + 12876 / (0.19 \cdot \pi \cdot 1174^2 \cdot 42)$$

$$= 663.15 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 94872 (690 - 2 \cdot 127) / (690 + (4 \cdot 0 / 3))$$

$$= 59810.9 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 59811 / (1174 \cdot 42)$$

$$= 142.01 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1174 \cdot 42)}$$

$$= 173.202 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 94872 \cdot 0.1 / (42(204 + 173 + 173))$$

$$= -31.20 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -94872 / (4 \cdot 42(204 + 173 + 173)) - 12 \cdot 0.053 \cdot 94872 \cdot 1174 / (690 \cdot 42^2)$$

$$= -682.97 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{(1174 \cdot 42)}, 2 \cdot 127)$$

$$= 550.40 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 3900.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.13139\text{E-}04 \cdot 3900(270 - 21.1)$$

$$= 12.753 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length BpLen 2100.0000 mm



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Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	340.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2671.5	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$
 $= ((1.56 \cdot \sqrt{(1153 \cdot 42) + 306})^3) / 12$
 $= 400.873 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 401 + 273 (105 - 21)^2$
 $= 19614.586 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	649.3	42.0	21.0	272.7	572675.6	0.196E+05
Wearplate	306.0	10.0	47.0	30.6	143820.0	0.103E+04
Web	10.0	289.0	196.5	28.9	567885.1	0.443E+04
BasePlate	340.0	26.0	354.0	88.4	3129360.0	0.549E+05
Totals	...	...	...	420.6	4413741.0	0.800E+05

Distance to Centroid [C1]:
 $= AY / A$
 $= 4413741 / 421$
 $= 104.938 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 94872$
 $= 19308.5977 \text{ kgf}$

Tension Stress, St = (Fh/As) = 130.5519 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1602.9094 kgf/cm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1520 - 1153 \cdot \sin(120/2) / 1.05$
 $= 566.477 \text{ mm}$

Bending Moment, M = Fh * d = 10937.8838 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 143.5394 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1781.0104 kgf/cm²



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Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(94872 + 425)340}{4 \cdot 2100 \cdot 1784}}$$

$$= 25.466 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1150 + 45 + 10)$$

$$= 2087.121 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2087 / (6 - 1)$$

$$= 417.424 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 417 \cdot 340 / 2$$

$$= 709.621 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 95297 / (2100 \cdot 340)$$

$$= 13.347 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 710 \cdot 13.3$$

$$= 9471.271 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19.4 + 20.9$$

$$= 40.271 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 9471 / 40.3$$

$$= 235.187 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	204.0	...	20.4	...	707.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204 - 10$$

$$= 194.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 40.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:



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= Saddle Width / 2
 = 204/2
 = 102.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{707/40.3}$
 = 41.914 mm

Length of Outer Rib [L]:
 = Saddle Height - $\cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 = 1520 - $\cos(120/2) (1150 + 45 + 10) - 26$
 = 891.500 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 2038903/2672}$
 = 122.739

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 892/41.9
 = 21.270

Bending Moment [Rm]:
 $= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
 $= 8573 / (2 \cdot 2100) \cdot 417 \cdot 892/2$
 = 379.802 kgf-m.

Compressive Allowable, $KL/r < Cc$ (21.3 < 123) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (21.3)^2 / (2 \cdot 123^2)) 2672 / (5/3 + 3 \cdot (21.3) / (8 \cdot 123) - (21.3)^3 / (8 \cdot 123^3))$
 = 1520 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 235/1520 + (380 \cdot 102/7074720) / 1781$
 = 0.462

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	194.0	0.0	19.4	0.0	707.
Web	417.4	10.0	0.0	41.7	0.0	3.48
Totals	...	...	...	61.1	...	711.

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0/61.1$
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 204/2
 = 102.000 mm

Length of Inner Rib [L]:
 $= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$
 $= 1520 - \sqrt{(1205 + 10)^2 - (417/2)^2} - 26$
 = 307.213 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{711/61.1}$$

$$= 34.098 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 307/34.1$$

$$= 9.010$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 8573 / (2 \cdot 2100)$$

$$= 2.041 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.04 \cdot 417 \cdot 307$$

$$= 261.761 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 262/69693$$

$$= 375.592 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($9.01 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.01)^2 / (2 \cdot 123^2)) 2672 / (5/3 + 3 \cdot (9.01) / (8 \cdot 123) - (9.01^3) / (8 \cdot 123^3))$$

$$= 1573 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 262/1573 + (262 \cdot 102/711) / 1781$$

$$= 0.378$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1399.95 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	6.0062 cm ²
Saddle Load Q0 (Weight)	Q0	85931.8 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	3341.3 kgf
Maximum Transverse Force	Ft	4286.5 kgf
Maximum Longitudinal Force	F1	8573.1 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 8573 / (6.01 \cdot 4)$$

$$= 356.8 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4287 / (6.01 \cdot 4)$$

$$= 178.4 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
 = $8573 / (1100 \cdot 4)$
 = 1.9485 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1520 \cdot 4287 + 0$
 = 6515.55 kgf-m.

Eccentricity (e):
 = R_{mom} / QO
 = $6516 / 85932$
 = 75.82 mm < $B_{\text{plen}} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0, 1.95, 0]$
 = 1.9485 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: D

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -48 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : D
 From Figure 3.8M- with PWHT, the MDMT is : -48 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:
 = $(t_r \cdot E^*) / (t_{\text{nom}} - c_i - c_{\text{ext}})$
 = $(33.3 \cdot 1) / (10 - 3 - 0)$
 = 1.000

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (306.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 = $\min(204 + 1.56 \cdot \sqrt{1174 \cdot 42}, 2 \cdot 1300)$
 = 550.4043 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1174.00	mm
Shell Thickness used in this Case	t	45.000	mm
Stiffened Vessel Length per 4.15.6	L	690.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1300.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees



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Inside or Mean Depth of Head	Hi or hm	59.69	cm
Shell Allowable Stress used in Calculation		1784.41	kgf/cm ²
Head Allowable Stress used in Calculation		1784.41	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1520.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		21719.10	kgf
Pressure used in Analysis	P	50.082	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		696.22	1784.41
Long. Stress at Bottom of Midspan		703.70	1784.41
Long. Stress at Top of Saddles		732.18	1784.41
Long. Stress at Bottom of Saddles		682.10	1784.41
Tangential Shear in Shell		28.81	1423.92
Circ. Stress at Horn of Saddle		156.35	2230.52
Circ. Compressive Stress in Shell		7.14	1784.41

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (3831/2 + 0) \cdot 1520/2087$
 $= 4184.8 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(1297, 0) \cdot 1520/3900$
 $= 505.6 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(8573, 0) \cdot 1520/3900$
 $= 3341.3 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (8573/2 + 0) \cdot 1520/2087$
 $= 9365.3 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 12354 + \max(506, 4185, 3341, 9365)$
 $= 21719.1 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1790 (0 + \pi/4 (2.39 \cdot 1.2)^2)$
 $= 12721.669 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	22143.82	kgf
Transverse Shear Load	4286.54	kgf
Longitudinal Shear Load	8573.08	kgf



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Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2300/2 + 45 + 10)\sin(120/2), 2 \cdot 1174)$$

$$= 2087.121 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -21719 \cdot 130 [1 - (1 - 130/690 + (117^2 - 59.7^2)/(2 \cdot 130 \cdot 690))/(1 + (4 \cdot 59.7)/(3 \cdot 690))]$$

$$= -6247.3 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L]$$

$$= 21719 \cdot 690/4 [(1 + 2(117^2 - 59.7^2)/(690^2))/(1 + (4 \cdot 59.7)/(3 \cdot 690)) - 4 \cdot 130/690]$$

$$= 6798.3 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 50.1 \cdot 1174/(2 \cdot 42) - 6798/(\pi \cdot 1174^2 \cdot 42)$$

$$= 696.22 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 50.1 \cdot 1174/(2 \cdot 42) + 6798/(\pi \cdot 1174^2 \cdot 42)$$

$$= 703.70 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 50.1 \cdot 1174/(2 \cdot 42) - 6247/(0.11 \cdot \pi \cdot 1174^2 \cdot 42)$$

$$= 732.18 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m/(2t) + M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 50.1 \cdot 1174/(2 \cdot 42) + 6247/(0.19 \cdot \pi \cdot 1174^2 \cdot 42)$$

$$= 682.10 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m)/3)$$

$$= 21719(690 - 2 \cdot 130)/(690 + (4 \cdot 59.7/3))$$

$$= 12135.4 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 12135 / (1174 \cdot 42)$$

$$= 28.81 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1174 \cdot 42}$$

$$= 173.202 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 21719 \cdot 0.1 / (42(204 + 173 + 173))$$



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$$= -7.14 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) $[\sigma_7^*]$:

$$\begin{aligned} &= -Q / (4 \cdot t \cdot (b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2) \\ &= -21719 / (4 \cdot 42 \cdot (204 + 173 + 173)) - \\ &\quad 12 \cdot 0.053 \cdot 21719 \cdot 1174 / (690 \cdot 42^2) \\ &= -156.35 \text{ kgf/cm}^2 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned} &= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ &= \min(204 + 1.56 \cdot \sqrt{1174 \cdot 42}, 2 \cdot 130) \\ &= 550.40 \text{ mm} \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2100.0000	mm
Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	340.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2671.5	kgf/cm ²
Height of Web at Center	Hw, c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned} &= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})t^3) / 12 \\ &= ((1.56 \cdot \sqrt{1153 \cdot 42} + 306)42^3) / 12 \\ &= 400.873 \text{ cm}^4 \end{aligned}$$

Resolved Inertia for the Shell [I,shell]:

$$\begin{aligned} &= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2 \\ &= 401 + 273 (105 - 21)^2 \\ &= 19614.586 \text{ cm}^4 \end{aligned}$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	649.3	42.0	21.0	272.7	572675.6	0.196E+05
Wearplate	306.0	10.0	47.0	30.6	143820.0	0.103E+04
Web	10.0	289.0	196.5	28.9	567885.1	0.443E+04
BasePlate	340.0	26.0	354.0	88.4	3129360.0	0.549E+05
Totals	...	...	...	420.6	4413741.0	0.800E+05

Distance to Centroid [C1]:

$$\begin{aligned} &= AY / A \\ &= 4413741 / 421 \\ &= 104.938 \text{ mm} \end{aligned}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle} / 2 \\ &= 180 - 120 / 2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$



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$$= 0.2 \cdot 21719$$

$$= 4420.3081 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 29.8872 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1602.9094 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1520 - 1153 \cdot \sin(120/2) / 1.05$$

$$= 566.477 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2504.0042 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 32.8604 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1781.0104 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(21719 + 425) 340 / (4 \cdot 2100 \cdot 1784)}$$

$$= 12.276 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1150 + 45 + 10)$$

$$= 2087.121 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N\text{ribs} - 1)$$

$$= 2087 / (6 - 1)$$

$$= 417.424 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 417 \cdot 340 / 2$$

$$= 709.621 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 22144 / (2100 \cdot 340)$$

$$= 3.101 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 710 \cdot 3.1$$

$$= 2200.802 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19.4 + 20.9$$

$$= 40.271 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 2201 / 40.3$$

$$= 54.650 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction



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	B	D	Y	A	AY	Io
Rib+Web	10.0	204.0	...	20.4	...	707.

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 204 - 10
 = 194.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/40.3
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 204/2
 = 102.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{707/40.3}$
 = 41.914 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1520 - cos(120/2)(1150 + 45 + 10) - 26
 = 891.500 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 2038903/2672}$
 = 122.739

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 892/41.9
 = 21.270

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 8573 / (2 • 2100) • 417 • 892/2
 = 379.802 kgf-m.

Compressive Allowable, KL/r < Cc (21.3 < 123) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (21.3)^2 / (2 \cdot 123^2)) 2672 / (5/3 + 3 \cdot (21.3) / (8 \cdot 123) - (21.3^3) / (8 \cdot 123^3))$
 = 1520 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I)/Sba
 = 54.6/1520 + (380 • 102/7074720)/1781
 = 0.343

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	194.0	0.0	19.4	0.0	707.
Web	417.4	10.0	0.0	41.7	0.0	3.48
Totals	...	...	...	61.1	...	711.



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Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/61.1$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204/2$$

$$= 102.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1520 - \sqrt{(1205 + 10)^2 - (417/2)^2} - 26$$

$$= 307.213 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(711/61.1)}$$

$$= 34.098 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 307/34.1$$

$$= 9.010$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 8573 / (2 \cdot 2100)$$

$$= 2.041 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.04 \cdot 417 \cdot 307$$

$$= 261.761 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 262/69693$$

$$= 375.592 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($9.01 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.01)^2 / (2 \cdot 123^2)) 2672 / (5/3 + 3 \cdot (9.01) / (8 \cdot 123) - (9.01^3) / (8 \cdot 123^3))$$

$$= 1573 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 61/1573 + (262 \cdot 102/711) / 1781$$

$$= 0.250$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1399.95 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	6.0062 cm ²
Saddle Load Q0 (Weight)	Q0	12778.5 kgf



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Saddle Load QL (Wind/Seismic contribution)	QL	3341.3 kgf
Maximum Transverse Force	Ft	4286.5 kgf
Maximum Longitudinal Force	F1	8573.1 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:
 = F_l / (Bolt Area • Number of Bolts)
 = 8573 / (6.01 • 4)
 = 356.8 kgf/cm². Must be less than 1100.0 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:
 = F_t / (Bolt Area • Number of Bolts)
 = 4287 / (6.01 • 4)
 = 178.4 kgf/cm². Must be less than 1100.0 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = F_l / (BoltShearAllowable • Nbolts)
 = 8573 / (1100 • 4)
 = 1.9485 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • F_t + Sum of X Moments
 = 1520 • 4287 + 0
 = 6515.55 kgf-m.

Eccentricity (e):
 = Rmom / QO
 = 6516 / 12778
 = 509.88 mm > Bplen/6 --> Uplift in Transverse direction

f = Bplen / 2 - Edgedis
 = 2100 / 2 - 50
 = 1000.00 mm

Modular Ratio Of Steel/Concrete (n1):
 = E_S / E_C
 = 2074056 / 219500
 = 9.45

K1 = 3 (e - 0.5 • Bplen)
 = 3 (510 - 0.5 • 2100)
 = -1620.35 mm

K2 = 6 • n1 • A_t / Bpwid • (f + e)
 = 6 • 9.45 • 12/340 • (1000 + 510)
 = 302434.78 mm²

K3 = -K2 • (0.5 • Bplen + f)
 = -302435 • (0.5 • 2100 + 1000)
 = -619991353.67 mm³

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -1620 \cdot Y^2 + 302435 \cdot Y + -619991296 = 0$
 Y = 1662.71 mm

Num = (Bplen / 2 - Y / 3 - e)
 = (2100 / 2 - 1663 / 3 - 510)



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$$= -14.12$$

$$\begin{aligned} \text{Denom} &= (\text{Bplen} / 2 - Y / 3 + f) \\ &= (2100/2 - 1663/3 + 1000) \\ &= 1495.76 \end{aligned}$$

$$\begin{aligned} \text{Total Bolt Tension Force [Tforce]:} \\ &= - \text{QO} \cdot \text{Num} / \text{Denom} \\ &= - 12778 \cdot -14.1/1496 \\ &= 120.65 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Bolt Area Required due to Transverse Load [Bltareart]:} \\ &= \text{Tforce} / (\text{Stba} \cdot \text{Nbt}) \\ &= 121 / (1400 \cdot 2) \\ &= 0.0431 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Required Area of a Single Bolt [Bltarear]:} \\ &= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}] \\ &= \max[0, 1.95, 0.043] \\ &= 1.9485 \text{ cm}^2 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

$$\begin{aligned} \text{Bearing Pressure (fc)} \\ &= 2(\text{QO} + \text{Tforce}) / (Y \cdot \text{Bpwid}) \\ &= 2(12778 + 121) / (1663 \cdot 340) \\ &= 4.56 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Distance from Baseplate Edge to the Web [ADIST]:} \\ &= (\text{Bplen} - \text{WebIngh}) / 2 \\ &= (2100 - 2049) / 2 \\ &= 25.4000 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Overturning Moment due To Bolt Tension [Mt]:} \\ &= \text{Tforce} \cdot \text{Adist} \\ &= 121 \cdot 2.54 \\ &= 3.06 \text{ kgf-m.} \end{aligned}$$

$$\begin{aligned} \text{Equivalent Bearing Pressure (f1):} \\ &= \text{fc} \cdot (Y - \text{Adist}) / Y \\ &= 4.56 \cdot (1663 - 25.4) / 1663 \\ &= 4.49 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Overturning Moment due to Bearing Pressure [Mc]:} \\ &= (\text{Adist}^2 \cdot \text{Bpwid} / 6) \cdot (f1 + 2 \cdot \text{fc}) \\ &= (25.4^2 \cdot 340 / 6) \cdot (4.49 + 2 \cdot 4.56) \\ &= 4.98 \text{ kgf-m.} \end{aligned}$$

$$\begin{aligned} \text{Baseplate Required Thickness [Treq]:} \\ &= \sqrt{(6 \cdot \max(\text{Mt}, \text{Mc}) / (\text{Bpwid} \cdot \text{Sba}))} \\ &= \sqrt{(6 \cdot \max(3.06, 4.98) / (340 \cdot 2677))} \\ &= 1.8119 \text{ mm} \end{aligned}$$

MDMT Calculations for: Right saddle, Saddle to Wearplate Junction, Curve: D

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -48 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: D
From Figure 3.8M- with PWHT, the MDMT is	: -48 °C
Temperature reduction per Graph 3.13	: 0 °C
The computed stress ratio is	: 1.000



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Saddle Calcs: Operating Case: Step: 19 9:32am Aug 12,2026

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (33.3 \cdot 1) / (10 - 3 - 0)$$

$$= 1.000$$

Review notes about nozzle loadings and supports in the Warnings report.

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 Saddle Calcs: Test Case: Step: 20 9:32am Aug 12,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

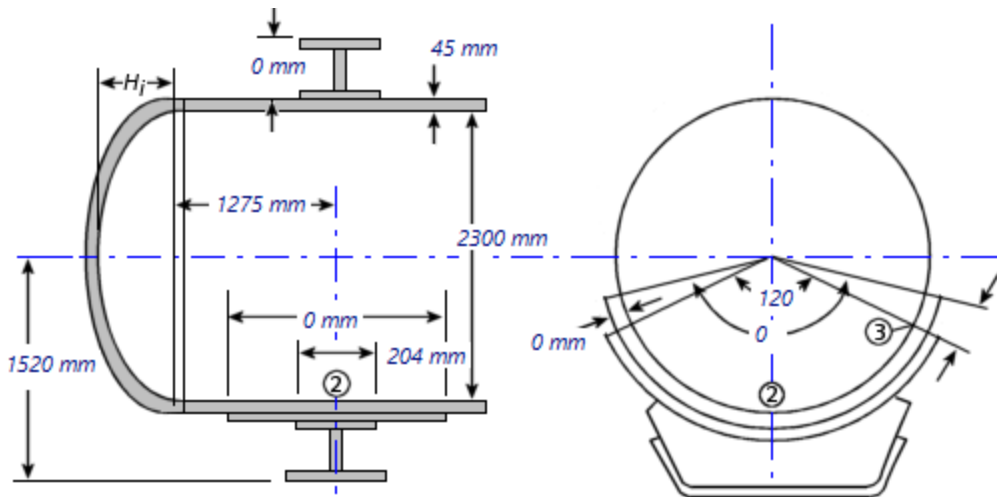
Horizontal Vessel Stress Calculations : Test Case

Note:
 Wear Pad Width (306.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(204 + 1.56 \cdot \sqrt{1172 \cdot 45}, 2 \cdot 1275)$
 $= 562.3334 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	1172.50	mm
Shell Thickness used in this Case	t	45.000	mm
Stiffened Vessel Length per 4.15.6	L	690.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1275.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		2518.57	kgf/cm ²
Head Allowable Stress used in Calculation		2518.57	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1520.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		89245.56	kgf
Pressure used in Analysis	P	63.617	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	803.55	2518.57
Long. Stress at Bottom of Midspan	854.02	2518.57
Long. Stress at Top of Saddles	887.35	2518.57
Long. Stress at Bottom of Saddles	796.32	2518.57



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 Saddle Calcs: Test Case: Step: 20 9:32am Aug 12,2026

Tangential Shear in Shell	124.84	1413.05
Circ. Stress at Horn of Saddle	563.14	3777.85
Circ. Compressive Stress in Shell	26.81	2518.57

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (1264/2 + 0) \cdot 1520/2087$
 $= 1381.0 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(428, 0) \cdot 1520/3900$
 $= 166.8 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 87865 + \max(167, 1381, 0, 0)$
 $= 89245.6 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1790 (0 + \pi/4 (2.39 \cdot 1.2)^2)$
 $= 12721.669 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	89670.28	kgf
Transverse Shear Load	632.08	kgf
Longitudinal Shear Load	428.09	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2300/2 + 45 + 10) \sin(120/2), 2 \cdot 1172)$
 $= 2087.121 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K11* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -89246 \cdot 127 [1 - (1 - 127/690 + (117^2 - 0^2)/(2 \cdot 127 \cdot 690)) / (1 + (4 \cdot 0)/(3 \cdot 690))]$
 $= -12135.4 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 89246 \cdot 690/4 ((1 + 2(117^2 - 0^2)/(690^2)) / (1 + (4 \cdot 0)/(3 \cdot 690)) - 4 \cdot 127/690)$
 $= 49051.2 \text{ kgf-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$
 $= 63.6 \cdot 1172 / (2 \cdot 45) - 49051 / (\pi \cdot 1172^2 \cdot 45)$
 $= 803.55 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



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$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 63.6 \cdot 1172 / (2 \cdot 45) + 49051 / (\pi \cdot 1172^2 \cdot 45)$$

$$= 854.02 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 63.6 \cdot 1172 / (2 \cdot 45) - 12135 / (0.11 \cdot \pi \cdot 1172^2 \cdot 45)$$

$$= 887.35 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 63.6 \cdot 1172 / (2 \cdot 45) + 12135 / (0.19 \cdot \pi \cdot 1172^2 \cdot 45)$$

$$= 796.32 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 89246 (690 - 2 \cdot 127) / (690 + (4 \cdot 0 / 3))$$

$$= 56263.5 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 56264 / (1172 \cdot 45)$$

$$= 124.84 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1172 \cdot 45}$$

$$= 179.167 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 89246 \cdot 0.1 / (45(204 + 179 + 179))$$

$$= -26.81 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ^*7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -89246 / (4 \cdot 45(204 + 179 + 179)) - 12 \cdot 0.053 \cdot 89246 \cdot 1172 / (690 \cdot 45^2)$$

$$= -563.14 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{1172 \cdot 45}, 2 \cdot 127)$$

$$= 562.33 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2100.0000	mm
Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	340.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw, c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t}) + \text{WearPlateWidth})^3 / 12$$

$$= ((1.56 \cdot \sqrt{1150 \cdot 45}) + 306)^3 / 12$$

$$= 501.854 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:



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$$\begin{aligned}
 &= \text{ShellInertia} + A\text{Shell}(C1 - Y)^2 \\
 &= 502 + 297 (102 - 22.5)^2 \\
 &= 19431.607 \text{ cm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	660.9	45.0	22.5	297.4	669139.5	0.194E+05
Wearplate	306.0	10.0	50.0	30.6	153000.0	839.
Web	10.0	289.0	199.5	28.9	576555.1	0.474E+04
BasePlate	340.0	26.0	357.0	88.4	3155880.0	0.574E+05
Totals	...	...	...	445.3	4554574.5	0.824E+05

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 4554574/445 \\
 &= 102.282 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.2 \cdot 89246 \\
 &= 18163.4082 \text{ kgf}
 \end{aligned}$$

Tension Stress, St = (Fh/As) = 122.8089 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1590.6736 kgf/cm²

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\
 &= 1520 - 1150 \cdot \sin(120/2) / 1.05 \\
 &= 568.958 \text{ mm}
 \end{aligned}$$

Bending Moment, M = Fh * d = 10334.2207 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 128.2490 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1767.4150 kgf/cm²

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{(3(89246 + 425)340) / (4 \cdot 2100 \cdot 1784)} \\
 &= 24.702 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120/2) (1150 + 45 + 10) \\
 &= 2087.121 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 2087 / (6 - 1)
 \end{aligned}$$



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= 417.424 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2
 = 417 • 340/2
 = 709.621 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 89670/(2100 • 340)
 = 12.559 kgf/cm²

Axial Load [P]:

= Ap • Bp
 = 710 • 12.6
 = 8912.036 kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 19.4 + 20.9
 = 40.271 cm²

Compressive Stress [Sc]:

= P/Ar
 = 8912/40.3
 = 221.301 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	204.0	...	20.4	...	707.

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 204 - 10
 = 194.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/40.3
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 204/2
 = 102.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{707/40.3}$
 = 41.914 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1520 - cos(120/2)(1150 + 45 + 10) - 26
 = 891.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 2038903/2651}$
 = 123.211



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Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 892/41.9$$

$$= 21.270$$

Bending Moment [Rm]:

$$= Fl / (2 \cdot Bpln) \cdot e \cdot L / 2$$

$$= 428 / (2 \cdot 2100) \cdot 417 \cdot 892/2$$

$$= 18.965 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ (21.3 < 123) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (21.3)^2 / (2 \cdot 123^2)) 2651 /$$

$$(5/3 + 3 \cdot (21.3) / (8 \cdot 123) - (21.3^3) / (8 \cdot 123^3))$$

$$= 1509 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 221/1509 + (19 \cdot 102/7074720) / 1767$$

$$= 0.162$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	194.0	0.0	19.4	0.0	707.
Web	417.4	10.0	0.0	41.7	0.0	3.48
Totals	...	...	...	61.1	...	711.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/61.1$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204/2$$

$$= 102.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1520 - \sqrt{ (1205 + 10)^2 - (417/2)^2 } - 26$$

$$= 307.213 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 711/61.1 }$$

$$= 34.098 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 307/34.1$$

$$= 9.010$$

Unit Force [Force,u]:

$$= Fl / (2 \cdot \text{Baseplate Length})$$

$$= 428 / (2 \cdot 2100)$$

$$= 0.102 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.1 \cdot 417 \cdot 307$$

$$= 13.071 \text{ kgf-m.}$$



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Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = 13.1/69693
 = 18.755 kgf/cm²

Compressive Allowable, $KL/r < C_c$ (9.01 < 123) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 = $(1 - (9.01)^2 / (2 \cdot 123^2)) 2651 / (5/3 + 3 \cdot (9.01) / (8 \cdot 123) - (9.01^3) / (8 \cdot 123^3))$
 = 1561 kgf/cm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 = $S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$
 = $247 / 1561 + (13.1 \cdot 102 / 711) / 1767$
 = 0.169

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	6.0062 cm ²
Saddle Load QO (Weight)	QO	88289.3 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	166.8 kgf
Maximum Transverse Force	Ft	632.1 kgf
Maximum Longitudinal Force	F1	428.1 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 = $F_1 / (Bolt Area \cdot Number of Bolts)$
 = $428 / (6.01 \cdot 4)$
 = 17.8 kgf/cm². Must be less than 1052.3 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 = $F_t / (Bolt Area \cdot Number of Bolts)$
 = $632 / (6.01 \cdot 4)$
 = 26.3 kgf/cm². Must be less than 1052.3 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_1 / (BoltShearAllowable \cdot Nbolts)$
 = $428 / (1052 \cdot 4)$
 = 0.1017 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1520 \cdot 632 + 0$
 = 960.76 kgf-m.

Eccentricity (e):
 = R_{mom} / QO
 = $961 / 88289$
 = 10.88 mm < Bplen/6 --> No Uplift in Transverse direction



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Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.1, 0]
 = 0.1017 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (306.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 $= \min(204 + 1.56 \cdot \sqrt{1172 \cdot 45}, 2 \cdot 1300)$
 = 562.3334 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1172.50	mm
Shell Thickness used in this Case	t	45.000	mm
Stiffened Vessel Length per 4.15.6	L	690.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1300.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	59.54	cm
Shell Allowable Stress used in Calculation		2518.57	kgf/cm ²
Head Allowable Stress used in Calculation		2518.57	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1520.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		9016.00	kgf
Pressure used in Analysis	P	63.617	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		827.33	2518.57
Long. Stress at Bottom of Midspan		830.24	2518.57
Long. Stress at Top of Saddles		841.30	2518.57
Long. Stress at Bottom of Saddles		821.85	2518.57
Tangential Shear in Shell		11.18	1413.05
Circ. Stress at Horn of Saddle		56.89	3777.85
Circ. Compressive Stress in Shell		2.71	2518.57

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = $Ftr (Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 = $3 (1264/2 + 0) \cdot 1520/2087$
 = 1381.0 kgf



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Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(428, 0) \cdot 1520/3900$
 $= 166.8 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 7635 + \max(167, 1381, 0, 0)$
 $= 9016.0 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1790 (0 + \pi/4 (2.39 \cdot 1.2)^2)$
 $= 12721.669 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	9440.72	kgf
Transverse Shear Load	632.08	kgf
Longitudinal Shear Load	428.09	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2300/2 + 45 + 10) \sin(120/2), 2 \cdot 1172)$
 $= 2087.121 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -9016 \cdot 130 [1 - (1 - 130/690 + (117^2 - 59.5^2)/(2 \cdot 130 \cdot 690)) / (1 + (4 \cdot 59.5)/(3 \cdot 690))]$
 $= -2592.0 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 9016 \cdot 690/4 ((1 + 2(117^2 - 59.5^2)/(690^2)) / (1 + (4 \cdot 59.5)/(3 \cdot 690)) - 4 \cdot 130/690)$
 $= 2824.8 \text{ kgf-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 63.6 \cdot 1172/(2 \cdot 45) - 2825/(\pi \cdot 1172^2 \cdot 45)$
 $= 827.33 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 63.6 \cdot 1172/(2 \cdot 45) + 2825/(\pi \cdot 1172^2 \cdot 45)$
 $= 830.24 \text{ kgf/cm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:
 $= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$
 $= 63.6 \cdot 1172/(2 \cdot 45) - 2592/(0.11 \cdot \pi \cdot 1172^2 \cdot 45)$
 $= 841.30 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:
 $= P \cdot R_m/(2t) + M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t)$
 $= 63.6 \cdot 1172/(2 \cdot 45) + 2592/(0.19 \cdot \pi \cdot 1172^2 \cdot 45)$



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$$= 821.85 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 9016 (690 - 2 \cdot 130)/(690 + (4 \cdot 59.5/3))$$

$$= 5038.9 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 5039 / (1172 \cdot 45)$$

$$= 11.18 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1172 \cdot 45)}$$

$$= 179.167 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 9016 \cdot 0.1 / (45(204 + 179 + 179))$$

$$= -2.71 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q/(4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -9016 / (4 \cdot 45(204 + 179 + 179)) -$$

$$12 \cdot 0.053 \cdot 9016 \cdot 1172 / (690 \cdot 45^2)$$

$$= -56.89 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{(1172 \cdot 45)}, 2 \cdot 130)$$

$$= 562.33 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2100.0000	mm
Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	340.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1150 \cdot 45)} + 306)45^3)/12$$

$$= 501.854 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 502 + 297(102 - 22.5)^2$$

$$= 19431.607 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	660.9	45.0	22.5	297.4	669139.5	0.194E+05
Wearplate	306.0	10.0	50.0	30.6	153000.0	839.
Web	10.0	289.0	199.5	28.9	576555.1	0.474E+04
BasePlate	340.0	26.0	357.0	88.4	3155880.0	0.574E+05



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Totals	...	...	...	445.3	4554574.5	0.824E+05
--------	-----	-----	-----	-------	-----------	-----------

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4554574/445$$

$$= 102.282 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 9016$$

$$= 1834.9510 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 12.4067 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1520 - 1150 \cdot \sin(120/2) / 1.05$$

$$= 568.958 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1044.0105 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 12.9563 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(9016 + 425)340 / (4 \cdot 2100 \cdot 1784)}$$

$$= 8.015 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1150 + 45 + 10)$$

$$= 2087.121 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2087 / (6 - 1)$$

$$= 417.424 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 417 \cdot 340 / 2$$

$$= 709.621 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 9441 / (2100 \cdot 340)$$

$$= 1.322 \text{ kgf/cm}^2$$



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Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 710 \cdot 1.32$$

$$= 938.282 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19.4 + 20.9$$

$$= 40.271 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 938/40.3$$

$$= 23.299 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	204.0	...	20.4	...	707.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204 - 10$$

$$= 194.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/40.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204/2$$

$$= 102.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(707/40.3)}$$

$$= 41.914 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1520 - \cos(120/2) (1150 + 45 + 10) - 26$$

$$= 891.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038903/2651)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 892/41.9$$

$$= 21.270$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 428 / (2 \cdot 2100) \cdot 417 \cdot 892/2$$

$$= 18.965 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($21.3 < 123$) per AISC E2-1 [Sca]:



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$$= (1 - (Kl/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot Cc) - (Kl/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (21.3)^2 / (2 \cdot 123^2)) 2651 /$$

$$(5/3 + 3 \cdot (21.3) / (8 \cdot 123) - (21.3)^3 / (8 \cdot 123^3))$$

$$= 1509 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 23.3 / 1509 + (19 \cdot 102 / 7074720) / 1767$$

$$= 0.031$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	194.0	0.0	19.4	0.0	707.
Web	417.4	10.0	0.0	41.7	0.0	3.48
Totals	...	...	...	61.1	...	711.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 61.1$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204 / 2$$

$$= 102.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2} - Bpthk$$

$$= 1520 - \sqrt{(1205 + 10)^2 - (417/2)^2} - 26$$

$$= 307.213 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{711 / 61.1}$$

$$= 34.098 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 307 / 34.1$$

$$= 9.010$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 428 / (2 \cdot 2100)$$

$$= 0.102 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.1 \cdot 417 \cdot 307$$

$$= 13.071 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 13.1 / 69693$$

$$= 18.755 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ (9.01 < 123) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot Cc) - (Kl/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (9.01)^2 / (2 \cdot 123^2)) 2651 /$$

$$(5/3 + 3 \cdot (9.01) / (8 \cdot 123) - (9.01)^3 / (8 \cdot 123^3))$$

$$= 1561 \text{ kgf/cm}^2$$



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AISC Unity Check of Inside Ribs (must be ≤ 1)

= $Sc/Sca + (M_{base,c} \cdot C1/I)/Sba$
 = $26/1561 + (13.1 \cdot 102/711)/1767$
 = 0.027

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	6.0062 cm ²
Saddle Load QO (Weight)	QO	8059.7 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	166.8 kgf
Maximum Transverse Force	Ft	632.1 kgf
Maximum Longitudinal Force	Fl	428.1 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $Fl / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = $428 / (6.01 \cdot 4)$
 = 17.8 kgf/cm². Must be less than 1052.3 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $Ft / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = $632 / (6.01 \cdot 4)$
 = 26.3 kgf/cm². Must be less than 1052.3 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $Fl / (BoltShearAllowable \cdot Nbolts)$
 = $428 / (1052 \cdot 4)$
 = 0.1017 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + Sum\ of\ X\ Moments$
 = $1520 \cdot 632 + 0$
 = 960.76 kgf-m.

Eccentricity (e):

= $Rmom / QO$
 = 961/8060
 = 119.20 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0, 0.1, 0]$
 = 0.1017 cm²

Review notes about nozzle loadings and supports in the Warnings report.



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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

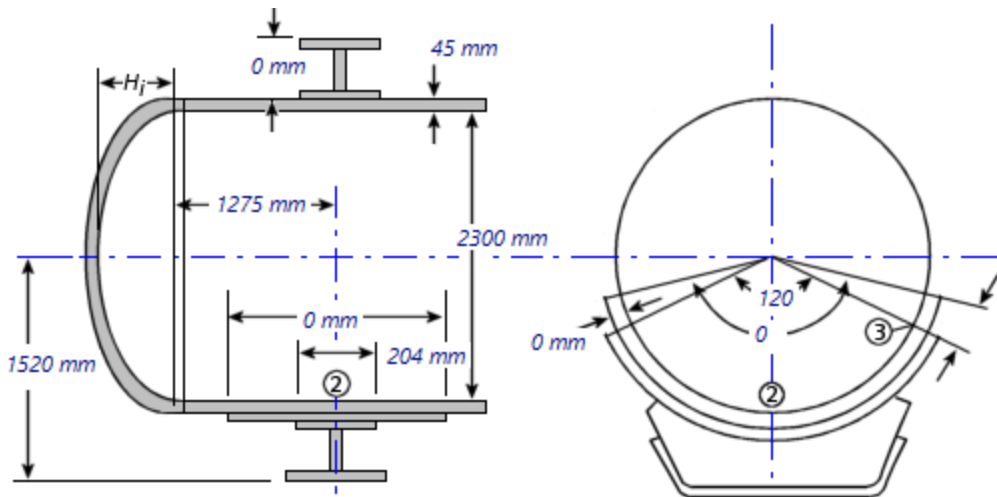
Horizontal Vessel Stress Calculations : External Pressure

Note:
 Wear Pad Width (306.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$
 $= \min(204 + 1.56 \cdot \sqrt{ 1174 \cdot 42 }, 2 \cdot 1275)$
 $= 550.4043 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	1174.00	mm
Shell Thickness used in this Case	t	45.000	mm
Stiffened Vessel Length per 4.15.6	L	690.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1275.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		1784.41	kgf/cm ²
Head Allowable Stress used in Calculation		1570.34	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1520.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	94872.44	kgf
Pressure used in Analysis	P	-1.033 kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top	of Midspan	-43.12	-2404.36
Long. Stress at Top	of Midspan	43.12	1784.41
Long. Stress at Bottom	of Midspan	14.25	1784.41
Long. Stress at Top	of Saddles	51.98	1784.41



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Long. Stress at Bottom of Saddles	-51.25	-2404.36
Long. Stress at Bottom of Saddles	51.25	1784.41
Tangential Shear in Shell	142.01	1423.92
Circ. Stress at Horn of Saddle	682.97	2230.52
Circ. Compressive Stress in Shell	31.20	1784.41

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (3831/2 + 0) \cdot 1520/2087$$

$$= 4184.8 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(1297, 0) \cdot 1520/3900$$

$$= 505.6 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(8573, 0) \cdot 1520/3900$$

$$= 3341.3 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (8573/2 + 0) \cdot 1520/2087$$

$$= 9365.3 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 85507 + \max(506, 4185, 3341, 9365)$$

$$= 94872.4 \text{ kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1790 (0 + \pi/4 (2.39 \cdot 1.2)^2)$$

$$= 12721.669 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	95297.16	kgf
Transverse Shear Load	4286.54	kgf
Longitudinal Shear Load	8573.08	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2300/2 + 45 + 10)\sin(120/2), 2 \cdot 1174)$$

$$= 2087.121 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L)) / (1 + (4\text{hm})/3L)]$$

$$= -94872 \cdot 127 [1 - (1 - 127/690 + (117^2 - 0^2)/(2 \cdot 127 \cdot 690)) / (1 + (4 \cdot 0)/(3 \cdot 690))]$$

$$= -12876.4 \text{ kgf-m.}$$



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Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L)$$

$$= 94872 \cdot 690 / 4 \cdot ((1 + 2(117^2 - 0^2) / (690^2)) / (1 + (4 \cdot 0) / (3 \cdot 690)) - 4 \cdot 127 / 690)$$

$$= 52168.1 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -1.03 \cdot 1174 / (2 \cdot 42) - 52168 / (\pi \cdot 1174^2 \cdot 42)$$

$$= -43.12 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -1.03 \cdot 1174 / (2 \cdot 42) + 52168 / (\pi \cdot 1174^2 \cdot 42)$$

$$= 14.25 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -1.03 \cdot 1174 / (2 \cdot 42) - 12876 / (0.11 \cdot \pi \cdot 1174^2 \cdot 42)$$

$$= 51.98 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -1.03 \cdot 1174 / (2 \cdot 42) + 12876 / (0.19 \cdot \pi \cdot 1174^2 \cdot 42)$$

$$= -51.25 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 94872 (690 - 2 \cdot 127) / (690 + (4 \cdot 0 / 3))$$

$$= 59810.9 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 59811 / (1174 \cdot 42)$$

$$= 142.01 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{Rm \cdot t}$$

$$= 0.78 \cdot \sqrt{1174 \cdot 42}$$

$$= 173.202 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 94872 \cdot 0.1 / (42(204 + 173 + 173))$$

$$= -31.20 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -94872 / (4 \cdot 42(204 + 173 + 173)) - 12 \cdot 0.053 \cdot 94872 \cdot 1174 / (690 \cdot 42^2)$$

$$= -682.97 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{1174 \cdot 42}, 2 \cdot 127)$$

$$= 550.40 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2100.0000	mm
Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	340.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	



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Saddle Design Temperature 21.1 °C
 Saddle Yield Stress Sy 2671.5 kgf/cm²
 Height of Web at Center Hw,c 306.0 mm
 Friction Coefficient mu 0.000

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$
 $= ((1.56 \cdot \sqrt{(1153 \cdot 42) + 306})^3) / 12$
 $= 400.873 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 401 + 273 (105 - 21)^2$
 $= 19614.586 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	649.3	42.0	21.0	272.7	572675.6	0.196E+05
Wearplate	306.0	10.0	47.0	30.6	143820.0	0.103E+04
Web	10.0	289.0	196.5	28.9	567885.1	0.443E+04
BasePlate	340.0	26.0	354.0	88.4	3129360.0	0.549E+05
Totals	...	...	...	420.6	4413741.0	0.800E+05

Distance to Centroid [C1]:
 $= AY / A$
 $= 4413741 / 421$
 $= 104.938 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 94872$
 $= 19308.5977 \text{ kgf}$

Tension Stress, St = (Fh/As) = 130.5519 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1602.9094 kgf/cm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1520 - 1153 \cdot \sin(120/2) / 1.05$
 $= 566.477 \text{ mm}$

Bending Moment, M = Fh * d = 10937.8838 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 143.5394 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1781.0104 kgf/cm²

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$



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$$= \sqrt{3(94872 + 425)340 / (4 \cdot 2100 \cdot 1784)}$$

$$= 25.466 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1150 + 45 + 10)$$

$$= 2087.121 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2087 / (6 - 1)$$

$$= 417.424 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 417 \cdot 340 / 2$$

$$= 709.621 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 95297 / (2100 \cdot 340)$$

$$= 13.347 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 710 \cdot 13.3$$

$$= 9471.271 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19.4 + 20.9$$

$$= 40.271 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 9471 / 40.3$$

$$= 235.187 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	204.0	...	20.4	...	707.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204 - 10$$

$$= 194.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 40.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204 / 2$$

$$= 102.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$



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$$= \sqrt{707/40.3}$$

$$= 41.914 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1520 - \cos(120/2) (1150 + 45 + 10) - 26$$

$$= 891.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 2038903 / 2672}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 892 / 41.9$$

$$= 21.270$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 8573 / (2 \cdot 2100) \cdot 417 \cdot 892 / 2$$

$$= 379.802 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($21.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (21.3)^2 / (2 \cdot 123^2)) 2672 / (5/3 + 3 \cdot (21.3) / (8 \cdot 123) - (21.3)^3 / (8 \cdot 123^3))$$

$$= 1520 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 235 / 1520 + (380 \cdot 102 / 7074720) / 1781$$

$$= 0.462$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	194.0	0.0	19.4	0.0	707.
Web	417.4	10.0	0.0	41.7	0.0	3.48
Totals	...	...	...	61.1	...	711.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 61.1$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204 / 2$$

$$= 102.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1520 - \sqrt{(1205 + 10)^2 - (417/2)^2} - 26$$

$$= 307.213 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{711 / 61.1}$$

$$= 34.098 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$



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$$= 1 \cdot 307/34.1$$

$$= 9.010$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 8573 / (2 \cdot 2100)$$

$$= 2.041 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.04 \cdot 417 \cdot 307$$

$$= 261.761 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 262/69693$$

$$= 375.592 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($9.01 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.01)^2 / (2 \cdot 123^2)) 2672 / (5/3 + 3 \cdot (9.01) / (8 \cdot 123) - (9.01^3) / (8 \cdot 123^3))$$

$$= 1573 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 262/1573 + (262 \cdot 102/711) / 1781$$

$$= 0.378$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (306.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{1174 \cdot 42}, 2 \cdot 1300)$$

$$= 550.4043 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1174.00	mm
Shell Thickness used in this Case	t	45.000	mm
Stiffened Vessel Length per 4.15.6	L	690.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1300.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	59.69	cm
Shell Allowable Stress used in Calculation		1784.41	kgf/cm ²
Head Allowable Stress used in Calculation		1784.41	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1520.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		21719.10	kgf
Pressure used in Analysis	P	-1.033	kgf/cm ²



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Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-18.18	-2404.36
Long. Stress at Top of Midspan	18.18	1784.41
Long. Stress at Bottom of Midspan	-10.70	-2404.36
Long. Stress at Bottom of Midspan	10.70	1784.41
Long. Stress at Top of Saddles	17.78	1784.41
Long. Stress at Bottom of Saddles	-32.30	-2404.36
Long. Stress at Bottom of Saddles	32.30	1784.41
Tangential Shear in Shell	28.81	1423.92
Circ. Stress at Horn of Saddle	156.35	2230.52
Circ. Compressive Stress in Shell	7.14	1784.41

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (3831/2 + 0) \cdot 1520/2087$
 $= 4184.8\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(1297, 0) \cdot 1520/3900$
 $= 505.6\ kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(8573, 0) \cdot 1520/3900$
 $= 3341.3\ kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (8573/2 + 0) \cdot 1520/2087$
 $= 9365.3\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 12354 + \max(506, 4185, 3341, 9365)$
 $= 21719.1\ kgf$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1790 (0 + \pi/4(2.39 \cdot 1.2)^2)$
 $= 12721.669\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	22143.82	kgf
Transverse Shear Load	4286.54	kgf
Longitudinal Shear Load	8573.08	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(2300/2 + 45 + 10) \sin(120/2), 2 \cdot 1174)$
 $= 2087.121\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	



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Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -21719 \cdot 130 [1 - (1 - 130/690 + (117^2 - 59.7^2)/(2 \cdot 130 \cdot 690))/(1 + (4 \cdot 59.7)/(3 \cdot 690))]$$

$$= -6247.3 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L]$$

$$= 21719 \cdot 690/4 [(1 + 2(117^2 - 59.7^2)/(690^2))/(1 + (4 \cdot 59.7)/(3 \cdot 690)) - 4 \cdot 130/690]$$

$$= 6798.3 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= -1.03 \cdot 1174/(2 \cdot 42) - 6798/(\pi \cdot 1174^2 \cdot 42)$$

$$= -18.18 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= -1.03 \cdot 1174/(2 \cdot 42) + 6798/(\pi \cdot 1174^2 \cdot 42)$$

$$= -10.70 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$$

$$= -1.03 \cdot 1174/(2 \cdot 42) - 6247/(0.11 \cdot \pi \cdot 1174^2 \cdot 42)$$

$$= 17.78 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -1.03 \cdot 1174/(2 \cdot 42) + 6247/(0.19 \cdot \pi \cdot 1174^2 \cdot 42)$$

$$= -32.30 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 21719 (690 - 2 \cdot 130)/(690 + (4 \cdot 59.7/3))$$

$$= 12135.4 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (\pi \cdot Rm \cdot t)$$

$$= 1.17 \cdot 12135/(\pi \cdot 1174 \cdot 42)$$

$$= 28.81 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1174 \cdot 42)}$$

$$= 173.202 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 21719 \cdot 0.1/(42(204 + 173 + 173))$$

$$= -7.14 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.23) [σ_{7*}]:

$$= -Q/(4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm/(L \cdot t^2)$$

$$= -21719/(4 \cdot 42(204 + 173 + 173)) - 12 \cdot 0.053 \cdot 21719 \cdot 1174/(690 \cdot 42^2)$$

$$= -156.35 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{(1174 \cdot 42)}, 2 \cdot 130)$$

$$= 550.40 \text{ mm}$$



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Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2100.0000	mm
Baseplate Thickness	Bpthk	26.0000	mm
Baseplate Width	Bpwid	340.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2671.5	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1153 \cdot 42)} + 306)^3) / 12$$

$$= 400.873 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}}(C1 - Y)^2$$

$$= 401 + 273(105 - 21)^2$$

$$= 19614.586 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	649.3	42.0	21.0	272.7	572675.6	0.196E+05
Wearplate	306.0	10.0	47.0	30.6	143820.0	0.103E+04
Web	10.0	289.0	196.5	28.9	567885.1	0.443E+04
BasePlate	340.0	26.0	354.0	88.4	3129360.0	0.549E+05
Totals	...	...	...	420.6	4413741.0	0.800E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4413741 / 421$$

$$= 104.938 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 21719$$

$$= 4420.3081 \text{ kgf}$$

Tension Stress, St = (Fh/As) = 29.8872 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1602.9094 kgf/cm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1520 - 1153 \cdot \sin(120 / 2) / 1.05$$

$$= 566.477 \text{ mm}$$

Bending Moment, M = Fh * d = 2504.0042 kgf-m.



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Bending Stress, $S_b = (M * C_1 / I) = 32.8604 \text{ kgf/cm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 1781.0104 \text{ kgf/cm}^2$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 * \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{ \frac{3(Q + \text{SaddleWt}) \text{BasePlateWidth}}{4 * \text{BasePlateLength} * \text{AllStress}} } \\
= \sqrt{ \frac{3(21719 + 425)340}{4 * 2100 * 1784} } \\
= 12.276 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2)(\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120/2)(1150 + 45 + 10)$
 $= 2087.121 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 2087 / (6 - 1)$
 $= 417.424 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 417 * 340 / 2$
 $= 709.621 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 22144 / (2100 * 340)$
 $= 3.101 \text{ kgf/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 710 * 3.1$
 $= 2200.802 \text{ kgf}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 19.4 + 20.9$
 $= 40.271 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 2201 / 40.3$
 $= 54.650 \text{ kgf/cm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	204.0	...	20.4	...	707.

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 204 - 10$
 $= 194.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 40.3$



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= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 204/2
 = 102.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{707/40.3}$
 = 41.914 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1520 - cos(120/2)(1150 + 45 + 10) - 26
 = 891.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 2038903/2672}$
 = 122.739

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 892/41.9
 = 21.270

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 8573 / (2 • 2100) • 417 • 892/2
 = 379.802 kgf-m.

Compressive Allowable, KL/r < Cc (21.3 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (21.3)^2 / (2 \cdot 123^2)) 2672 / (5/3 + 3 \cdot (21.3) / (8 \cdot 123) - (21.3^3) / (8 \cdot 123^3))$
 = 1520 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 54.6/1520 + (380 • 102/7074720) / 1781
 = 0.343

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	194.0	0.0	19.4	0.0	707.
Web	417.4	10.0	0.0	41.7	0.0	3.48
Totals	...	...	...	61.1	...	711.

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/61.1
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 204/2
 = 102.000 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk
 = 1520 - $\sqrt{(1205 + 10)^2 - (417/2)^2}$ - 26



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= 307.213 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{711/61.1}$
 = 34.098 mm

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 307/34.1
 = 9.010

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 8573/(2 • 2100)
 = 2.041 kgf/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 2.04 • 417 • 307
 = 261.761 kgf-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = 262/69693
 = 375.592 kgf/cm²

Compressive Allowable, KL/r < Cc (9.01 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (9.01)^2 / (2 \cdot 123^2)) 2672 /$
 $(5/3 + 3 \cdot (9.01) / (8 \cdot 123) - (9.01^3) / (8 \cdot 123^3))$
 = 1573 kgf/cm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 61/1573 + (262 • 102/711)/1781
 = 0.250

Review notes about nozzle loadings and supports in the Warnings report.

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Conical Calculations per ASME VIII-2, 4.3.11

Input Values:

Shell cone

Design Internal Pressure, Small End	50.197	kgf/cm ²
Design Internal Pressure, Large End	50.197	kgf/cm ²
Temperature for Internal Pressure	270.0	°C
Design External Pressure	1.033	kgf/cm ²
Temperature for External Pressure	270.0	°C

CONE Properties:

Cone Material		SA-516 70 +N
Cone Allowable Stress at Temperature	Ss	1784.41 kgf/cm ²
Cone Allowable Stress At Ambient		1784.41 kgf/cm ²
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00
Circumferential Joint Efficiency of Cone		1.00
Actual Thickness of Cone	Tc	52.0000 mm
Corrosion Allowance for Cone	Cac	3.0000 mm
Diameter Basis for Cone and Cylinders		ID
Diameter of Small End of Cone	Ds	1300.000 mm
Diameter of Large End of Cone	Dl	2300.000 mm
Half Apex Angle for Cone		29.74 degrees
Axial Length of Cone	Lc	1750.000 mm

SMALL-END Properties:

Attached Small End Material Specification		SA-516 70
Operating Allowable Stress	Ss	1784.41 kgf/cm ²
Ambient Allowable Stress		1784.41 kgf/cm ²
Longitudinal Joint Efficiency	Es,E1	1.00
Actual Thickness	Ts	30.0000 mm
Corrosion Allowance	Cas	3.0000 mm
Axial Length of Attached Element	Ls	300.030 mm

LARGE-END Properties:

Attached Large End Material Specification		SA-516 70 +N
Operating Allowable Stress	Sl	1784.41 kgf/cm ²
Ambient Allowable Stress		1784.41 kgf/cm ²
Longitudinal Joint Efficiency	El,E1	1.00
Actual Thickness	Tl	45.0000 mm
Corrosion Allowance	Cal	3.0000 mm
Axial Length of Attached Element	Ll	5041.667 mm

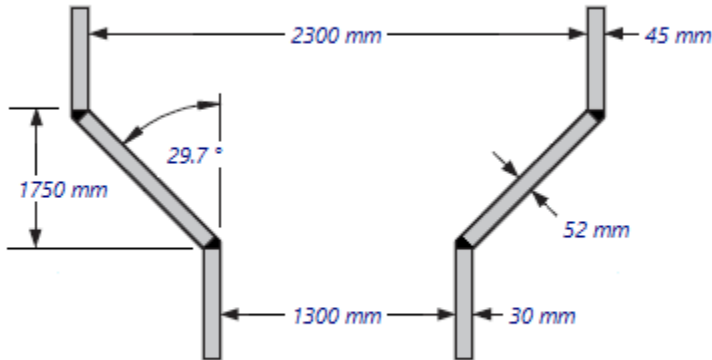
Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: None

Warning: rtk -3.000mm, must be > Tl, 42.000mm, see 4.3.12



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Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 270 °C

Elastic Modulus of Cone Material	0.191E+07 kgf/cm ² at 270 °C
Elastic Modulus of Small Cylinder Material	0.191E+07 kgf/cm ² at 270 °C
Elastic Modulus of Large Cylinder Material	0.191E+07 kgf/cm ² at 270 °C
Elastic Modulus of Large End Reinforcement	0.191E+07 kgf/cm ² at 270 °C
Elastic Modulus of Small End Reinforcement	0.191E+07 kgf/cm ² at 270 °C

Elastic Modulus Data from ASME Section II Part D at 270 °C

Elastic Modulus of Cone Material	0.191E+07 kgf/cm ² at 270 °C
Elastic Modulus of Small Cylinder Material	0.191E+07 kgf/cm ² at 270 °C
Elastic Modulus of Large Cylinder Material	0.191E+07 kgf/cm ² at 270 °C
Elastic Modulus of Large End Reinforcement	0.191E+07 kgf/cm ² at 270 °C
Elastic Modulus of Small End Reinforcement	0.191E+07 kgf/cm ² at 270 °C

Axial forces and moments are not computed for Horizontal geometries.
 Compute them manually and enter the loads using the Force/Moment dialog.
 Place the loads at the ends of the cone as needed.

Maximum Centroid Reinforcement Distance Large End	25.4000 mm
Maximum Centroid Reinforcement Distance Small End	25.4000 mm

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Junction Calculations per 4.3.11:

Calculations for the Large End Cone/Shell Junction - Without Knuckle

Fhe per 4.1.3 or 4.3.11.2 Step 6 [Fhe]:

$$= 0.4 \cdot E(tc/D)$$

$$= 0.4 \cdot 1909280 (49/2390)$$

$$= 15657.7 \text{ kgf/cm}^2$$

Factor of Safety [FS]:

Since Fic [2672] >= Fy[2672], FS = 1.667

Allowable Stress per 4.3.11 Step 6 [Fha]:



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= Fic/FS
 = 2672/1.67
 = 1602.589 kgf/cm²

Large End under Internal Pressure:

Cone Length along Cone surface per 4.3.46 [ConeMinLen]:
 $= 2 \cdot \sqrt{(RL \cdot Tc/\cos(\alpha))} + 1.4 \cdot \sqrt{(Rs \cdot tc/\cos(\alpha))}$
 $= 2 \cdot \sqrt{(1153 \cdot 49/\cos(29.7))} + 1.4 \cdot \sqrt{(653 \cdot 49/\cos(29.7))}$
 = 779.037 mm

Results for the Large End Cylinder Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	689.292	2676.619	Pass
Sigma sm-	689.292	2676.619	Pass
Sigma thetam+	208.307	2676.619	Pass
Sigma thetam-	208.307	2676.619	Pass
Sigma sm+ + Sigm	-1532.413	5353.238	Pass
Sigma sm+ - Sigm	2910.997	5353.238	Pass
Sigma sm- + Sigm	-1532.413	5353.238	Pass
Sigma sm- - Sigm	2910.997	5353.238	Pass
Sigma thetam+ +	-458.205	5353.238	Pass
Sigma thetam+ -	874.818	5353.238	Pass
Sigma thetam- +	-458.205	5353.238	Pass
Sigma thetam- -	874.818	5353.238	Pass

Results for the Large End Cone Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	588.541	2676.619	Pass
Sigma sm-	588.541	2676.619	Pass
Sigma thetam+	106.230	2676.619	Pass
Sigma thetam-	106.230	2676.619	Pass
Sigma sm+ + Sigm	-1043.732	5353.238	Pass
Sigma sm+ - Sigm	2220.814	5353.238	Pass
Sigma sm- + Sigm	-1043.732	5353.238	Pass
Sigma sm- - Sigm	2220.814	5353.238	Pass
Sigma thetam+ +	-383.452	5353.238	Pass
Sigma thetam+ -	595.912	5353.238	Pass
Sigma thetam- +	-383.452	5353.238	Pass
Sigma thetam- -	595.912	5353.238	Pass

Cone Large end without a Knuckle Cylinder Section:

Net Axial Force ML +ve [XL+]:
 $= FL / (2 \cdot \pi \cdot RL) + ML / (\pi \cdot RL^2)$
 $= 0 / (2 \cdot 3.141 \cdot 1153) + 0 / (3.14 \cdot 1153^2)$
 = 0 kgf/mm

Net Axial Force ML -ve [XL-]:
 $= FL / (2 \cdot \pi \cdot RL) - ML / (\pi \cdot RL^2)$
 $= 0 / (2 \cdot 3.14 \cdot 1153) - 0 / (3.141 \cdot 1153^2)$
 = 0 kgf/mm

Thickness Ratio [n]:
 = tc/tL = 49/42 = 1.17

Curve Fitting Geometry [H and B]:
 $H = \sqrt{0.5 \cdot D/tL} = \sqrt{0.5 \cdot 2307/42} = 5.24$
 $B = \tan(29.7) = 0.57$

This analysis is preformed per Table 4.3.1 for the Large End:



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Using Data from Table 4.3.3 and 4.3.4 knowing n
 Table 4.3.3 - n: 1 < (n=1.17) < 1.25

Use n to construct a table: C1, C2, C3..... by interpolation

Table 4.3.3 [MsN1]:

C1 = -3.097512 C2 = 3.686273 C3 = 0.761135 C4 = -0.233165
 C5 = -0.077764 C6 = 0.067131 C7 = 0.026880 C8 = -0.015577
 C9 = 0.001363 C10 = -0.010718 C11 = 0.000000

Compute using Table 4.3.3 [MsN1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) + C11 \cdot \ln(H) \cdot (\ln(B))^2$$

$$= C1 + C2 \cdot \ln(5.24) + C3 \cdot \ln(0.57) + C4 \cdot (\ln(5.24))^2 + C5 \cdot (\ln(0.57))^2 + C6 \cdot \ln(5.24) \cdot \ln(0.57) + C7 \cdot (\ln(5.24))^3 + C8 \cdot (\ln(0.57))^3 + C9 \cdot \ln(5.24) \cdot (\ln(0.57))^2 + C10 \cdot (\ln(5.24))^2 \cdot \ln(0.57) + C11 \cdot \ln(5.24) \cdot (\ln(0.57))^2$$

$$X = 2, \text{ MsN1} = -\exp(X) = -7.38$$

Table 4.3.3 [Qn1]:

C1 = -1.935534 C2 = 2.331614 C3 = 0.527501 C4 = -0.092494
 C5 = -0.088386 C6 = 0.187162 C7 = 0.008887 C8 = -0.020067
 C9 = 0.001094 C10 = -0.028094 C11 = 0.000000

Compute using Table 4.3.3 [Qn1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) + C11 \cdot \ln(H) \cdot (\ln(B))^2$$

$$= C1 + C2 \cdot \ln(5.24) + C3 \cdot \ln(0.57) + C4 \cdot (\ln(5.24))^2 + C5 \cdot (\ln(0.57))^2 + C6 \cdot \ln(5.24) \cdot \ln(0.57) + C7 \cdot (\ln(5.24))^3 + C8 \cdot (\ln(0.57))^3 + C9 \cdot \ln(5.24) \cdot (\ln(0.57))^2 + C10 \cdot (\ln(5.24))^2 \cdot \ln(0.57) + C11 \cdot \ln(5.24) \cdot (\ln(0.57))^2$$

$$X = 1.26, \text{ Qn} = -\exp(X) = -3.54$$

Table 4.3.4 MsN2:

C1 = -5.717372 C2 = 0.005787 C3 = 0.473328 C4 = -0.213055
 C5 = 2.250262 C6 = 0.000017 C7 = 0.003074 C8 = -0.001940
 C9 = -0.215397 C10 = -0.000026 C11 = -0.106247

Compute using Table 4.3.4 [MsN2]:

$$X1 = C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A)$$

$$= C1 + C3 \cdot \ln(5.24^2) + C5 \cdot \ln(0.52) + C7 \cdot (\ln(5.24^2))^2 + C9 \cdot (\ln(0.52))^2 + C11 \cdot \ln(5.24^2) \cdot \ln(0.52) = -0.15$$

$$X2 = 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR)$$

$$= 1 + C2 \cdot \ln(5.24^2) + C4 \cdot \ln(0.52) + C6 \cdot (\ln(5.24^2))^2 + C8 \cdot (\ln(0.52))^2 + C10 \cdot \ln(5.24^2) \cdot \ln(0.52) = 0.27$$

$$\text{MsN2} = \exp(X1/X2) = \exp(-0.15/0.27) = -0.57$$

Table 4.3.4 QN2:

C1 = -5.008318 C2 = 0.014737 C3 = -0.038229 C4 = -0.196938
 C5 = 2.098516 C6 = -0.000277 C7 = 0.000047 C8 = -0.004739
 C9 = -0.216746 C10 = -0.001647 C11 = 0.008920

Compute using Table 4.3.4 [Qn2]:

$$X3 = C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A)$$

$$= C1 + C3 \cdot \ln(5.24^2) + C5 \cdot \ln(0.52) + C7 \cdot (\ln(5.24^2))^2 + C9 \cdot (\ln(0.52))^2 + C11 \cdot \ln(5.24^2) \cdot \ln(0.52) = -0.15$$

$$X4 = 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR)$$

$$= 1 + C2 \cdot \ln(5.24^2) + C4 \cdot \ln(0.52) + C6 \cdot (\ln(5.24^2))^2 + C8 \cdot (\ln(0.52))^2 + C10 \cdot \ln(5.24^2) \cdot \ln(0.52) = 0.27$$

$$\text{Qn2} = \exp(X3/X4) = \exp(-0.41/0.3) = -0.26$$

Stresses for the Large End Cylinder per Table 4.3.1

Stress Resultant Value [Msp]:

$$= P \cdot t \cdot L^2 \cdot \text{MsN1} = 50.2 \cdot 42^2 \cdot -7.38 = -6532 \text{ kgf}$$

Compute with XL+ [Msx+]:

$$= XL \cdot t \cdot L \cdot \text{MsN2} = 0 \cdot 42 \cdot -0.57 = -0 \text{ kgf}$$



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Compute wiht XL- [Msx-]:

$$= XL \cdot tL \cdot MsN2 = 0 \cdot 42 \cdot -0.57 = -0 \text{ kgf}$$

Compute with XL+ [Ms+]:

$$= MsP + Msx = -6532 + -0 = -6532 \text{ kgf}$$

Compute with XL- [Ms-]:

$$= MsP + Msx = -6532 + -0 = -6532 \text{ kgf}$$

Compute [Qp]:

$$= P \cdot tL \cdot Qn1 = 50.2 \cdot 42 \cdot -3.54 = -74.6 \text{ kgf/mm}$$

Compute with XL+ [Qx+]:

$$= XL \cdot QN2 = 0 \cdot -0.0047 = -0 \text{ kgf/mm}$$

Compute with XL- [Qx-]:

$$= XL \cdot QN2 = 0 \cdot -0.0047 = -0 \text{ kgf/mm}$$

Compute with XL+ [Q+]:

$$= Qp + Qx = -74.6 + -0 = -74.6 \text{ kgf/mm}$$

Compute with XL- [Q-]:

$$= Qp + Qx = -74.6 + -0 = -74.6 \text{ kgf/mm}$$

Compute [β_{cy}]:

$$= 3 \cdot (1 - v^2) / (RL^2 \cdot tL^2) \cdot 0.25 \\ = (3 \cdot (1 - 0.3^2) / (1153^2 \cdot 42^2)) \cdot 0.25 = 0.0058 \text{ 1/mm}$$

Compute with XL+ [Ns+]:

$$= P \cdot RL/2 + XL = 50.2 \cdot 1153/2 + 0 = 290 \text{ kgf/mm}$$

Compute with XL- [Ns-]:

$$= P \cdot RL/2 + XL = 50.2 \cdot 1153/2 + 0 = 290 \text{ kgf/mm}$$

Compute with XL+ [Ntheta+]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q) \\ = 50.2 \cdot 1153 + 2 \cdot 0.0058 \cdot 1153 \cdot \\ (-6532 \cdot 0.0058 + -74.6) = 87.5 \text{ kgf/mm}$$

Compute with XL- [Ntheta-]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q) \\ = 50.2 \cdot 1153 + 2 \cdot 0.0058 \cdot 1153 \cdot \\ (-6532 \cdot 0.0058 + -74.6) = 87.5 \text{ kgf/mm}$$

Note: Kpc has been set to 1

σ_{Sm} with XL+ [σ_{sm+}]:

$$= Ns / tL \\ = 290/42 \\ = 689.292 \text{ kgf/cm}^2$$

σ_{Sm} with XL- [σ_{sm-}]:

$$= Ns / tL \\ = 290/42 \\ = 689.292 \text{ kgf/cm}^2$$

σ_{Sb} with XL+ [σ_{sb+}]:

$$= 6 \cdot Ms / (tL^2 \cdot Kpc) \\ = 6 \cdot -6532 / (42^2 \cdot 1) \\ = -2221.705 \text{ kgf/cm}^2$$

σ_{Sb} with XL- [σ_{sb-}]:

$$= 6 \cdot Ms / (tL^2 \cdot Kpc) \\ = 6 \cdot -6532 / (42^2 \cdot 1) \\ = -2221.705 \text{ kgf/cm}^2$$



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σ_{Thetam} with XL+ [σ_{thetam}]:

$$\begin{aligned} &= N_{\text{theta}} / tL \\ &= 87.5/42 \text{ kgf/cm}^2 \\ &= 208.307 \text{ kgf/cm}^2 \end{aligned}$$

σ_{Thetam} with XL- [σ_{thetam}]:

$$\begin{aligned} &= N_{\text{theta}} / tL \\ &= 87.5/42 \\ &= 208.307 \text{ kgf/cm}^2 \end{aligned}$$

σ_{Thetab} with XL+ [σ_{thetab}]:

$$\begin{aligned} &= 6 \cdot v \cdot M_s / (tL^2 \cdot K_{pc}) \\ &= 6 \cdot 0.3 \cdot -6532 / (42^2 \cdot 1) \\ &= -666.512 \text{ kgf/cm}^2 \end{aligned}$$

σ_{Thetab} with XL- [σ_{thetab}]:

$$\begin{aligned} &= 6 \cdot v \cdot M_s / (tL^2 \cdot K_{pc}) \\ &= 6 \cdot 0.3 \cdot -6532 / (42^2 \cdot 1) \\ &= -666.512 \text{ kgf/cm}^2 \end{aligned}$$

For the Large End Cone Junction per Table 4.3.1

Compute [McsP]:

$$= M_{sp} = M_{sp} = -6532 \text{ kgf}$$

McsX with XL+ [McsX+]:

$$= M_{sX} = -0 \text{ kgf}$$

McsX with XL- [McsX-]:

$$= M_{sX} = -0 \text{ kgf}$$

Mcs with XL+ [Mcs+]:

$$\begin{aligned} &= M_{csP} + M_{sX} = -6532 + -0 \\ &= -6531.803 \text{ kgf} \end{aligned}$$

Mcs with XL- [Mcs-]:

$$\begin{aligned} &= M_{csP} + M_{sX} = -6532 + -0 \\ &= -6531.803 \text{ kgf} \end{aligned}$$

Compute with XL+ [McsX+]:

$$= M_{sX} = -0 \text{ kgf}$$

Compute with XL- [McsX-]:

$$= M_{sX} = -0 \text{ kgf}$$

Compute with XL+ [Qc+]:

$$\begin{aligned} &= Q \cdot \cos(a) + N_s \cdot \sin(a) \\ &= -74.6 \cdot \cos(29.7) + 290 \cdot \sin(29.7) \\ &= 78.837 \text{ kgf/mm} \end{aligned}$$

Compute with XL- [Qc-]:

$$\begin{aligned} &= Q \cdot \cos(a) + N_s \cdot \sin(a) \\ &= -74.6 \cdot \cos(29.7) + 290 \cdot \sin(29.7) \\ &= 78.837 \text{ kgf/mm} \end{aligned}$$

Compute [Rc]:

$$= RL / \cos(a) = 1153 / \cos(29.7) = 1328 \text{ mm}$$

Compute [β -co]:

$$\begin{aligned} &= (3 \cdot (1 - v^2) / (R_c^2 \cdot tC^2))^{1/4} \\ &= (3 \cdot (1 - 0.3^2) / (1328^2 \cdot 49^2))^{1/4} \\ &= 0.005 \text{ 1/mm} \end{aligned}$$

Compute with XL +ve [Ncs+]:



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$$\begin{aligned}
 &= N_s \cdot \cos(a) - Q \cdot \sin(a) \\
 &= 290 \cdot \cos(29.7) - 74.6 \cdot \sin(29.7) \\
 &= 288.385 \text{ kgf/mm}
 \end{aligned}$$

Compute with XL -ve [Ncs-]:

$$\begin{aligned}
 &= N_s \cdot \cos(a) - Q \cdot \sin(a) \\
 &= 290 \cdot \cos(29.7) - 74.6 \cdot \sin(29.7) \\
 &= 288.385 \text{ kgf/mm}
 \end{aligned}$$

Compute with XL +ve [NcTheta+]:

$$\begin{aligned}
 &= P \cdot RL / \cos(a) + 2 \cdot \beta_{co} \cdot Rc (-Mcs \cdot \beta_{co} - Qc) \\
 &= 50.2 \cdot 1153 / \cos(29.7) + 2 \cdot 0.005 \cdot 1328 \cdot \\
 &\quad (-6532 \cdot 0.005 - 78.8) \\
 &= 52.053 \text{ kgf/mm}
 \end{aligned}$$

Compute with XL -ve [NcTheta-]:

$$\begin{aligned}
 &= P \cdot RL / \cos(A) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc) \\
 &= 50.2 \cdot 1153 / \cos(29.7) + 2 \cdot 0.005 \cdot 1328 \cdot \\
 &\quad (-6532 \cdot 0.005 - 78.8) \\
 &= 52.053 \text{ kgf/mm}
 \end{aligned}$$

Note: Kcpc = 1

Compute with XL +ve [σsm+]:

$$= Ncs / tC = 288 / 49 = 589 \text{ kgf/cm}^2$$

Compute with XL -ve [σsm-]:

$$= Ncs / tC = 288 / 49 = 589 \text{ kgf/cm}^2$$

Compute with XL +ve [σsb+]:

$$\begin{aligned}
 &= 6 \cdot Mcs / (tC^2 \cdot Kcpc) \\
 &= 6 \cdot -6532 / (49^2 \cdot 1) \\
 &= -1632.273 \text{ kgf/cm}^2
 \end{aligned}$$

Compute with XL -ve [σsb-]:

$$\begin{aligned}
 &= 6 \cdot Mcs / (tC^2 \cdot Kcpc) \\
 &= 6 \cdot -6532 / (49^2 \cdot 1) \\
 &= -1632.273 \text{ kgf/cm}^2
 \end{aligned}$$

Compute with XL +ve [σThetam+]:

$$= NCTheta / tC = 52.1 / 49 = 106 \text{ kgf/cm}^2$$

Compute with XL -ve [σThetam-]:

$$= NCTheta / tC = 52.1 / 49 = 106 \text{ kgf/cm}^2$$

Compute with XL +ve [σthetab+]:

$$\begin{aligned}
 &= 6 \cdot v \cdot Mcs / (tC^2 \cdot Kcpc) \\
 &= 6 \cdot 0.3 \cdot -6532 / (49^2 \cdot 1) \\
 &= -489.682 \text{ kgf/cm}^2
 \end{aligned}$$

Compute with XL -ve [σthetab-]:

$$\begin{aligned}
 &= 6 \cdot v \cdot Mcs / (tC^2 \cdot Kcpc) \\
 &= 6 \cdot 0.3 \cdot -6532 / (49^2 \cdot 1) \\
 &= -489.682 \text{ kgf/cm}^2
 \end{aligned}$$

Large End under External Pressure:

Cone Length along Cone surface per 4.3.46 [ConeMinLen]:

$$\begin{aligned}
 &= 2 \cdot \sqrt{(RL \cdot Tc / \cos(\alpha)) + 1.4 \cdot \sqrt{(Rs \cdot tc / \cos(\alpha))}} \\
 &= 2 \cdot \sqrt{(1153 \cdot 49 / \cos(29.7)) + 1.4 \cdot \sqrt{(653 \cdot 49 / \cos(29.7))}} \\
 &= 779.037 \text{ mm}
 \end{aligned}$$

Results for the Large End Cylinder Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
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Sigma sm+	-14.185	2676.619	Pass
Sigma sm-	14.185	2676.619	Pass
Sigma thetam+	-4.287	2676.619	Pass
Sigma thetam-	-4.287	2676.619	Pass
Sigma sm+ + Sigma	31.535	5353.238	Pass
Sigma sm+ - Sigma	-59.904	5353.238	Pass
Sigma sm- + Sigma	31.535	5353.238	Pass
Sigma sm- - Sigma	-59.904	5353.238	Pass
Sigma thetam+ +	9.429	5353.238	Pass
Sigma thetam+ -	-18.003	5353.238	Pass
Sigma thetam- +	9.429	5353.238	Pass
Sigma thetam- -	-18.003	5353.238	Pass

Results for the Large End Cone Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	-12.111	2676.619	Pass
Sigma sm-	12.111	2676.619	Pass
Sigma thetam+	-2.186	2676.619	Pass
Sigma thetam-	-2.186	2676.619	Pass
Sigma sm+ + Sigma	21.479	5353.238	Pass
Sigma sm+ - Sigma	-45.701	5353.238	Pass
Sigma sm- + Sigma	21.479	5353.238	Pass
Sigma sm- - Sigma	-45.701	5353.238	Pass
Sigma thetam+ +	7.891	5353.238	Pass
Sigma thetam+ -	-12.263	5353.238	Pass
Sigma thetam- +	7.891	5353.238	Pass
Sigma thetam- -	-12.263	5353.238	Pass

Average Compressive Stresses and comparisons with Fha and Fxa:

SigmaSmCyl+ + S	-13.148	-1858.103	Pass
SigmaSmCyl- + S	-13.148	-1858.103	Pass
SigmaThetamCyl+	-3.236	-1602.589	Pass
SigmaThetamCyl-	-3.236	-1602.589	Pass

Cone Large end without a Knuckle Cylinder Section:

Net Axial Force ML +ve [XL+]:
 $= FL / (2 \cdot \pi \cdot RL) + ML / (\pi \cdot RL^2)$
 $= 0 / (2 \cdot 3.141 \cdot 1153) + 0 / (3.14 \cdot 1153^2)$
 $= 0 \text{ kgf/mm}$

Net Axial Force ML -ve [XL-]:
 $= FL / (2 \cdot \pi \cdot RL) - ML / (\pi \cdot RL^2)$
 $= 0 / (2 \cdot 3.14 \cdot 1153) - 0 / (3.141 \cdot 1153^2)$
 $= 0 \text{ kgf/mm}$

Thickness Ratio [n]:
 $= tc/tL = 49/42 = 1.17$

Curve Fitting Geometry [H and B]:
 $H = \text{Sqrt}(0.5 \cdot D/tL) = \text{Sqrt}(0.5 \cdot 2307/42) = 5.24$
 $B = \tan(29.7) = 0.57$

This analysis is preformed per Table 4.3.1 for the Large End:

Using Data from Table 4.3.3 and 4.3.4 knowing n
 Table 4.3.3 - n: $1 < (n=1.17) < 1.25$

Use n to construct a table: C1, C2, C3..... by interpolation

Table 4.3.3 [MsN1]:
 C1 = -3.097512 C2 = 3.686273 C3 = 0.761135 C4 = -0.233165



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C5 = -0.077764 C6 = 0.067131 C7 = 0.026880 C8 = -0.015577
C9 = 0.001363 C10 = -0.010718 C11 = 0.000000

Compute using Table 4.3.3 [MsN1]:

$$\begin{aligned} X &= C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + \\ &C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) \\ &= C1 + C2 \cdot \ln(5.24) + C3 \cdot \ln(0.57) + C4 \cdot (\ln(5.24))^2 + C5 \cdot (\ln(0.57))^2 + C6 \cdot \ln(5.24) \cdot \ln(0.57) + \\ &C7 \cdot (\ln(5.24))^3 + C8 \cdot (\ln(0.57))^3 + C9 \cdot \ln(5.24) \cdot (\ln(0.57))^2 + C10 \cdot (\ln(5.24))^2 \cdot \ln(0.57) \\ X &= 2, \text{ MsN1} = -\exp(X) = -7.38 \end{aligned}$$

Table 4.3.3 [Qn1]:

C1 = -1.935534 C2 = 2.331614 C3 = 0.527501 C4 = -0.092494
C5 = -0.088386 C6 = 0.187162 C7 = 0.008887 C8 = -0.020067
C9 = 0.001094 C10 = -0.028094 C11 = 0.000000

Compute using Table 4.3.3 [Qn1]:

$$\begin{aligned} X &= C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + \\ &C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) \\ &= C1 + C2 \cdot \ln(5.24) + C3 \cdot \ln(0.57) + C4 \cdot (\ln(5.24))^2 + C5 \cdot (\ln(0.57))^2 + C6 \cdot \ln(5.24) \cdot \ln(0.57) + \\ &C7 \cdot (\ln(5.24))^3 + C8 \cdot (\ln(0.57))^3 + C9 \cdot \ln(5.24) \cdot (\ln(0.57))^2 + C10 \cdot (\ln(5.24))^2 \cdot \ln(0.57) \\ X &= 1.26, \text{ Qn} = -\exp(X) = -3.54 \end{aligned}$$

Table 4.3.4 MsN2:

C1 = -5.717372 C2 = 0.005787 C3 = 0.473328 C4 = -0.213055
C5 = 2.250262 C6 = 0.000017 C7 = 0.003074 C8 = -0.001940
C9 = -0.215397 C10 = -0.000026 C11 = -0.106247

Compute using Table 4.3.4 [MsN2]:

$$\begin{aligned} X1 &= C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) \\ &= C1 + C3 \cdot \ln(5.24^2) + C5 \cdot \ln(0.52) + C7 \cdot (\ln(5.24^2))^2 + C9 \cdot (\ln(0.52))^2 + \\ &C11 \cdot \ln(5.24^2) \cdot \ln(0.52) = -0.15 \\ X2 &= 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) \\ &= 1 + C2 \cdot \ln(5.24) + C4 \cdot \ln(0.52) + C6 \cdot (\ln(5.24^2))^2 + C8 \cdot (\ln(0.52))^2 + \\ &C10 \cdot \ln(5.24^2) \cdot \ln(0.52) = 0.27 \\ \text{MsN2} &= \exp(X1/X2) = \exp(-0.15/0.27) = -0.57 \end{aligned}$$

Table 4.3.4 QN2:

C1 = -5.008318 C2 = 0.014737 C3 = -0.038229 C4 = -0.196938
C5 = 2.098516 C6 = -0.000277 C7 = 0.000047 C8 = -0.004739
C9 = -0.216746 C10 = -0.001647 C11 = 0.008920

Compute using Table 4.3.4 [QN2]:

$$\begin{aligned} X3 &= C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) \\ &= C1 + C3 \cdot \ln(5.24^2) + C5 \cdot \ln(0.52) + C7 \cdot (\ln(5.24^2))^2 + C9 \cdot (\ln(0.52))^2 + \\ &C11 \cdot \ln(5.24^2) \cdot \ln(0.52) = -0.15 \\ X4 &= 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) \\ &= 1 + C2 \cdot \ln(5.24) + C4 \cdot \ln(0.52) + C6 \cdot (\ln(5.24^2))^2 + C8 \cdot (\ln(0.52))^2 + \\ &C10 \cdot \ln(5.24^2) \cdot \ln(0.52) = 0.27 \\ \text{Qn2} &= \exp(X3/X4) = \exp(-0.41/0.3) = -0.26 \end{aligned}$$

Stresses for the Large End Cylinder per Table 4.3.1

Stress Resultant Value [Msp]:

$$= P \cdot tL^2 \cdot \text{MsN1} = -1.03 \cdot 42^2 \cdot -7.38 = 134 \text{ kgf}$$

Compute with XL+ [Msx+]:

$$= XL \cdot tL \cdot \text{MsN2} = 0 \cdot 42 \cdot -0.57 = -0 \text{ kgf}$$

Compute with XL- [Msx-]:

$$= XL \cdot tL \cdot \text{MsN2} = 0 \cdot 42 \cdot -0.57 = -0 \text{ kgf}$$

Compute with XL+ [Ms+]:

$$= \text{MsP} + \text{Msx} = 134 + -0 = 134 \text{ kgf}$$

Compute with XL- [Ms-]:



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$$= MsP + Msx = 134 + -0 = 134 \text{ kgf}$$

Compute [Qp]:

$$= P \cdot tL \cdot Qn1 = -1.03 \cdot 42 \cdot -3.54 = 1.54 \text{ kgf/mm}$$

Compute with XL+ [Qx+]:

$$= XL \cdot QN2 = 0 \cdot -0.0047 = -0 \text{ kgf/mm}$$

Compute with XL- [Qx-]:

$$= XL \cdot QN2 = 0 \cdot -0.0047 = -0 \text{ kgf/mm}$$

Compute with XL+ [Q+]:

$$= Qp + Qx = 1.54 + -0 = 1.54 \text{ kgf/mm}$$

Compute with XL- [Q-]:

$$= Qp + Qx = 1.54 + -0 = 1.54 \text{ kgf/mm}$$

Compute [βcy]:

$$= 3 \cdot (1 - v^2) / (RL^2 \cdot tL^2) \cdot 0.25 \\ = (3 \cdot (1 - 0.3^2) / (1153^2 \cdot 42^2)) \cdot 0.25 = 0.0058 \text{ 1/mm}$$

Compute with XL+ [Ns+]:

$$= P \cdot RL/2 + XL = -1.03 \cdot 1153/2 + 0 = -5.96 \text{ kgf/mm}$$

Compute with XL- [Ns-]:

$$= P \cdot RL/2 + XL = -1.03 \cdot 1153/2 + 0 = -5.96 \text{ kgf/mm}$$

Compute with XL+ [Ntheta+]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q) \\ = -1.03 \cdot 1153 + 2 \cdot 0.0058 \cdot 1153 \cdot \\ (-134 \cdot 0.0058 + 1.54) = -1.8 \text{ kgf/mm}$$

Compute with XL- [Ntheta-]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q) \\ = -1.03 \cdot 1153 + 2 \cdot 0.0058 \cdot 1153 \cdot \\ (-134 \cdot 0.0058 + 1.54) = -1.8 \text{ kgf/mm}$$

Note: Kpc has been set to 1

σSm with XL+ [σsm+]:

$$= Ns / tL \\ = -5.96/42 \\ = -14.185 \text{ kgf/cm}^2$$

σSm with XL- [σsm-]:

$$= Ns / tL \\ = -5.96/42 \\ = -14.185 \text{ kgf/cm}^2$$

σSb with XL+ [σsb+]:

$$= 6 \cdot Ms / (tL^2 \cdot Kpc) \\ = 6 \cdot 134 / (42^2 \cdot 1) \\ = 45.720 \text{ kgf/cm}^2$$

σSb with XL- [σsb-]:

$$= 6 \cdot Ms / (tL^2 \cdot Kpc) \\ = 6 \cdot 134 / (42^2 \cdot 1) \\ = 45.720 \text{ kgf/cm}^2$$

σThetam with XL+ [σthetam+]:

$$= Ntheta / tL \\ = -1.8/42 \text{ kgf/cm}^2 \\ = -4.287 \text{ kgf/cm}^2$$

σThetam with XL- [σthetam-]:



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$$= N_{\theta} / tL$$

$$= -1.8/42$$

$$= -4.287 \text{ kgf/cm}^2$$

$$\sigma_{\theta} \text{ with XL+ } [\sigma_{\theta} \text{ with XL+}]:$$

$$= 6 \cdot v \cdot M_s / (tL^2 \cdot K_{pc})$$

$$= 6 \cdot 0.3 \cdot 134 / (42^2 \cdot 1)$$

$$= 13.716 \text{ kgf/cm}^2$$

$$\sigma_{\theta} \text{ with XL- } [\sigma_{\theta} \text{ with XL-}]:$$

$$= 6 \cdot v \cdot M_s / (tL^2 \cdot K_{pc})$$

$$= 6 \cdot 0.3 \cdot 134 / (42^2 \cdot 1)$$

$$= 13.716 \text{ kgf/cm}^2$$

For the Large End Cone Junction per Table 4.3.1

$$\text{Compute } [M_{csP}]:$$

$$= M_{sp} = M_{sp} = 134 \text{ kgf}$$

$$M_{csX} \text{ with XL+ } [M_{csX+}]:$$

$$= M_{sX} = -0 \text{ kgf}$$

$$M_{csX} \text{ with XL- } [M_{csX-}]:$$

$$= M_{sX} = -0 \text{ kgf}$$

$$M_{cs} \text{ with XL+ } [M_{cs+}]:$$

$$= M_{csP} + M_{sX} = 134 + -0$$

$$= 134.416 \text{ kgf}$$

$$M_{cs} \text{ with XL- } [M_{cs-}]:$$

$$= M_{csP} + M_{sX} = 134 + -0$$

$$= 134.416 \text{ kgf}$$

$$\text{Compute with XL+ } [M_{csX+}]:$$

$$= M_{sX} = -0 \text{ kgf}$$

$$\text{Compute with XL- } [M_{csX-}]:$$

$$= M_{sX} = -0 \text{ kgf}$$

$$\text{Compute with XL+ } [Q_{c+}]:$$

$$= Q \cdot \cos(a) + N_s \cdot \sin(a)$$

$$= 1.54 \cdot \cos(29.7) + -5.96 \cdot \sin(29.7)$$

$$= -1.622 \text{ kgf/mm}$$

$$\text{Compute with XL- } [Q_{c-}]:$$

$$= Q \cdot \cos(a) + N_s \cdot \sin(a)$$

$$= 1.54 \cdot \cos(29.7) + -5.96 \cdot \sin(29.7)$$

$$= -1.622 \text{ kgf/mm}$$

$$\text{Compute } [R_c]:$$

$$= RL / \cos(a) = 1153 / \cos(29.7) = 1328 \text{ mm}$$

$$\text{Compute } [\beta - \text{co}]:$$

$$= (3 \cdot (1 - v^2) / (R_c^2 \cdot tC^2))^{1/4}$$

$$= (3 \cdot (1 - 0.3^2) / (1328^2 \cdot 49^2))^{1/4}$$

$$= 0.005 \text{ 1/mm}$$

$$\text{Compute with XL +ve } [N_{cs+}]:$$

$$= N_s \cdot \cos(a) - Q \cdot \sin(a)$$

$$= -5.96 \cdot \cos(29.7) - 1.54 \cdot \sin(29.7)$$

$$= -5.935 \text{ kgf/mm}$$

$$\text{Compute with XL -ve } [N_{cs-}]:$$

$$= N_s \cdot \cos(a) - Q \cdot \sin(a)$$

$$= -5.96 \cdot \cos(29.7) - 1.54 \cdot \sin(29.7)$$



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= -5.935 kgf/mm

Compute with XL +ve [NcTheta+]:

= $P \cdot RL / \cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$

= $-1.03 \cdot 1153 / \cos(29.7) + 2 \cdot 0.005 \cdot 1328 \cdot$

$(-134 \cdot 0.005 - -1.62)$

= -1.071 kgf/mm

Compute with XL -ve [NcTheta-]:

= $P \cdot RL / \cos(A) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$

= $-1.03 \cdot 1153 / \cos(29.7) + 2 \cdot 0.005 \cdot 1328 \cdot$

$(-134 \cdot 0.005 - -1.62)$

= -1.071 kgf/mm

Note: Kcpc = 1

Compute with XL +ve [σ_{sm}]:

= $Ncs/tC = -5.93/49 = -12.1 \text{ kgf/cm}^2$

Compute with XL -ve [σ_{sm}]:

= $Ncs/tC = -5.93/49 = -12.1 \text{ kgf/cm}^2$

Compute with XL +ve [σ_{sb}]

= $6 \cdot Mcs / (tC^2 \cdot Kcpc)$

= $6 \cdot 134 / (49^2 \cdot 1)$

= 33.590 kgf/cm²

Compute with XL -ve [σ_{sb}]

= $6 \cdot Mcs / (tC^2 \cdot Kcpc)$

= $6 \cdot 134 / (49^2 \cdot 1)$

= 33.590 kgf/cm²

Compute with XL +ve [$\sigma_{Thetam+}$]:

= $NCTheta/tC = -1.07/49 = -2.19 \text{ kgf/cm}^2$

Compute with XL -ve [$\sigma_{Thetam-}$]:

= $NCTheta/tC = -1.07/49 = -2.19 \text{ kgf/cm}^2$

Compute with XL +ve [$\sigma_{thetab+}$]:

= $6 \cdot v \cdot Mcs / (tC^2 \cdot Kcpc)$

= $6 \cdot 0.3 \cdot 134 / (49^2 \cdot 1)$

= 10.077 kgf/cm²

Compute with XL -ve [$\sigma_{thetab-}$]:

= $6 \cdot v \cdot Mcs / (tC^2 \cdot Kcpc)$

= $6 \cdot 0.3 \cdot 134 / (49^2 \cdot 1)$

= 10.077 kgf/cm²

Calculations for the Small End Cone/Shell Junction

Fhe per 4.1.3 or 4.3.11.2 Step 6 [Fhe]:

= $0.4 \cdot E(tc/D)$

= $0.4 \cdot 1909280 (49/1360)$

= 27516.1 kgf/cm²

Factor of Safety [FS]:

Since $Fic [2672] \geq Fy[2672]$, FS = 1.667

Allowable Stress per 4.3.11 Step 6 [Fha]:

= Fic/FS

= $2672/1.67$

= 1602.589 kgf/cm²

Small End under Internal Pressure:



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Results for Cylinder Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	607.441	2676.614	Pass
Sigma sm-	607.441	2676.614	Pass
Sigma thetam+	1741.783	2676.614	Pass
Sigma thetam-	1741.783	2676.614	Pass
Sigma sm+ + Sigma	2138.672	5353.228	Pass
Sigma sm- + Sigma	2138.672	5353.228	Pass
Sigma sm+ - Sigma	-923.791	5353.228	Pass
Sigma sm- - Sigma	-923.791	5353.228	Pass
Sigma thetam+ +	2201.153	5353.228	Pass
Sigma thetam- +	2201.153	5353.228	Pass
Sigma thetam+ -	1282.414	5353.228	Pass
Sigma thetam- -	1282.414	5353.228	Pass

Results for the Cone Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	320.231	2676.619	Pass
Sigma sm-	320.231	2676.619	Pass
Sigma thetam+	1665.865	2676.619	Pass
Sigma thetam-	1665.865	2676.619	Pass
Sigma sm+ + Sigma	785.148	5353.238	Pass
Sigma sm- + Sigma	785.148	5353.238	Pass
Sigma sm+ - Sigma	-144.687	5353.238	Pass
Sigma sm- - Sigma	-144.687	5353.238	Pass
Sigma thetam+ +	1805.341	5353.238	Pass
Sigma thetam- +	1805.341	5353.238	Pass
Sigma thetam+ -	1526.390	5353.238	Pass
Sigma thetam- -	1526.390	5353.238	Pass

Small End Cylinder Without a Flare

Compute the ratio equation 4.3.60 [n]:
 $= t_c/t_s = 49/27 = 1.81$

Compute the ratio equation 4.3.61 [H]:
 $= \sqrt{(R_s/t_s)} = \sqrt{(653/27)} = 4.92$

Compute the ratio equation 4.3.62 [B]:
 $= \tan(a) = \tan(29.7) = 0.57$

Compute Ms+ from equation 4.3.59 [Xs+]:
 $= F_s / (2 \cdot \pi \cdot R_s) + M_s / (\pi \cdot R_s^2)$
 $= 0 / (2 \cdot 3.14 \cdot 653) + 0 / (3.141 \cdot 653^2)$
 $= 0.0000 \text{ kgf/mm}$

Compute Ms- from equation 4.3.59 [Xs-]:
 $= F_s / (2 \cdot \pi \cdot R_s) - M_s / (\pi \cdot R_s^2) + X_s'$
 $= 0 / (2 \cdot 3.14 \cdot 653) - 0 / (3.141 \cdot 653^2) + 0$
 $= 0.0000 \text{ kgf/mm}$

From Table 4.3.5 MsN:

C1 = -15.185586 C2 = 3.046973 C3 = 6.478770 C4 = -0.156801
 C5 = -1.465606 C6 = -0.372439 C7 = 0.007770 C8 = 0.143329
 C9 = 0.041178 C10 = 0.007300 C11 = 0.000000

Compute from Table 4.3.5 [MsN]:
 $= \exp[C1 + C2 \cdot \ln(H^2) + C3 \cdot \ln(a) + C4 \cdot (\ln(H^2))^2 + C5 \cdot \ln(a)^2 + C6 \cdot \ln(H^2) \cdot \ln(a) +$
 $C7 \cdot \ln(H^2)^3 + C8 \cdot \ln(a)^3 + C9 \cdot \ln(H^2) \cdot \ln(a)^2 + C10 \cdot \ln(H^2)^2 \cdot \ln(a)]$
 $= \exp[-15.2 + 3.05 \cdot \ln(4.92^2) + 6.48 \cdot \ln(29.7) +$



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$$-0.16 \cdot (\ln(4.92^2))^2 + -1.47 \cdot \ln(29.7)^2 +$$

$$-0.37 \cdot \ln(4.92^2) \cdot \ln(29.7) + 0.0078 \cdot \ln(4.92^2)^3$$

$$0.14 \cdot \ln(29.7)^3 + 0.041 \cdot \ln(4.92^2) \cdot \ln(29.7)^2 +$$

$$0.0073 \cdot \ln(4.92^2)^2 \cdot \ln(29.7)] = 5.08$$

Compute from Table 4.3.2 [Msp]:
 $= p \cdot tL^2 \cdot Msn = 50.2 \cdot 27^2 \cdot 5.08 = 1860 \text{ kgf}$

From table 4.3.6 MsN:

C1 =	0.006829	C2 =	0.000292	C3 =	-0.000927	C4 =	0.121601
C5 =	0.010585	C6 =	-0.000011	C7 =	-0.000008	C8 =	0.005945
C9 =	0.001298	C10 =	0.000186	C11 =	0.194335		

Compute from Table 4.3.6 [MsN]:
 $= (C1 + C3 \cdot H + C5 \cdot B + C7 \cdot H^2 + C9 \cdot B^2 + C11 \cdot H \cdot B) /$
 $(1 + C2 \cdot H + C4 \cdot B + C6 \cdot H^2 + C8 \cdot B^2 + C10 \cdot H \cdot B)$
 $= (0.0068 + -0.00093 \cdot 4.92 + 0.011 \cdot 29.7 +$
 $-0.80000E-05 \cdot 4.92^2 + 0.0013 \cdot 29.7^2 +$
 $0.19 \cdot 4.92 \cdot 29.7) / (1 + 0.00029 \cdot 4.92 +$
 $0.12 \cdot 29.7 + -0.10556E-04 \cdot 4.92^2 + 0.0059 \cdot 29.7^2 +$
 $0.00019 \cdot 4.92 \cdot 29.7)$
 $= 0.5170$

Compute from Table 4.3.2 [Msx+]:
 $= Xs \cdot ts \cdot MsN = 0 \cdot 27 \cdot 0.52 = 0 \text{ kgf}$

Compute from Table 4.3.2 [Msx-]:
 $= Xs \cdot ts \cdot MsN = 0 \cdot 27 \cdot 0.52 = 0 \text{ kgf}$

Compute from Table 4.3.2 [Ms+]:
 $= Ms + Msx+ = 1860 + 0 = 1860 \text{ kgf}$

Compute from Table 4.3.2 [Ms-]:
 $= Ms + Msx- = 1860 + 0 = 1860 \text{ kgf}$

From table 4.3.5 Qn:

C1 =	0.571280	C2 =	-0.000027	C3 =	0.008448	C4 =	0.002689
C5 =	-0.002892	C6 =	0.000000	C7 =	-0.000005	C8 =	-0.000117
C9 =	-0.000087	C10 =	0.000001	C11 =	-0.003775		

Compute from Table 4.3.5 [Qn]:
 $= (C1 + C3 \cdot H^2 + C5 \cdot \alpha + C7 \cdot H^4 + C9 \cdot \alpha^2 + C11 \cdot H^2 \cdot \alpha) /$
 $(1 + C2 \cdot H^2 + C4 \cdot \alpha + C6 \cdot H^4 + C8 \cdot \alpha^2 + C10 \cdot H^2 \cdot \alpha)$
 $= (0.57 + 0.0084 \cdot 4.92^2 + -0.0029 \cdot 29.7$
 $-0.52593E-05 \cdot 4.92^4 + -0.87148E-04 \cdot 29.7^2 + -0.0038 \cdot 4.92^2$
 $\cdot 29.7) / (1 + -0.27407E-04 \cdot 4.92^2 + 0.0027 \cdot 29.7 +$
 $0 \cdot 4.92^4 + -0.00012 \cdot 29.7^2 + 0.10000E-05 \cdot 4.92^2 \cdot 29.7)$
 $= -2.158$

Compute from Table 4.3.2 [Qp]:
 $= P \cdot ts \cdot Qn = 50.2 \cdot 27 \cdot -2.16 = -29.3 \text{ kgf/mm}$

From table 4.3.6 Qn:

C1 =	-0.407896	C2 =	0.021262	C3 =	-0.325315	C4 =	-0.004004
C5 =	-0.111190	C6 =	0.002205	C7 =	0.000256	C8 =	-0.014421
C9 =	0.000820	C10 =	0.000106	C11 =	0.000000		

Compute from Table 4.3.6 [Qn]:
 $= C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot \ln(H)^2 + C5 \cdot \ln(B)^2 + C6 \cdot \ln(H) \cdot \ln(B) +$
 $C7 \cdot \ln(H)^3 + C8 \cdot \ln(B)^3 + C9 \cdot \ln(H) \cdot \ln(B)^2 + C10 \cdot \ln(H)^2 \cdot \ln(B)$
 $= -0.41 + 0.021 \cdot \ln(4.92) + -0.33 \cdot \ln(0.57) +$
 $-0.004 \cdot \ln(4.92)^2 + -0.11 \cdot (\ln(0.57))^2 +$
 $0.0022 \cdot \ln(4.92) \cdot \ln(0.57) + 0.00026 \cdot \ln(4.92)^3 +$
 $-0.014 \cdot \ln(0.57)^3 + 0.00082 \cdot \ln(4.92) \cdot \ln(0.57)^2 +$
 $0.00011 \cdot \ln(4.92)^2 \cdot \ln(0.57)$



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$$= -0.2351$$

Compute with Xs+ [Qx+]:
 $= Xs+ \cdot Qn = 0 \cdot -0.24 = -0 \text{ kgf/mm}$

Compute with Xs- [Qx-]:
 $= Xs- \cdot Qn = 0 \cdot -0.24 = -0 \text{ kgf/mm}$

Compute with Xs+ [Q+]:
 $= Qp + Qx+ = -29.3 + -0 = -29.3 \text{ kgf/mm}$

Compute with Xs- [Q-]:
 $= Qp + Qx- = -29.3 + -0 = -29.3 \text{ kgf/mm}$

Compute angle [β_{cy}]:
 $= (3 \cdot (1-v^2) / (Rs^2 \cdot ts^2))^{0.25} = (3 \cdot (1-0.3^2) / (653^2 \cdot 27^2))^{0.25}$
 $= 0.0097 \text{ 1/mm}$

Compute with Xs+ [Ns+]:
 $= P \cdot Rs / 2 + Xs+ = 50.2 \cdot 653 / 2 + 0 = 164 \text{ kgf/mm}$

Compute with Xs- [Ns-]:
 $= P \cdot Rs / 2 + Xs- = 50.2 \cdot 653 / 2 + 0 = 164 \text{ kgf/mm}$

Compute with Xs+ [NTheta+]:
 $= P \cdot Rs + 2 \cdot \beta_{cy} \cdot Rs \cdot (-Ms \cdot \beta_{cy} - Q)$
 $= 50.2 \cdot 653 + 2 \cdot 0.0097 \cdot 653 \cdot (-1860 \cdot 0.0097 - 1.54) = 470 \text{ kgf/mm}$

Compute with Xs- [NTheta-]:
 $= P \cdot Rs + 2 \cdot \beta_{cy} \cdot Rs \cdot (-Ms \cdot \beta_{cy} - Q)$
 $= 50.2 \cdot 653 + 2 \cdot 0.0097 \cdot 653 \cdot (-1860 \cdot 0.0097 - 1.54) = 470 \text{ kgf/mm}$

Compute Stress with Xs+ [σ_{sm+}]:
 $= Ns(+)/ts = 164/27 = 607 \text{ kgf/cm}^2$

Compute Stress with Xs- [σ_{sm-}]:
 $= Ns(-)/ts = 164/27 = 607 \text{ kgf/cm}^2$

Compute stress with Xs+ [σ_{sb+}]:
 $= 6 \cdot Ms(+)/(ts^2 \cdot K_{pc}) = 6 \cdot 1860 / (27^2 \cdot 1) = 1531 \text{ kgf/cm}^2$

Compute stress with Xs- [σ_{sb-}]:
 $= 6 \cdot Ms(-)/(ts^2 \cdot K_{pc}) = 6 \cdot 1860 / (27^2 \cdot 1) = 1531 \text{ kgf/cm}^2$

Compute stress with Xs+ [$\sigma_{\theta\theta am+}$]:
 $= N_{\theta\theta}(+)/ts = 470/27 = 1742 \text{ kgf/cm}^2$

Compute stress with Xs- [$\sigma_{\theta\theta am-}$]:
 $= N_{\theta\theta}(-)/ts = 470/27 = 1742 \text{ kgf/cm}^2$

Compute stress with Xs+ [$\sigma_{\theta\theta ab+}$]:
 $= 6 \cdot v \cdot Ms(+)/(ts^2 \cdot K_{pc}) = 6 \cdot 0.3 \cdot 1860 / (27^2 \cdot 1) = 459 \text{ kgf/cm}^2$

Compute stress with Xs- [$\sigma_{\theta\theta ab-}$]:
 $= 6 \cdot v \cdot Ms(-)/(ts^2 \cdot K_{pc}) = 6 \cdot 0.3 \cdot 1860 / (27^2 \cdot 1) = 459 \text{ kgf/cm}^2$

Small End Cone Without a Flare

Compute with Xs+ [Qc+]:
 $= Q+ \cdot \cos(a) + Ns+ \cdot \sin(a) = -29.3 \cdot \cos(29.7) + 164 \cdot \sin(29.7)$
 $= 55.9729 \text{ kgf/mm}$

Compute with Xs- [Qc-]:



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$$= Q \cdot \cos(a) + N_s \cdot \sin(a) = -29.3 \cdot \cos(29.7) + 164 \cdot \sin(29.7) \\ = 55.9729 \text{ kgf/mm}$$

$$\text{Compute effective radius [Rc]:} \\ = R_s / \cos(a) = 653 / \cos(29.7) = 753 \text{ mm}$$

$$\text{Compute the angle } [\beta_{co}]: \\ = (3 \cdot (1 - v^2) / (R_c^2 \cdot t_c^2))^{0.25} = (3 \cdot (1 - 0.3^2) / (753^2 \cdot 49^2))^{0.25} \\ = 0.0067 \text{ 1/mm}$$

$$\text{Compute with Ms+ [Ncs+]:} \\ = N_s \cdot \cos(a) - Q \cdot \sin(a) \\ = 164 \cdot \cos(29.7) - -29.3 \cdot \sin(29.7) = 157 \text{ kgf/mm}$$

$$\text{Compute with Ms- [Ncs-]:} \\ = N_s \cdot \cos(a) - Q \cdot \sin(a) \\ = 164 \cdot \cos(29.7) - -29.3 \cdot \sin(29.7) = 157 \text{ kgf/mm}$$

$$\text{Compute with Ms+ [Nctheta+]:} \\ = P \cdot R_s / \cos(a) + 2 \cdot \beta_{co} \cdot R_c \cdot (-M_{cs} \cdot \beta_{co} + Q_c) \\ = 50.2 \cdot 653 / \cos(29.7) + 2 \cdot 0.0067 \cdot 753 \cdot \\ (-1860 \cdot 0.0067 + 56) = 816 \text{ kgf/mm}$$

$$\text{Compute with Ms- [Nctheta-]:} \\ = P \cdot R_s / \cos(a) + 2 \cdot \beta_{co} \cdot R_c \cdot (-M_{cs} \cdot \beta_{co} + Q_c) \\ = 50.2 \cdot 653 / \cos(29.7) + 2 \cdot 0.0067 \cdot \\ (-1860 \cdot 0.0067 + 56) = 816 \text{ kgf/mm}$$

$$\text{Compute the stress with Ms+ } [\sigma_{sm+}]: \\ = N_{cs} / t_c = 157 / 49 = 320 \text{ kgf/cm}^2$$

$$\text{Compute the stress with Ms- } [\sigma_{sm-}]: \\ = N_{cs} / t_c = 157 / 49 = 320 \text{ kgf/cm}^2$$

$$\text{Compute the stress with Ms+ } [\sigma_{sb+}]: \\ = 6 \cdot M_{cs} / (t_c^2 \cdot K_{cpc}) = 6 \cdot 1860 / (49^2 \cdot 1) = 465 \text{ kgf/cm}^2$$

$$\text{Compute the stress with Ms- } [\sigma_{sb-}]: \\ = 6 \cdot M_{cs} / (t_c^2 \cdot K_{cpc}) = 6 \cdot 1860 / (49^2 \cdot 1) = 465 \text{ kgf/cm}^2$$

$$\text{Compute stress with Ms+ } [\sigma_{thetam+}]: \\ = N_{ctheta} / t_c = 816 / 49 = 1666 \text{ kgf/cm}^2$$

$$\text{Compute stress with Ms- } [\sigma_{thetam-}]: \\ = N_{ctheta} / t_c = 816 / 49 = 1666 \text{ kgf/cm}^2$$

$$\text{Compute stress with Ms+ } [\sigma_{thetab+}]: \\ = 6 \cdot v \cdot M_{cs} / (t_c^2 \cdot K_{cpc}) = 6 \cdot 0.3 \cdot 1860 / (49^2 \cdot 1) = 139 \text{ kgf/cm}^2$$

$$\text{Compute stress with Ms- } [\sigma_{thetab-}]: \\ = 6 \cdot v \cdot M_{cs} / (t_c^2 \cdot K_{cpc}) = 6 \cdot 0.3 \cdot 1860 / (49^2 \cdot 1) = 139 \text{ kgf/cm}^2$$

Small End under External Pressure:

Results for Cylinder Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	-12.500	2676.614	Pass
Sigma sm-	-12.500	2676.614	Pass
Sigma thetam+	-35.844	2676.614	Pass
Sigma thetam-	-35.844	2676.614	Pass
Sigma sm+ + Sigma	-44.011	5353.228	Pass
Sigma sm- + Sigma	-44.011	5353.228	Pass
Sigma sm+ - Sigma	19.010	5353.228	Pass



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Sigma sm- - Sigm	19.010	5353.228	Pass
Sigma thetam+ +	-45.297	5353.228	Pass
Sigma thetam- +	-45.297	5353.228	Pass
Sigma thetam+ -	-26.390	5353.228	Pass
Sigma thetam- -	-26.390	5353.228	Pass

Results for the Cone Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	-6.590	2676.619	Pass
Sigma sm-	-6.590	2676.619	Pass
Sigma thetam+	-34.281	2676.619	Pass
Sigma thetam-	-34.281	2676.619	Pass
Sigma sm+ + Sigm	-16.157	5353.238	Pass
Sigma sm- + Sigm	-16.157	5353.238	Pass
Sigma sm+ - Sigm	2.977	5353.238	Pass
Sigma sm- - Sigm	2.977	5353.238	Pass
Sigma thetam+ +	-37.152	5353.238	Pass
Sigma thetam- +	-37.152	5353.238	Pass
Sigma thetam+ -	-31.411	5353.238	Pass
Sigma thetam- -	-31.411	5353.238	Pass

Average Compressive Stresses and comparisons with Fha and Fxa:

Sigma _{smCyl} + + S	-9.545	-1858.103	Pass
Sigma _{smCyl} - + S	-9.545	-1858.103	Pass
Sigma _{thetamCyl} +	-35.062	-1602.589	Pass
Sigma _{thetamCyl} -	-35.062	-1602.589	Pass

Small End Cylinder Without a Flare

Compute the ratio equation 4.3.60 [n]:
 $= t_c/t_s = 49/27 = 1.81$

Compute the ratio equation 4.3.61 [H]:
 $= \sqrt{(R_s/t_s)} = \sqrt{(653/27)} = 4.92$

Compute the ratio equation 4.3.62 [B]:
 $= \tan(a) = \tan(29.7) = 0.57$

Compute Ms+ from equation 4.3.59 [Xs+]:
 $= F_s / (2 \cdot \pi \cdot R_s) + M_s / (\pi \cdot R_s^2)$
 $= 0 / (2 \cdot 3.14 \cdot 653) + 0 / (3.141 \cdot 653^2)$
 $= 0.0000 \text{ kgf/mm}$

Compute Ms- from equation 4.3.59 [Xs-]:
 $= F_s / (2 \cdot \pi \cdot R_s) - M_s / (\pi \cdot R_s^2) + X_s'$
 $= 0 / (2 \cdot 3.14 \cdot 653) - 0 / (3.141 \cdot 653^2) + 0$
 $= 0.0000 \text{ kgf/mm}$

From Table 4.3.5 MsN:

C1 = -15.185586 C2 = 3.046973 C3 = 6.478770 C4 = -0.156801
 C5 = -1.465606 C6 = -0.372439 C7 = 0.007770 C8 = 0.143329
 C9 = 0.041178 C10 = 0.007300 C11 = 0.000000

Compute from Table 4.3.5 [MsN]:
 $= \exp[C1+C2 \cdot \ln(H^2)+C3 \cdot \ln(a)+C4 \cdot (\ln(H^2))^2+C5 \cdot \ln(a)^2+C6 \cdot \ln(H^2) \cdot \ln(a)+$
 $C7 \cdot \ln(H^2)^3+C8 \cdot \ln(a)^3+C9 \cdot \ln(H^2) \cdot \ln(a)^2+C10 \cdot \ln(H^2)^2 \cdot \ln(a)]$
 $= \exp[-15.2 + 3.05 \cdot \ln(4.92^2) + 6.48 \cdot \ln(29.7) +$
 $-0.16 \cdot (\ln(4.92^2))^2 + -1.47 \cdot \ln(29.7)^2 +$
 $-0.37 \cdot \ln(4.92^2) \cdot \ln(29.7) + 0.0078 \cdot \ln(4.92^2)^3$
 $0.14 \cdot \ln(29.7)^3 + 0.041 \cdot \ln(4.92^2) \cdot \ln(29.7)^2 +$
 $0.0073 \cdot \ln(4.92^2)^2 \cdot \ln(29.7)] = 5.08$



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Compute from Table 4.3.2 [Msp]:

$$= p \cdot tL^2 \cdot Msn = -1.03 \cdot 27^2 \cdot 5.08 = -38.3 \text{ kgf}$$

From table 4.3.6 MsN:

$$\begin{aligned} C1 &= 0.006829 & C2 &= 0.000292 & C3 &= -0.000927 & C4 &= 0.121601 \\ C5 &= 0.010585 & C6 &= -0.000011 & C7 &= -0.000008 & C8 &= 0.005945 \\ C9 &= 0.001298 & C10 &= 0.000186 & C11 &= 0.194335 \end{aligned}$$

Compute from Table 4.3.6 [MsN]:

$$\begin{aligned} &= (C1 + C3 \cdot H + C5 \cdot B + C7 \cdot H^2 + C9 \cdot B^2 + C11 \cdot H \cdot B) / \\ &\quad (1 + C2 \cdot H + C4 \cdot B + C6 \cdot H^2 + C8 \cdot B^2 + C10 \cdot H \cdot B) \\ &= (0.0068 + -0.00093 \cdot 4.92 + 0.011 \cdot 29.7 + \\ &\quad -0.80000E-05 \cdot 4.92^2 + 0.0013 \cdot 29.7^2 + \\ &\quad 0.19 \cdot 4.92 \cdot 29.7) / (1 + 0.0029 \cdot 4.92 + \\ &\quad 0.12 \cdot 29.7 + -0.10556E-04 \cdot 4.92^2 + 0.0059 \cdot 29.7^2 + \\ &\quad 0.00019 \cdot 4.92 \cdot 29.7) \\ &= 0.5170 \end{aligned}$$

Compute from Table 4.3.2 [Msx+]:

$$= Xs \cdot ts \cdot MsN = 0 \cdot 27 \cdot 0.52 = 0 \text{ kgf}$$

Compute from Table 4.3.2 [Msx-]:

$$= Xs \cdot ts \cdot MsN = 0 \cdot 27 \cdot 0.52 = 0 \text{ kgf}$$

Compute from Table 4.3.2 [Ms+]:

$$= Ms + Msx+ = -38.3 + 0 = -38.3 \text{ kgf}$$

Compute from Table 4.3.2 [Ms-]:

$$= Ms + Msx- = -38.3 + 0 = -38.3 \text{ kgf}$$

From table 4.3.5 Qn:

$$\begin{aligned} C1 &= 0.571280 & C2 &= -0.000027 & C3 &= 0.008448 & C4 &= 0.002689 \\ C5 &= -0.002892 & C6 &= 0.000000 & C7 &= -0.000005 & C8 &= -0.000117 \\ C9 &= -0.000087 & C10 &= 0.000001 & C11 &= -0.003775 \end{aligned}$$

Compute from Table 4.3.5 [Qn]:

$$\begin{aligned} &= (C1 + C3 \cdot H^2 + C5 \cdot \alpha + C7 \cdot H^4 + C9 \cdot \alpha^2 + C11 \cdot H^2 \cdot \alpha) / \\ &\quad (1 + C2 \cdot H^2 + C4 \cdot \alpha + C6 \cdot H^4 + C8 \cdot \alpha^2 + C10 \cdot H^2 \cdot \alpha) \\ &= (0.57 + 0.0084 \cdot 4.92^2 + -0.0029 \cdot 29.7 \\ &\quad -0.52593E-05 \cdot 4.92^4 + -0.87148E-04 \cdot 29.7^2 + -0.0038 \cdot 4.92^2 \\ &\quad \cdot 29.7) / (1 + -0.27407E-04 \cdot 4.92^2 + 0.0027 \cdot 29.7 + \\ &\quad 0 \cdot 4.92^4 + -0.00012 \cdot 29.7^2 + 0.10000E-05 \cdot 4.92^2 \cdot 29.7) \\ &= -2.158 \end{aligned}$$

Compute from Table 4.3.2 [Qp]:

$$= P \cdot ts \cdot Qn = -1.03 \cdot 27 \cdot -2.16 = 0.6 \text{ kgf/mm}$$

From table 4.3.6 Qn:

$$\begin{aligned} C1 &= -0.407896 & C2 &= 0.021262 & C3 &= -0.325315 & C4 &= -0.004004 \\ C5 &= -0.111190 & C6 &= 0.002205 & C7 &= 0.000256 & C8 &= -0.014421 \\ C9 &= 0.000820 & C10 &= 0.000106 & C11 &= 0.000000 \end{aligned}$$

Compute from Table 4.3.6 [Qn]:

$$\begin{aligned} &= C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot \ln(H)^2 + C5 \cdot \ln(B)^2 + C6 \cdot \ln(H) \cdot \ln(B) + \\ &\quad C7 \cdot \ln(H)^3 + C8 \cdot \ln(B)^3 + C9 \cdot \ln(H) \cdot \ln(B)^2 + C10 \cdot \ln(H)^2 \cdot \ln(B) \\ &= -0.41 + 0.021 \cdot \ln(4.92) + -0.33 \cdot \ln(0.57) + \\ &\quad -0.004 \cdot \ln(4.92)^2 + -0.11 \cdot (\ln(0.57))^2 + \\ &\quad 0.0022 \cdot \ln(4.92) \cdot \ln(0.57) + 0.00026 \cdot \ln(4.92)^3 + \\ &\quad -0.014 \cdot \ln(0.57)^3 + 0.00082 \cdot \ln(4.92) \cdot \ln(0.57)^2 + \\ &\quad 0.00011 \cdot \ln(4.92)^2 \cdot \ln(0.57) \\ &= -0.2351 \end{aligned}$$

Compute with Xs+ [Qx+]:

$$= Xs+ \cdot Qn = 0 \cdot -0.24 = -0 \text{ kgf/mm}$$



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Compute with Xs- [Qx-]:
 $= Xs- \cdot Qn = 0 \cdot -0.24 = -0 \text{ kgf/mm}$

Compute with Xs+ [Q+]:
 $= Qp + Qx+ = 0.6 + -0 = 0.6 \text{ kgf/mm}$

Compute with Xs- [Q-]:
 $= Qp + Qx- = 0.6 + -0 = 0.6 \text{ kgf/mm}$

Compute angle [β_{cy}]:
 $= (3 \cdot (1 - v^2) / (Rs^2 \cdot ts^2))^{0.25} = (3 \cdot (1 - 0.3^2) / (653^2 \cdot 27^2))^{0.25}$
 $= 0.0097 \text{ 1/mm}$

Compute with Xs+ [Ns+]:
 $= P \cdot Rs / 2 + Xs+ = -1.03 \cdot 653 / 2 + 0 = -3.38 \text{ kgf/mm}$

Compute with Xs- [Ns-]:
 $= P \cdot Rs / 2 + Xs- = -1.03 \cdot 653 / 2 + 0 = -3.38 \text{ kgf/mm}$

Compute with Xs+ [NTheta+]:
 $= P \cdot Rs + 2 \cdot \beta_{cy} \cdot Rs \cdot (-Ms \cdot \beta_{cy} - Q)$
 $= -1.03 \cdot 653 + 2 \cdot 0.0097 \cdot 653 \cdot (-38.3 \cdot 0.0097 - 1.54) = -9.68 \text{ kgf/mm}$

Compute with Xs- [NTheta-]:
 $= P \cdot Rs + 2 \cdot \beta_{cy} \cdot Rs \cdot (-Ms \cdot \beta_{cy} - Q)$
 $= -1.03 \cdot 653 + 2 \cdot 0.0097 \cdot 653 \cdot (-38.3 \cdot 0.0097 - 1.54) = -9.68 \text{ kgf/mm}$

Compute Stress with Xs+ [σ_{sm+}]:
 $= Ns(+)/ts = -3.38/27 = -12.5 \text{ kgf/cm}^2$

Compute Stress with Xs- [σ_{sm-}]:
 $= Ns(-)/ts = -3.38/27 = -12.5 \text{ kgf/cm}^2$

Compute stress with Xs+ [σ_{sb+}]:
 $= 6 \cdot Ms(+)/(ts^2 \cdot Kpc) = 6 \cdot -38.3/(27^2 \cdot 1) = -31.5 \text{ kgf/cm}^2$

Compute stress with Xs- [σ_{sb-}]:
 $= 6 \cdot Ms(-)/(ts^2 \cdot Kpc) = 6 \cdot -38.3/(27^2 \cdot 1) = -31.5 \text{ kgf/cm}^2$

Compute stress with Xs+ [$\sigma_{\theta\theta\theta\theta+}$]:
 $= N\theta\theta\theta\theta(+)/ts = -9.68/27 = -35.8 \text{ kgf/cm}^2$

Compute stress with Xs- [$\sigma_{\theta\theta\theta\theta-}$]:
 $= N\theta\theta\theta\theta(-)/ts = -9.68/27 = -35.8 \text{ kgf/cm}^2$

Compute stress with Xs+ [$\sigma_{\theta\theta\theta\theta\theta+}$]:
 $= 6 \cdot v \cdot Ms(+)/(ts^2 \cdot Kpc) = 6 \cdot 0.3 \cdot -38.3/(27^2 \cdot 1) = -9.45 \text{ kgf/cm}^2$

Compute stress with Xs- [$\sigma_{\theta\theta\theta\theta\theta-}$]:
 $= 6 \cdot v \cdot Ms(-)/(ts^2 \cdot Kpc) = 6 \cdot 0.3 \cdot -38.3/(27^2 \cdot 1) = -9.45 \text{ kgf/cm}^2$

Small End Cone Without a Flare

Compute with Xs+ [Qc+]:
 $= Q+ \cdot \cos(a) + Ns \cdot \sin(a) = 0.6 \cdot \cos(29.7) + -3.38 \cdot \sin(29.7)$
 $= -1.1518 \text{ kgf/mm}$

Compute with Xs- [Qc-]:
 $= Q- \cdot \cos(a) + Ns \cdot \sin(a) = 0.6 \cdot \cos(29.7) + -3.38 \cdot \sin(29.7)$
 $= -1.1518 \text{ kgf/mm}$

Compute effective radius [Rc]:
 $= Rs / \cos(a) = 653 / \cos(29.7) = 753 \text{ mm}$



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Compute the angle $[\beta_{co}]$:
 $= (3 \cdot (1 - v^2) / (Rc^2 \cdot tc^2))^{0.25} = (3 \cdot (1 - 0.3^2) / (753^2 \cdot 49^2))^{0.25}$
 $= 0.0067 \text{ 1/mm}$

Compute with Ms+ $[Ncs+]$:
 $= Ns \cdot \cos(a) - Q \cdot \sin(a)$
 $= -3.38 \cdot \cos(29.7) - 0.6 \cdot \sin(29.7) = -3.23 \text{ kgf/mm}$

Compute with Ms- $[Ncs-]$:
 $= Ns \cdot \cos(a) - Q \cdot \sin(a)$
 $= -3.38 \cdot \cos(29.7) - 0.6 \cdot \sin(29.7) = -3.23 \text{ kgf/mm}$

Compute with Ms+ $[Nc\theta_{theta+}]$:
 $= P \cdot Rs / \cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} + Qc)$
 $= -1.03 \cdot 653 / \cos(29.7) + 2 \cdot 0.0067 \cdot 753 \cdot$
 $(-38.3 \cdot 0.0067 + 1.15) = -16.8 \text{ kgf/mm}$

Compute with Ms- $[Nc\theta_{theta-}]$:
 $= P \cdot Rs / \cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} + Qc)$
 $= -1.03 \cdot 653 / \cos(29.7) + 2 \cdot 0.0067 \cdot$
 $(-38.3 \cdot 0.0067 + 1.15) = -16.8 \text{ kgf/mm}$

Compute the stress with Ms+ $[\sigma_{sm+}]$:
 $= Ncs / tc = -3.23 / 49 = -6.59 \text{ kgf/cm}^2$

Compute the stress with Ms- $[\sigma_{sm-}]$:
 $= Ncs / tc = -3.23 / 49 = -6.59 \text{ kgf/cm}^2$

Compute the stress with Ms+ $[\sigma_{sb+}]$:
 $= 6 \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot -38.3 / (49^2 \cdot 1) = -9.57 \text{ kgf/cm}^2$

Compute the stress with Ms- $[\sigma_{sb-}]$:
 $= 6 \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot -38.3 / (49^2 \cdot 1) = -9.57 \text{ kgf/cm}^2$

Compute stress with Ms+ $[\sigma_{\theta_{theta}+}]$:
 $= Nc\theta_{theta} / tc = -16.8 / 49 = -34.3 \text{ kgf/cm}^2$

Compute stress with Ms- $[\sigma_{\theta_{theta}-}]$:
 $= Nc\theta_{theta} / tc = -16.8 / 49 = -34.3 \text{ kgf/cm}^2$

Compute stress with Ms+ $[\sigma_{\theta_{theta}b+}]$:
 $= 6 \cdot v \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 0.3 \cdot -38.3 / (49^2 \cdot 1) = -2.87 \text{ kgf/cm}^2$

Compute stress with Ms- $[\sigma_{\theta_{theta}b-}]$:
 $= 6 \cdot v \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 0.3 \cdot -38.3 / (49^2 \cdot 1) = -2.87 \text{ kgf/cm}^2$



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 Nozzle Calcs.: N1- Channel inl Nozl: 25 9:32am Aug 12,2026

Input, Nozzle Desc: N1- Channel inlet

From: 30

Pressure for Reinforcement Calculations	P	163.158	kgf/cm ²
Temperature for Internal Pressure	Temp	450	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	1570.34	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1300.00	mm
Shell Finished (Minimum) Thickness	t	74.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		139.10	cm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

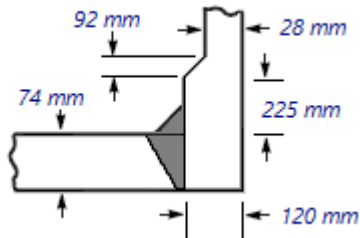
Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1570.34	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		17.7953	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	28.0000	mm
Flange Material		SA-350 LF2 +IT	
Hub Height of Integral Nozzle	h	225.0000	mm
Height of Beveled Transition	L'	92.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		120.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	280.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	74.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		2500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N1- Channel inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	17.795	in.
Actual Thickness Used in Calculation	1.102	in.



Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 455 \cdot (\exp(163/(1570 \cdot 1)) - 1) + 1.5 + 0$$

$$= 24.9 + 1.5 + 0 = 26.4 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1570 \cdot 1 \cdot \ln((2 \cdot 26.5 + 455) / 455) - 0$$

$$= 173 - 0 = 173 \text{ kgf/cm}^2$$

Nozzle/Tubesheet Proximity Warning:

This nozzle opening seems to be too close to the tubesheet (see 4.18.3(h) and Figure 4.18.19).
 Nozzle Dimension d, 524.000 mm, is < $1.8 \cdot \sqrt{D \cdot t}$: 553.240 mm.

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 181.93 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 563.33 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	9.840 mm
Min. required shell thickness, internal p + c :	72.900 mm
Nozzle neck minimum required thickness :	9.840 mm

Minimum nozzle neck thickness 27.991 mm , is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	941.9,	Allowable :	1570.3 kgf/cm ²	Passed
Expansion :	0.0,	Allowable :	3659.4 kgf/cm ²	Passed
Occasional :	661.8,	Allowable :	2088.6 kgf/cm ²	Passed
Shear :	193.5,	Allowable :	1099.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:



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Nozzle Calcs.: N1- Channel inl Nozl: 25 9:32am Aug 12,2026

= min[Sn/S, 1]
= min[1570/1570, 1]
= 1.000

Thickness of Nozzle at Shell [tn]:
= hub thickness - corrosion allowance
= 120 - 1.5
= 118.500 mm

Thickness of Nozzle at Top [tn2]:
= thickness - corrosion allowance
= 28 - 1.5
= 26.500 mm

Shell Diameter to Thickness ratio [D/t]:
= Di/t
= 1303/72.5
= 17.972 must be less than 400.

Effective Pressure Radius [Reff]:
= Di/2 + corrosion allowance
= 1300/2 + 1.5
= 651.500 mm

Effective Length of Vessel Wall [LR]:
= min($\sqrt{\text{Reff} \cdot t}$, $2 \cdot R_n$)
= min($\sqrt{652 \cdot 72.5}$, $2 \cdot 227$)
= 217.333 mm

Thickness Limit Candidate [LH1]:
= min[1.5t, te] + $\sqrt{R_n \cdot t_n}$
= min[1.5•72.5, 0] + $\sqrt{227 \cdot 118}$
= 164.191 mm

Thickness Limit Candidate [LH2]:
= Lpr1
= 280
= 280.000 mm

Thickness Limit Candidate [LH3]:
= 8(t + te)
= 8(72.5 + 0)
= 580.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:
= min[LH1, LH2, LH3] + t for set-in nozzles
= min[164, 280, 580] + 72.5
= 236.691 mm

Effective Vessel Thickness [teff]:
= t + ((A5 • frp) / LR)
= 72.5 + ((0•1)/217)
= 72.500 mm

Parameter [λ]:
= min(12, (2•Rn + tn) / ($\sqrt{(Di + teff) \cdot teff}$))
= min(12, (2•227 + 118) / ($\sqrt{(1303 + 72.5) \cdot 72.5}$))
= 1.816

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
= t • LR • max((λ/5)^{0.85}, 1)
= 72.5 • 217 • max((1.82/5)^{0.85}, 1)
= 157.567 cm²



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Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= t_n \cdot L_H \\
 &= 118 \cdot 237 \\
 &= 280.479 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot \text{Leg}^2 \\
 &= 0.5 \cdot 20^2 \\
 &= 2.000 \text{ cm}^2
 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned}
 &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\
 &= 158 + 1(280 + 0) + 2 + 0 + 0 \\
 &= 440.046 \text{ cm}^2
 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= t_n / \ln[1 + t_n/R_n] \\
 &= 118 / \ln[1 + 118/227] \\
 &= 282.622 \text{ mm}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\
 &= 72.5 / \ln[1 + 72.5/652] \\
 &= 687.113 \text{ mm}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \cdot E \\
 &= 1.5 \cdot 1570 \cdot 1 \\
 &= 2355.5 \text{ kgf/cm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot L_H \\
 &= 163 \cdot 283 \cdot 237 \\
 &= 109143.1 \text{ kgf}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} (L_R + t_n) \\
 &= 163 \cdot 687 (217 + 118) \\
 &= 376496.2 \text{ kgf}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot R_{nc} \\
 &= 163 \cdot 687 \cdot 227 \\
 &= 255045.9 \text{ kgf}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (f_S + f_Y + f_N) / P \\
 &= (376496 + 255046 + 109143) / 163 \\
 &= 4539.7 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\
 &= 2356 / (2 \cdot 4540 / 440 - 687 / 72.5) \\
 &= 211.2 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S / [t / R_{xs}] \\
 &= 1570 [72.5 / 687] \\
 &= 165.7 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(P_{max1}, P_{max2}) \\
 &= \min(211, 166) \\
 &= 165.693 \text{ kgf/cm}^2
 \end{aligned}$$



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Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (109143 + 376496 + 255046) / 440$$

$$= 1683.203 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / t_{eff}$$

$$= 163 \cdot 687 / 72.5$$

$$= 1546.3 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1683 - 1546, 1546)$$

$$= 1820.1 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2355.51	kgf/cm ²
Local Primary Membrane Stress	PL	1820.09	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	165.69	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (227 + 118) / 227$$

$$= 1.521 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (227 + 118)$$

$$= 543.496 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 14.142, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 14.142, \text{ must be } \geq \min(0.7 \cdot 118.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot ky, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot ky^2)$$

$$= \min(255046 \cdot 1.52, 1.5 \cdot 1570(280+0), \pi/4 \cdot 163 \cdot 227^2 \cdot 1.52^2)$$

$$= 153409.047 \text{ kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (L_t(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 153409 / (543(0.49 \cdot 14.1 + 0.6 \cdot 72.5 + 0.49 \cdot 0))$$

$$= 559.719 < \text{or} = \text{to } 1570.341 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-19 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	-2 °C
Temperature reduction per Graph 3.13	:	17 °C
The computed stress ratio is	:	0.690

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (18.3 \cdot 1) / (28 - 1.5 - 0)$$

$$= 0.690$$



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Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	22 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	37 °C
Temperature reduction per Graph 3.13	:	15 °C
The computed stress ratio is	:	0.722

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (52.4 \cdot 1) / (74 - 1.5 - 0)$$

$$= 0.722$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 22 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 99.7419 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 453.7419 mm

Input Echo, WRC297 Item 1, Description: N1- Channel in:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1300.000	mm
Shell Thickness	74.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 22	
Vessel Cold S.I. Allowable	Smc	2110.70 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1570.34 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	452.000	mm
Nozzle Thickness	120.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F22	
Nozzle Cold S.I. Allowable	SNmc	2110.70 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1570.34 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	163.16	kgf/cm ²
Radial Load	P	-7393.32	kgf
Circumferential Shear	Vc	6118.59	kgf
Longitudinal Shear	Vl	6118.59	kgf
Circumferential Moment	Mc	9683.86	kgf-m.
Longitudinal Moment	Ml	9683.86	kgf-m.
Torsional Moment	Mt	12741.90	kgf-m.



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Nozzle Calcs.: N1- Channel inl Nozl: 25 9:32am Aug 12,2026

Include Axial Pressure Thrust No
Include Pressure Stress Indices per Div. 2 No
Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, D_v/T_v (19.568) is < 20
check the limitations of WRC bulletin 297.*
*Warning - The ratio, D_n/T_n (5.767) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 692.0 + 2 \cdot 1.65 \cdot \sqrt{687.75(74.0 - 1.5)}$$

$$= 1428.883 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1375.500 mm
Nozzle Outside Diameter (d) = 692.000 mm
Vessel Thickness used (T) = 72.500 mm
Nozzle Thickness used (t) = 118.500 mm
T / t = 0.612
d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.191$$

Nr/P = 0.070
Mr/P = 0.129
M0/P = 0.039
N0/P = 0.097
MrD/Mc = 0.272
NrDL/Mc = 0.110
M0d/Mc = 0.087
N0DL/Mc = 0.069
MrD/Ml = 0.109
NrDL/Ml = 0.063
M0D/Ml = 0.035
N0DL/Ml = 0.120

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	33	-33	33	-33
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-31	-31	31	31
Outplane Bending (Ml)	-56	56	56	-56
Normal Pressure Stress	1388	1552	1388	1552
Outplane Stress Summary	1347	1557	1523	1507

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	9	9	9	9
Inplane Bending (P)	108	-108	108	-108
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



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Nozzle Calcs.: N1- Channel inl Nozl: 25 9:32am Aug 12,2026

Inplane Membrane (ML)	-16	-16	16	16
Inplane Bending (ML)	-173	173	173	-173
Inplane Pressure Stress	694	694	694	694

Inplane Stress Summary	622	752	1003	438
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	23	23	23	23

Shear Stress Summary	31	31	15	15
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	1348	1558	1524	1508
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	33	-33	33	-33
Outplane Membrane (Mc)	-18	-18	18	18
Outplane Bending (Mc)	-138	138	138	-138
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	694	694	694	694

Outplane Stress Summary	584	794	897	555
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	9	9	9	9
Inplane Bending (P)	108	-108	108	-108
Inplane Membrane (Mc)	-29	-29	29	29
Inplane Bending (Mc)	-434	434	434	-434
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1388	1552	1388	1552

Inplane Stress Summary	1043	1858	1971	1047
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-7	-7	7	7
Torsional Shear (Mt)	23	23	23	23

Shear Stress Summary	15	15	31	31
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Nozzle Calcs.: N1- Channel inl Nozl: 25 9:32am Aug 12,2026

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1044	1858	1972	1049

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-31	-31	31	31
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	298	493	298	493
Hoop Stress Summary	279	475	343	539

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	29	-29	29	-29
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-36	-36	36	36
Axial Bending (ML)	-46	46	46	-46
Axial Pressure Stress	149	149	149	149
Axial Stress Summary	99	132	264	113

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	14	14	14	14
Shear Stress Summary	18	18	9	9

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	281	476	344	539

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-18	-18	18	18
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	298	493	298	493



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Hoop Stress Summary	293	489	330	525
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	29	-29	29	-29
Axial Membrane (Mc)	-36	-36	36	36
Axial Bending (Mc)	-129	129	129	-129
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	149	149	149	149

Axial Stress Summary	15	215	348	29
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	14	14	14	14

Shear Stress Summary	9	9	18	18
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	293	489	360	526

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1388	1552	1388	1552	1388	1552	1388	1552
Circ. Pl (SUS)	-18	-18	44	44	-20	-20	38	38
Circ. Q (SUS)	-23	23	89	-89	-326	326	542	-542
Long. Pm (SUS)	694	694	694	694	694	694	694	694
Long. Pl (SUS)	-7	-7	25	25	-5	-5	31	31
Long. Q (SUS)	-65	65	281	-281	-105	105	171	-171
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	23	23	23	23	23	23	23	23
Pm (SUS)	1388	1552	1388	1552	1388	1552	1388	1552
Pm+Pl (SUS)	1370	1534	1432	1596	1368	1532	1426	1590
Pm+Pl+Q (Total)	1348	1558	1521	1507	1042	1858	1968	1049

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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 Nozzle Calcs.: N1- Channel inl Nozl: 25 9:32am Aug 12,2026

Pm (SUS)	1552.00	1570.34	Passed
Pm+Pl (SUS)	1596.06	2355.51	Passed
Pm+Pl+Q (TOTAL)	1968.84	4711.02	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	298	493	298	493	298	493	298	493
Circ. Pl (SUS)	-18	-18	44	44	-5	-5	31	31
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	149	149	149	149	149	149	149	149
Long. Pl (SUS)	-33	-33	39	39	-33	-33	39	39
Long. Q (SUS)	-17	17	75	-75	-100	100	158	-158
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	14	14	14	14	14	14	14	14
Pm (SUS)	298.0	493.0	298.0	493.0	298.0	493.0	298.0	493.0
Pm+Pl (SUS)	280.1	475.0	342.1	537.0	293.1	488.0	329.1	524.0
Pm+Pl+Q (Total)	281.8	475.9	343.2	537.2	293.4	488.4	357.4	524.7

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	493.00	1570.34	Passed
Pm+Pl (SUS)	537.05	2355.51	Passed
Pm+Pl+Q (TOTAL)	537.24	4711.02	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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 Nozzle Calcs.: N2- Channel out Nozl: 26 9:32am Aug 12,2026

Input, Nozzle Desc: N2- Channel outlet

From: 30

Pressure for Reinforcement Calculations	P	163.162	kgf/cm ²
Temperature for Internal Pressure	Temp	450	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	1570.34	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1300.00	mm
Shell Finished (Minimum) Thickness	t	74.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		139.10	cm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

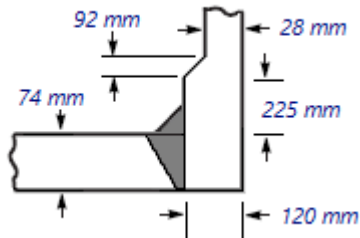
Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1570.34	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		17.7953	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	28.0000	mm
Hub Height of Integral Nozzle	h	225.0000	mm
Height of Beveled Transition	L'	92.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		120.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	280.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		74.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N2- Channel outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	17.795	in.
Actual Thickness Used in Calculation	1.102	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 455 \cdot (\exp(163/(1570 \cdot 1)) - 1) + 1.5 + 0$
 $= 24.9 + 1.5 + 0 = 26.4 \text{ mm}$ Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1570 \cdot 1 \cdot \ln((2 \cdot 26.5 + 455) / 455) - 0$
 $= 173 - 0 = 173 \text{ kgf/cm}^2$

Nozzle/Tubesheet Proximity Warning:

This nozzle opening seems to be too close to the tubesheet (see 4.18.3(h) and Figure 4.18.19).
 Nozzle Dimension d, 524.000 mm, is $< 1.8 \cdot \sqrt{D \cdot t}$: 553.240 mm.

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm
 Min. required shell thickness, internal p + c : 72.902 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 27.991 mm, is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 941.9,	Allowable	: 1570.3 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3659.4 kgf/cm ²	Passed
Occasional	: 661.8,	Allowable	: 2088.6 kgf/cm ²	Passed
Shear	: 193.5,	Allowable	: 1099.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1570/1570, 1]$
 $= 1.000$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 120 - 1.5$
 $= 118.500 \text{ mm}$



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Nozzle Calcs.: N2- Channel out Nozl: 26 9:32am Aug 12,2026

Thickness of Nozzle at Top [tn2]:
 = thickness - corrosion allowance
 = 28 - 1.5
 = 26.500 mm

Shell Diameter to Thickness ratio [D/t]:
 = Di/t
 = 1303/72.5
 = 17.972 must be less than 400.

Effective Pressure Radius [Reff]:
 = Di/2 + corrosion allowance
 = 1300/2 + 1.5
 = 651.500 mm

Effective Length of Vessel Wall [LR]:
 = $\min(\sqrt{Reff \cdot t}, 2 \cdot Rn)$
 = $\min(\sqrt{652 \cdot 72.5}, 2 \cdot 227)$
 = 217.333 mm

Thickness Limit Candidate [LH1]:
 = $\min[1.5t, te] + \sqrt{Rn \cdot tn}$
 = $\min[1.5 \cdot 72.5, 0] + \sqrt{227 \cdot 118}$
 = 164.191 mm

Thickness Limit Candidate [LH2]:
 = Lpr1
 = 280
 = 280.000 mm

Thickness Limit Candidate [LH3]:
 = $8(t + te)$
 = $8(72.5 + 0)$
 = 580.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:
 = $\min[LH1, LH2, LH3] + t$ for set-in nozzles
 = $\min[164, 280, 580] + 72.5$
 = 236.691 mm

Effective Vessel Thickness [teff]:
 = $t + ((A5 \cdot frp) / LR)$
 = $72.5 + ((0 \cdot 1) / 217)$
 = 72.500 mm

Parameter [λ]:
 = $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$
 = $\min(12, (2 \cdot 227 + 118) / (\sqrt{(1303 + 72.5) \cdot 72.5}))$
 = 1.816

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
 = $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$
 = $72.5 \cdot 217 \cdot \max((1.82/5)^{0.85}, 1)$
 = 157.567 cm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:
 = $tn \cdot LH$
 = $118 \cdot 237$
 = 280.479 cm²

Area Contributed by the Outside Fillet Weld [A41]:
 = $0.5 \cdot Leg41^2$



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Nozzle Calcs.: N2- Channel out Nozl: 26 9:32am Aug 12,2026

$$= 0.5 \cdot 20^2$$

$$= 2.000 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 158 + 1(280 + 0) + 2 + 0 + 0$$

$$= 440.046 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 118 / \ln[1 + 118/227]$$

$$= 282.622 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 72.5 / \ln[1 + 72.5/652]$$

$$= 687.113 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1570 \cdot 1$$

$$= 2355.5 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 163 \cdot 283 \cdot 237$$

$$= 109145.7 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 163 \cdot 687 (217 + 118)$$

$$= 376505.2 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 163 \cdot 687 \cdot 227$$

$$= 255052.0 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (376505 + 255052 + 109146) / 163$$

$$= 4539.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 2356 / (2 \cdot 4540 / 440 - 687 / 72.5)$$

$$= 211.2 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / R_{xs}]$$

$$= 1570 [72.5 / 687]$$

$$= 165.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(211, 166)$$

$$= 165.693 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (109146 + 376505 + 255052) / 440$$

$$= 1683.243 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$



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 Nozzle Calcs.: N2- Channel out Nozl: 26 9:32am Aug 12,2026

$$= 163 \cdot 687/72.5$$

$$= 1546.4 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:
 $= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$
 $= \max(2 \cdot 1683 - 1546, 1546)$
 $= 1820.1 \text{ kgf/cm}^2$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2355.51	kgf/cm ²
Local Primary Membrane Stress	PL	1820.13	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	165.69	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
 $= (R_{nc} + t_n) / R_{nc}$
 $= (227 + 118) / 227$
 $= 1.521 \text{ For set-in Nozzles}$

Weld Length of Nozzle to Shell Weld [Lt]:
 $= \pi/2 \cdot (R_n + t_n)$
 $= \pi/2 \cdot (227 + 118)$
 $= 543.496 \text{ mm}$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
 $= 14.142, 0.000, 0.000, \text{ mm}$

Required Weld Throat Dimensions :
 $t_c, L41T = 14.142, \text{ must be } \geq \min(0.7 \cdot 118.500, 6.000) \text{ Passed}$

Weld Load Value [fwelds]:
 $= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 $= \min(255052 \cdot 1.52, 1.5 \cdot 1570(280+0), \pi/4 \cdot 163 \cdot 227^2 \cdot 1.52^2)$
 $= 153412.703 \text{ kgf}$

Weld Stress Value [τ]:
 $= \text{fwelds} / (L_t(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$
 $= 153413 / (543(0.49 \cdot 14.1 + 0.6 \cdot 72.5 + 0.49 \cdot 0))$
 $= 559.733 < \text{or to } 1570.341 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	22 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	37 °C
Temperature reduction per Graph 3.13	:	15 °C
The computed stress ratio is	:	0.722

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (52.4 \cdot 1) / (74 - 1.5 - 0)$
 $= 0.722$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 22 °C

The Drop for this Nozzle is : 99.7419 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 453.7419 mm



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 Nozzle Calcs.: N2- Channel out Nozl: 26 9:32am Aug 12,2026

Input Echo, WRC297 Item 1, Description: N2- Channel ou:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1300.000	mm
Shell Thickness	74.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 22	
Vessel Cold S.I. Allowable	Smc 2110.70	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1570.34	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	452.000	mm
Nozzle Thickness	120.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F22	
Nozzle Cold S.I. Allowable	SNmc 2110.70	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1570.34	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	163.16	kgf/cm ²
Radial Load	P	-7393.32	kgf
Circumferential Shear	Vc	6118.59	kgf
Longitudinal Shear	Vl	6118.59	kgf
Circumferential Moment	Mc	9683.86	kgf-m.
Longitudinal Moment	Ml	9683.86	kgf-m.
Torsional Moment	Mt	12741.93	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (19.568) is < 20
 check the limitations of WRC bulletin 297.
Warning - The ratio, D_n/T_n (5.767) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 692.0 + 2 \cdot 1.65 \cdot \sqrt{687.75(74.0 - 1.5)}$
 $= 1428.883 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1375.500	mm
Nozzle Outside Diameter	(d) =	692.000	mm
Vessel Thickness used	(T) =	72.500	mm
Nozzle Thickness used	(t) =	118.500	mm
T / t	=	0.612	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.191$$



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Nozzle Calcs.: N2- Channel out Nozl: 26 9:32am Aug 12,2026

Nr/P = 0.070
 Mr/P = 0.129
 MØ/P = 0.039
 NØ/P = 0.097
 MrD/Mc = 0.272
 NrDL/Mc = 0.110
 MØd/Mc = 0.087
 NØDL/Mc = 0.069
 MrD/Ml = 0.109
 NrDL/Ml = 0.063
 MØD/Ml = 0.035
 NØDL/Ml = 0.120

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	33	-33	33	-33
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-31	-31	31	31
Outplane Bending (ML)	-56	56	56	-56
Normal Pressure Stress	1388	1552	1388	1552
Outplane Stress Summary	1347	1557	1523	1507

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	9	9	9	9
Inplane Bending (P)	108	-108	108	-108
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-16	-16	16	16
Inplane Bending (ML)	-173	173	173	-173
Inplane Pressure Stress	694	694	694	694
Inplane Stress Summary	622	752	1003	438

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	23	23	23	23
Shear Stress Summary	31	31	15	15

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	1348	1558	1524	1508

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	D1 Right
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Nozzle Calcs.: N2- Channel out Nozl: 26 9:32am Aug 12,2026

circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	33	-33	33	-33
Outplane Membrane (Mc)	-18	-18	18	18
Outplane Bending (Mc)	-138	138	138	-138
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	694	694	694	694
Outplane Stress Summary	584	794	897	555

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	9	9	9	9
Inplane Bending (P)	108	-108	108	-108
Inplane Membrane (Mc)	-29	-29	29	29
Inplane Bending (Mc)	-434	434	434	-434
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1388	1552	1388	1552
Inplane Stress Summary	1043	1858	1971	1047

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-7	-7	7	7
Torsional Shear (Mt)	23	23	23	23
Shear Stress Summary	15	15	31	31

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1044	1858	1972	1049

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-31	-31	31	31
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	298	493	298	493
Hoop Stress Summary	279	475	343	539

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Nozzle Calcs.: N2- Channel out Nozl: 26 9:32am Aug 12,2026

Axial Membrane (P)	3	3	3	3
Axial Bending (P)	29	-29	29	-29
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-36	-36	36	36
Axial Bending (ML)	-46	46	46	-46
Axial Pressure Stress	149	149	149	149
Axial Stress Summary	99	132	264	113

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	14	14	14	14
Shear Stress Summary	18	18	9	9

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	281	476	344	539

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-18	-18	18	18
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	298	493	298	493
Hoop Stress Summary	293	489	330	525

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	29	-29	29	-29
Axial Membrane (Mc)	-36	-36	36	36
Axial Bending (Mc)	-129	129	129	-129
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	149	149	149	149
Axial Stress Summary	15	215	348	29

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	14	14	14	14
Shear Stress Summary	9	9	18	18



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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	293	489	360	526

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1388	1552	1388	1552	1388	1552	1388	1552
Circ. Pl (SUS)	-18	-18	44	44	-20	-20	38	38
Circ. Q (SUS)	-23	23	89	-89	-326	326	542	-542
Long. Pm (SUS)	694	694	694	694	694	694	694	694
Long. Pl (SUS)	-7	-7	25	25	-5	-5	31	31
Long. Q (SUS)	-65	65	281	-281	-105	105	171	-171
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	23	23	23	23	23	23	23	23
Pm (SUS)	1388	1552	1388	1552	1388	1552	1388	1552
Pm+Pl (SUS)	1370	1534	1432	1596	1368	1532	1426	1590
Pm+Pl+Q (Total)	1348	1558	1521	1507	1042	1858	1968	1049

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1552.00	1570.34	Passed
Pm+Pl (SUS)	1596.06	2355.51	Passed
Pm+Pl+Q (TOTAL)	1968.84	4711.02	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	298	493	298	493	298	493	298	493
Circ. Pl (SUS)	-18	-18	44	44	-5	-5	31	31
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	149	149	149	149	149	149	149	149
Long. Pl (SUS)	-33	-33	39	39	-33	-33	39	39
Long. Q (SUS)	-17	17	75	-75	-100	100	158	-158
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	14	14	14	14	14	14	14	14
Pm (SUS)	298.0	493.0	298.0	493.0	298.0	493.0	298.0	493.0



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Nozzle Calcs.: N2- Channel out Nozl: 26 9:32am Aug 12,2026

Pm+Pl (SUS)	280.1	475.0	342.1	537.0	293.1	488.0	329.1	524.0
Pm+Pl+Q (Total)	281.8	475.9	343.2	537.2	293.4	488.4	357.4	524.7

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	493.00	1570.34	Passed
Pm+Pl (SUS)	537.05	2355.51	Passed
Pm+Pl+Q (TOTAL)	537.24	4711.02	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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 Nozzle Calcs.: N4- Demister Nozl: 27 9:32am Aug 12,2026

Input, Nozzle Desc: N4- Demister

From: 60

Pressure for Reinforcement Calculations	P	49.967	kgf/cm ²
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	1.03	kgf/cm ²
Temperature for External Pressure	Tempex	270	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1784.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	5041.6665	mm
Shell Finished (Minimum) Thickness	t	45.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		653.60	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1435.72	kgf/cm ²
Allowable Stress At Ambient	Sna	1784.41	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		55.1100	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	92.0000	mm

Corrosion Allowance	can	3.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	700.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	30.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1435.72	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1883.7900	mm
Thickness of Pad	te	38.0000	mm
Weld leg size between Pad and Shell	Wp	30.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	38.0000	mm
Reinforcing Pad Width		149.9980	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

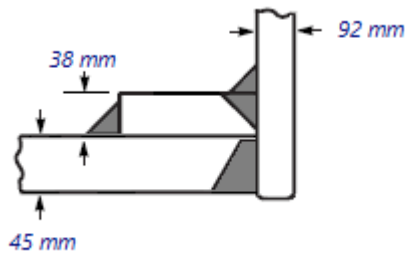
Reinforcement CALCULATION, Description: N4- Demister

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	55.110	in.
Actual Thickness Used in Calculation	3.622	in.



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 Nozzle Calcs.: N4- Demister Nozl: 27 9:32am Aug 12,2026



Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1406 \cdot (\exp(50/(1436 \cdot 1)) - 1) + 3 + 0$
 $= 24.9 + 3 + 0 = 27.9 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1436 \cdot 1 \cdot \ln((2 \cdot 89 + 1406) / 1406) - 0$
 $= 171 - 0 = 171 \text{ kgf/cm}^2$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 885.43 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 1101.09 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c	: 11.340 mm
Min. required shell thickness, internal p + c	: 35.745 mm
Min. required shell thickness, external p + c	: 15.973 mm

Nozzle neck minimum required thickness	: 11.340 mm
--	-------------

Minimum nozzle neck thickness 91.999 mm , is greater than required thickness 11.340 mm

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1436/1784, 1]$
 $= 0.805$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[1436/1784, 1]$
 $= 0.805$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 92 - 3$
 $= 89.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:



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Nozzle Calcs.: N4- Demister

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= Di/t
 = 2306/42
 = 54.905 must be less than 400.

Effective Pressure Radius [Reff]:
 = Di/2 + corrosion allowance
 = 2300/2 + 3
 = 1153.000 mm

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(Reff \cdot t) + W}$$

$$= \sqrt{(1153 \cdot 42) + 150}$$
 = 370.057 mm

Limit [LR2]:

$$= \sqrt{((Reff + t) \cdot (t + te))}$$

$$= \sqrt{((1153 + 42) \cdot (42 + 38))}$$
 = 309.192 mm

Limit [LR3]:
 = 2 • Rn
 = 2 • 703
 = 1405.794 mm

Final value of [LR]:
 = min(LR1, LR2, LR3)
 = min(370, 309, 1406)
 = 309.192 mm

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, te] + \sqrt{(Rn \cdot tn)}$$

$$= \min[1.5 \cdot 42, 38] + \sqrt{(703 \cdot 89)}$$
 = 288.116 mm

Thickness Limit Candidate [LH2]:
 = Lpr1
 = 700
 = 700.000 mm

Thickness Limit Candidate [LH3]:
 = 8(t + te)
 = 8(42 + 38)
 = 640.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:
 = min[LH1, LH2, LH3] + t for set-in nozzles
 = min[288, 700, 640] + 42
 = 330.116 mm

Effective Vessel Thickness [teff]:
 = t + ((A5 • frp) / LR)
 = 42 + ((57•0.8)/309)
 = 56.833 mm

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{((Di + teff) \cdot teff)}))$$

$$= \min(12, (2 \cdot 703 + 89) / (\sqrt{((2306 + 56.8) \cdot 56.8)}))$$
 = 4.079

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :



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Nozzle Calcs.: N4- Demister

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Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((\lambda/5)0.85, 1) \\
 &= 42 \cdot 309 \cdot \max((4.08/5)0.85, 1) \\
 &= 129.861 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= t_n \cdot LH \\
 &= 89 \cdot 330 \\
 &= 293.803 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned}
 &= 0.5 \cdot \text{Leg}42^2 \\
 &= 0.5 \cdot 30^2 \\
 &= 4.500 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot \text{Leg}41^2 \\
 &= 0.5 \cdot 30^2 \\
 &= 4.500 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned}
 &= \min(W \cdot t_e, LR \cdot t_e) \\
 &= \min(150 \cdot 38, 309 \cdot 38) \\
 &= 56.999 \text{ cm}^2
 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned}
 &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\
 &= 130 + 0.8(294 + 0) + 4.5 + 4.5 + 0 + 0.8(57) \\
 &= 421.113 \text{ cm}^2
 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= t_n / \ln[1 + t_n/R_n] \\
 &= 89 / \ln[1 + 89/703] \\
 &= 746.513 \text{ mm}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\
 &= 56.8 / \ln[1 + 56.8/1153] \\
 &= 1181.188 \text{ mm}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \cdot E \\
 &= 1.5 \cdot 1784 \cdot 1 \\
 &= 2676.6 \text{ kgf/cm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot LH \\
 &= 50 \cdot 747 \cdot 330 \\
 &= 123136.9 \text{ kgf}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} (LR + t_n) \\
 &= 50 \cdot 1181 (309 + 89) \\
 &= 235015.8 \text{ kgf}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot R_{nc} \\
 &= 50 \cdot 1181 \cdot 703 \\
 &= 414854.4 \text{ kgf}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (fS + fY + fN)/P \\
 &= (235016 + 414854 + 123137)/50 \\
 &= 15470.3 \text{ cm}^2
 \end{aligned}$$



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Maximum Allowable Working Pressure Candidate [Pmax1]:
 = Sallow / (2 • Ap/AT - Rxs/teff)
 = 2677 / (2 • 15470/421 - 1181/56.8)
 = 50.8 kgf/cm²

Maximum Allowable Working Pressure Candidate [Pmax2]:
 = S[t/Rxs]
 = 1784 [42/1181]
 = 63.4 kgf/cm²

Maximum Allowable Working Pressure [Pmax]:
 = min(Pmax1, Pmax2)
 = min(50.8, 63.4)
 = 50.800 kgf/cm²

Average Primary Membrane Stress [σAvg]:
 = (fN + fS + fY) / AT
 = (123137 + 235016 + 414854)/421
 = 1835.631 kgf/cm²

General Primary Membrane Stress [σCirc]:
 = P • Rxs / teff
 = 50 • 1181/56.8
 = 1038.5 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:
 = max(2 • σAvg - σCirc, σCirc)
 = max(2 • 1836 - 1039, 1039)
 = 2632.8 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	2632.76	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	50.80	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
 = (Rnc + tn) / Rnc
 = (703 + 89)/703
 = 1.127 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lτ]:
 = π/2 • (Rn + tn)
 = π/2 • (703 + 89)
 = 1243.909 mm

Weld Length of Pad to Shell Weld [LτP]:
 = π/2 • (Rn + tn + W)
 = π/2 • (703 + 89 + 150)
 = 1479.525 mm

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
 = 21.213, 21.213, 0.000, mm

Required Weld Throat Dimensions :
 tc, L41T = 21.213, must be >= min(0.7 * 89.000, 6.000) Passed
 tf1, L42T = 21.213, must be >= min(0.6 * 38.000, 0.6 * 42.000) Failed

Weld Load Value [fwelds]:
 = min(fy • ky, 1.5 • Sn(A2 + A3), π/4•P•Rn²•ky²)
 = min(414854•1.13, 1.5•1436(294+0), π/4•50•703²•1.13²)
 = 246100.359 kgf

Discontinuity Force [fws]:



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$$\begin{aligned}
 &= fwelds \cdot t \cdot S / (t \cdot S + te \cdot Sp) \\
 &= 246100 \cdot 42 \cdot 1784 / (42 \cdot 1784 + 38 \cdot 1436) \\
 &= 142422.219 \text{ kgf}
 \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned}
 &= fwelds \cdot te \cdot Sp / (t \cdot S + te \cdot Sp) \\
 &= 246100 \cdot 38 \cdot 1436 / (42 \cdot 1784 + 38 \cdot 1436) \\
 &= 103678.156 \text{ kgf}
 \end{aligned}$$

Shear Stress [τ1]:

$$\begin{aligned}
 &= fws / (Lt \cdot (0.6 \cdot tw1 + 0.49 \cdot L43T)) \\
 &= 142422 / (1244 \cdot (0.6 \cdot 38 + 0.49 \cdot 0)) \\
 &= 502.175 \text{ kgf/cm}^2
 \end{aligned}$$

Shear Stress [τ2]:

$$\begin{aligned}
 &= fwp / (Lt \cdot (0.6 \cdot tw2 + 0.49 \cdot L41T)) \\
 &= 103678 / (1244 \cdot (0.6 \cdot 38 + 0.49 \cdot 21.2)) \\
 &= 251.093 \text{ kgf/cm}^2
 \end{aligned}$$

Shear Stress [τ3]:

$$\begin{aligned}
 &= fwp / (LtP \cdot (0.49 \cdot L42T)) \\
 &= 103678 / (1480 \cdot (0.49 \cdot 21.2)) \\
 &= 674.167 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned}
 &= \max(\tau_1, \tau_2, \tau_3) \\
 &= \max(502, 251, 674) \\
 &= 674.2 \text{ must be less than or equal to } 1784.4 \text{ kgf/cm}^2
 \end{aligned}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= tn / \ln[1 + tn/Rn] \\
 &= 89 / \ln[1 + 89/703] \\
 &= 746.513 \text{ mm}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= teff / \ln[1 + teff/Reff] \\
 &= 56.8 / \ln[1 + 56.8/1153] \\
 &= 1181.188 \text{ mm}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \\
 &= 1.5 \cdot 1784 \\
 &= 2676.6 \text{ kgf/cm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot Rxn \cdot LH \\
 &= 1.03 \cdot 747 \cdot 330 \\
 &= 2545.7 \text{ kgf}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot (LR + tn) \\
 &= 1.03 \cdot 1181 \cdot (309 + 89) \\
 &= 4858.6 \text{ kgf}
 \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot Rnc \\
 &= 1.03 \cdot 1181 \cdot 703 \\
 &= 8576.5 \text{ kgf}
 \end{aligned}$$

Area Resisting External Pressure [Ap]:



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$$\begin{aligned}
 &= (fS + fY + fN) / P \\
 &= (4859 + 8576 + 2546) / 1.03 \\
 &= 15470.3 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= Sallow / (2 \cdot Ap / AT - Rxs / teff) \\
 &= 2677 / (2 \cdot 15470 / 421 - 1181 / 56.8) \\
 &= 50.8 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S[t / Rxs] \\
 &= 1784 [42 / 1181] \\
 &= 63.4 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(Pmax1, Pmax2) \\
 &= \min(50.8, 63.4) \\
 &= 50.800 \text{ kgf/cm}^2
 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned}
 &= (fN + fS + fY) / AT \\
 &= (2546 + 4859 + 8576) / 421 \\
 &= 37.949 \text{ kgf/cm}^2
 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned}
 &= P \cdot Rxs / teff \\
 &= 1.03 \cdot 1181 / 56.8 \\
 &= 21.5 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\
 &= \max(2 \cdot 37.9 - 21.5, 21.5) \\
 &= 54.4 \text{ kgf/cm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	54.43	kgf/cm ²
Maximum Allowable External Pressure	Pmax	50.80	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	1 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	4 °C
Temperature reduction per Graph 3.13	:	3 °C
The computed stress ratio is	:	0.936

The computed thickness ratio [sr]:

$$\begin{aligned}
 &= (tr \cdot E^*) / (tnom - ci - cext) \\
 &= (32.7 \cdot 1) / (38 - 3 - 0) \\
 &= 0.936
 \end{aligned}$$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	1 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	4 °C
Temperature reduction per Graph 3.13	:	3 °C
The computed stress ratio is	:	0.936



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The computed thickness ratio [sr]:

$$= (tr \cdot E) / (tnom - ci - cext)$$

$$= (32.7 \cdot 1) / (38 - 3 - 0)$$

$$= 0.936$$

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : 1 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 4 °C
 Temperature reduction per Graph 3.13 : 3 °C
 The computed stress ratio is : 0.936

The computed thickness ratio [sr]:

$$= (tr \cdot E) / (tnom - ci - cext)$$

$$= (32.7 \cdot 1) / (38 - 3 - 0)$$

$$= 0.936$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -5 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 7 °C
 Temperature reduction per Graph 3.13 : 12 °C
 The computed stress ratio is : 0.780

The computed thickness ratio [sr]:

$$= (tr \cdot E) / (tnom - ci - cext)$$

$$= (32.7 \cdot 1) / (45 - 3 - 0)$$

$$= 0.780$$

Gov. MDMT of the Nozzle : 1 °C
 Gov. MDMT of the Reinforcement Pad : 1 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 1 °C

The Drop for this Nozzle is : 316.0940 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 1061.0940 mm

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Nozzle Calcs.: N3- Shell inlet Nozl: 28 9:32am Aug 12,2026

Input, Nozzle Desc: N3- Shell inlet

From: 60

Pressure for Reinforcement Calculations	P	50.197	kgf/cm ²
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	1.03	kgf/cm ²
Temperature for External Pressure	Tempex	270	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1784.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	5041.6665	mm
Shell Finished (Minimum) Thickness	t	45.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		533.60	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1321.51	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1435.72	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	368.2750	mm
Thickness of Pad	te	20.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm
Reinforcing Pad Width		100.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

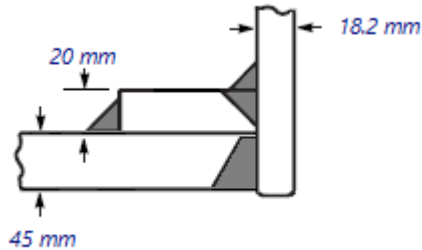
Reinforcement CALCULATION, Description: N3- Shell inlet



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ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation 5.189 in.
 Actual Thickness Used in Calculation 0.718 in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 138 \cdot (\exp(50.2/(1322 \cdot 1)) - 1) + 3 + 0$
 $= 2.67 + 3 + 0 = 5.67 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1322 \cdot 1 \cdot \ln((2 \cdot 15.2 + 138) / 138) - 0$
 $= 264 - 0 = 264 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.220 mm
 Min. required shell thickness, internal p + c : 35.898 mm
 Min. required shell thickness, external p + c : 15.973 mm

Nozzle neck minimum required thickness : 9.220 mm

Minimum nozzle neck thickness 15.951 mm , is greater than required thickness 9.220 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 378.8,	Allowable	: 1321.5 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3325.2 kgf/cm ²	Passed
Occasional	: 122.0,	Allowable	: 1757.6 kgf/cm ²	Passed
Shear	: 186.3,	Allowable	: 925.1 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1322/1784, 1]$
 $= 0.741$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[1436/1784, 1]$
 $= 0.805$

Thickness of Nozzle [tn]:



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Nozzle Calcs.: N3- Shell inlet Nozl: 28 9:32am Aug 12,2026

= thickness - corrosion allowance
= 18.2 - 3
= 15.237 mm

Shell Diameter to Thickness ratio [D/t]:

= Di/t
= 2306/42
= 54.905 must be less than 400.

Effective Pressure Radius [Reff]:

= Di/2 + corrosion allowance
= 2300/2 + 3
= 1153.000 mm

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

= $\sqrt{(Reff \cdot t) + W}$
= $\sqrt{(1153 \cdot 42) + 100}$
= 320.059 mm

Limit [LR2]:

= $\sqrt{((Reff + t) \cdot (t + te))}$
= $\sqrt{((1153 + 42) \cdot (42 + 20))}$
= 272.195 mm

Limit [LR3]:

= 2 • Rn
= 2 • 68.9
= 137.801 mm

Final value of [LR]:

= min(LR1, LR2, LR3)
= min(320, 272, 138)
= 137.801 mm

Thickness Limit Candidate [LH1]:

= min[1.5t, te] + $\sqrt{(Rn \cdot tn)}$
= min[1.5•42, 20] + $\sqrt{(68.9 \cdot 15.2)}$
= 52.401 mm

Thickness Limit Candidate [LH2]:

= Lpr1
= 300
= 300.000 mm

Thickness Limit Candidate [LH3]:

= 8(t + te)
= 8(42 + 20)
= 496.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:

= min[LH1, LH2, LH3] + t for set-in nozzles
= min[52.4, 300, 496] + 42
= 94.401 mm

Effective Vessel Thickness [teff]:

= t + ((A5 • frp) / LR)
= 42 + ((20•0.8)/138)
= 53.678 mm

Parameter [λ]:

= min(12, (2•Rn + tn) / ($\sqrt{((Di + teff) \cdot teff)}$))



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$$= \min(12, (2 \cdot 68.9 + 15.2) / (\sqrt{(2306 + 53.7) \cdot 53.7}))$$

$$= 0.430$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((N/5)0.85, 1)$$

$$= 42 \cdot 138 \cdot \max((0.43/5)0.85, 1)$$

$$= 57.876 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 15.2 \cdot 94.4$$

$$= 14.384 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg}42^2$$

$$= 0.5 \cdot 20^2$$

$$= 2.000 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}41^2$$

$$= 0.5 \cdot 20^2$$

$$= 2.000 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$

$$= \min(100 \cdot 20, 138 \cdot 20)$$

$$= 20.000 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$

$$= 57.9 + 0.74(14.4 + 0) + 2 + 2 + 0 + 0.8(20)$$

$$= 88.621 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 15.2 / \ln[1 + 15.2/68.9]$$

$$= 76.265 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 53.7 / \ln[1 + 53.7/1153]$$

$$= 1179.636 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1784 \cdot 1$$

$$= 2676.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 50.2 \cdot 76.3 \cdot 94.4$$

$$= 3614.0 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 50.2 \cdot 1180 (138 + 15.2)$$

$$= 90620.3 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 50.2 \cdot 1180 \cdot 68.9$$

$$= 40798.8 \text{ kgf}$$



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Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (90620 + 40799 + 3614) / 50.2$$

$$= 2690.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 2677 / (2 \cdot 2690 / 88.6 - 1180 / 53.7)$$

$$= 69.1 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 1784 [42 / 1180]$$

$$= 63.5 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(69.1, 63.5)$$

$$= 63.533 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (3614 + 90620 + 40799) / 88.6$$

$$= 1523.721 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 50.2 \cdot 1180 / 53.7$$

$$= 1103.1 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 1524 - 1103, 1103)$$

$$= 1944.3 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	1944.29	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	63.53	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (68.9 + 15.2) / 68.9$$

$$= 1.221 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lτ]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (68.9 + 15.2)$$

$$= 132.163 \text{ mm}$$

Weld Length of Pad to Shell Weld [LτP]:

$$= \pi / 2 \cdot (Rn + tn + W)$$

$$= \pi / 2 \cdot (68.9 + 15.2 + 100)$$

$$= 289.243 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 14.142, 14.142, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 14.142, \text{ must be } \geq \min(0.7 \cdot 15.237, 6.000) \text{ Passed}$$

$$tf1, L42T = 14.142, \text{ must be } \geq \min(0.6 \cdot 20.000, 0.6 \cdot 42.000) \text{ Passed}$$

Weld Load Value [fwelds]:



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Nozzle Calcs.: N3- Shell inlet Nozl: 28 9:32am Aug 12, 2026

$$\begin{aligned}
 &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\
 &= \min(40799 \cdot 1.22, 1.5 \cdot 1322(14.4+0), \pi/4 \cdot 50.2 \cdot 68.9^2 \cdot 1.22^2) \\
 &= 2790.921 \text{ kgf}
 \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned}
 &= f_{welds} \cdot t \cdot S / (t \cdot S + t_e \cdot S_p) \\
 &= 2791 \cdot 42 \cdot 1784 / (42 \cdot 1784 + 20 \cdot 1436) \\
 &= 2017.818 \text{ kgf}
 \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned}
 &= f_{welds} \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p) \\
 &= 2791 \cdot 20 \cdot 1436 / (42 \cdot 1784 + 20 \cdot 1436) \\
 &= 773.104 \text{ kgf}
 \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned}
 &= f_{ws} / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\
 &= 2018 / (132 \cdot (0.6 \cdot 38 + 0.49 \cdot 0)) \\
 &= 66.964 \text{ kgf/cm}^2
 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned}
 &= f_{wp} / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T})) \\
 &= 773 / (132 \cdot (0.6 \cdot 20 + 0.49 \cdot 14.1)) \\
 &= 30.902 \text{ kgf/cm}^2
 \end{aligned}$$

Shear Stress [τ_3]:

$$\begin{aligned}
 &= f_{wp} / (L_{TP} \cdot (0.49 \cdot L_{42T})) \\
 &= 773 / (289 \cdot (0.49 \cdot 14.1)) \\
 &= 38.572 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned}
 &= \max(\tau_1, \tau_2, \tau_3) \\
 &= \max(67, 30.9, 38.6) \\
 &= 67.0 \text{ must be less than or equal to } 1784.4 \text{ kgf/cm}^2
 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= t_n / \ln[1 + t_n/R_n] \\
 &= 15.2 / \ln[1 + 15.2/68.9] \\
 &= 76.265 \text{ mm}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\
 &= 53.7 / \ln[1 + 53.7/1153] \\
 &= 1179.636 \text{ mm}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \\
 &= 1.5 \cdot 1784 \\
 &= 2676.6 \text{ kgf/cm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot L_H \\
 &= 1.03 \cdot 76.3 \cdot 94.4 \\
 &= 74.4 \text{ kgf}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} (L_R + t_n) \\
 &= 1.03 \cdot 1180 (138 + 15.2) \\
 &= 1864.9 \text{ kgf}
 \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot R_{nc} \\
 &= 1.03 \cdot 1180 \cdot 68.9 \\
 &= 839.6 \text{ kgf}
 \end{aligned}$$



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Area Resisting External Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (1865 + 840 + 74.4) / 1.03$$

$$= 2690.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 2677 / (2 \cdot 2690 / 88.6 - 1180 / 53.7)$$

$$= 69.1 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 1784 [42 / 1180]$$

$$= 63.5 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(69.1, 63.5)$$

$$= 63.533 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (74.4 + 1865 + 840) / 88.6$$

$$= 31.356 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 1.03 \cdot 1180 / 53.7$$

$$= 22.7 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 31.4 - 22.7, 22.7)$$

$$= 40.0 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	40.01	kgf/cm ²
Maximum Allowable External Pressure	Pmax	63.53	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.165

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$

$$= (2.14 \cdot 1) / (16 - 3 - 0)$$

$$= 0.165$$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B



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As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.165

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (2.14 \cdot 1) / (16 - 3 - 0)$$

$$= 0.165$$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.165

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (2.14 \cdot 1) / (16 - 3 - 0)$$

$$= 0.165$$

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -9 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.8M- with PWHT, the MDMT is : -9 °C
Temperature reduction per Graph 3.13 : 0 °C
The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (32.9 \cdot 1) / (20 - 3 - 0)$$

$$= 1.000$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.165

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (2.14 \cdot 1) / (16 - 3 - 0)$$

$$= 0.165$$

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -9 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -9 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 3.0820 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 348.0819 mm

Input Echo, WRC297 Item 1, Description: N3- Shell inle:



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Nozzle Calcs.: N3- Shell inlet Nozl: 28 9:32am Aug 12,2026

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2300.000	mm
Shell Thickness	45.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1784.41	kgf/cm ²

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	131.801	mm
Nozzle Thickness	18.2372	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1641.66	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1321.51	kgf/cm ²

Diameter of Reinforcing Pad	368.275	mm
Thickness of Reinforcing Pad	20.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	50.20	kgf/cm ²
Radial Load	P	-509.88	kgf
Circumferential Shear	Vc	764.82	kgf
Longitudinal Shear	Vl	764.82	kgf
Circumferential Moment	Mc	336.39	kgf-m.
Longitudinal Moment	Ml	458.71	kgf-m.
Torsional Moment	Mt	570.84	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (3.739) is < 5
check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (9.227) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{1184.0(45.0 - 3.0)}$$

$$= 904.168 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2368.000	mm
Nozzle Outside Diameter	(d) =	168.275	mm
Vessel Thickness used	(T) =	62.000	mm
Nozzle Thickness used	(t) =	15.237	mm
T / t	=	4.069	
d / t	=	11.044	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.439$$

Nr/P	=	0.188
Mr/P	=	0.140
M0/P	=	0.221



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Nozzle Calcs.: N3- Shell inlet Nozl: 28 9:32am Aug 12,2026

NØ/P = 0.239
 MrD/Mc = 0.148
 NrDL/Mc = 0.166
 MØd/Mc = 0.261
 NØDL/Mc = -0.111
 MrD/Ml = 0.143
 NrDL/Ml = 0.166
 MØD/Ml = 0.252
 NØDL/Ml = 0.104

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-107	107	107	-107
Normal Pressure Stress	908	959	908	959
Outplane Stress Summary	815	1044	1044	845

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	11	-11	11	-11
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-11	-11	11	11
Inplane Bending (ML)	-60	60	60	-60
Inplane Pressure Stress	454	454	454	454
Inplane Stress Summary	395	494	540	396

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	20	20	20	20
Shear Stress Summary	25	25	16	16

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	816	1045	1044	845

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	17	-17	17	-17



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Nozzle Calcs.: N3- Shell inlet Nozl: 28 9:32am Aug 12,2026

Outplane Membrane (Mc)	5	5	-5	-5
Outplane Bending (Mc)	-81	81	81	-81
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	454	454	454	454

Outplane Stress Summary	399	527	550	352
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	11	-11	11	-11
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-45	45	45	-45
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	908	959	908	959

Inplane Stress Summary	868	987	977	913
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4
Torsional Shear (Mt)	20	20	20	20

Shear Stress Summary	16	16	25	25
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	868	988	978	914
------------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	245	305	245	305

Hoop Stress Summary	241	301	255	315
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	6	6	6	6
Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0



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Nozzle Calcs.: N3- Shell inlet Nozl: 28 9:32am Aug 12,2026

Axial Membrane (ML)	-178	-178	178	178
Axial Bending (ML)	-422	422	422	-422
Axial Pressure Stress	122	122	122	122

Axial Stress Summary	-410	313	790	-175
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	18	18	-18	-18
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	84	84	84	84

Shear Stress Summary	103	103	65	65
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	683	410	798	508
------------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	5	5	-5	-5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	245	305	245	305

Hoop Stress Summary	254	314	242	302
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	6	6	6	6
Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	-130	-130	130	130
Axial Bending (Mc)	-332	332	332	-332
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	122	122	122	122

Axial Stress Summary	-273	271	653	-133
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-18	-18	18	18
Shear due to Torsion	84	84	84	84

Shear Stress Summary	65	65	103	103
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside



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Two * Max Shear Stress	543	361	677	482
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{\text{Rmean}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{1174.0(45.0 - 3.0)}$$

$$= 901.053 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2348.000 mm
Nozzle Outside Diameter	(d) =	368.275 mm
Vessel Thickness used	(T) =	42.000 mm
Nozzle Thickness used	(t) =	115.237 mm
T / t	=	0.364
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.173$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.097
Mr/P	=	0.204
M0/P	=	0.060
N0/P	=	0.117
MrD/Mc	=	0.306
NrDL/Mc	=	0.085
M0d/Mc	=	0.094
N0DL/Mc	=	0.050
MrD/Ml	=	0.174
NrDL/Ml	=	0.090
M0D/Ml	=	0.055
N0DL/Ml	=	0.121

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-23	23	23	-23
Normal Pressure Stress	1353	1403	1353	1403
Outplane Stress Summary	1335	1411	1398	1381

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	35	-35	35	-35
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-73	73	73	-73
Inplane Pressure Stress	676	676	676	676



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Inplane Stress Summary	634	711	795	576
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	3	3	-3	-3
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6

Shear Stress Summary	9	9	3	3
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	1335	1411	1398	1381
------------------------	------	------	------	------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-29	29	29	-29
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	676	676	676	676

Outplane Stress Summary	658	696	722	642
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	35	-35	35	-35
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-95	95	95	-95
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1353	1403	1353	1403

Inplane Stress Summary	1292	1461	1490	1280
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	6	6	6	6

Shear Stress Summary	3	3	9	9
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



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Two * Max Shear Stress 1292 1461 1491 1280

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	908	959	908	959	908	959	908	959
Circ. Pl (SUS)	-4	-4	10	10	-6	-6	10	10
Circ. Q (SUS)	-90	90	124	-124	-34	34	56	-56
Long. Pm (SUS)	454	454	454	454	454	454	454	454
Long. Pl (SUS)	-9	-9	13	13	8	8	-2	-2
Long. Q (SUS)	-49	49	71	-71	-64	64	98	-98
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	20	20	20	20	20	20	20	20
Pm (SUS)	908.0	959.0	908.0	959.0	908.0	959.0	908.0	959.0
Pm+Pl (SUS)	904.0	955.0	918.0	969.0	902.0	953.0	918.0	969.0
Pm+Pl+Q (Total)	815	1046	1042	845	868	987	975	914

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	959.00	1784.41	Passed
Pm+Pl (SUS)	969.03	2676.62	Passed
Pm+Pl+Q (TOTAL)	1046.04	5353.24	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1353	1403	1353	1403	1353	1403	1353	1403
Circ. Pl (SUS)	-5	-5	11	11	-2	-2	6	6
Circ. Q (SUS)	-13	13	33	-33	-60	60	130	-130
Long. Pm (SUS)	676	676	676	676	676	676	676	676
Long. Pl (SUS)	-4	-4	8	8	1	1	5	5
Long. Q (SUS)	-38	38	108	-108	-19	19	39	-39
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	1353	1403	1353	1403	1353	1403	1353	1403
Pm+Pl (SUS)	1348	1398	1364	1414	1351	1401	1359	1409
Pm+Pl+Q (Total)	1335	1411	1397	1381	1291	1461	1489	1279



Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1403.00	1784.41	Passed
Pm+Pl (SUS)	1414.01	2676.62	Passed
Pm+Pl+Q (TOTAL)	1489.11	5353.24	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	245	305	245	305	245	305	245	305
Circ. Pl (SUS)	-4	-4	10	10	8	8	-2	-2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	122	122	122	122	122	122	122	122
Long. Pl (SUS)	-172	-172	184	184	-124	-124	136	136
Long. Q (SUS)	-362	362	482	-482	-272	272	392	-392
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	18	18	-18	-18	-18	-18	18	18
Shear Q (SUS)	84	84	84	84	84	84	84	84
Pm (SUS)	245.0	305.0	245.0	305.0	245.0	305.0	245.0	305.0
Pm+Pl (SUS)	293.2	352.8	311.7	329.1	257.5	317.1	270.0	309.3
Pm+Pl+Q (Total)	684.1	408.6	796.1	508.4	543.3	360.9	674.1	482.3

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	305.00	1321.51	Passed
Pm+Pl (SUS)	352.84	1982.27	Passed
Pm+Pl+Q (TOTAL)	796.05	3964.53	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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 Nozzle Calcs.: N4- Shell outlet Nozl: 29 9:32am Aug 12,2026

Input, Nozzle Desc: N4- Shell outlet

From: 60

Pressure for Reinforcement Calculations	P	49.967	kgf/cm ²
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	1.03	kgf/cm ²
Temperature for External Pressure	Tempex	270	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1784.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	5041.6665	mm
Shell Finished (Minimum) Thickness	t	45.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		493.60	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1321.51	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		10.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1435.72	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	373.0500	mm
Thickness of Pad	te	20.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm
Reinforcing Pad Width		50.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

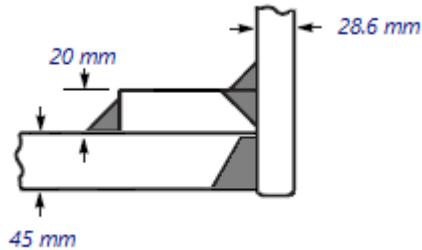
Reinforcement CALCULATION, Description: N4- Shell outlet



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 Nozzle Calcs.: N4- Shell outle Nozl: 29 9:32am Aug 12,2026

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation 8.500 in.
 Actual Thickness Used in Calculation 1.125 in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 222 \cdot (\exp(50/(1322 \cdot 1)) - 1) + 3 + 0$
 $= 4.28 + 3 + 0 = 7.28 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1322 \cdot 1 \cdot \ln((2 \cdot 25.6 + 222) / 222) - 0$
 $= 274 - 0 = 274 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.110 mm
 Min. required shell thickness, internal p + c : 35.745 mm
 Min. required shell thickness, external p + c : 15.973 mm

Nozzle neck minimum required thickness : 11.110 mm

Minimum nozzle neck thickness 24.994 mm , is greater than required thickness 11.110 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 276.6,	Allowable	: 1321.5 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3427.4 kgf/cm ²	Passed
Occasional	: 114.6,	Allowable	: 1757.6 kgf/cm ²	Passed
Shear	: 116.9,	Allowable	: 925.1 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1322/1784, 1]$
 $= 0.741$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[1436/1784, 1]$
 $= 0.805$

Thickness of Nozzle [tn]:



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= thickness - corrosion allowance
= 28.6 - 3
= 25.575 mm

Shell Diameter to Thickness ratio [D/t]:

= Di/t
= 2306/42
= 54.905 must be less than 400.

Effective Pressure Radius [Reff]:

= Di/2 + corrosion allowance
= 2300/2 + 3
= 1153.000 mm

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

= $\sqrt{(Reff \cdot t) + W}$
= $\sqrt{(1153 \cdot 42) + 50}$
= 270.059 mm

Limit [LR2]:

= $\sqrt{((Reff + t) \cdot (t + te))}$
= $\sqrt{((1153 + 42) \cdot (42 + 20))}$
= 272.195 mm

Limit [LR3]:

= 2 • Rn
= 2 • 111
= 221.900 mm

Final value of [LR]:

= min(LR1, LR2, LR3)
= min(270, 272, 222)
= 221.900 mm

Thickness Limit Candidate [LH1]:

= min[1.5t, te] + $\sqrt{(Rn \cdot tn)}$
= min[1.5•42, 20] + $\sqrt{(111 \cdot 25.6)}$
= 73.269 mm

Thickness Limit Candidate [LH2]:

= Lpr1
= 300
= 300.000 mm

Thickness Limit Candidate [LH3]:

= 8(t + te)
= 8(42 + 20)
= 496.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:

= min[LH1, LH2, LH3] + t for set-in nozzles
= min[73.3, 300, 496] + 42
= 115.269 mm

Effective Vessel Thickness [teff]:

= t + ((A5 • frp) / LR)
= 42 + ((10•0.8)/222)
= 45.626 mm

Parameter [λ]:

= min(12, (2•Rn + tn) / ($\sqrt{(Di + teff) \cdot teff}$))



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Nozzle Calcs.: N4- Shell outle Nozl: 29 9:32am Aug 12,2026

$$= \min(12, (2 \cdot 111 + 25.6) / (\sqrt{(2306 + 45.6) \cdot 45.6}))$$

$$= 0.756$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((N/5)^{0.85}, 1)$$

$$= 42 \cdot 222 \cdot \max((0.76/5)^{0.85}, 1)$$

$$= 93.198 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 25.6 \cdot 115$$

$$= 29.480 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 20^2$$

$$= 2.000 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 20^2$$

$$= 2.000 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$

$$= \min(50 \cdot 20, 222 \cdot 20)$$

$$= 10.000 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$

$$= 93.2 + 0.74(29.5 + 0) + 2 + 2 + 0 + 0.8(10)$$

$$= 127.076 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 25.6 / \ln[1 + 25.6/111]$$

$$= 123.296 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 45.6 / \ln[1 + 45.6/1153]$$

$$= 1175.664 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1784 \cdot 1$$

$$= 2676.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 50 \cdot 123 \cdot 115$$

$$= 7101.4 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 50 \cdot 1176 (222 + 25.6)$$

$$= 145378.3 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 50 \cdot 1176 \cdot 111$$

$$= 65177.2 \text{ kgf}$$



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 Nozzle Calcs.: N4- Shell outle Nozl: 29 9:32am Aug 12,2026

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (145378 + 65177 + 7101) / 50$$

$$= 4356.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 2677 / (2 \cdot 4356 / 127 - 1176 / 45.6)$$

$$= 62.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 1784 [42 / 1176]$$

$$= 63.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(62.6, 63.7)$$

$$= 62.553 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (7101 + 145378 + 65177) / 127$$

$$= 1712.806 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 50 \cdot 1176 / 45.6$$

$$= 1287.5 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 1713 - 1288, 1288)$$

$$= 2138.1 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	2138.08	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	62.55	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (111 + 25.6) / 111$$

$$= 1.231 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lτ]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (111 + 25.6)$$

$$= 214.453 \text{ mm}$$

Weld Length of Pad to Shell Weld [LτP]:

$$= \pi / 2 \cdot (Rn + tn + W)$$

$$= \pi / 2 \cdot (111 + 25.6 + 50)$$

$$= 292.993 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 14.142, 14.142, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 14.142, \text{ must be } \geq \min(0.7 \cdot 25.575, 6.000) \text{ Passed}$$

$$tf1, L42T = 14.142, \text{ must be } \geq \min(0.6 \cdot 20.000, 0.6 \cdot 42.000) \text{ Passed}$$

Weld Load Value [fwelds]:



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Nozzle Calcs.: N4- Shell outle Nozl: 29 9:32am Aug 12, 2026

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(65177 \cdot 1.23, 1.5 \cdot 1322(29.5+0), \pi/4 \cdot 50 \cdot 111^2 \cdot 1.23^2)$$

$$= 7314.745 \text{ kgf}$$

Discontinuity Force [fws]:

$$= f_{welds} \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 7315 \cdot 42 \cdot 1784 / (42 \cdot 1784 + 20 \cdot 1436)$$

$$= 5288.512 \text{ kgf}$$

Discontinuity Force [fwp]:

$$= f_{welds} \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 7315 \cdot 20 \cdot 1436 / (42 \cdot 1784 + 20 \cdot 1436)$$

$$= 2026.233 \text{ kgf}$$

Shear Stress [τ_1]:

$$= f_{ws} / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 5289 / (214 \cdot (0.6 \cdot 38 + 0.49 \cdot 0))$$

$$= 108.160 \text{ kgf/cm}^2$$

Shear Stress [τ_2]:

$$= f_{wp} / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T}))$$

$$= 2026 / (214 \cdot (0.6 \cdot 20 + 0.49 \cdot 14.1))$$

$$= 49.913 \text{ kgf/cm}^2$$

Shear Stress [τ_3]:

$$= f_{wp} / (L_{TP} \cdot (0.49 \cdot L_{42T}))$$

$$= 2026 / (293 \cdot (0.49 \cdot 14.1))$$

$$= 99.799 \text{ kgf/cm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(108, 49.9, 99.8)$$

$$= 108.2 \text{ must be less than or equal to } 1784.4 \text{ kgf/cm}^2$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 25.6 / \ln[1 + 25.6/111]$$

$$= 123.296 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 45.6 / \ln[1 + 45.6/1153]$$

$$= 1175.664 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1784$$

$$= 2676.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.03 \cdot 123 \cdot 115$$

$$= 146.8 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.03 \cdot 1176 (222 + 25.6)$$

$$= 3005.5 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.03 \cdot 1176 \cdot 111$$

$$= 1347.4 \text{ kgf}$$



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Area Resisting External Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (3005 + 1347 + 147) / 1.03$$

$$= 4356.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 2677 / (2 \cdot 4356 / 127 - 1176 / 45.6)$$

$$= 62.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 1784 [42 / 1176]$$

$$= 63.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(62.6, 63.7)$$

$$= 62.553 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (147 + 3005 + 1347) / 127$$

$$= 35.410 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 1.03 \cdot 1176 / 45.6$$

$$= 26.6 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 35.4 - 26.6, 26.6)$$

$$= 44.2 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	44.20	kgf/cm ²
Maximum Allowable External Pressure	Pmax	62.55	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.156

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$

$$= (3.43 \cdot 1) / (25 - 3 - 0)$$

$$= 0.156$$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -9 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B
From Figure 3.8M- with PWHT, the MDMT is	: -9 °C



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Nozzle Calcs.: N4- Shell outle Nozl: 29 9:32am Aug 12,2026

Temperature reduction per Graph 3.13 : 0 °C
The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (32.7 \cdot 1) / (20 - 3 - 0)$$

$$= 1.000$$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -9 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.8M- with PWHT, the MDMT is : -9 °C
Temperature reduction per Graph 3.13 : 0 °C
The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (32.7 \cdot 1) / (20 - 3 - 0)$$

$$= 1.000$$

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -9 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B
From Figure 3.8M- with PWHT, the MDMT is : -9 °C
Temperature reduction per Graph 3.13 : 0 °C
The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (32.7 \cdot 1) / (20 - 3 - 0)$$

$$= 1.000$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 11 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.156

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (3.43 \cdot 1) / (25 - 3 - 0)$$

$$= 0.156$$

Gov. MDMT of the Nozzle : -9 °C
Gov. MDMT of the Reinforcement Pad : -9 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -9 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 8.1326 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.1326 mm

Input Echo, WRC297 Item 1, Description: N4- Shell outl:



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 Nozzle Calcs.: N4- Shell outle Nozl: 29 9:32am Aug 12,2026

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2300.000	mm
Shell Thickness	45.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1784.41	kgf/cm ²

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	215.900	mm
Nozzle Thickness	28.5750	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1641.66	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1321.51	kgf/cm ²

Diameter of Reinforcing Pad	373.050	mm
Thickness of Reinforcing Pad	20.0000	mm

Note:
External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	49.97	kgf/cm ²
Radial Load	P	-856.60	kgf
Circumferential Shear	Vc	1274.71	kgf
Longitudinal Shear	Vl	1274.71	kgf
Circumferential Moment	Mc	948.00	kgf-m.
Longitudinal Moment	Ml	1274.19	kgf-m.
Torsional Moment	Mt	1590.20	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (9.556) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{1184.0(45.0 - 3.0)}$$

$$= 1008.943 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2368.000 mm
Nozzle Outside Diameter	(d) =	273.050 mm
Vessel Thickness used	(T) =	62.000 mm
Nozzle Thickness used	(t) =	25.575 mm
T / t	=	2.424
d / t	=	10.676

$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.713$

Nr/P	=	0.167
Mr/P	=	0.147
M0/P	=	0.113
N0/P	=	0.167
MrD/Mc	=	0.206



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Nozzle Calcs.: N4- Shell outle Nozl: 29 9:32am Aug 12,2026

NrDL/Mc = 0.188
MØd/Mc = 0.182
NØDL/Mc = 0.124
MrD/Ml = 0.177
NrDL/Ml = 0.168
MØD/Ml = 0.159
NØDL/Ml = 0.098

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-11	-11	11	11
Outplane Bending (ML)	-115	115	115	-115
Normal Pressure Stress	904	954	904	954
Outplane Stress Summary	796	1047	1051	839

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-20	-20	20	20
Inplane Bending (ML)	-129	129	129	-129
Inplane Pressure Stress	452	452	452	452
Inplane Stress Summary	326	545	625	327

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	21	21	21	21
Shear Stress Summary	26	26	17	17

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	797	1048	1051	840

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-98	98	98	-98



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Nozzle Calcs.: N4- Shell outle Nozl: 29 9:32am Aug 12,2026

Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	452	452	452	452

Outplane Stress Summary	361	528	581	353
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	-16	-16	16	16
Inplane Bending (Mc)	-111	111	111	-111
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	904	954	904	954

Inplane Stress Summary	799	1033	1056	844
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4
Torsional Shear (Mt)	21	21	21	21

Shear Stress Summary	17	17	26	26
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	800	1034	1058	845
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-11	-11	11	11
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	233	293	233	293

Hoop Stress Summary	225	285	248	308
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Axial Membrane (P)	4	4	4	4
Axial Bending (P)	49	-49	49	-49
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-113	-113	113	113
Axial Bending (ML)	-400	400	400	-400



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Nozzle Calcs.: N4- Shell outle Nozl: 29 9:32am Aug 12,2026

Axial Pressure Stress	116	116	116	116
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Axial Stress Summary	-342	358	684	-216
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	11	11	-11	-11
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Shear due to (Vl)	0	0	0	0
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Shear due to Torsion	53	53	53	53
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Shear Stress Summary	64	64	41	41
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	582	396	688	531
------------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	3	3	3	3
--------------------	---	---	---	---

Hoop Bending (P)	0	0	0	0
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Hoop Membrane (Mc)	-11	-11	11	11
--------------------	-----	-----	----	----

Hoop Bending (Mc)	0	0	0	0
-------------------	---	---	---	---

Hoop Membrane (ML)	0	0	0	0
--------------------	---	---	---	---

Hoop Bending (ML)	0	0	0	0
-------------------	---	---	---	---

Hoop Pressure Stress	233	293	233	293
----------------------	-----	-----	-----	-----

Hoop Stress Summary	225	285	248	308
---------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	4	4	4	4
---------------------	---	---	---	---

Axial Bending (P)	49	-49	49	-49
--------------------	----	-----	----	-----

Axial Membrane (Mc)	-84	-84	84	84
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Axial Bending (Mc)	-356	356	356	-356
--------------------	------	-----	-----	------

Axial Membrane (ML)	0	0	0	0
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Axial Bending (ML)	0	0	0	0
--------------------	---	---	---	---

Axial Pressure Stress	116	116	116	116
-----------------------	-----	-----	-----	-----

Axial Stress Summary	-269	343	611	-201
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
-------------------	---	---	---	---

Shear due to (Vl)	-11	-11	11	11
-------------------	-----	-----	----	----

Shear due to Torsion	53	53	53	53
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Shear Stress Summary	41	41	64	64
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	502	364	622	525
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(1174.0(45.0 - 3.0))}$$

$$= 1005.828 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2348.000 mm
 Nozzle Outside Diameter (d) = 373.050 mm
 Vessel Thickness used (T) = 42.000 mm
 Nozzle Thickness used (t) = 75.575 mm
 T / t = 0.556
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.188$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.097
 Mr/P = 0.197
 M0/P = 0.059
 N0/P = 0.123
 MrD/Mc = 0.301
 NrDL/Mc = 0.100
 M0d/Mc = 0.096
 N0DL/Mc = 0.047
 MrD/Ml = 0.174
 NrDL/Ml = 0.086
 M0D/Ml = 0.057
 N0DL/Ml = 0.128

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-24	-24	24	24
Outplane Bending (ML)	-66	66	66	-66
Normal Pressure Stress	1347	1397	1347	1397
Outplane Stress Summary	1278	1427	1461	1344

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	57	-57	57	-57
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-16	-16	16	16
Inplane Bending (ML)	-202	202	202	-202
Inplane Pressure Stress	673	673	673	673
Inplane Stress Summary	516	806	954	434



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Nozzle Calcs.: N4- Shell outle Nozl: 29 9:32am Aug 12,2026

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			5	5	-5	-5
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			17	17	17	17
Shear Stress Summary			22	22	12	12

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			1279	1428	1461	1344

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			5	5	5	5
Outplane Bending (P)			17	-17	17	-17
Outplane Membrane (Mc)			-6	-6	6	6
Outplane Bending (Mc)			-82	82	82	-82
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			673	673	673	673
Outplane Stress Summary			607	738	785	586

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			4	4	4	4
Inplane Bending (P)			57	-57	57	-57
Inplane Membrane (Mc)			-14	-14	14	14
Inplane Bending (Mc)			-260	260	260	-260
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			1347	1397	1347	1397
Inplane Stress Summary			1134	1590	1683	1098

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-5	-5	5	5
Torsional Shear (Mt)			17	17	17	17
Shear Stress Summary			12	12	22	22

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress			1134	1590	1684	1099



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	904	954	904	954	904	954	904	954
Circ. Pl (SUS)	-8	-8	14	14	-13	-13	19	19
Circ. Q (SUS)	-100	100	130	-130	-92	92	130	-130
Long. Pm (SUS)	452	452	452	452	452	452	452	452
Long. Pl (SUS)	-17	-17	23	23	-8	-8	14	14
Long. Q (SUS)	-110	110	148	-148	-83	83	113	-113
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	21	21	21	21	21	21	21	21
Pm (SUS)	904.0	954.0	904.0	954.0	904.0	954.0	904.0	954.0
Pm+Pl (SUS)	896.0	946.0	918.0	968.0	891.0	941.0	923.0	973.0
Pm+Pl+Q (Total)	797	1047	1048	838	799	1033	1054	844

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	954.00	1784.41	Passed
Pm+Pl (SUS)	973.03	2676.62	Passed
Pm+Pl+Q (TOTAL)	1054.31	5353.24	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1347	1397	1347	1397	1347	1397	1347	1397
Circ. Pl (SUS)	-19	-19	29	29	-10	-10	18	18
Circ. Q (SUS)	-49	49	83	-83	-203	203	317	-317
Long. Pm (SUS)	673	673	673	673	673	673	673	673
Long. Pl (SUS)	-12	-12	20	20	-1	-1	11	11
Long. Q (SUS)	-145	145	259	-259	-65	65	99	-99
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	17	17	17	17	17	17	17	17
Pm (SUS)	1347	1397	1347	1397	1347	1397	1347	1397
Pm+Pl (SUS)	1328	1378	1376	1426	1337	1387	1365	1415
Pm+Pl+Q (Total)	1279	1427	1459	1343	1134	1590	1682	1098



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Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1397.00	1784.41	Passed
Pm+Pl (SUS)	1426.03	2676.62	Passed
Pm+Pl+Q (TOTAL)	1682.54	5353.24	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	233	293	233	293	233	293	233	293
Circ. Pl (SUS)	-8	-8	14	14	-8	-8	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	116	116	116	116	116	116	116	116
Long. Pl (SUS)	-109	-109	117	117	-80	-80	88	88
Long. Q (SUS)	-351	351	449	-449	-307	307	405	-405
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	11	11	-11	-11	-11	-11	11	11
Shear Q (SUS)	53	53	53	53	53	53	53	53
Pm (SUS)	233.0	293.0	233.0	293.0	233.0	293.0	233.0	293.0
Pm+Pl (SUS)	225.6	285.4	253.0	308.6	225.6	285.5	249.7	308.2
Pm+Pl+Q (Total)	583.2	395.2	686.0	529.7	503.1	365.0	620.0	523.9

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	293.00	1321.51	Passed
Pm+Pl (SUS)	308.60	1982.27	Passed
Pm+Pl+Q (TOTAL)	686.02	3964.53	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N5- Blow down

Nozl: 30

9:32am

Aug 12,2026

Input, Nozzle Desc: N5- Blow down

From: 60

Pressure for Reinforcement Calculations	P	50.197	kgf/cm ²
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	1.03	kgf/cm ²
Temperature for External Pressure	Tempex	270	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1784.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	5041.6665	mm
Shell Finished (Minimum) Thickness	t	45.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		583.60	cm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1321.51	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1435.72	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	373.0500	mm
Thickness of Pad	te	20.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm
Reinforcing Pad Width		156.3625	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

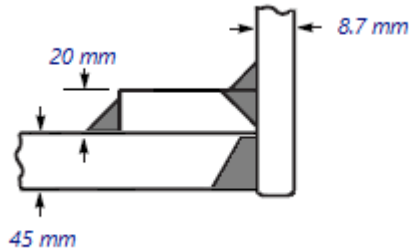
Reinforcement CALCULATION, Description: N5- Blow down



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 Nozzle Calcs.: N5- Blow down Nozl: 30 9:32am Aug 12,2026

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation 1.687 in.
 Actual Thickness Used in Calculation 0.344 in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 48.8 \cdot (\exp(50.2/(1322 \cdot 1)) - 1) + 3 + 0$
 $= 0.95 + 3 + 0 = 3.95 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1322 \cdot 1 \cdot \ln((2 \cdot 5.74 + 48.8) / 48.8) - 0$
 $= 279 - 0 = 279 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 6.420 mm
 Min. required shell thickness, internal p + c : 35.898 mm
 Min. required shell thickness, external p + c : 15.973 mm

Nozzle neck minimum required thickness : 6.420 mm

Minimum nozzle neck thickness 7.645 mm , is greater than required thickness 6.420 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 713.7,	Allowable	: 1321.5 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2990.3 kgf/cm ²	Passed
Occasional	: 120.5,	Allowable	: 1757.6 kgf/cm ²	Passed
Shear	: 452.4,	Allowable	: 925.1 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1322/1784, 1]$
 $= 0.741$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[1436/1784, 1]$
 $= 0.805$

Thickness of Nozzle [tn]:



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Nozzle Calcs.: N5- Blow down Nozl: 30 9:32am Aug 12,2026

= thickness - corrosion allowance
= 8.74 - 3
= 5.738 mm

Shell Diameter to Thickness ratio [D/t]:

= Di/t
= 2306/42
= 54.905 must be less than 400.

Effective Pressure Radius [Reff]:

= Di/2 + corrosion allowance
= 2300/2 + 3
= 1153.000 mm

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

= $\sqrt{(Reff \cdot t) + W}$
= $\sqrt{(1153 \cdot 42) + 156}$
= 376.422 mm

Limit [LR2]:

= $\sqrt{((Reff + t) \cdot (t + te))}$
= $\sqrt{((1153 + 42) \cdot (42 + 20))}$
= 272.195 mm

Limit [LR3]:

= 2 • Rn
= 2 • 24.4
= 48.850 mm

Final value of [LR]:

= min(LR1, LR2, LR3)
= min(376, 272, 48.8)
= 48.850 mm

Thickness Limit Candidate [LH1]:

= min[1.5t, te] + $\sqrt{(Rn \cdot tn)}$
= min[1.5•42, 20] + $\sqrt{(24.4 \cdot 5.74)}$
= 31.838 mm

Thickness Limit Candidate [LH2]:

= Lpr1
= 300
= 300.000 mm

Thickness Limit Candidate [LH3]:

= 8(t + te)
= 8(42 + 20)
= 496.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:

= min[LH1, LH2, LH3] + t for set-in nozzles
= min[31.8, 300, 496] + 42
= 73.838 mm

Effective Vessel Thickness [teff]:

= t + ((A5 • frp) / LR)
= 42 + ((9.77•0.8)/48.8)
= 58.092 mm

Parameter [λ]:

= min(12, (2•Rn + tn) / $\sqrt{((Di + teff) \cdot teff)}$))



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Nozzle Calcs.: N5- Blow down Nozl: 30 9:32am Aug 12,2026

$$= \min(12, (2 \cdot 24.4 + 5.74) / (\sqrt{(2306 + 58.1) \cdot 58.1}))$$

$$= 0.147$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((N/5)^{0.85}, 1)$$

$$= 42 \cdot 48.8 \cdot \max((0.15/5)^{0.85}, 1)$$

$$= 20.517 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 5.74 \cdot 73.8$$

$$= 4.237 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2 - \text{Area cut by thickness limit}$$

$$= 0.5 \cdot 20^2 - 0.33$$

$$= 1.667 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$

$$= \min(156 \cdot 20, 48.8 \cdot 20)$$

$$= 9.770 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$

$$= 20.5 + 0.74(4.24 + 0) + 1.67 + 0 + 0 + 0.8(9.77)$$

$$= 33.182 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 5.74 / \ln[1 + 5.74/24.4]$$

$$= 27.193 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 58.1 / \ln[1 + 58.1/1153]$$

$$= 1181.807 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1784 \cdot 1$$

$$= 2676.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 50.2 \cdot 27.2 \cdot 73.8$$

$$= 1007.9 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 50.2 \cdot 1182 (48.8 + 5.74)$$

$$= 32383.0 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 50.2 \cdot 1182 \cdot 24.4$$

$$= 14489.7 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (32383 + 14490 + 1008) / 50.2$$

$$= 953.9 \text{ cm}^2$$



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 Nozzle Calcs.: N5- Blow down Nozl: 30 9:32am Aug 12,2026

Maximum Allowable Working Pressure Candidate [Pmax1]:
 = Sallow / (2 • Ap/AT - Rxs/teff)
 = 2677 / (2 • 954/33.2 - 1182/58.1)
 = 72.1 kgf/cm²

Maximum Allowable Working Pressure Candidate [Pmax2]:
 = S[t/Rxs]
 = 1784 [42/1182]
 = 63.4 kgf/cm²

Maximum Allowable Working Pressure [Pmax]:
 = min(Pmax1, Pmax2)
 = min(72.1, 63.4)
 = 63.416 kgf/cm²

Average Primary Membrane Stress [σAvg]:
 = (fN + fS + fY) / AT
 = (1008 + 32383 + 14490) / 33.2
 = 1442.964 kgf/cm²

General Primary Membrane Stress [σCirc]:
 = P • Rxs / teff
 = 50.2 • 1182/58.1
 = 1021.2 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:
 = max(2 • σAvg - σCirc, σCirc)
 = max(2 • 1443 - 1021, 1021)
 = 1864.7 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	1864.73	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	63.42	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
 = (Rnc + tn) / Rnc
 = (24.4 + 5.74) / 24.4
 = 1.235 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lτ]:
 = π/2 • (Rn + tn)
 = π/2 • (24.4 + 5.74)
 = 47.379 mm

Weld Length of Pad to Shell Weld [LτP]:
 = π/2 • (Rn + tn + W)
 = π/2 • (24.4 + 5.74 + 156)
 = 292.993 mm

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
 = 14.142, 14.142, 0.000, mm

Required Weld Throat Dimensions :
 tc, L41T = 14.142, must be >= min(0.7 * 5.738, 6.000) Passed
 tf1, L42T = 14.142, must be >= min(0.6 * 20.000, 0.6 * 42.000) Passed

Weld Load Value [fwelds]:
 = min(fy • ky, 1.5 • Sn(A2 + A3), π/4•P•Rn²•ky²)
 = min(14490•1.23, 1.5•1322(4.24+0), π/4•50.2•24.4²•1.23²)
 = 358.677 kgf

Discontinuity Force [fws]:



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Nozzle Calcs.: N5- Blow down Nozl: 30 9:32am Aug 12,2026

$$\begin{aligned}
 &= fwelds \cdot t \cdot S / (t \cdot S + te \cdot Sp) \\
 &= 359 \cdot 42 \cdot 1784 / (42 \cdot 1784 + 20 \cdot 1436) \\
 &= 259.321 \text{ kgf}
 \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned}
 &= fwelds \cdot te \cdot Sp / (t \cdot S + te \cdot Sp) \\
 &= 359 \cdot 20 \cdot 1436 / (42 \cdot 1784 + 20 \cdot 1436) \\
 &= 99.356 \text{ kgf}
 \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned}
 &= fws / (Lt \cdot (0.6 \cdot tw1 + 0.49 \cdot L43T)) \\
 &= 259 / (47.4 \cdot (0.6 \cdot 38 + 0.49 \cdot 0)) \\
 &= 24.006 \text{ kgf/cm}^2
 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned}
 &= fwp / (Lt \cdot (0.6 \cdot tw2 + 0.49 \cdot L41T)) \\
 &= 99.4 / (47.4 \cdot (0.6 \cdot 20 + 0.49 \cdot 14.1)) \\
 &= 11.078 \text{ kgf/cm}^2
 \end{aligned}$$

Shear Stress [τ_3]:

$$\begin{aligned}
 &= fwp / (LtP \cdot (0.49 \cdot L42T)) \\
 &= 99.4 / (293 \cdot (0.49 \cdot 14.1)) \\
 &= 4.894 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned}
 &= \max(\tau_1, \tau_2, \tau_3) \\
 &= \max(24, 11.1, 4.89) \\
 &= 24.0 \text{ must be less than or equal to } 1784.4 \text{ kgf/cm}^2
 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= tn / \ln[1 + tn/Rn] \\
 &= 5.74 / \ln[1 + 5.74/24.4] \\
 &= 27.193 \text{ mm}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= teff / \ln[1 + teff/Reff] \\
 &= 58.1 / \ln[1 + 58.1/1153] \\
 &= 1181.807 \text{ mm}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \\
 &= 1.5 \cdot 1784 \\
 &= 2676.6 \text{ kgf/cm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot Rxn \cdot LH \\
 &= 1.03 \cdot 27.2 \cdot 73.8 \\
 &= 20.7 \text{ kgf}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot (LR + tn) \\
 &= 1.03 \cdot 1182 \cdot (48.8 + 5.74) \\
 &= 666.4 \text{ kgf}
 \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot Rnc \\
 &= 1.03 \cdot 1182 \cdot 24.4 \\
 &= 298.2 \text{ kgf}
 \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned}
 &= (fS + fY + fN) / P \\
 &= (666 + 298 + 20.7) / 1.03 \\
 &= 953.9 \text{ cm}^2
 \end{aligned}$$



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Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 2677 / (2 \cdot 954/33.2 - 1182/58.1)$$

$$= 72.1 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 1784 [42/1182]$$

$$= 63.4 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(72.1, 63.4)$$

$$= 63.416 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (20.7 + 666 + 298) / 33.2$$

$$= 29.694 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / teff$$

$$= 1.03 \cdot 1182/58.1$$

$$= 21.0 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 29.7 - 21, 21)$$

$$= 38.4 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	38.37	kgf/cm ²
Maximum Allowable External Pressure	Pmax	63.42	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.163

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$

$$= (0.76 \cdot 1) / (7.65 - 3 - 0)$$

$$= 0.163$$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.163

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$



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Nozzle Calcs.: N5- Blow down Nozl: 30 9:32am Aug 12,2026

$$= (0.76 \cdot 1) / (7.65 - 3 - 0)$$

$$= 0.163$$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.163

The computed thickness ratio [sr]:

$$= (tr \cdot E) / (tnom - ci - cext)$$

$$= (0.76 \cdot 1) / (7.65 - 3 - 0)$$

$$= 0.163$$

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -9 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : -9 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

$$= (tr \cdot E) / (tnom - ci - cext)$$

$$= (32.9 \cdot 1) / (20 - 3 - 0)$$

$$= 1.000$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.163

The computed thickness ratio [sr]:

$$= (tr \cdot E) / (tnom - ci - cext)$$

$$= (0.76 \cdot 1) / (7.65 - 3 - 0)$$

$$= 0.163$$

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -9 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -9 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 0.3956 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 345.3956 mm

Input Echo, WRC297 Item 1, Description: N5- Blow down :

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	3.0000 mm
Shell External Corrosion Allowance	0.0000 mm
Shell Diameter	2300.000 mm



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Nozzle Calcs.: N5- Blow down Nozl: 30 9:32am Aug 12,2026

Shell Thickness		45.0000	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1784.41	kgf/cm ²
Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		3.0000	mm
Nozzle Diameter		42.850	mm
Nozzle Thickness		8.7376	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	1641.66	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1321.51	kgf/cm ²
Diameter of Reinforcing Pad		373.050	mm
Thickness of Reinforcing Pad		20.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	50.20	kgf/cm ²
Radial Load	P	-163.16	kgf
Circumferential Shear	Vc	254.94	kgf
Longitudinal Shear	Vl	254.94	kgf
Circumferential Moment	Mc	30.58	kgf-m.
Longitudinal Moment	Ml	50.97	kgf-m.
Torsional Moment	Mt	61.16	kgf-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

Warning - The ratio, Dn/Tv, (1.341) is < 5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (6.904) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{1184.0(45.0 - 3.0)}$$

$$= 796.218 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2368.000	mm
Nozzle Outside Diameter	(d) =	60.325	mm
Vessel Thickness used	(T) =	62.000	mm
Nozzle Thickness used	(t) =	5.738	mm
T / t	=	10.000	
d / t	=	10.514	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.157$$

Nr/P	=	0.231
Mr/P	=	0.144
M0/P	=	0.372
N0/P	=	0.231
MrD/Mc	=	0.110
NrDL/Mc	=	0.153
M0d/Mc	=	0.307
N0DL/Mc	=	-0.136



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Nozzle Calcs.: N5- Blow down Nozl: 30 9:32am Aug 12,2026

MrD/M1 = 0.111
NrDL/M1 = 0.157
MØD/M1 = 0.305
NØDL/M1 = -0.157

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	3	3	-3	-3
Outplane Bending (ML)	-40	40	40	-40
Normal Pressure Stress	908	959	908	959
Outplane Stress Summary	882	994	956	906

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-14	14	14	-14
Inplane Pressure Stress	454	454	454	454
Inplane Stress Summary	440	462	477	440

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	17	17	17	17
Shear Stress Summary	21	21	12	12

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	883	995	956	907

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	1	1	-1	-1
Outplane Bending (Mc)	-24	24	24	-24
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	454	454	454	454



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Nozzle Calcs.: N5- Blow down

Nozl: 30 9:32am Aug 12,2026

Outplane Stress Summary	442	471	487	419
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-8	8	8	-8
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	908	959	908	959

Inplane Stress Summary	902	963	924	949
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4
Torsional Shear (Mt)	17	17	17	17

Shear Stress Summary	12	12	21	21
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	903	963	925	950
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	3	3	-3	-3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	229	289	229	289

Hoop Stress Summary	233	294	226	287
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	16	16	16	16
Axial Bending (P)	83	-83	83	-83
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-414	-414	414	414
Axial Bending (ML)	-495	495	495	-495
Axial Pressure Stress	114	114	114	114

Axial Stress Summary	-695	128	1124	-32
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Nozzle Calcs.: N5- Blow down

Nozl: 30 9:32am Aug 12,2026

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	46	46	-46	-46
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	186	186	186	186
Shear Stress Summary	233	233	139	139

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	1039	495	1145	424

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	1	1	-1	-1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	229	289	229	289
Hoop Stress Summary	232	292	228	288

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	16	16	16	16
Axial Bending (P)	83	-83	83	-83
Axial Membrane (Mc)	-248	-248	248	248
Axial Bending (Mc)	-315	315	315	-315
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	114	114	114	114
Axial Stress Summary	-350	114	779	-19

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-46	-46	46	46
Shear due to Torsion	186	186	186	186
Shear Stress Summary	139	139	233	233

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	645	368	864	559

Stress Computations at the Edge of the Reinforcing Pad:



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 Nozzle Calcs.: N5- Blow down Nozl: 30 9:32am Aug 12,2026

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{1174.0(45.0 - 3.0)}$$

$$= 793.103 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2348.000 mm
 Nozzle Outside Diameter (d) = 373.050 mm
 Vessel Thickness used (T) = 42.000 mm
 Nozzle Thickness used (t) = 162.100 mm
 T / t = 0.259
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.188$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.097
 Mr/P = 0.204
 M0/P = 0.059
 N0/P = 0.110
 MrD/Mc = 0.307
 NrDL/Mc = 0.079
 M0d/Mc = 0.094
 N0DL/Mc = 0.053
 MrD/Ml = 0.172
 NrDL/Ml = 0.095
 M0D/Ml = 0.054
 N0DL/Ml = 0.115

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-2	2	2	-2
Normal Pressure Stress	1353	1403	1353	1403
Outplane Stress Summary	1354	1402	1360	1399

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	11	-11	11	-11
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-7	7	7	-7
Inplane Pressure Stress	676	676	676	676
Inplane Stress Summary	680	673	697	658

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
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Nozzle Calcs.: N5- Blow down Nozl: 30 9:32am Aug 12,2026

longitudinal plane)			Outside	Inside	Outside	Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			0	0	0	0
Shear Stress Summary			1	1	0	0

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
(Stress Intensities			Top	Top	Bottom	Bottom
			Outside	Inside	Outside	Inside
Two * Max Shear Stress			1354	1402	1360	1399

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Outplane Membrane (P)			0	0	0	0
Outplane Bending (P)			3	-3	3	-3
Outplane Membrane (Mc)			0	0	0	0
Outplane Bending (Mc)			-2	2	2	-2
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			676	676	676	676
Outplane Stress Summary			678	676	683	671

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Inplane Membrane (P)			0	0	0	0
Inplane Bending (P)			11	-11	11	-11
Inplane Membrane (Mc)			0	0	0	0
Inplane Bending (Mc)			-8	8	8	-8
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			1353	1403	1353	1403
Inplane Stress Summary			1356	1401	1374	1384

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			0	0	0	0
Shear Stress Summary			0	0	1	1

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stress Intensities			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Two * Max Shear Stress			1356	1401	1374	1384



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 Nozzle Calcs.: N5- Blow down Nozl: 30 9:32am Aug 12,2026

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	908	959	908	959	908	959	908	959
Circ. Pl (SUS)	3	3	-3	-3	-1	-1	1	1
Circ. Q (SUS)	-31	31	49	-49	-5	5	11	-11
Long. Pm (SUS)	454	454	454	454	454	454	454	454
Long. Pl (SUS)	-3	-3	3	3	1	1	-1	-1
Long. Q (SUS)	-11	11	17	-17	-15	15	33	-33
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	17	17	17	17	17	17	17	17
Pm (SUS)	908.0	959.0	908.0	959.0	908.0	959.0	908.0	959.0
Pm+Pl (SUS)	911.0	962.0	905.0	956.0	907.0	958.0	909.0	960.0
Pm+Pl+Q (Total)	881.0	993.8	954.4	907.4	902.4	963.3	921.0	949.8

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	959.00	1784.41	Passed
Pm+Pl (SUS)	962.03	2676.62	Passed
Pm+Pl+Q (TOTAL)	993.83	5353.24	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1353	1403	1353	1403	1353	1403	1353	1403
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	1	-1	5	-5	3	-3	19	-19
Long. Pm (SUS)	676	676	676	676	676	676	676	676
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	4	-4	18	-18	1	-1	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1353	1403	1353	1403	1353	1403	1353	1403
Pm+Pl (SUS)	1353	1403	1353	1403	1353	1403	1353	1403
Pm+Pl+Q (Total)	1354	1402	1358	1398	1356	1400	1372	1384

Vessel Stress Summation Comparison (kgf/cm²):

Type of	Max. S.I.	S.I. Allowable	Result
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 Nozzle Calcs.: N5- Blow down Nozl: 30 9:32am Aug 12,2026

Stress Int.			
Pm (SUS)	1403.00	1784.41	Passed
Pm+Pl (SUS)	1403.00	2676.62	Passed
Pm+Pl+Q (TOTAL)	1402.00	5353.24	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	229	289	229	289	229	289	229	289
Circ. Pl (SUS)	3	3	-3	-3	1	1	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	114	114	114	114	114	114	114	114
Long. Pl (SUS)	-398	-398	430	430	-232	-232	264	264
Long. Q (SUS)	-412	412	578	-578	-232	232	398	-398
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	46	46	-46	-46	-46	-46	46	46
Shear Q (SUS)	186	186	186	186	186	186	186	186
Pm (SUS)	229.0	289.0	229.0	289.0	229.0	289.0	229.0	289.0
Pm+Pl (SUS)	524.1	583.3	550.5	552.0	360.0	418.2	391.0	397.4
Pm+Pl+Q (Total)	1037	492	1143	425	644	367	861	556

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	289.00	1321.51	Passed
Pm+Pl (SUS)	583.30	1982.27	Passed
Pm+Pl+Q (TOTAL)	1143.37	3964.53	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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 Nozzle Calcs.: N8- MP steam Nozl: 31 9:32am Aug 12,2026

Input, Nozzle Desc: N8- MP steam

From: 60

Pressure for Reinforcement Calculations	P	50.197	kgf/cm ²
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	1.03	kgf/cm ²
Temperature for External Pressure	Tempex	270	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1784.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	5041.6665	mm
Shell Finished (Minimum) Thickness	t	45.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		643.60	cm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1321.51	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1435.72	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	288.9000	mm
Thickness of Pad	te	20.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm
Reinforcing Pad Width		100.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

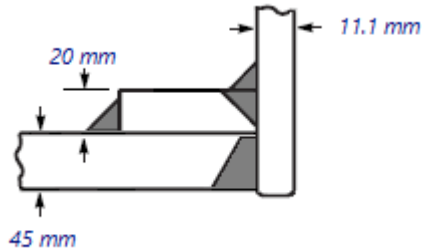
Reinforcement CALCULATION, Description: N8- MP steam



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 Nozzle Calcs.: N8- MP steam Nozl: 31 9:32am Aug 12,2026

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation 2.624 in.
 Actual Thickness Used in Calculation 0.438 in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 72.6 \cdot (\exp(50.2/(1322 \cdot 1)) - 1) + 3 + 0$
 $= 1.41 + 3 + 0 = 4.41 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1322 \cdot 1 \cdot \ln((2 \cdot 8.13 + 72.6) / 72.6) - 0$
 $= 267 - 0 = 267 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 7.800 mm
 Min. required shell thickness, internal p + c : 35.898 mm
 Min. required shell thickness, external p + c : 15.973 mm

Nozzle neck minimum required thickness : 7.800 mm

Minimum nozzle neck thickness 9.728 mm , is greater than required thickness 7.800 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 556.7,	Allowable	: 1321.5 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3147.2 kgf/cm ²	Passed
Occasional	: 123.9,	Allowable	: 1757.6 kgf/cm ²	Passed
Shear	: 325.0,	Allowable	: 925.1 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1322/1784, 1]$
 $= 0.741$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[1436/1784, 1]$
 $= 0.805$

Thickness of Nozzle [tn]:



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Nozzle Calcs.: N8- MP steam Nozl: 31 9:32am Aug 12,2026

= thickness - corrosion allowance
= 11.1 - 3
= 8.125 mm

Shell Diameter to Thickness ratio [D/t]:

= Di/t
= 2306/42
= 54.905 must be less than 400.

Effective Pressure Radius [Reff]:

= Di/2 + corrosion allowance
= 2300/2 + 3
= 1153.000 mm

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

= $\sqrt{(Reff \cdot t) + W}$
= $\sqrt{(1153 \cdot 42) + 100}$
= 320.059 mm

Limit [LR2]:

= $\sqrt{((Reff + t) \cdot (t + te))}$
= $\sqrt{((1153 + 42) \cdot (42 + 20))}$
= 272.195 mm

Limit [LR3]:

= 2 • Rn
= 2 • 36.3
= 72.650 mm

Final value of [LR]:

= min(LR1, LR2, LR3)
= min(320, 272, 72.6)
= 72.650 mm

Thickness Limit Candidate [LH1]:

= min[1.5t, te] + $\sqrt{(Rn \cdot tn)}$
= min[1.5•42, 20] + $\sqrt{(36.3 \cdot 8.13)}$
= 37.180 mm

Thickness Limit Candidate [LH2]:

= Lpr1
= 300
= 300.000 mm

Thickness Limit Candidate [LH3]:

= 8(t + te)
= 8(42 + 20)
= 496.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:

= min[LH1, LH2, LH3] + t for set-in nozzles
= min[37.2, 300, 496] + 42
= 79.180 mm

Effective Vessel Thickness [teff]:

= t + ((A5 • frp) / LR)
= 42 + ((14.5•0.8)/72.6)
= 58.092 mm

Parameter [λ]:

= min(12, (2•Rn + tn) / ($\sqrt{(Di + teff) \cdot teff}$))



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 Nozzle Calcs.: N8- MP steam Nozl: 31 9:32am Aug 12,2026

$$= \min(12, (2 \cdot 36.3 + 8.13) / (\sqrt{(2306 + 58.1) \cdot 58.1}))$$

$$= 0.218$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((N/5)^{0.85}, 1)$$

$$= 42 \cdot 72.6 \cdot \max((0.22/5)^{0.85}, 1)$$

$$= 30.513 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 8.13 \cdot 79.2$$

$$= 6.434 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2 - \text{Area cut by thickness limit}$$

$$= 0.5 \cdot 20^2 - 0.04$$

$$= 1.960 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$

$$= \min(100 \cdot 20, 72.6 \cdot 20)$$

$$= 14.530 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$

$$= 30.5 + 0.74(6.43 + 0) + 1.96 + 0 + 0 + 0.8(14.5)$$

$$= 48.928 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 8.13 / \ln[1 + 8.13/36.3]$$

$$= 40.251 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 58.1 / \ln[1 + 58.1/1153]$$

$$= 1181.807 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1784 \cdot 1$$

$$= 2676.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 50.2 \cdot 40.3 \cdot 79.2$$

$$= 1599.8 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 50.2 \cdot 1182 (72.6 + 8.13)$$

$$= 47918.3 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 50.2 \cdot 1182 \cdot 36.3$$

$$= 21549.1 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (47918 + 21549 + 1600) / 50.2$$

$$= 1415.8 \text{ cm}^2$$



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Maximum Allowable Working Pressure Candidate [Pmax1]:
 = Sallow / (2 • Ap/AT - Rxs/teff)
 = 2677 / (2 • 1416/48.9 - 1182/58.1)
 = 71.3 kgf/cm²

Maximum Allowable Working Pressure Candidate [Pmax2]:
 = S[t/Rxs]
 = 1784 [42/1182]
 = 63.4 kgf/cm²

Maximum Allowable Working Pressure [Pmax]:
 = min(Pmax1, Pmax2)
 = min(71.3, 63.4)
 = 63.416 kgf/cm²

Average Primary Membrane Stress [σAvg]:
 = (fN + fS + fY) / AT
 = (1600 + 47918 + 21549) / 48.9
 = 1452.478 kgf/cm²

General Primary Membrane Stress [σCirc]:
 = P • Rxs / teff
 = 50.2 • 1182/58.1
 = 1021.2 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:
 = max(2 • σAvg - σCirc, σCirc)
 = max(2 • 1452 - 1021, 1021)
 = 1883.8 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	1883.76	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	63.42	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
 = (Rnc + tn) / Rnc
 = (36.3 + 8.13) / 36.3
 = 1.224 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lτ]:
 = π/2 • (Rn + tn)
 = π/2 • (36.3 + 8.13)
 = 69.822 mm

Weld Length of Pad to Shell Weld [LτP]:
 = π/2 • (Rn + tn + W)
 = π/2 • (36.3 + 8.13 + 100)
 = 226.902 mm

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
 = 14.142, 14.142, 0.000, mm

Required Weld Throat Dimensions :
 tc, L41T = 14.142, must be >= min(0.7 * 8.125, 6.000) Passed
 tf1, L42T = 14.142, must be >= min(0.6 * 20.000, 0.6 * 42.000) Passed

Weld Load Value [fwelds]:
 = min(fy • ky, 1.5 • Sn(A2 + A3), π/4•P•Rn²•ky²)
 = min(21549•1.22, 1.5•1322(6.43+0), π/4•50.2•36.3²•1.22²)
 = 778.954 kgf

Discontinuity Force [fws]:



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$$\begin{aligned}
 &= f_{welds} \cdot t \cdot S / (t \cdot S + t_e \cdot S_p) \\
 &= 779 \cdot 42 \cdot 1784 / (42 \cdot 1784 + 20 \cdot 1436) \\
 &= 563.179 \text{ kgf}
 \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned}
 &= f_{welds} \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p) \\
 &= 779 \cdot 20 \cdot 1436 / (42 \cdot 1784 + 20 \cdot 1436) \\
 &= 215.776 \text{ kgf}
 \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned}
 &= f_{ws} / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\
 &= 563 / (69.8 \cdot (0.6 \cdot 38 + 0.49 \cdot 0)) \\
 &= 35.377 \text{ kgf/cm}^2
 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned}
 &= f_{wp} / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T})) \\
 &= 216 / (69.8 \cdot (0.6 \cdot 20 + 0.49 \cdot 14.1)) \\
 &= 16.326 \text{ kgf/cm}^2
 \end{aligned}$$

Shear Stress [τ_3]:

$$\begin{aligned}
 &= f_{wp} / (L_{TP} \cdot (0.49 \cdot L_{42T})) \\
 &= 216 / (227 \cdot (0.49 \cdot 14.1)) \\
 &= 13.723 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned}
 &= \max(\tau_1, \tau_2, \tau_3) \\
 &= \max(35.4, 16.3, 13.7) \\
 &= 35.4 \text{ must be less than or equal to } 1784.4 \text{ kgf/cm}^2
 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= t_n / \ln[1 + t_n/R_n] \\
 &= 8.13 / \ln[1 + 8.13/36.3] \\
 &= 40.251 \text{ mm}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\
 &= 58.1 / \ln[1 + 58.1/1153] \\
 &= 1181.807 \text{ mm}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \\
 &= 1.5 \cdot 1784 \\
 &= 2676.6 \text{ kgf/cm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot L_H \\
 &= 1.03 \cdot 40.3 \cdot 79.2 \\
 &= 32.9 \text{ kgf}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} (L_R + t_n) \\
 &= 1.03 \cdot 1182 (72.6 + 8.13) \\
 &= 986.1 \text{ kgf}
 \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot R_{nc} \\
 &= 1.03 \cdot 1182 \cdot 36.3 \\
 &= 443.5 \text{ kgf}
 \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned}
 &= (f_S + f_Y + f_N) / P \\
 &= (986 + 443 + 32.9) / 1.03 \\
 &= 1415.8 \text{ cm}^2
 \end{aligned}$$



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Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 2677 / (2 \cdot 1416/48.9 - 1182/58.1)$$

$$= 71.3 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 1784 [42/1182]$$

$$= 63.4 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(71.3, 63.4)$$

$$= 63.416 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (32.9 + 986 + 443) / 48.9$$

$$= 29.890 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / teff$$

$$= 1.03 \cdot 1182 / 58.1$$

$$= 21.0 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 29.9 - 21, 21)$$

$$= 38.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	38.77	kgf/cm ²
Maximum Allowable External Pressure	Pmax	63.42	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.167

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$

$$= (1.13 \cdot 1) / (9.73 - 3 - 0)$$

$$= 0.167$$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.167

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$



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 Nozzle Calcs.: N8- MP steam Nozl: 31 9:32am Aug 12,2026

$$= (1.13 \cdot 1) / (9.73 - 3 - 0)$$

$$= 0.167$$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.167

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (1.13 \cdot 1) / (9.73 - 3 - 0)$
 $= 0.167$

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -9 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : -9 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (32.9 \cdot 1) / (20 - 3 - 0)$
 $= 1.000$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.167

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (1.13 \cdot 1) / (9.73 - 3 - 0)$
 $= 0.167$

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -9 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -9 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 0.8593 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 345.8593 mm

Input Echo, WRC297 Item 1, Description: N8- MP steam :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2300.000	mm



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Nozzle Calcs.: N8- MP steam Nozl: 31 9:32am Aug 12,2026

Shell Thickness		45.0000	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1784.41	kgf/cm ²
Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		3.0000	mm
Nozzle Diameter		66.650	mm
Nozzle Thickness		11.1252	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	1641.66	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1321.51	kgf/cm ²
Diameter of Reinforcing Pad		288.900	mm
Thickness of Reinforcing Pad		20.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	50.20	kgf/cm ²
Radial Load	P	-254.94	kgf
Circumferential Shear	Vc	377.31	kgf
Longitudinal Shear	Vl	377.31	kgf
Circumferential Moment	Mc	81.55	kgf-m.
Longitudinal Moment	Ml	112.13	kgf-m.
Torsional Moment	Mt	142.71	kgf-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

Warning - The ratio, Dn/Tv, (1.976) is < 5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (7.991) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{1184.0(45.0 - 3.0)}$$

$$= 824.793 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2368.000	mm
Nozzle Outside Diameter	(d) =	88.900	mm
Vessel Thickness used	(T) =	62.000	mm
Nozzle Thickness used	(t) =	8.125	mm
T / t	=	7.631	
d / t	=	10.941	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.232$$

Nr/P	=	0.198
Mr/P	=	0.123
M0/P	=	0.313
N0/P	=	0.260
MrD/Mc	=	0.110
NrDL/Mc	=	0.151
M0d/Mc	=	0.305
N0DL/Mc	=	-0.131



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Nozzle Calcs.: N8- MP steam

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MrD/M1 = 0.110
NrDL/M1 = 0.155
MØD/M1 = 0.300
NØDL/M1 = -0.122

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	3	3	-3	-3
Outplane Bending (ML)	-58	58	58	-58
Normal Pressure Stress	908	959	908	959
Outplane Stress Summary	868	1011	978	885

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-21	21	21	-21
Inplane Pressure Stress	454	454	454	454
Inplane Stress Summary	433	467	487	434

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	18	18	18	18
Shear Stress Summary	22	22	14	14

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	869	1012	978	886

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	3	3	-3	-3
Outplane Bending (Mc)	-43	43	43	-43
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	454	454	454	454



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Nozzle Calcs.: N8- MP steam

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Outplane Stress Summary	428	490	509	396
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-15	15	15	-15
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	908	959	908	959

Inplane Stress Summary	895	967	934	943
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4
Torsional Shear (Mt)	18	18	18	18

Shear Stress Summary	14	14	22	22
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	896	968	935	944
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	3	3	-3	-3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	242	302	242	302

Hoop Stress Summary	247	308	239	300
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	12	12	12	12
Axial Bending (P)	53	-53	53	-53
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-293	-293	293	293
Axial Bending (ML)	-370	370	370	-370
Axial Pressure Stress	121	121	121	121

Axial Stress Summary	-476	156	851	2
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Nozzle Calcs.: N8- MP steam

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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	33	33	-33	-33
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	141	141	141	141
Shear Stress Summary	174	174	108	108

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	803	422	869	368

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	3	3	-3	-3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	242	302	242	302
Hoop Stress Summary	246	307	240	300

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	12	12	12	12
Axial Bending (P)	53	-53	53	-53
Axial Membrane (Mc)	-213	-213	213	213
Axial Bending (Mc)	-287	287	287	-287
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	121	121	121	121
Axial Stress Summary	-313	153	687	5

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-33	-33	33	33
Shear due to Torsion	141	141	141	141
Shear Stress Summary	108	108	174	174

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	600	362	748	457

Stress Computations at the Edge of the Reinforcing Pad:



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Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 88.9 + 2 \cdot 1.65 \cdot \sqrt{1174.0(45.0 - 3.0)}$
 $= 821.678 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2348.000 mm
 Nozzle Outside Diameter (d) = 288.900 mm
 Vessel Thickness used (T) = 42.000 mm
 Nozzle Thickness used (t) = 108.125 mm
 T / t = 0.388
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.920$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.109
 Mr/P = 0.237
 M0/P = 0.069
 N0/P = 0.132
 MrD/Mc = 0.310
 NrDL/Mc = 0.080
 M0d/Mc = 0.096
 N0DL/Mc = 0.037
 MrD/Ml = 0.206
 NrDL/Ml = 0.090
 M0D/Ml = 0.066
 N0DL/Ml = 0.109

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	1353	1403	1353	1403
Outplane Stress Summary	1350	1405	1372	1393

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-27	27	27	-27
Inplane Pressure Stress	676	676	676	676
Inplane Stress Summary	669	682	727	632

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
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longitudinal plane)			Outside	Inside	Outside	Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			2	2	2	2
Shear Stress Summary			4	4	0	0

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
(Stress Intensities			Top	Top	Bottom	Bottom
			Outside	Inside	Outside	Inside
Two * Max Shear Stress			1350	1405	1372	1393

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stresses Normal to			Left	Left	Right	Right
circumferential plane)			Outside	Inside	Outside	Inside
Outplane Membrane (P)			1	1	1	1
Outplane Bending (P)			5	-5	5	-5
Outplane Membrane (Mc)			0	0	0	0
Outplane Bending (Mc)			-9	9	9	-9
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			676	676	676	676
Outplane Stress Summary			674	681	694	663

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stresses parallel to			Left	Left	Right	Right
circumferential plane)			Outside	Inside	Outside	Inside
Inplane Membrane (P)			1	1	1	1
Inplane Bending (P)			20	-20	20	-20
Inplane Membrane (Mc)			-1	-1	1	1
Inplane Bending (Mc)			-29	29	29	-29
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			1353	1403	1353	1403
Inplane Stress Summary			1344	1412	1406	1356

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Shear stress normal to			Left	Left	Right	Right
circumferential plane)			Outside	Inside	Outside	Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			2	2	2	2
Shear Stress Summary			0	0	4	4

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stress Intensities			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Two * Max Shear Stress			1344	1412	1406	1356



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Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	908	959	908	959	908	959	908	959
Circ. Pl (SUS)	4	4	-2	-2	-2	-2	4	4
Circ. Q (SUS)	-46	46	70	-70	-11	11	19	-19
Long. Pm (SUS)	454	454	454	454	454	454	454	454
Long. Pl (SUS)	-4	-4	6	6	4	4	-2	-2
Long. Q (SUS)	-17	17	25	-25	-31	31	55	-55
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	18	18	18	18	18	18	18	18
Pm (SUS)	908.0	959.0	908.0	959.0	908.0	959.0	908.0	959.0
Pm+Pl (SUS)	912.0	963.0	906.0	957.0	906.0	957.0	912.0	963.0
Pm+Pl+Q (Total)	867	1009	976	887	895	968	932	944

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	959.00	1784.41	Passed
Pm+Pl (SUS)	963.03	2676.62	Passed
Pm+Pl+Q (TOTAL)	1009.89	5353.24	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1353	1403	1353	1403	1353	1403	1353	1403
Circ. Pl (SUS)	-1	-1	3	3	0	0	2	2
Circ. Q (SUS)	-3	3	13	-13	-9	9	49	-49
Long. Pm (SUS)	676	676	676	676	676	676	676	676
Long. Pl (SUS)	0	0	2	2	1	1	1	1
Long. Q (SUS)	-7	7	47	-47	-4	4	14	-14
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	1353	1403	1353	1403	1353	1403	1353	1403
Pm+Pl (SUS)	1352	1402	1356	1406	1353	1403	1355	1405
Pm+Pl+Q (Total)	1349	1405	1369	1393	1344	1412	1404	1356

Vessel Stress Summation Comparison (kgf/cm²):

Type of	Max. S.I.	S.I. Allowable	Result
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Stress Int.			
Pm (SUS)	1403.00	1784.41	Passed
Pm+Pl (SUS)	1406.00	2676.62	Passed
Pm+Pl+Q (TOTAL)	1412.00	5353.24	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	242	302	242	302	242	302	242	302
Circ. Pl (SUS)	4	4	-2	-2	4	4	-2	-2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	121	121	121	121	121	121	121	121
Long. Pl (SUS)	-281	-281	305	305	-201	-201	225	225
Long. Q (SUS)	-317	317	423	-423	-234	234	340	-340
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	33	33	-33	-33	-33	-33	33	33
Shear Q (SUS)	141	141	141	141	141	141	141	141
Pm (SUS)	242.0	302.0	242.0	302.0	242.0	302.0	242.0	302.0
Pm+Pl (SUS)	411.3	470.7	431.7	434.1	332.6	391.6	355.4	363.2
Pm+Pl+Q (Total)	802.4	420.8	867.6	367.2	600.2	362.1	745.9	455.6

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	302.00	1321.51	Passed
Pm+Pl (SUS)	470.65	1982.27	Passed
Pm+Pl+Q (TOTAL)	867.59	3964.53	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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 Nozzle Calcs.: K1- PSV Nozl: 32 9:32am Aug 12,2026

Input, Nozzle Desc: K1- PSV

From: 60

Pressure for Reinforcement Calculations	P	49.967	kgf/cm ²
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	1.03	kgf/cm ²
Temperature for External Pressure	Tempex	270	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1784.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	5041.6665	mm
Shell Finished (Minimum) Thickness	t	45.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		793.60	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1321.51	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1435.72	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	160.3250	mm
Thickness of Pad	te	15.0000	mm
Weld leg size between Pad and Shell	Wp	15.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	15.0000	mm
Reinforcing Pad Width		50.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

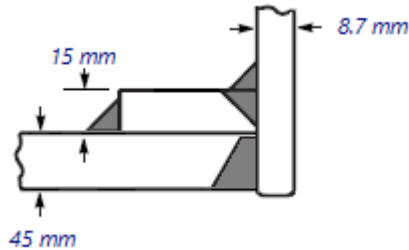
Reinforcement CALCULATION, Description: K1- PSV



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ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation 1.687 in.
 Actual Thickness Used in Calculation 0.344 in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 48.8 \cdot (\exp(50/(1322 \cdot 1)) - 1) + 3 + 0$
 $= 0.94 + 3 + 0 = 3.94 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1322 \cdot 1 \cdot \ln((2 \cdot 5.74 + 48.8) / 48.8) - 0$
 $= 279 - 0 = 279 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 6.420 mm
 Min. required shell thickness, internal p + c : 35.745 mm
 Min. required shell thickness, external p + c : 15.973 mm

Nozzle neck minimum required thickness : 6.420 mm

Minimum nozzle neck thickness 7.645 mm , is greater than required thickness 6.420 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 713.1,	Allowable	: 1321.5 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2990.8 kgf/cm ²	Passed
Occasional	: 120.0,	Allowable	: 1757.6 kgf/cm ²	Passed
Shear	: 452.4,	Allowable	: 925.1 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1322/1784, 1]$
 $= 0.741$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[1436/1784, 1]$
 $= 0.805$

Thickness of Nozzle [tn]:



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Nozzle Calcs.: K1- PSV

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= thickness - corrosion allowance
 = 8.74 - 3
 = 5.738 mm

Shell Diameter to Thickness ratio [D/t]:
 = Di/t
 = 2306/42
 = 54.905 must be less than 400.

Effective Pressure Radius [Reff]:
 = Di/2 + corrosion allowance
 = 2300/2 + 3
 = 1153.000 mm

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:
 $= \sqrt{(Reff \cdot t)} + W$
 $= \sqrt{(1153 \cdot 42)} + 50$
 = 270.059 mm

Limit [LR2]:
 $= \sqrt{((Reff + t) \cdot (t + te))}$
 $= \sqrt{((1153 + 42) \cdot (42 + 15))}$
 = 260.988 mm

Limit [LR3]:
 = 2 • Rn
 = 2 • 24.4
 = 48.850 mm

Final value of [LR]:
 = min(LR1, LR2, LR3)
 = min(270, 261, 48.8)
 = 48.850 mm

Thickness Limit Candidate [LH1]:
 $= \min[1.5t, te] + \sqrt{(Rn \cdot tn)}$
 $= \min[1.5 \cdot 42, 15] + \sqrt{(24.4 \cdot 5.74)}$
 = 26.838 mm

Thickness Limit Candidate [LH2]:
 = Lpr1
 = 300
 = 300.000 mm

Thickness Limit Candidate [LH3]:
 = 8(t + te)
 = 8(42 + 15)
 = 456.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:
 = min[LH1, LH2, LH3] + t for set-in nozzles
 = min[26.8, 300, 456] + 42
 = 68.838 mm

Effective Vessel Thickness [teff]:
 = t + ((A5 • frp) / LR)
 = 42 + ((7.33 • 0.8) / 48.8)
 = 54.069 mm

Parameter [λ]:
 $= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$



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$$= \min(12, (2 \cdot 24.4 + 5.74) / (\sqrt{(2306 + 54.1) \cdot 54.1}))$$

$$= 0.153$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((N/5)^{0.85}, 1)$$

$$= 42 \cdot 48.8 \cdot \max((0.15/5)^{0.85}, 1)$$

$$= 20.517 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 5.74 \cdot 68.8$$

$$= 3.950 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2 - \text{Area cut by thickness limit}$$

$$= 0.5 \cdot 20^2 - 0.33$$

$$= 1.667 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$

$$= \min(50 \cdot 15, 48.8 \cdot 15)$$

$$= 7.327 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$

$$= 20.5 + 0.74(3.95 + 0) + 1.67 + 0 + 0 + 0.8(7.33)$$

$$= 31.005 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 5.74 / \ln[1 + 5.74/24.4]$$

$$= 27.193 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 54.1 / \ln[1 + 54.1/1153]$$

$$= 1179.828 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1784 \cdot 1$$

$$= 2676.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 50 \cdot 27.2 \cdot 68.8$$

$$= 935.3 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 50 \cdot 1180 (48.8 + 5.74)$$

$$= 32180.7 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 50 \cdot 1180 \cdot 24.4$$

$$= 14399.1 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (32181 + 14399 + 935) / 50$$

$$= 950.9 \text{ cm}^2$$



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Maximum Allowable Working Pressure Candidate [Pmax1]:
 = Sallow / (2 • Ap/AT - Rxs/teff)
 = 2677 / (2 • 951/31 - 1180/54.1)
 = 67.7 kgf/cm²

Maximum Allowable Working Pressure Candidate [Pmax2]:
 = S[t/Rxs]
 = 1784 [42/1180]
 = 63.5 kgf/cm²

Maximum Allowable Working Pressure [Pmax]:
 = min(Pmax1, Pmax2)
 = min(67.7, 63.5)
 = 63.522 kgf/cm²

Average Primary Membrane Stress [σAvg]:
 = (fN + fS + fY) / AT
 = (935 + 32181 + 14399) / 31
 = 1532.528 kgf/cm²

General Primary Membrane Stress [σCirc]:
 = P • Rxs / teff
 = 50 • 1180/54.1
 = 1090.3 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:
 = max(2 • σAvg - σCirc, σCirc)
 = max(2 • 1533 - 1090, 1090)
 = 1974.7 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	1974.73	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	63.52	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
 = (Rnc + tn) / Rnc
 = (24.4 + 5.74) / 24.4
 = 1.235 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lτ]:
 = π/2 • (Rn + tn)
 = π/2 • (24.4 + 5.74)
 = 47.379 mm

Weld Length of Pad to Shell Weld [LτP]:
 = π/2 • (Rn + tn + W)
 = π/2 • (24.4 + 5.74 + 50)
 = 125.919 mm

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
 = 14.142, 10.606, 0.000, mm

Required Weld Throat Dimensions :
 tc, L41T = 14.142, must be >= min(0.7 * 5.738, 6.000) Passed
 tf1, L42T = 10.606, must be >= min(0.6 * 15.000, 0.6 * 42.000) Passed

Weld Load Value [fwelds]:
 = min(fy • ky, 1.5 • Sn(A2 + A3), π/4•P•Rn²•ky²)
 = min(14399•1.23, 1.5•1322(3.95+0), π/4•50•24.4²•1.23²)
 = 357.034 kgf

Discontinuity Force [fws]:



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$$\begin{aligned}
 &= f_{welds} \cdot t \cdot S / (t \cdot S + t_e \cdot S_p) \\
 &= 357 \cdot 42 \cdot 1784 / (42 \cdot 1784 + 15 \cdot 1436) \\
 &= 277.339 \text{ kgf}
 \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned}
 &= f_{welds} \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p) \\
 &= 357 \cdot 15 \cdot 1436 / (42 \cdot 1784 + 15 \cdot 1436) \\
 &= 79.695 \text{ kgf}
 \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned}
 &= f_{ws} / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\
 &= 277 / (47.4 \cdot (0.6 \cdot 38 + 0.49 \cdot 0)) \\
 &= 25.674 \text{ kgf/cm}^2
 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned}
 &= f_{wp} / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T})) \\
 &= 79.7 / (47.4 \cdot (0.6 \cdot 15 + 0.49 \cdot 14.1)) \\
 &= 10.559 \text{ kgf/cm}^2
 \end{aligned}$$

Shear Stress [τ_3]:

$$\begin{aligned}
 &= f_{wp} / (L_{TP} \cdot (0.49 \cdot L_{42T})) \\
 &= 79.7 / (126 \cdot (0.49 \cdot 10.6)) \\
 &= 12.178 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned}
 &= \max(\tau_1, \tau_2, \tau_3) \\
 &= \max(25.7, 10.6, 12.2) \\
 &= 25.7 \text{ must be less than or equal to } 1784.4 \text{ kgf/cm}^2
 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= t_n / \ln[1 + t_n/R_n] \\
 &= 5.74 / \ln[1 + 5.74/24.4] \\
 &= 27.193 \text{ mm}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\
 &= 54.1 / \ln[1 + 54.1/1153] \\
 &= 1179.828 \text{ mm}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \\
 &= 1.5 \cdot 1784 \\
 &= 2676.6 \text{ kgf/cm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot L_H \\
 &= 1.03 \cdot 27.2 \cdot 68.8 \\
 &= 19.3 \text{ kgf}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} (L_R + t_n) \\
 &= 1.03 \cdot 1180 (48.8 + 5.74) \\
 &= 665.3 \text{ kgf}
 \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot R_{nc} \\
 &= 1.03 \cdot 1180 \cdot 24.4 \\
 &= 297.7 \text{ kgf}
 \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned}
 &= (f_S + f_Y + f_N) / P \\
 &= (665 + 298 + 19.3) / 1.03 \\
 &= 950.9 \text{ cm}^2
 \end{aligned}$$



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Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 2677 / (2 \cdot 951/31 - 1180/54.1)$$

$$= 67.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 1784 [42/1180]$$

$$= 63.5 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(67.7, 63.5)$$

$$= 63.522 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (19.3 + 665 + 298) / 31$$

$$= 31.683 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / teff$$

$$= 1.03 \cdot 1180/54.1$$

$$= 22.5 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 31.7 - 22.5, 22.5)$$

$$= 40.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	40.82	kgf/cm ²
Maximum Allowable External Pressure	Pmax	63.52	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.162

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$

$$= (0.75 \cdot 1) / (7.65 - 3 - 0)$$

$$= 0.162$$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.162

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$



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$$= (0.75 \cdot 1) / (7.65 - 3 - 0)$$

$$= 0.162$$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.162

The computed thickness ratio [sr]:
 $= (tr \cdot E) / (tnom - ci - cext)$
 $= (0.75 \cdot 1) / (7.65 - 3 - 0)$
 $= 0.162$

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -14 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : -14 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:
 $= (tr \cdot E) / (tnom - ci - cext)$
 $= (32.7 \cdot 1) / (15 - 3 - 0)$
 $= 1.000$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.162

The computed thickness ratio [sr]:
 $= (tr \cdot E) / (tnom - ci - cext)$
 $= (0.75 \cdot 1) / (7.65 - 3 - 0)$
 $= 0.162$

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -14 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -14 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 0.3956 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 345.3956 mm

Input Echo, WRC297 Item 1, Description: K1- PSV :

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	3.0000 mm
Shell External Corrosion Allowance	0.0000 mm
Shell Diameter	2300.000 mm



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Nozzle Calcs.: K1- PSV

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Shell Thickness		45.0000	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1784.41	kgf/cm ²
Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		3.0000	mm
Nozzle Diameter		42.850	mm
Nozzle Thickness		8.7376	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	1641.66	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1321.51	kgf/cm ²
Diameter of Reinforcing Pad		160.325	mm
Thickness of Reinforcing Pad		15.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	49.97	kgf/cm ²
Radial Load	P	-163.16	kgf
Circumferential Shear	Vc	254.94	kgf
Longitudinal Shear	Vl	254.94	kgf
Circumferential Moment	Mc	30.58	kgf-m.
Longitudinal Moment	Ml	50.97	kgf-m.
Torsional Moment	Mt	61.16	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv, (1.341) is < 5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (6.904) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{1181.5(45.0 - 3.0)}$$

$$= 795.440 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2363.000	mm
Nozzle Outside Diameter	(d) =	60.325	mm
Vessel Thickness used	(T) =	57.000	mm
Nozzle Thickness used	(t) =	5.738	mm
T / t	=	9.934	
d / t	=	10.514	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.164$$

Nr/P	=	0.228
Mr/P	=	0.141
M0/P	=	0.365
N0/P	=	0.234
MrD/Mc	=	0.110
NrDL/Mc	=	0.153
M0d/Mc	=	0.306
N0DL/Mc	=	-0.135



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Nozzle Calcs.: K1- PSV

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MrD/M1 = 0.111
NrDL/M1 = 0.157
MØD/M1 = 0.305
NØDL/M1 = -0.154

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	3	3	-3	-3
Outplane Bending (ML)	-47	47	47	-47
Normal Pressure Stress	986	1036	986	1036
Outplane Stress Summary	954	1077	1042	975

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-17	17	17	-17
Inplane Pressure Stress	493	493	493	493
Inplane Stress Summary	477	503	519	476

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	18	18	18	18
Shear Stress Summary	23	23	13	13

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	955	1078	1042	975

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-28	28	28	-28
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	493	493	493	493



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Nozzle Calcs.: K1- PSV

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Outplane Stress Summary	478	514	531	452
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-10	10	10	-10
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	986	1036	986	1036

Inplane Stress Summary	979	1041	1004	1025
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4
Torsional Shear (Mt)	18	18	18	18

Shear Stress Summary	13	13	23	23
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	979	1041	1005	1026
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	3	3	-3	-3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	228	288	228	288

Hoop Stress Summary	233	293	225	285
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	16	16	16	16
Axial Bending (P)	81	-81	81	-81
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-414	-414	414	414
Axial Bending (ML)	-494	494	494	-494
Axial Pressure Stress	114	114	114	114

Axial Stress Summary	-697	128	1121	-30
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Nozzle Calcs.: K1- PSV

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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	46	46	-46	-46
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	186	186	186	186
Shear Stress Summary	233	233	139	139

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	1040	494	1143	421

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	228	288	228	288
Hoop Stress Summary	231	291	227	287

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	16	16	16	16
Axial Bending (P)	81	-81	81	-81
Axial Membrane (Mc)	-248	-248	248	248
Axial Bending (Mc)	-315	315	315	-315
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	114	114	114	114
Axial Stress Summary	-352	116	777	-17

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-46	-46	46	46
Shear due to Torsion	186	186	186	186
Shear Stress Summary	139	139	233	233

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	647	368	862	557

Stress Computations at the Edge of the Reinforcing Pad:



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Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{1174.0(45.0 - 3.0)}$$

$$= 793.103 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2348.000 mm
 Nozzle Outside Diameter (d) = 160.325 mm
 Vessel Thickness used (T) = 42.000 mm
 Nozzle Thickness used (t) = 55.738 mm
 T / t = 0.754
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.511$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.156
 Mr/P = 0.299
 M0/P = 0.097
 N0/P = 0.172
 MrD/Mc = 0.302
 NrDL/Mc = 0.112
 M0d/Mc = 0.109
 N0DL/Mc = -0.065
 MrD/Ml = 0.264
 NrDL/Ml = 0.113
 M0D/Ml = 0.094
 N0DL/Ml = 0.019

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	1347	1397	1347	1397
Outplane Stress Summary	1343	1403	1364	1383

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	16	-16	16	-16
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-28	28	28	-28
Inplane Pressure Stress	673	673	673	673
Inplane Stress Summary	661	684	721	631

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
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Nozzle Calcs.: K1- PSV

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longitudinal plane)			Outside	Inside	Outside	Inside
Outplane Shear (Vc)			2	2	-2	-2
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			3	3	3	3
Shear Stress Summary			5	5	1	1

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
(Stress Intensities			Top	Top	Bottom	Bottom
			Outside	Inside	Outside	Inside
Two * Max Shear Stress			1343	1403	1364	1383

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stresses Normal to			Left	Left	Right	Right
circumferential plane)			Outside	Inside	Outside	Inside
Outplane Membrane (P)			1	1	1	1
Outplane Bending (P)			5	-5	5	-5
Outplane Membrane (Mc)			0	0	0	0
Outplane Bending (Mc)			-7	7	7	-7
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			673	673	673	673
Outplane Stress Summary			674	677	686	662

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stresses parallel to			Left	Left	Right	Right
circumferential plane)			Outside	Inside	Outside	Inside
Inplane Membrane (P)			1	1	1	1
Inplane Bending (P)			16	-16	16	-16
Inplane Membrane (Mc)			-1	-1	1	1
Inplane Bending (Mc)			-19	19	19	-19
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			1347	1397	1347	1397
Inplane Stress Summary			1344	1400	1385	1363

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Shear stress normal to			Left	Left	Right	Right
circumferential plane)			Outside	Inside	Outside	Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-2	-2	2	2
Torsional Shear (Mt)			3	3	3	3
Shear Stress Summary			1	1	5	5

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stress Intensities			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Two * Max Shear Stress			1344	1400	1385	1363



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Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	986	1036	986	1036	986	1036	986	1036
Circ. Pl (SUS)	4	4	-2	-2	-1	-1	3	3
Circ. Q (SUS)	-37	37	57	-57	-6	6	14	-14
Long. Pm (SUS)	493	493	493	493	493	493	493	493
Long. Pl (SUS)	-3	-3	5	5	3	3	-1	-1
Long. Q (SUS)	-13	13	21	-21	-18	18	38	-38
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	18	18	18	18	18	18	18	18
Pm (SUS)	986	1036	986	1036	986	1036	986	1036
Pm+Pl (SUS)	990	1040	984	1034	985	1035	989	1039
Pm+Pl+Q (Total)	954	1077	1041	977	979	1041	1004	1025

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1036.00	1784.41	Passed
Pm+Pl (SUS)	1040.03	2676.62	Passed
Pm+Pl+Q (TOTAL)	1077.84	5353.24	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1347	1397	1347	1397	1347	1397	1347	1397
Circ. Pl (SUS)	1	1	1	1	0	0	2	2
Circ. Q (SUS)	-5	5	15	-15	-3	3	35	-35
Long. Pm (SUS)	673	673	673	673	673	673	673	673
Long. Pl (SUS)	0	0	2	2	1	1	1	1
Long. Q (SUS)	-12	12	44	-44	-2	2	12	-12
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	1347	1397	1347	1397	1347	1397	1347	1397
Pm+Pl (SUS)	1348	1398	1348	1398	1347	1397	1349	1399
Pm+Pl+Q (Total)	1343	1403	1363	1383	1344	1400	1384	1364

Vessel Stress Summation Comparison (kgf/cm²):

Type of	Max. S.I.	S.I. Allowable	Result
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Stress Int.			
Pm (SUS)	1397.00	1784.41	Passed
Pm+Pl (SUS)	1399.01	2676.62	Passed
Pm+Pl+Q (TOTAL)	1403.03	5353.24	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	228	288	228	288	228	288	228	288
Circ. Pl (SUS)	4	4	-2	-2	3	3	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	114	114	114	114	114	114	114	114
Long. Pl (SUS)	-398	-398	430	430	-232	-232	264	264
Long. Q (SUS)	-413	413	575	-575	-234	234	396	-396
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	46	46	-46	-46	-46	-46	46	46
Shear Q (SUS)	186	186	186	186	186	186	186	186
Pm (SUS)	228.0	288.0	228.0	288.0	228.0	288.0	228.0	288.0
Pm+Pl (SUS)	524.1	583.3	550.5	552.0	360.9	419.2	390.9	397.2
Pm+Pl+Q (Total)	1038	491	1140	422	646	368	859	555

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	288.00	1321.51	Passed
Pm+Pl (SUS)	583.30	1982.27	Passed
Pm+Pl+Q (TOTAL)	1140.43	3964.53	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K2- PSV

From: 60

Pressure for Reinforcement Calculations	P	49.967	kgf/cm ²
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	1.03	kgf/cm ²
Temperature for External Pressure	Tempex	270	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1784.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	5041.6665	mm
Shell Finished (Minimum) Thickness	t	45.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		843.60	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1321.51	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1435.72	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	368.2750	mm
Thickness of Pad	te	15.0000	mm
Weld leg size between Pad and Shell	Wp	15.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	15.0000	mm
Reinforcing Pad Width		100.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

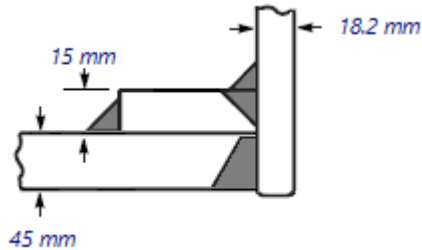
Reinforcement CALCULATION, Description: K2- PSV



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ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation 5.189 in.
 Actual Thickness Used in Calculation 0.718 in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 138 \cdot (\exp(50/(1322 \cdot 1)) - 1) + 3 + 0$
 $= 2.66 + 3 + 0 = 5.66 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1322 \cdot 1 \cdot \ln((2 \cdot 15.2 + 138) / 138) - 0$
 $= 264 - 0 = 264 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.220 mm
 Min. required shell thickness, internal p + c : 35.745 mm
 Min. required shell thickness, external p + c : 15.973 mm

Nozzle neck minimum required thickness : 9.220 mm

Minimum nozzle neck thickness 15.951 mm , is greater than required thickness 9.220 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 378.2,	Allowable	: 1321.5 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3325.7 kgf/cm ²	Passed
Occasional	: 121.4,	Allowable	: 1757.6 kgf/cm ²	Passed
Shear	: 186.3,	Allowable	: 925.1 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1322/1784, 1]$
 $= 0.741$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[1436/1784, 1]$
 $= 0.805$

Thickness of Nozzle [tn]:



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Nozzle Calcs.: K2- PSV

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= thickness - corrosion allowance
 = 18.2 - 3
 = 15.237 mm

Shell Diameter to Thickness ratio [D/t]:
 = Di/t
 = 2306/42
 = 54.905 must be less than 400.

Effective Pressure Radius [Reff]:
 = Di/2 + corrosion allowance
 = 2300/2 + 3
 = 1153.000 mm

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(Reff \cdot t) + W}$$

$$= \sqrt{(1153 \cdot 42) + 100}$$
 = 320.059 mm

Limit [LR2]:

$$= \sqrt{((Reff + t) \cdot (t + te))}$$

$$= \sqrt{((1153 + 42) \cdot (42 + 15))}$$
 = 260.988 mm

Limit [LR3]:
 = 2 • Rn
 = 2 • 68.9
 = 137.801 mm

Final value of [LR]:
 = min(LR1, LR2, LR3)
 = min(320, 261, 138)
 = 137.801 mm

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, te] + \sqrt{(Rn \cdot tn)}$$

$$= \min[1.5 \cdot 42, 15] + \sqrt{(68.9 \cdot 15.2)}$$
 = 47.401 mm

Thickness Limit Candidate [LH2]:
 = Lpr1
 = 300
 = 300.000 mm

Thickness Limit Candidate [LH3]:
 = 8(t + te)
 = 8(42 + 15)
 = 456.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:
 = min[LH1, LH2, LH3] + t for set-in nozzles
 = min[47.4, 300, 456] + 42
 = 89.401 mm

Effective Vessel Thickness [teff]:
 = t + ((A5 • frp) / LR)
 = 42 + ((15•0.8)/138)
 = 50.758 mm

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$$



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$$= \min(12, (2 \cdot 68.9 + 15.2) / (\sqrt{(2306 + 50.8) \cdot 50.8}))$$

$$= 0.442$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((N/5)0.85, 1)$$

$$= 42 \cdot 138 \cdot \max((0.44/5)0.85, 1)$$

$$= 57.876 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 15.2 \cdot 89.4$$

$$= 13.622 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg}42^2$$

$$= 0.5 \cdot 15^2$$

$$= 1.125 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}41^2$$

$$= 0.5 \cdot 20^2$$

$$= 2.000 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$

$$= \min(100 \cdot 15, 138 \cdot 15)$$

$$= 15.000 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$

$$= 57.9 + 0.74(13.6 + 0) + 2 + 1.12 + 0 + 0.8(15)$$

$$= 83.159 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 15.2 / \ln[1 + 15.2/68.9]$$

$$= 76.265 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 50.8 / \ln[1 + 50.8/1153]$$

$$= 1178.198 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1784 \cdot 1$$

$$= 2676.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 50 \cdot 76.3 \cdot 89.4$$

$$= 3406.9 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 50 \cdot 1178 (138 + 15.2)$$

$$= 90095.3 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 50 \cdot 1178 \cdot 68.9$$

$$= 40562.5 \text{ kgf}$$



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Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (90095 + 40562 + 3407) / 50$$

$$= 2683.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 2677 / (2 \cdot 2683 / 83.2 - 1178 / 50.8)$$

$$= 64.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 1784 [42 / 1178]$$

$$= 63.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(64.8, 63.6)$$

$$= 63.610 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (3407 + 90095 + 40562) / 83.2$$

$$= 1612.159 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 50 \cdot 1178 / 50.8$$

$$= 1159.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 1612 - 1160, 1160)$$

$$= 2064.5 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	2064.48	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	63.61	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (68.9 + 15.2) / 68.9$$

$$= 1.221 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (68.9 + 15.2)$$

$$= 132.163 \text{ mm}$$

Weld Length of Pad to Shell Weld [LtP]:

$$= \pi / 2 \cdot (Rn + tn + W)$$

$$= \pi / 2 \cdot (68.9 + 15.2 + 100)$$

$$= 289.243 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 14.142, 10.606, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 14.142, \text{ must be } \geq \min(0.7 \cdot 15.237, 6.000) \text{ Passed}$$

$$tf1, L42T = 10.606, \text{ must be } \geq \min(0.6 \cdot 15.000, 0.6 \cdot 42.000) \text{ Passed}$$

Weld Load Value [fwelds]:



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Nozzle Calcs.: K2- PSV Nozl: 33 9:32am Aug 12, 2026

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(40562 \cdot 1.22, 1.5 \cdot 1322(13.6+0), \pi/4 \cdot 50 \cdot 68.9^2 \cdot 1.22^2)$$

$$= 2778.140 \text{ kgf}$$

Discontinuity Force [fws]:

$$= f_{welds} \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 2778 \cdot 42 \cdot 1784 / (42 \cdot 1784 + 15 \cdot 1436)$$

$$= 2158.023 \text{ kgf}$$

Discontinuity Force [fwp]:

$$= f_{welds} \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 2778 \cdot 15 \cdot 1436 / (42 \cdot 1784 + 15 \cdot 1436)$$

$$= 620.116 \text{ kgf}$$

Shear Stress [τ_1]:

$$= f_{ws} / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 2158 / (132 \cdot (0.6 \cdot 38 + 0.49 \cdot 0))$$

$$= 71.616 \text{ kgf/cm}^2$$

Shear Stress [τ_2]:

$$= f_{wp} / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T}))$$

$$= 620 / (132 \cdot (0.6 \cdot 15 + 0.49 \cdot 14.1))$$

$$= 29.455 \text{ kgf/cm}^2$$

Shear Stress [τ_3]:

$$= f_{wp} / (L_{TP} \cdot (0.49 \cdot L_{42T}))$$

$$= 620 / (289 \cdot (0.49 \cdot 10.6))$$

$$= 41.252 \text{ kgf/cm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(71.6, 29.5, 41.3)$$

$$= 71.6 \text{ must be less than or equal to } 1784.4 \text{ kgf/cm}^2$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 15.2 / \ln[1 + 15.2/68.9]$$

$$= 76.265 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 50.8 / \ln[1 + 50.8/1153]$$

$$= 1178.198 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1784$$

$$= 2676.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.03 \cdot 76.3 \cdot 89.4$$

$$= 70.4 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.03 \cdot 1178 (138 + 15.2)$$

$$= 1862.6 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.03 \cdot 1178 \cdot 68.9$$

$$= 838.6 \text{ kgf}$$



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Area Resisting External Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (1863 + 839 + 70.4) / 1.03$$

$$= 2683.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 2677 / (2 \cdot 2683 / 83.2 - 1178 / 50.8)$$

$$= 64.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 1784 [42 / 1178]$$

$$= 63.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(64.8, 63.6)$$

$$= 63.610 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (70.4 + 1863 + 839) / 83.2$$

$$= 33.329 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 1.03 \cdot 1178 / 50.8$$

$$= 24.0 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 33.3 - 24, 24)$$

$$= 42.7 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	42.68	kgf/cm ²
Maximum Allowable External Pressure	Pmax	63.61	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.164

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$

$$= (2.13 \cdot 1) / (16 - 3 - 0)$$

$$= 0.164$$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -14 °C
User entered Minimum Design Temperature	: 11 °C
The Curve for this Material is	: B
From Figure 3.8M- with PWHT, the MDMT is	: -14 °C



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Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (32.7 \cdot 1) / (15 - 3 - 0)$
 $= 1.000$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -14 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : -14 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (32.7 \cdot 1) / (15 - 3 - 0)$
 $= 1.000$

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -14 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : -14 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (32.7 \cdot 1) / (15 - 3 - 0)$
 $= 1.000$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.164

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (2.13 \cdot 1) / (16 - 3 - 0)$
 $= 0.164$

Gov. MDMT of the Nozzle : -14 °C
 Gov. MDMT of the Reinforcement Pad : -14 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -14 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 3.0820 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 348.0819 mm

Input Echo, WRC297 Item 1, Description: K2- PSV :



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Nozzle Calcs.: K2- PSV

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Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2300.000	mm
Shell Thickness	45.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1784.41	kgf/cm ²

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	131.801	mm
Nozzle Thickness	18.2372	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1641.66	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1321.51	kgf/cm ²

Diameter of Reinforcing Pad	368.275	mm
Thickness of Reinforcing Pad	15.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	49.97	kgf/cm ²
Radial Load	P	-509.88	kgf
Circumferential Shear	Vc	764.82	kgf
Longitudinal Shear	Vl	764.82	kgf
Circumferential Moment	Mc	336.39	kgf-m.
Longitudinal Moment	Ml	458.71	kgf-m.
Torsional Moment	Mt	570.84	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (3.739) is < 5
check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (9.227) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{1181.5(45.0 - 3.0)}$$

$$= 903.390 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2363.000	mm
Nozzle Outside Diameter	(d) =	168.275	mm
Vessel Thickness used	(T) =	57.000	mm
Nozzle Thickness used	(t) =	15.237	mm
T / t	=	3.741	
d / t	=	11.044	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.459$$

Nr/P	=	0.189
Mr/P	=	0.148
M0/P	=	0.204



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Nozzle Calcs.: K2- PSV

Nozl: 33 9:32am Aug 12,2026

NØ/P = 0.228
MrD/Mc = 0.160
NrDL/Mc = 0.171
MØd/Mc = 0.247
NØDL/Mc = -0.108
MrD/Ml = 0.154
NrDL/Ml = 0.169
MØD/Ml = 0.237
NØDL/Ml = 0.101

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-119	119	119	-119
Normal Pressure Stress	986	1036	986	1036
Outplane Stress Summary	881	1131	1136	909

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-14	-14	14	14
Inplane Bending (ML)	-77	77	77	-77
Inplane Pressure Stress	493	493	493	493
Inplane Stress Summary	418	545	601	419

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	5	5	-5	-5
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	22	22	22	22
Shear Stress Summary	27	27	17	17

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	882	1132	1137	910

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	19	-19	19	-19



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Nozzle Calcs.: K2- PSV Nozl: 33 9:32am Aug 12,2026

Outplane Membrane (Mc)	6	6	-6	-6
Outplane Bending (Mc)	-91	91	91	-91
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	493	493	493	493

Outplane Stress Summary	431	575	600	379
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	-10	-10	10	10
Inplane Bending (Mc)	-59	59	59	-59
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	986	1036	986	1036

Inplane Stress Summary	933	1073	1072	976
------------------------	-----	------	------	-----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-5	-5	5	5
Torsional Shear (Mt)	22	22	22	22

Shear Stress Summary	17	17	27	27
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	934	1074	1074	978
------------------------	-----	------	------	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-8	-8	8	8
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	244	304	244	304

Hoop Stress Summary	239	299	256	316
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	6	6	6	6
Axial Bending (P)	70	-70	70	-70
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0



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Nozzle Calcs.: K2- PSV Nozl: 33 9:32am Aug 12,2026

Axial Membrane (ML)	-178	-178	178	178
Axial Bending (ML)	-486	486	486	-486
Axial Pressure Stress	122	122	122	122

Axial Stress Summary	-465	367	864	-250
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	18	18	-18	-18
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	84	84	84	84

Shear Stress Summary	103	103	65	65
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	734	441	871	580
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	6	6	-6	-6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	244	304	244	304

Hoop Stress Summary	254	314	241	300
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	6	6	6	6
Axial Bending (P)	70	-70	70	-70
Axial Membrane (Mc)	-130	-130	130	130
Axial Bending (Mc)	-386	386	386	-386
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	122	122	122	122

Axial Stress Summary	-317	314	716	-197
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-18	-18	18	18
Shear due to Torsion	84	84	84	84

Shear Stress Summary	65	65	103	103
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside



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Nozzle Calcs.: K2- PSV

Nozl: 33 9:32am Aug 12,2026

Two * Max Shear Stress	586	379	737	539
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{\text{Rmean}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{1174.0(45.0 - 3.0)}$$

$$= 901.053 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2348.000 mm
Nozzle Outside Diameter	(d) =	368.275 mm
Vessel Thickness used	(T) =	42.000 mm
Nozzle Thickness used	(t) =	115.237 mm
T / t	=	0.364
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.173$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.097
Mr/P	=	0.204
M0/P	=	0.060
N0/P	=	0.117
MrD/Mc	=	0.306
NrDL/Mc	=	0.085
M0d/Mc	=	0.094
N0DL/Mc	=	0.050
MrD/Ml	=	0.174
NrDL/Ml	=	0.090
M0D/Ml	=	0.055
N0DL/Ml	=	0.121

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-23	23	23	-23
Normal Pressure Stress	1347	1397	1347	1397
Outplane Stress Summary	1328	1405	1392	1375

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	35	-35	35	-35
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-73	73	73	-73
Inplane Pressure Stress	673	673	673	673



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Nozzle Calcs.: K2- PSV

Nozl: 33 9:32am Aug 12,2026

Inplane Stress Summary	631	708	791	573
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	3	3	-3	-3
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6

Shear Stress Summary	9	9	3	3
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	1328	1405	1392	1375
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-29	29	29	-29
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	673	673	673	673

Outplane Stress Summary	655	693	719	639
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	35	-35	35	-35
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-95	95	95	-95
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1347	1397	1347	1397

Inplane Stress Summary	1285	1455	1484	1273
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	6	6	6	6

Shear Stress Summary	3	3	9	9
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



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 Nozzle Calcs.: K2- PSV Nozl: 33 9:32am Aug 12,2026

Two * Max Shear Stress 1285 1455 1484 1273

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	986	1036	986	1036	986	1036	986	1036
Circ. Pl (SUS)	-5	-5	11	11	-8	-8	12	12
Circ. Q (SUS)	-100	100	138	-138	-46	46	72	-72
Long. Pm (SUS)	493	493	493	493	493	493	493	493
Long. Pl (SUS)	-12	-12	16	16	9	9	-3	-3
Long. Q (SUS)	-64	64	90	-90	-72	72	110	-110
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	22	22	22	22	22	22	22	22
Pm (SUS)	986	1036	986	1036	986	1036	986	1036
Pm+Pl (SUS)	981	1031	997	1047	978	1028	998	1048
Pm+Pl+Q (Total)	882	1132	1135	909	932	1074	1071	977

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1036.00	1784.41	Passed
Pm+Pl (SUS)	1048.04	2676.62	Passed
Pm+Pl+Q (TOTAL)	1135.54	5353.24	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1347	1397	1347	1397	1347	1397	1347	1397
Circ. Pl (SUS)	-5	-5	11	11	-2	-2	6	6
Circ. Q (SUS)	-13	13	33	-33	-60	60	130	-130
Long. Pm (SUS)	673	673	673	673	673	673	673	673
Long. Pl (SUS)	-4	-4	8	8	1	1	5	5
Long. Q (SUS)	-38	38	108	-108	-19	19	39	-39
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	1347	1397	1347	1397	1347	1397	1347	1397
Pm+Pl (SUS)	1342	1392	1358	1408	1345	1395	1353	1403
Pm+Pl+Q (Total)	1329	1405	1391	1375	1285	1455	1483	1273



Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1397.00	1784.41	Passed
Pm+Pl (SUS)	1408.01	2676.62	Passed
Pm+Pl+Q (TOTAL)	1483.11	5353.24	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	244	304	244	304	244	304	244	304
Circ. Pl (SUS)	-5	-5	11	11	9	9	-3	-3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	122	122	122	122	122	122	122	122
Long. Pl (SUS)	-172	-172	184	184	-124	-124	136	136
Long. Q (SUS)	-416	416	556	-556	-316	316	456	-456
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	18	18	-18	-18	-18	-18	18	18
Shear Q (SUS)	84	84	84	84	84	84	84	84
Pm (SUS)	244.0	304.0	244.0	304.0	244.0	304.0	244.0	304.0
Pm+Pl (SUS)	291.2	350.9	311.7	329.1	257.5	317.1	269.4	307.5
Pm+Pl+Q (Total)	733.9	439.9	869.1	580.2	586.1	379.5	735.1	539.1

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	304.00	1321.51	Passed
Pm+Pl (SUS)	350.85	1982.27	Passed
Pm+Pl+Q (TOTAL)	869.09	3964.53	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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Nozzle Calcs.: K9- Temperature Nozl: 34 9:32am Aug 12,2026

Input, Nozzle Desc: K9- Temperature guage

From: 60

Pressure for Reinforcement Calculations	P	50.197	kgf/cm ²
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	1.03	kgf/cm ²
Temperature for External Pressure	Tempex	270	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1784.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	5041.6665	mm
Shell Finished (Minimum) Thickness	t	45.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		793.60	cm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1321.51	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		1.5000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1435.72	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	148.2600	mm
Thickness of Pad	te	15.0000	mm
Weld leg size between Pad and Shell	Wp	15.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	15.0000	mm
Reinforcing Pad Width		50.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

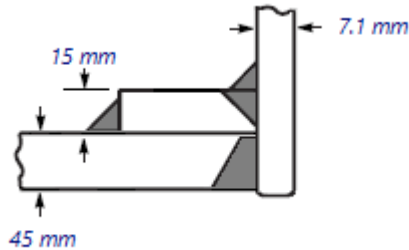
Reinforcement CALCULATION, Description: K9- Temperature guage



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 Nozzle Calcs.: K9- Temperature Nozl: 34 9:32am Aug 12,2026

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation 1.338 in.
 Actual Thickness Used in Calculation 0.281 in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 40 \cdot (\exp(50.2/(1322 \cdot 1)) - 1) + 3 + 0$
 $= 0.77 + 3 + 0 = 3.77 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1322 \cdot 1 \cdot \ln((2 \cdot 4.14 + 40) / 40) - 0$
 $= 249 - 0 = 249 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 6.220 mm
 Min. required shell thickness, internal p + c : 35.898 mm
 Min. required shell thickness, external p + c : 15.973 mm

Nozzle neck minimum required thickness : 6.220 mm

Minimum nozzle neck thickness 6.248 mm , is greater than required thickness 6.220 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1322/1784, 1]$
 $= 0.741$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[1436/1784, 1]$
 $= 0.805$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 7.14 - 3$
 $= 4.137 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 2306/42$
 $= 54.905 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$



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Nozzle Calcs.: K9- Temperature Nozl: 34 9:32am Aug 12, 2026

$$= 2300/2 + 3$$

$$= 1153.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(R_{eff} \cdot t)} + W$$

$$= \sqrt{(1153 \cdot 42)} + 50$$

$$= 270.059 \text{ mm}$$

Limit [LR2]:

$$= \sqrt{((R_{eff} + t) \cdot (t + t_e))}$$

$$= \sqrt{((1153 + 42) \cdot (42 + 15))}$$

$$= 260.988 \text{ mm}$$

Limit [LR3]:

$$= 2 \cdot R_n$$

$$= 2 \cdot 20$$

$$= 39.985 \text{ mm}$$

Final value of [LR]:

$$= \min(LR1, LR2, LR3)$$

$$= \min(270, 261, 40)$$

$$= 39.985 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 42, 15] + \sqrt{(20 \cdot 4.14)}$$

$$= 24.095 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300$$

$$= 300.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(42 + 15)$$

$$= 456.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[24.1, 300, 456] + 42$$

$$= 66.095 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 42 + ((6 \cdot 0.8) / 40)$$

$$= 54.069 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 20 + 4.14) / (\sqrt{(2306 + 54.1) \cdot 54.1}))$$

$$= 0.124$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 42 \cdot 40 \cdot \max((0.12/5)^{0.85}, 1)$$

$$= 16.794 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:



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$$= t_n \cdot L_H$$

$$= 4.14 \cdot 66.1$$

$$= 2.735 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2 - \text{Area cut by thickness limit}$$

$$= 0.5 \cdot 20^2 - 0.59$$

$$= 1.405 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$

$$= \min(50 \cdot 15, 40 \cdot 15)$$

$$= 5.998 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$

$$= 16.8 + 0.74(2.73 + 0) + 1.41 + 0 + 0 + 0.8(6)$$

$$= 25.050 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 4.14 / \ln[1 + 4.14/20]$$

$$= 21.996 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 54.1 / \ln[1 + 54.1/1153]$$

$$= 1179.828 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1784 \cdot 1$$

$$= 2676.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 50.2 \cdot 22 \cdot 66.1$$

$$= 729.8 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 50.2 \cdot 1180 (40 + 4.14)$$

$$= 26131.1 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 50.2 \cdot 1180 \cdot 20$$

$$= 11840.4 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (26131 + 11840 + 730) / 50.2$$

$$= 771.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 2677 / (2 \cdot 771 / 25.1 - 1180 / 54.1)$$

$$= 67.4 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / (t / R_{xs})$$

$$= 1784 [42 / 1180]$$

$$= 63.5 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:



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$$\begin{aligned}
 &= \min(P_{max1}, P_{max2}) \\
 &= \min(67.4, 63.5) \\
 &= 63.522 \text{ kgf/cm}^2
 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned}
 &= (f_N + f_S + f_Y) / A_T \\
 &= (730 + 26131 + 11840) / 25.1 \\
 &= 1544.956 \text{ kgf/cm}^2
 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned}
 &= P \cdot R_{xs} / t_{eff} \\
 &= 50.2 \cdot 1180 / 54.1 \\
 &= 1095.3 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\
 &= \max(2 \cdot 1545 - 1095, 1095) \\
 &= 1994.6 \text{ kgf/cm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	1994.57	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	63.52	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned}
 &= (R_{nc} + t_n) / R_{nc} \\
 &= (20 + 4.14) / 20 \\
 &= 1.207 \text{ For set-in Nozzles}
 \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned}
 &= \pi / 2 \cdot (R_n + t_n) \\
 &= \pi / 2 \cdot (20 + 4.14) \\
 &= 37.903 \text{ mm}
 \end{aligned}$$

Weld Length of Pad to Shell Weld [LTp]:

$$\begin{aligned}
 &= \pi / 2 \cdot (R_n + t_n + W) \\
 &= \pi / 2 \cdot (20 + 4.14 + 50) \\
 &= 116.443 \text{ mm}
 \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 14.142, 10.606, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned}
 &t_c, L41T = 14.142, \text{ must be } \geq \min(0.7 \cdot 4.137, 6.000) \text{ Passed} \\
 &t_f1, L42T = 10.606, \text{ must be } \geq \min(0.6 \cdot 15.000, 0.6 \cdot 42.000) \text{ Passed}
 \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned}
 &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi / 4 \cdot P \cdot R_n^2 \cdot k_y^2) \\
 &= \min(11840 \cdot 1.21, 1.5 \cdot 1322(2.73+0), \pi / 4 \cdot 50.2 \cdot 20^2 \cdot 1.21^2) \\
 &= 229.553 \text{ kgf}
 \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned}
 &= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p) \\
 &= 230 \cdot 42 \cdot 1784 / (42 \cdot 1784 + 15 \cdot 1436) \\
 &= 178.314 \text{ kgf}
 \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned}
 &= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p) \\
 &= 230 \cdot 15 \cdot 1436 / (42 \cdot 1784 + 15 \cdot 1436) \\
 &= 51.239 \text{ kgf}
 \end{aligned}$$

Shear Stress [τ_1]:

$$= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$



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$$= 178 / (37.9 \cdot (0.6 \cdot 38 + 0.49 \cdot 0))$$

$$= 20.634 \text{ kgf/cm}^2$$

Shear Stress [τ_2]:

$$= fwp / (L_T \cdot (0.6 \cdot tw_2 + 0.49 \cdot L_{41T}))$$

$$= 51.2 / (37.9 \cdot (0.6 \cdot 15 + 0.49 \cdot 14.1))$$

$$= 8.486 \text{ kgf/cm}^2$$

Shear Stress [τ_3]:

$$= fwp / (L_{TP} \cdot (0.49 \cdot L_{42T}))$$

$$= 51.2 / (116 \cdot (0.49 \cdot 10.6))$$

$$= 8.467 \text{ kgf/cm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(20.6, 8.49, 8.47)$$

$$= 20.6 \text{ must be less than or equal to } 1784.4 \text{ kgf/cm}^2$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 4.14 / \ln[1 + 4.14/20]$$

$$= 21.996 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 54.1 / \ln[1 + 54.1/1153]$$

$$= 1179.828 \text{ mm}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1784$$

$$= 2676.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.03 \cdot 22 \cdot 66.1$$

$$= 15.0 \text{ kgf}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.03 \cdot 1180 (40 + 4.14)$$

$$= 537.7 \text{ kgf}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.03 \cdot 1180 \cdot 20$$

$$= 243.7 \text{ kgf}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$

$$= (538 + 244 + 15) / 1.03$$

$$= 771.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2677 / (2 \cdot 771 / 25.1 - 1180 / 54.1)$$

$$= 67.4 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S / t / R_{xs}$$

$$= 1784 [42 / 1180]$$

$$= 63.5 \text{ kgf/cm}^2$$



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Maximum Allowable Working Pressure [Pmax]:
 = min(Pmax1, Pmax2)
 = min(67.4, 63.5)
 = 63.522 kgf/cm²

Average Primary Membrane Stress [σAvg]:
 = (fN + fS + fY) / AT
 = (15 + 538 + 244)/25.1
 = 31.793 kgf/cm²

General Primary Membrane Stress [σCirc]:
 = P • Rxs / teff
 = 1.03 • 1180/54.1
 = 22.5 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:
 = max(2 • σAvg - σCirc, σCirc)
 = max(2 • 31.8 - 22.5, 22.5)
 = 41.0 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	41.05	kgf/cm ²
Maximum Allowable External Pressure	Pmax	63.52	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.191

The computed thickness ratio [sr]:
 = (tr • E*)/(tnom - ci - cext)
 = (0.62 • 1)/(6.25 - 3 - 0)
 = 0.191

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.191

The computed thickness ratio [sr]:
 = (tr • E*)/(tnom - ci - cext)
 = (0.62 • 1)/(6.25 - 3 - 0)
 = 0.191

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C



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 Nozzle Calcs.: K9- Temperature Nozl: 34 9:32am Aug 12,2026

The computed stress ratio is : 0.191

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (0.62 \cdot 1) / (6.25 - 3 - 0)$
 $= 0.191$

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -14 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : -14 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (32.9 \cdot 1) / (15 - 3 - 0)$
 $= 1.000$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.191

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (0.62 \cdot 1) / (6.25 - 3 - 0)$
 $= 0.191$

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -14 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -14 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 0.2532 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 345.2531 mm

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 Nozzle Calcs.: N10- Chemical a Nozl: 35 9:32am Aug 12,2026

Input, Nozzle Desc: N10- Chemical additive

From: 60

Pressure for Reinforcement Calculations	P	50.082	kgf/cm ²
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	1.03	kgf/cm ²
Temperature for External Pressure	Tempex	270	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1784.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	5041.6665	mm
Shell Finished (Minimum) Thickness	t	45.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		893.60	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1321.51	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		1.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1435.72	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	133.4010	mm
Thickness of Pad	te	15.0000	mm
Weld leg size between Pad and Shell	Wp	15.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	15.0000	mm
Reinforcing Pad Width		50.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

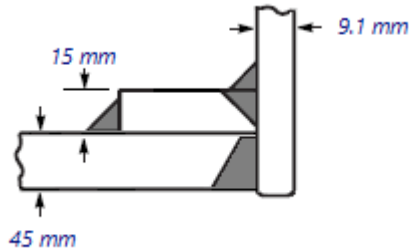
Reinforcement CALCULATION, Description: N10- Chemical additive



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 Nozzle Calcs.: N10- Chemical a Nozl: 35 9:32am Aug 12,2026

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation 0.599 in.
 Actual Thickness Used in Calculation 0.358 in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 21.2 \cdot (\exp(50.1/(1322 \cdot 1)) - 1) + 3 + 0$
 $= 0.41 + 3 + 0 = 3.41 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$
 $= 1322 \cdot 1 \cdot \ln((2 \cdot 6.09 + 21.2)/21.2) - 0$
 $= 600 - 0 = 600 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 5.960 mm
 Min. required shell thickness, internal p + c : 35.821 mm
 Min. required shell thickness, external p + c : 15.973 mm

Nozzle neck minimum required thickness : 5.960 mm

Minimum nozzle neck thickness 7.950 mm , is greater than required thickness 5.960 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1322/1784, 1]$
 $= 0.741$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[1436/1784, 1]$
 $= 0.805$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 9.09 - 3$
 $= 6.093 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D/t$
 $= 2306/42$
 $= 54.905 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$



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Nozzle Calcs.: N10- Chemical a Nozl: 35 9:32am Aug 12,2026

$$= 2300/2 + 3$$

$$= 1153.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(\text{Reff} \cdot t) + W}$$

$$= \sqrt{(1153 \cdot 42) + 50}$$

$$= 270.059 \text{ mm}$$

Limit [LR2]:

$$= \sqrt{((\text{Reff} + t) \cdot (t + t_e))}$$

$$= \sqrt{((1153 + 42) \cdot (42 + 15))}$$

$$= 260.988 \text{ mm}$$

Limit [LR3]:

$$= 2 \cdot R_n$$

$$= 2 \cdot 10.6$$

$$= 21.215 \text{ mm}$$

Final value of [LR]:

$$= \min(\text{LR1}, \text{LR2}, \text{LR3})$$

$$= \min(270, 261, 21.2)$$

$$= 21.215 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 42, 15] + \sqrt{(10.6 \cdot 6.09)}$$

$$= 23.039 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300$$

$$= 300.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(42 + 15)$$

$$= 456.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[\text{LH1}, \text{LH2}, \text{LH3}] + t \quad \text{for set-in nozzles}$$

$$= \min[23, 300, 456] + 42$$

$$= 65.039 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot \text{frp}) / \text{LR})$$

$$= 42 + ((3.18 \cdot 0.8) / 21.2)$$

$$= 54.069 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{((D_i + t_{eff}) \cdot t_{eff})}))$$

$$= \min(12, (2 \cdot 10.6 + 6.09) / (\sqrt{((2306 + 54.1) \cdot 54.1)}))$$

$$= 0.076$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot \text{LR} \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 42 \cdot 21.2 \cdot \max((0.076/5)^{0.85}, 1)$$

$$= 8.910 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:



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Nozzle Calcs.: N10- Chemical a Nozl: 35 9:32am Aug 12,2026

$$= t_n \cdot L_H$$

$$= 6.09 \cdot 65$$

$$= 3.963 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2 - \text{Area cut by thickness limit}$$

$$= 0.5 \cdot 20^2 - 0.72$$

$$= 1.285 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e , LR \cdot t_e)$$

$$= \min(50 \cdot 15, 21.2 \cdot 15)$$

$$= 3.182 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$

$$= 8.91 + 0.74(3.96 + 0) + 1.28 + 0 + 0 + 0.8(3.18)$$

$$= 15.690 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 6.09 / \ln[1 + 6.09/10.6]$$

$$= 13.424 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 54.1 / \ln[1 + 54.1/1153]$$

$$= 1179.828 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1784 \cdot 1$$

$$= 2676.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 50.1 \cdot 13.4 \cdot 65$$

$$= 437.3 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 50.1 \cdot 1180 (21.2 + 6.09)$$

$$= 16135.7 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 50.1 \cdot 1180 \cdot 10.6$$

$$= 6267.7 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (16136 + 6268 + 437) / 50.1$$

$$= 456.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 2677 / (2 \cdot 456 / 15.7 - 1180 / 54.1)$$

$$= 73.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / (t / R_{xs})$$

$$= 1784 [42 / 1180]$$

$$= 63.5 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:



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$$= \min(P_{max1}, P_{max2})$$

$$= \min(73.7, 63.5)$$

$$= 63.522 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (437 + 16136 + 6268) / 15.7$$

$$= 1455.735 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 50.1 \cdot 1180 / 54.1$$

$$= 1092.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1456 - 1093, 1093)$$

$$= 1818.6 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	1818.63	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	63.52	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (10.6 + 6.09) / 10.6$$

$$= 1.574 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (10.6 + 6.09)$$

$$= 26.233 \text{ mm}$$

Weld Length of Pad to Shell Weld [LTp]:

$$= \pi/2 \cdot (R_n + t_n + W)$$

$$= \pi/2 \cdot (10.6 + 6.09 + 50)$$

$$= 104.773 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 14.142, 10.606, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 14.142, \text{ must be } \geq \min(0.7 \cdot 6.093, 6.000) \text{ Passed}$$

$$t_f1, L42T = 10.606, \text{ must be } \geq \min(0.6 \cdot 15.000, 0.6 \cdot 42.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(6268 \cdot 1.57, 1.5 \cdot 1322(3.96+0), \pi/4 \cdot 50.1 \cdot 10.6^2 \cdot 1.57^2)$$

$$= 109.706 \text{ kgf}$$

Discontinuity Force [fws]:

$$= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 110 \cdot 42 \cdot 1784 / (42 \cdot 1784 + 15 \cdot 1436)$$

$$= 85.218 \text{ kgf}$$

Discontinuity Force [fwp]:

$$= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 110 \cdot 15 \cdot 1436 / (42 \cdot 1784 + 15 \cdot 1436)$$

$$= 24.488 \text{ kgf}$$

Shear Stress [τ_1]:

$$= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$



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$$= 85.2 / (26.2 \cdot (0.6 \cdot 38 + 0.49 \cdot 0))$$

$$= 14.248 \text{ kgf/cm}^2$$

Shear Stress [τ_2]:

$$= fwp / (L_T \cdot (0.6 \cdot tw_2 + 0.49 \cdot L_{41T}))$$

$$= 24.5 / (26.2 \cdot (0.6 \cdot 15 + 0.49 \cdot 14.1))$$

$$= 5.860 \text{ kgf/cm}^2$$

Shear Stress [τ_3]:

$$= fwp / (L_{TP} \cdot (0.49 \cdot L_{42T}))$$

$$= 24.5 / (105 \cdot (0.49 \cdot 10.6))$$

$$= 4.497 \text{ kgf/cm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(14.2, 5.86, 4.5)$$

$$= 14.2 \text{ must be less than or equal to } 1784.4 \text{ kgf/cm}^2$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n / R_n]$$

$$= 6.09 / \ln[1 + 6.09 / 10.6]$$

$$= 13.424 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff} / R_{eff}]$$

$$= 54.1 / \ln[1 + 54.1 / 1153]$$

$$= 1179.828 \text{ mm}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1784$$

$$= 2676.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.03 \cdot 13.4 \cdot 65$$

$$= 9.0 \text{ kgf}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.03 \cdot 1180 (21.2 + 6.09)$$

$$= 332.8 \text{ kgf}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.03 \cdot 1180 \cdot 10.6$$

$$= 129.3 \text{ kgf}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$

$$= (333 + 129 + 9.02) / 1.03$$

$$= 456.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 2677 / (2 \cdot 456 / 15.7 - 1180 / 54.1)$$

$$= 73.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S / t / R_{xs}$$

$$= 1784 [42 / 1180]$$

$$= 63.5 \text{ kgf/cm}^2$$



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Maximum Allowable Working Pressure [Pmax]:
 = min(Pmax1, Pmax2)
 = min(73.7, 63.5)
 = 63.522 kgf/cm²

Average Primary Membrane Stress [σAvg]:
 = (fN + fS + fY) / AT
 = (9.02 + 333 + 129)/15.7
 = 30.026 kgf/cm²

General Primary Membrane Stress [σCirc]:
 = P • Rxs / teff
 = 1.03 • 1180/54.1
 = 22.5 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:
 = max(2 • σAvg - σCirc, σCirc)
 = max(2 • 30 - 22.5, 22.5)
 = 37.5 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	37.51	kgf/cm ²
Maximum Allowable External Pressure	Pmax	63.52	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.066

The computed thickness ratio [sr]:
 = (tr • E*)/(tnom - ci - cext)
 = (0.33 • 1)/(7.96 - 3 - 0)
 = 0.066

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.066

The computed thickness ratio [sr]:
 = (tr • E*)/(tnom - ci - cext)
 = (0.33 • 1)/(7.96 - 3 - 0)
 = 0.066

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C



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 Nozzle Calcs.: N10- Chemical a Nozl: 35 9:32am Aug 12,2026

The computed stress ratio is : 0.066

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (0.33 \cdot 1) / (7.96 - 3 - 0)$
 $= 0.066$

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -14 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : -14 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (32.8 \cdot 1) / (15 - 3 - 0)$
 $= 1.000$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.066

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (0.33 \cdot 1) / (7.96 - 3 - 0)$
 $= 0.066$

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -14 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -14 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 0.1213 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 345.1213 mm

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 Nozzle Calcs.: MH1- Manhole Nozl: 36 9:32am Aug 12,2026

Input, Nozzle Desc: MH1- Manhole

From: 70

Pressure for Reinforcement Calculations	P	50.082	kgf/cm ²
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	1.03	kgf/cm ²
Temperature for External Pressure	Tempex	270	°C

Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	1784.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²

Inside Diameter of Elliptical Head	D	2300.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	40.8000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1321.51	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		120	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	34.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1435.72	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	909.6000	mm
Thickness of Pad	te	34.0000	mm
Weld leg size between Pad and Shell	Wp	26.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	34.0000	mm
Reinforcing Pad Width		150.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

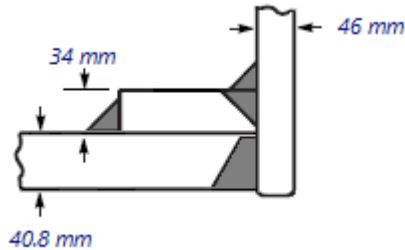
Reinforcement CALCULATION, Description: MH1- Manhole



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 Nozzle Calcs.: MH1- Manhole Nozl: 36 9:32am Aug 12,2026

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation 20.376 in.
 Actual Thickness Used in Calculation 1.812 in.



Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 524 \cdot (\exp(50.1/(1322 \cdot 1)) - 1) + 3 + 0$
 $= 10.1 + 3 + 0 = 13.1 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1322 \cdot 1 \cdot \ln((2 \cdot 43 + 524) / 524) - 0$
 $= 201 - 0 = 201 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
 Min. required shell thickness, internal p + c : 39.264 mm
 Min. required shell thickness, external p + c : 11.572 mm

Nozzle neck minimum required thickness : 11.340 mm

Minimum nozzle neck thickness 40.284 mm , is greater than required thickness 11.340 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1322/1784, 1]$
 $= 0.741$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[1436/1784, 1]$
 $= 0.805$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 46 - 3$
 $= 43.025 \text{ mm}$

Effective Pressure Radius [Reff]:
 $= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2]$
 $= 0.9/6 \cdot 2306 [2 + (2306/(2 \cdot 578))^2]$



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= 2068.228 mm

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(\text{Reff} \cdot t) + W}$$

$$= \sqrt{(2068 \cdot 37.8) + 150}$$

$$= 429.605 \text{ mm}$$

Limit [LR2]:

$$= \sqrt{((\text{Reff} + t) \cdot (t + te))}$$

$$= \sqrt{((2068 + 37.8) \cdot (37.8 + 34))}$$

$$= 388.861 \text{ mm}$$

Limit [LR3]:

$$= 2 \cdot Rn$$

$$= 2 \cdot 262$$

$$= 523.550 \text{ mm}$$

Final value of [LR]:

$$= \min(\text{LR1}, \text{LR2}, \text{LR3})$$

$$= \min(430, 389, 524)$$

$$= 388.861 \text{ mm}$$

Intermediate value [Cn]:

$$= \min(((t + te)/tn)^{0.35}, 1)$$

$$= \min(((37.8 + 34)/43)^{0.35}, 1)$$

$$= \min(1.2, 1)$$

$$= 1.000$$

Intermediate value [Cp]:

$$= \exp((0.35 \cdot di - Xo)/(16t))$$

$$= \exp((0.35 \cdot 2306 - 305)/(16 \cdot 37.8))$$

$$= 2.295$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + te + Fp \cdot \sqrt{(Rn \cdot tn)}, Lpr1 + t)$$

$$= \min(37.8 + 34 + 1 \cdot \sqrt{(262 \cdot 43)}, 200 + 37.8)$$

$$= 177.926 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 37.8 + ((51 \cdot 0.8) / 389)$$

$$= 48.352 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 37.8 \cdot 389$$

$$= 146.989 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH$$

$$= 43 \cdot 178$$

$$= 76.553 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 26^2$$

$$= 3.380 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:



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$$= 0.5 \cdot \text{Leg41}^2$$

$$= 0.5 \cdot 20^2$$

$$= 2.000 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$

$$= \min(150 \cdot 34, 389 \cdot 34)$$

$$= 51.000 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$

$$= 147 + 0.74(76.6 + 0) + 2 + 3.38 + 0 + 0.8(51)$$

$$= 250.097 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 43 / \ln[1 + 43/262]$$

$$= 282.742 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 48.4 / \ln[1 + 48.4/2068]$$

$$= 2092.315 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 1784 \cdot 1$$

$$= 2676.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 50.1 \cdot 283 \cdot 178$$

$$= 25195.0 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (L_R + t_n) / 2$$

$$= 50.1 \cdot 2092 \cdot (389 + 43) / 2$$

$$= 226281.4 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$

$$= (50.1 \cdot 2092 \cdot 262) / 2$$

$$= 137154.0 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (226281 + 137154 + 25195) / 50.1$$

$$= 7759.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$$

$$= 2677 / (2 \cdot 7760 / 250 - 2092 / (2 \cdot 48.4))$$

$$= 66.2 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S / [t / R_{xs}]$$

$$= 2 \cdot 1784 [37.8 / 2092]$$

$$= 64.5 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(66.2, 64.5)$$

$$= 64.475 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:



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$$= (fN + fS + fY) / AT$$

$$= (25195 + 226281 + 137154) / 250$$

$$= 1553.919 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / (2 \cdot t_{eff})$$

$$= 50.1 \cdot 2092 / (2 \cdot 48.4)$$

$$= 1083.6 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1554 - 1084, 1084)$$

$$= 2024.3 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	2024.25	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	64.47	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (262 + 43) / 262$$

$$= 1.164 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (262 + 43)$$

$$= 478.779 \text{ mm}$$

Weld Length of Pad to Shell Weld [LtP]:

$$= \pi/2 \cdot (R_n + t_n + W)$$

$$= \pi/2 \cdot (262 + 43 + 150)$$

$$= 714.398 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 14.142, 18.385, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 14.142, \text{ must be } \geq \min(0.7 \cdot 43.025, 6.000) \text{ Passed}$$

$$t_{f1}, L42T = 18.385, \text{ must be } \geq \min(0.6 \cdot 34.000, 0.6 \cdot 37.800) \text{ Failed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(137154 \cdot 1.16, 1.5 \cdot 1322(76.6+0), \pi/4 \cdot 50.1 \cdot 262^2 \cdot 1.16^2)$$

$$= 36542.875 \text{ kgf}$$

Discontinuity Force [fws]:

$$= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 36543 \cdot 37.8 \cdot 1784 / (37.8 \cdot 1784 + 34 \cdot 1436)$$

$$= 21200.176 \text{ kgf}$$

Discontinuity Force [fwp]:

$$= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 36543 \cdot 34 \cdot 1436 / (37.8 \cdot 1784 + 34 \cdot 1436)$$

$$= 15342.700 \text{ kgf}$$

Shear Stress [τ_1]:

$$= fws / (L_t \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 21200 / (479 \cdot (0.6 \cdot 34 + 0.49 \cdot 0))$$

$$= 217.058 \text{ kgf/cm}^2$$

Shear Stress [τ_2]:

$$= fwp / (L_t \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T}))$$



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Nozzle Calcs.: MH1- Manhole Nozl: 36 9:32am Aug 12,2026

$$= 15343 / (479 \cdot (0.6 \cdot 34 + 0.49 \cdot 14.1))$$

$$= 117.256 \text{ kgf/cm}^2$$

Shear Stress [τ_3]:

$$= fwp / (LTP \cdot (0.49 \cdot L42T))$$

$$= 15343 / (714 \cdot (0.49 \cdot 18.4))$$

$$= 238.403 \text{ kgf/cm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(217, 117, 238)$$

$$= 238.4 \text{ must be less than or equal to } 1784.4 \text{ kgf/cm}^2$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 43 / \ln[1 + 43/262]$$

$$= 282.742 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 48.4 / \ln[1 + 48.4/2068]$$

$$= 2092.315 \text{ mm}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 1784$$

$$= 2676.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 1.03 \cdot 283 \cdot 178$$

$$= 519.7 \text{ kgf}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs} \cdot (LR + t_n) / 2$$

$$= 1.03 \cdot 2092 \cdot (389 + 43) / 2$$

$$= 4667.3 \text{ kgf}$$

Discontinuity Force from External Pressure [f_Y]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$

$$= (1.03 \cdot 2092 \cdot 262) / 2$$

$$= 2828.9 \text{ kgf}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$

$$= (4667 + 2829 + 520) / 1.03$$

$$= 7759.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$$

$$= 2677 / (2 \cdot 7760/250 - 2092 / (2 \cdot 48.4))$$

$$= 66.2 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= 2 \cdot S / [t / R_{xs}]$$

$$= 2 \cdot 1784 / [37.8 / 2092]$$

$$= 64.5 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(66.2, 64.5)$$

$$= 64.475 \text{ kgf/cm}^2$$



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Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (520 + 4667 + 2829) / 250$$

$$= 32.051 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 1.03 \cdot 2092 / (2 \cdot 48.4)$$

$$= 22.4 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 32.1 - 22.4, 22.4)$$

$$= 41.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	41.75	kgf/cm ²
Maximum Allowable External Pressure	Pmax	64.47	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.218

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (8.11 \cdot 1) / (40.3 - 3 - 0)$$

$$= 0.218$$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -1 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 2 °C
 Temperature reduction per Graph 3.13 : 2 °C
 The computed stress ratio is : 0.945

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (29.3 \cdot 1) / (34 - 3 - 0)$$

$$= 0.945$$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -1 °C
 User entered Minimum Design Temperature : 11 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 2 °C
 Temperature reduction per Graph 3.13 : 2 °C
 The computed stress ratio is : 0.945

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (29.3 \cdot 1) / (34 - 3 - 0)$$



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 Nozzle Calcs.: MH1- Manhole Nozl: 36 9:32am Aug 12,2026

= 0.945

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-1 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	2 °C
Temperature reduction per Graph 3.13	:	2 °C
The computed stress ratio is	:	0.945

The computed thickness ratio [sr]:

= (tr • E*)/(tnom - ci - cext)
 = (29.3 • 1)/(34 - 3 - 0)
 = 0.945

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.218

The computed thickness ratio [sr]:

= (tr • E*)/(tnom - ci - cext)
 = (8.11 • 1)/(40.3 - 3 - 0)
 = 0.218

Gov. MDMT of the Nozzle	:	-1 °C
Gov. MDMT of the Reinforcement Pad	:	-1 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-1 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction	:	-48 °C
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The Drop for this Nozzle is : 22.2093 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 263.0093 mm

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Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	49.97 kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	49.97 kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	1.03 kgf/cm ²
Shell Thickness	ts	30.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1300.000 mm
Shell Temperature for Internal Pressure	Ts	270.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	1435.72 kgf/cm ²
Shell Allowable Stress at Ambient		1784.41 kgf/cm ²

Channel Data:

Channel Description: Channel

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	163.16 kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	163.16 kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00 kgf/cm ²
Channel Thickness	tc	74.0000 mm
Channel Internal Corrosion Allowance	cac	1.5000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1300.000 mm
Channel Design Temperature	TEMPC	450.00 °C
Channel Flange Matl		SA-387 22

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K21590
Channel Allowable Stress at Temperature	Sc	1570.34 kgf/cm ²
Channel Allowable Stress at Ambient		2110.70 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	888
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	25.0000 mm
Total Straight Tube Length	Lt	6600.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6355.00 mm
Design Temperature of the Tubes		450.00 °C
Tube Material		SA-213 T22
Tube Material UNS Number		K21590
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		1233.81 kgf/cm ²
Tube Allowable Stress At Ambient		1407.13 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1845.63 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	34.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	617.500 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm



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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	450.00 °C
Tubesheet Material		SA-336 F22

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

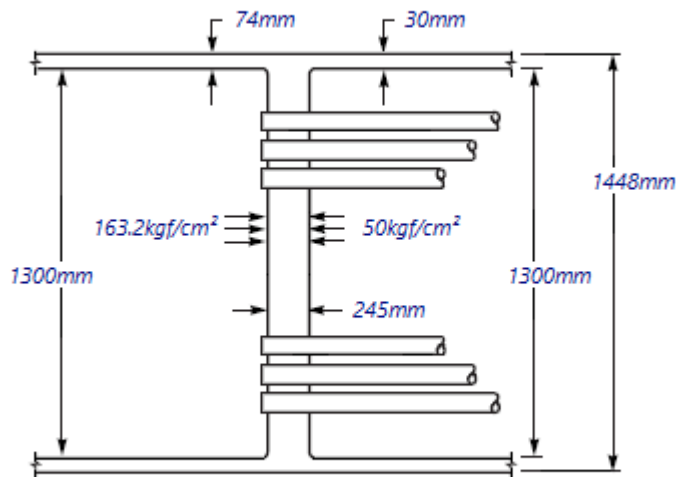
Tubesheet Material UNS Number		K21590
Tubesheet Allowable Stress at Temperature	S	1570.34 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	2110.70 kgf/cm ²
Thickness of Tubesheet	h	245.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	6.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	6.0000 mm
Tubesheet Outside Diameter	A	1448.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	270.0 °C	0.19093E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²



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Channel - TM-1 Material Group D

Elastic Mod. at Design Temperature	450.0 °C	0.18284E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21514E+07 kgf/cm ²

Tubes - TM-1 Material Group D

Elastic Mod. at Tubsht. Design Temp.	450.0 °C	0.18284E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21514E+07 kgf/cm ²

TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	450.0 °C	0.18284E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.21514E+07 kgf/cm ²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1
 $= 0.5 \cdot 20 \cdot (\exp(163/(1234 \cdot 1)) - 1) + 0$
 $= 1.41 + 0$
 $= 1.41 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 6972 / \sqrt{(12.5 \cdot 2.5)}$
 $= 1247.279$

Compute the Factor [Ch]:

Since $M_x > 2 \cdot (D_o/t)^{0.94}$, $Ch = 0.55 / (D_o/t)$
 $= 0.55 \cdot (25/2.5)$
 $= 0.0550$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y (t / D_o)$
 $= 1.6 \cdot 0.055 \cdot 1828437 (2.5/25)$
 $= 16090.243 \text{ kgf/cm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 16090.2/1845.6$
 $= 8.718$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 1846/4092$
 $= 0.451$

Ratio [AE]:

$= Fhe / E$
 $= 16090/1828437$
 $= 0.008800$

Factor [m1]:

$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.45) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.173$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$



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$$= 1846 (1 + 0.0020) / \ln(1 + 0.0020) 0.17335507$$

$$= 5432.114 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6 (1 - 0.45)$$

$$= 0.329$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (4092 \cdot \exp(0.33)) / (0.33 0.33)$$

$$= 8200.224 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.451.5 - 0.5 \cdot 0.452.5 - 0.453.5$$

$$= 0.324$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2045	-1.450	0.504E-05	0.355E-05	0.110E-04	-0.581E-05	0.700E+05	0.204E-01
2	1022	-4.258	0.185E-06	0.185E-06	0.267E-05	-0.267E-05	0.109E+07	0.786E-02
3	1534	-2.854	0.128E-05	0.126E-05	0.610E-05	-0.595E-05	0.309E+06	0.383E-02
4	1790	-2.152	0.267E-05	0.248E-05	0.835E-05	-0.742E-05	0.151E+06	0.306E-02
5	1662	-2.503	0.187E-05	0.181E-05	0.718E-05	-0.680E-05	0.217E+06	0.113E-02
6	1726	-2.328	0.224E-05	0.213E-05	0.776E-05	-0.716E-05	0.181E+06	0.732E-03
7	1694	-2.415	0.205E-05	0.197E-05	0.747E-05	-0.699E-05	0.198E+06	0.252E-03
8	1710	-2.371	0.215E-05	0.205E-05	0.761E-05	-0.708E-05	0.189E+06	0.226E-03
9	1702	-2.393	0.210E-05	0.201E-05	0.754E-05	-0.703E-05	0.194E+06	0.166E-04
10	1706	-2.382	0.212E-05	0.203E-05	0.757E-05	-0.705E-05	0.192E+06	0.104E-03
11	1704	-2.388	0.211E-05	0.202E-05	0.756E-05	-0.704E-05	0.193E+06	0.434E-04
12	1703	-2.391	0.210E-05	0.201E-05	0.755E-05	-0.704E-05	0.193E+06	0.133E-04
13	1702	-2.392	0.210E-05	0.201E-05	0.754E-05	-0.704E-05	0.194E+06	0.164E-05
14	1703	-2.391	0.210E-05	0.201E-05	0.754E-05	-0.704E-05	0.193E+06	0.585E-05
15	1702	-2.392	0.210E-05	0.201E-05	0.754E-05	-0.704E-05	0.193E+06	0.211E-05
16	1702	-2.392	0.210E-05	0.201E-05	0.754E-05	-0.704E-05	0.193E+06	0.236E-06

Factor of Safety [FS]:

Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 1703/1846$$

$$= 1.723$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic/FS$$

$$= 1703/1.72$$

$$= 988.112 \text{ kgf/cm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha(t / Do)$$

$$= 2 \cdot 988 (2.5/25)$$

$$= 197.622 \text{ kgf/cm}^2$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.4138 mm
Allowable Internal Pressure at Corroded thickness	275.32 kgf/cm²
Required Internal Design Pressure	163.16 kgf/cm²
Allowable External Pressure at Corroded thickness	197.62 kgf/cm²
Required External Design Pressure	49.97 kgf/cm²
Required Thickness due to Shell Side pressure	1.0295 mm



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Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 618 + 25 \\ &= 1260.000 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (34 - 25) / 34 \\ &= 0.265 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 1300 / 1260 \\ &= 1.0317 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 1300 / 1260 \\ &= 1.0317 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0 / 245 \\ &= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2tt(E_t/E) \cdot (S_t/S) \cdot (p); dt - 2tt) \\ &= \max(25 - 2 \cdot 2.5 (1828437 / 1828437) \cdot (1234 / 1570) \cdot (0); 25 - 2 \cdot 2.5) \\ &= 25.0000 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\ &= 34 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1260 \cdot 34) / (\pi \cdot 1260^2)) } \\ &= 34.0000 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (34 - 25) / 34 \\ &= 0.2647 \end{aligned}$$

$h/p = 7.205882$; $\mu^* = 0.264706$; $E^* = 430310 \text{ kgf/cm}^2$

$E^*/E = 0.235343$; $v^* = 0.382088$

Note: As h/p (7.206) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (1300 + 30) 30 } \end{aligned}$$



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$$= 0.0091 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0091 \cdot 1909280 \cdot 30^3 / (6(1 - 0.3^2))$$

$$= 85923104.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [ls]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1300 + 30) \cdot 85923104 / 245^3 \cdot (1 + 245 \cdot 0.0091 + (245 \cdot 0.0091)^2 / 2)$$

$$= 266473.1 \text{ kgf/cm}^2$$

Shell coefficient [ds]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1300^2 / (4 \cdot 1909280 \cdot 30) \cdot (1 - 1300 \cdot 0.3 / (2(1300 + 30)))$$

$$= 0.62948\text{E-}02 \text{ mm/kgf/cm}^2$$

Parameter [ws]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.03 \cdot 85923104 \cdot 0.0091 \cdot 0.006294772 (1 + 245 \cdot 0.0091)$$

$$= 16401.6113 \text{ mm}^2$$

Channel coefficient [bc]:

$$= ((12(1 - \nu_c^2))^{1/4} / \sqrt{(D_c + t_c)t_c})$$

$$= ((12(1 - 0.3^2))^{1/4} / \sqrt{(1300 + 74)74})$$

$$= 0.0057 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.0057 \cdot 1828437 \cdot 74^3 / (6(1 - 0.3^2))$$

$$= 773622144.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [lc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1300 + 74 / \cos(0)) \cdot 773622144 / 245^3 \cdot (1 + 245 \cdot 0.0057 \cdot (1 + 1) / 2 + (245 \cdot 0.0057)^2 / 2 \cdot 1)$$

$$= 1462434.8 \text{ kgf/cm}^2$$

Channel coefficient [dc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1300^2 / (4 \cdot 1828437 \cdot 74) \cdot (1 - 1300 \cdot 0.3 / (2(1300 + 74)))$$

$$= 0.26794\text{E-}02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [wc]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.03 \cdot 773622144 [0.002679428 (1 + 0.0057 \cdot 1 \cdot 245) - 0.000 (1/245 + 1 \cdot 0.0057/2)]$$

$$= 29221.9531 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1448 / 1260$$

$$= 1.1492$$

Coefficient [F]:



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$$\begin{aligned}
 &= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\
 &= (1 - 0.38) / 430310 \cdot (266473 + 1462434 + \\
 &\quad 1828437 \cdot \ln(1.15)) \\
 &= 2.8478
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= Do^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (pc - 1) \cdot (pc^2 + 1) \cdot Pt] \\
 &= 1260^2 / 16 \cdot [(1.03 - 1) \cdot (1.03^2 + 1) \cdot 50 - \\
 &\quad (1.03 - 1) \cdot (1.03^2 + 1) \cdot 163] \\
 &= -736099.5625 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps \\
 &= -736100 + 29222 \cdot 163 - 16402 \cdot 50 \\
 &= 3212153.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned}
 &= ((M^*) - Do^2 / 32 \cdot F (Ps - Pt)) / (1 + F) \\
 &= ((3212153) - 1260^2 / 32 \cdot 2.85 (50 - 163)) / (1 + 2.85) \\
 &= 4991030.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= Mp + Do^2 / 64 \cdot (3 + v^*) \cdot (Ps - Pt) \\
 &= 4991030 + 1260^2 / 64 \cdot (3 + 0.38) \cdot (50 - 163) \\
 &= -4505330.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= \max(| Mp |; | Mo |) \\
 &= \max(| 4991030 |; | -4505330 |) \\
 &= 4991030.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu^* (h - hg')^2) \\
 &= 6 \cdot 4991030 / (0.26 (245 - 8)^2) \\
 &= 2014.1002 \text{ kgf/cm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot S \\
 &= 2 \cdot 1570 \\
 &= 3140.68 \text{ kgf/cm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu^* (h - hg')^2) \\
 &= 6 \cdot 5066981 / (0.26 (200 - 8)^2) \\
 &= 3131.0898 \text{ kgf/cm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + Cats + Catc \\
 &= 200 + 0 + 0 \\
 &= 199.5229 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(HreqB; HreqS) \\
 &= \max(200; 107) \\
 &= 199.5229 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| Ps - Pt |$]:



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$$= | P_s - P_t |$$

$$= | 50 - 163 |$$

$$= 113.191 \text{ kgf/cm}^2$$

Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.26 \cdot 245 / 1260 \cdot \min(0.8 \cdot 1570; 0.533 \cdot 2366)$$

$$= 258.645 \text{ kgf/cm}^2$$

$| P_s - P_t |$ is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu) \cdot (D_o / \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.26) \cdot (1260 / 1256) \cdot | 50 - 163 | + 0$$

$$= 107.2195 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 30.000mm
 Min. Shell len. adjacent to the tubesheet is: 355.472 mm

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 1300^2 / (4 \cdot 30 (1300 + 30)) \cdot 50$$

$$= 529.1013 \text{ kgf/cm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 1894279 / 30^2 \cdot [0.0091 \cdot 0.31 + 6(1 - 0.38) / 430310 \cdot 1260 / 245^3 \cdot (1 + 245 \cdot 0.0091 / 2) \cdot (4991030 + 1260^2 / 32 \cdot (50 - 163))]$$

$$= 1081.062 \text{ kgf/cm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 1610.163 \text{ kgf/cm}^2 \text{ (Should be } \leq 2153.58)$$

Note: For a given Channel thickness of: 74.000 mm
 Min. Channel len. adjacent to the tubesheet is: 558.290mm

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1300^2 / (4 \cdot 74 (1300 + 74) \cdot \cos(0)) \cdot 163$$

$$= 677.980 \text{ kgf/cm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 17055438 / 74^2 \cdot [(0.002679428 \cdot 1 \cdot 0.0057 - 0.000 \cdot 1) \cdot 163 - 6((1 - 0.38) / 430310) \cdot 1260 / 245^3 \cdot (1 + 0.0057 \cdot 245 / 2 \cdot 1) \cdot (4991030 + 1260^2 / 32 \cdot (50 - 163))]$$

$$= 2776.405 \text{ kgf/cm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 3454.385 \text{ kgf/cm}^2 \text{ (Should be } \leq 2355.51)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.
 Performing a simplified Elastic-Plastic calculation
 (option 3) to reduce the overstress condition.
 Modify E_s and/or E_c and recompute from Step 4 onwards.



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Modified Elasticity [Ec]:

$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$

$$= 1828437 \cdot \sqrt{(1.5 \cdot 1570 / 3454)}$$

$$= 1509862.2 \text{ kgf/cm}^2$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 4925389 / (0.26 \cdot (245 - 8)^2)$$

$$= 1987.6 \text{ kgf/cm}^2 \text{ (Should be } \leq 3140.7 \text{)}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.
 The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + C_{ats} + C_{atc} = 200 + 0 + 0 = 200 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(H_{reqB}, H_{reqS}) = \max(200; 107) = 200 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25 - 2.5) 2.5$$

$$= 1.767 \text{ cm}^2$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.5 (25 - 2.5) 1234$$

$$= 2180.314 \text{ kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2 (25 + 0.67 \cdot 2) 1234$$

$$= 1735.6462 \text{ kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 1234 / 1234$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2 (25 + 0.67 \cdot 2) 1234$$

$$= 1123.0652 \text{ kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25)^2 + 1.07 \cdot 2.5 \cdot (25 - 2.5) \cdot 1 \cdot 1)} - 0.75 \cdot 25 \right)$$

$$= 3.083 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.5416 mm
 Minimum Required Groove Weld Leg [agr]: 1.5416 mm

Tube-Tubesheet Jt allowable, 2180 is $\geq$ tube strength 2180 kgf

Note: This tube-tubesheet joint is a Full Strength joint



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Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	1987.6 <=	3140.7 kgf/cm ²	Ok
Tubesheet shear stress	549.8 <=	1256.3 kgf/cm ²	Ok
Stress in Shell at Tubesheet	1610.2 <=	2153.6 kgf/cm ²	Ok
Stress in Channel at Tubesheet	3454.4 <=	4731.6 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	199.523	245.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.542	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.542	2.000	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	232.780 ...	2785	3141	792 1256	Ps+Pt D1	Ok
D2uc	135.689 ...	901	3141	243 1256	Ps+Pt D2	Ok
D3uc	199.523 ...	1988	3141	550 1256	Ps+Pt D3	Ok
D1c	238.326 ...	2963	3141	833 1256	Ps+Pt-c D1	Ok
D2c	141.741 ...	958	3141	255 1256	Ps+Pt-c D2	Ok
D3c	205.066 ...	2077	3141	578 1256	Ps+Pt-c D3	Ok
Max:	238.3260 ... mm	0.944	0.663	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Shell	Reqd. Thk. Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	1020	2154	4002	4732	Ok
D2uc	...	...	2630	4307	548	2356	Ok
D3uc	...	...	1610	2154	3454	4732	Ok
D1c	...	...	1164	2154	4277	4732	Ok
D2c	...	...	2953	4307	643	2356	Ok
D3c	...	...	1790	2154	3634	4732	Ok
Max	...	...	0.831	0.904			



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Note:
 Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	238.326	245.000	Ok
Tube thickness :	1.414	2.500	Ok
Tube-tubesheet fillet weld leg :	1.542	2.000	Ok
Tube-tubesheet groove weld leg :	1.542	2.000	Ok

Given Shell Thickness: 30.0000 mm
 Given Channel Thickness: 74.0000 mm

Min Shell length of thk, (30.000) adj. to tubesheet: 355.472 mm

Min Channel length of thk, (74.000) adj. to tubesheet: 558.290 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	172.358	0.999	164.318	1.000
Tubesheet Shear Stress	245.975	1.000	245.975	1.000
Shell Stress (Axial, Junction)	245.975	0.815	72.869	1.000
Tubesheet-Channel Junction	172.358	0.999	72.869	0.446
Tube Pressure Stress	275.315	1.000	197.621	1.000
Minimum MAWP	172.358		72.869	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	245.604	1.000	234.083	1.000
Tubesheet Shear Stress	346.867	1.000	346.867	1.000
Shell Stress (Axial, Junction)	346.867	0.751	102.540	0.995
Tubesheet-Channel Junction	245.604	1.000	102.540	0.441
Tube Pressure Stress	313.992	1.000	221.612	1.000
Minimum MAPnc	245.604		102.540	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

= max(tubesheet thk/4, min(tubesheet thk, shell thickness))
 = max(245/4, min(245, 30))
 = 61.250 mm

MDMT reduction ratio:
 = min(ps/Shellside MAPnc, 1)
 = min(50/103, 1)



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= 0.487

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	13 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	B
From Figure 3.7M- Non PWHT, the MDMT is	:	34 °C
Temperature reduction per Graph 3.12	:	21 °C
The computed stress ratio is	:	0.487

Warning: Impact Testing Is Mandatory!

Channel Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

= max(tubesheet thk/4, min(tubesheet thk, channel thickness))
= max(245/4, min(245, 74))
= 74.000 mm

MDMT reduction ratio:

= min(pt/Tubeside MAPnc, 1)
= min(163/246, 1)
= 0.664

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	18 °C
User entered Minimum Design Temperature	:	11 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	37 °C
Temperature reduction per Graph 3.13	:	19 °C
The computed stress ratio is	:	0.664

Warning: Impact Testing Is Mandatory!

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: The net force on one of the saddles putting it in tension!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: